STUDY PACKAGE FOUNDATION
CLASS X

CHEMISTRY
BOOK - 2

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ADDRESS : _____

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PHONE NO. : _____

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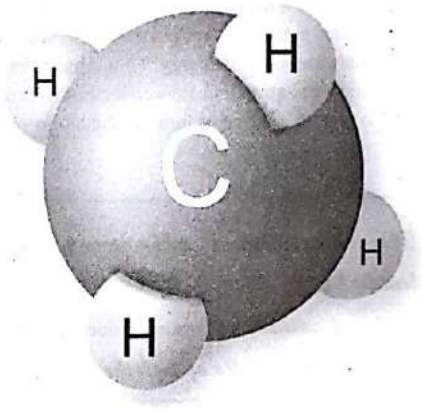


CARBON AND ITS COMPOUNDS

Introduction

Carbon has a unique place in our lives. Each living cell, food, wood, paper, petro-chemicals, cooking gas, perfumes, etc. are all made up of carbon. This abundance, together with the unique diversity of organic compounds and their unusual polymer-forming ability at the temperatures commonly encountered on earth, make this element the chemical basis of all known life. In fact more than 90% of all known compounds contain carbon. Carbon is the 15th most abundant element in the earth's crust, and the fourth most abundant element in the universe by mass after hydrogen, helium, and oxygen. Chemistry of carbon compounds is known as organic chemistry. Organic chemistry encompasses study of all carbon-hydrogen compounds. These are also called hydrocarbons. Inorganic carbon chemistry is the study of oxides, nitrides and allotropes of carbon. Because of the unique bonding properties of carbon, there are millions of different organic chemicals. Each one has unique properties. These are organic chemicals that make up your hair, your skin, even your fingernails.

Carbon has 6 protons, 6 neutrons and 6 electrons. Its chemical symbol is $^{12}C_6$. Its electronic configuration is 2 electrons in the K-shell, and 4 electrons in the L-shell. In principle it should either give up or borrow 4 electrons while forming compounds. But it doesn't form ionic bonds at all. It likes to share its four electrons with other atoms and form covalent bonds instead. Carbon covalent bonds are the strongest in nature.

TETRAVALENCY IN CARBON :

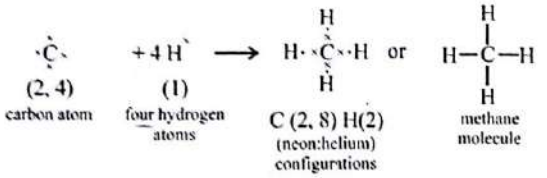
A carbon atom has a total of six electrons occupying the first two shells, i.e., the K-shell has two electrons and the L-shell has four electrons. This distribution indicates that in the outermost shell there are one completely filled 's' orbital and two half-filled 'p' orbitals, showing carbon to be a divalent atom. But in actuality, carbon displays tetravalency in the combined state. Therefore, a carbon atom has four valence electrons. It could gain four electrons to form C^{4-} anion or lose four electrons to form C^{4+} cation. Both these conditions require large amount of energy. Thus it has a very little tendency to form ionic compounds. To overcome this problem carbon undergoes bonding by sharing its valence electron with other carbon atoms or with atoms of other elements. This allows it to be covalently bonded to one, two, three or four carbon atoms or atoms of other elements or groups of atoms (molecule).

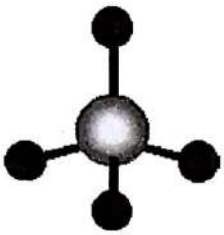
- *This characteristic of carbon atom by virtue of which it form four covalent bonds is generally referred to as tetracovalency of carbon.*
- *Not just carbon, but many other elements form molecules by sharing valence electrons. The shared electrons 'belong' to the outer shells of both the atoms and lead to both atoms attaining the noble gas configuration.*

Let us see how carbon forms the single, double and triple bonds in the following examples.

Methane Molecule :

Carbon atom has four electrons in its outermost shell. Thus, it requires four more electrons to acquire a stable noble gas configuration. Each of the hydrogen atoms has only one electron in its outermost shell and requires one more electron to complete its outermost shell (to acquire He configuration). This is done as follows.





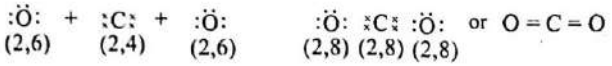
Methane is the simplest hydrocarbon and is known as marsh gas

Carbon Dioxide Molecule :

The electronic configurations of carbon and oxygen are:

C 2,4
O 2,6

Thus, each carbon atom requires four, and each oxygen atom requires two more electrons to acquire noble gas configurations. To achieve this, two oxygen atoms form a double covalent bond with carbon as follows.



Acetylene Molecule :

Carbon atom has four electrons in its outermost shell and hydrogen atoms have only one electron in its outermost shell. Carbon share one of its electrons with hydrogen to form a single bond each. Each carbon then requires three more electrons to acquire a stable

configuration of the nearest noble gas (neon). This is done by mutually sharing of three pairs of electrons between the two carbon atoms to form a triple bond as shown below.

$$\underset{(1)}{H^{\cdot}} + 2\,\underset{(2,6)}{\overset{\times}{\cdot}C\cdot} + \underset{(1)}{H^{\cdot}} \longrightarrow \underset{\text{helium configuration}}{\underset{(2)}{H:}\underset{(2,8)}{C}\vdots\underset{(2,8)}{C}:\underset{(2)}{H}} \longrightarrow H:C\equiv C:H \text{ or } CH\equiv CH$$

acetylene molecule

CHECK Point

Explain the formation of carbon double bond in ethylene molecule.

SOLUTION

Carbon atom has four electrons in its outermost shell and hydrogen atoms have only one electron in its outermost shell. Two carbon atoms share four electrons (two each atom) with four electrons of four hydrogen atoms. Each carbon then requires two more electrons to acquire a stable configuration which is done by mutual sharing of two pairs of electrons between two carbon atom to form a double as shown below.

$$\begin{matrix} H^{\cdot}_{(1)} & & H^{\cdot}_{(1)} \\ & 2\,\overset{\times}{\times}C^{\times}_{\times} & \\ H^{\cdot}_{(1)} & \overset{(2,6)}{} & H^{\cdot}_{(1)} \end{matrix} \longrightarrow \begin{matrix} 2(H) & & H(2) \\ & {}_{\times}C^{\times\times}_{\times\times}C^{\times}_{\times} & \\ 2(H) & {}_{(2,8)}\quad {}_{(2,8)} & H(2) \end{matrix} \longrightarrow \begin{matrix} H & & H \\ & C=C & \\ H & & H \end{matrix}$$

Ethylene molecule

ALLOTROPES OF CARBON :

Carbon (in Latin carbo means charcoal) is an element of prehistoric discovery. It is very widely distributed in nature. It is found in abundance in the sun, stars, comets, and atmospheres of most planets.

Carbon in the form of microscopic diamonds is found in some meteorites.

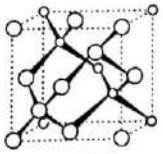
Carbon can either be in structural crystalline forms or may occur in a structure-less amorphous form. Some elements like carbon, sulphur, tin, oxygen, etc. are found in more than one structural forms. That is some elements can have several different structural forms while in the same physical state. These differing forms are called allotropes and the phenomena is called allotropy. The various allotropic forms of carbon can be broadly classified into two classes.

The phenomenon of existence of a chemical element in two or more forms differing in physical properties but having almost same chemical nature is known as allotropy. If compound exists in two or more forms, it is known as polymorphism. e.g., zinc blende and Wurtzite are polymorphs of ZnS.

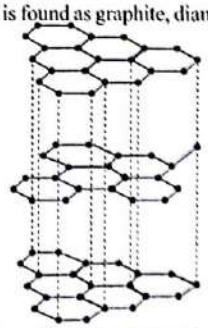
Crystalline form : Diamond, Graphite

Amorphous form : Coal, Coke, Charcoal (or wood charcoal), Animal Charcoal (or bone black), Lamp black, Carbon black, Gas carbon and Petroleum coke.

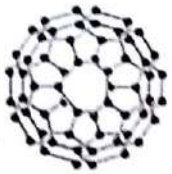
For example in crystalline form, pure carbon is found as graphite, diamond, Buckministerfullerene.



Structure of Diamond



Structure of Graphite



Structure of Buckministerfullerene

There is another form of crystalline carbon known as "white" carbon, but not much is known about it. Conditions such as temperature and pressure determine which allotrope of carbon occurs.

A **diamond** is hard, clear, and shiny substance. It is formed under very high temperatures and pressure. Graphite is black, with a slippery, greasy feel. Graphite is formed on heating coke or coal.

Though non-metallic, graphite possesses a metallic lustre. It is insoluble in ordinary solvents. Graphite is used in making electrodes in electric furnances and electric ores. It is also used in lead pencils.

CHECK Point

- **Why diamond is used for making cutting and grinding instruments?**
- **Why diamond is important material for making jewellary?**

SOLUTION

- Diamond has ability to reflect and refract light makes diamond an important material for making jewellary.
- ∞ As we know diamond is the hardest substance that's why it is used in making cutting and grinding instruments.

Comparison between Diamonds and Graphite :

Diamond	Graphite
1. Diamond is the hardest material known.	1. Graphite is one of the softest materials known.
2. Diamond is the ultimate abrasive.	2. Graphite is a very good lubricant.
3. Diamond is an excellent electrical insulator	3. Graphite is a conductor of electricity.
4. Diamond is the best known naturally occurring thermal conductor	4. Some forms of graphite are used for thermal insulation (i.e. firebreaks and heat shields)
5. Diamond is highly transparent.	5. Graphite is opaque.
6. Diamond crystallizes in the cubic system.	6. Graphite crystallizes in the hexagonal system.

These differences in the properties of diamond and graphite are due to the difference in their structures. In diamond, each C atom is linked to its neighbours by four single covalent bonds. This leads to a three-dimensional network of covalent bonds. In graphite, the carbon atoms are arranged in flat parallel layers as regular hexagons. Each carbon in these layers is bonded to three others by covalent bonds. Graphite thus acquires some double bond character. Each layer is bonded to adjacent layers by weak van der Waals forces. Moreover one electron is free at each carbon atom. This allows each layer to slide over the other easily. Due to this type of structure graphite is soft and slippery, and can act as a lubricant. Graphite is also a good conductor of electricity due to mobile electrons in it.

Graphite is also used as a industrial lubricant because it possesses high melting point. Thus there is no change in structure occur even in machineries working at high temperature and graphite there also serves as good lubricant.

CHECK *Point*

☛ **Why graphite is used as lubricant?**

SOLUTION

As structure of graphite contains flat hexagonal parallel layers joined together by weak vander waals forces. Thus these layers can slide over one another easily making graphite a good lubricant.

Buckministerfullerenes contain 60 carbon atoms arranged in a round molecule resembling a soccer ball. C60 molecule has marvelously symmetrical structure.

C60 molecule is a fused-ring of aromatic system containing 20 hexagons and 12 pentagons of C atoms. The five membered rings are connected to six membered rings while the six membered rings are fused to both five and six membered rings.

The structure bends around and closes to form a soccer ball shaped molecule. It is therefore, called buckyball also. Fullerenes were first prepared by the evaporation of graphite using a laser. These can be conveniently made by heating graphite in an electric arc in the presence of helium or argon. Fullerene looks different from diamond and graphite. It is a yellow powdery substance and are soluble in organic solvents and form coloured solutions. It turns pink on dissolution in solvents like toluene. It polymerizes on exposure to U.V. radiations. Fullerenes are fascinating because they show unusual characteristics and applications like: They are wonderful lubricants because the balls can roll between the surfaces.

CHECK *Point*

☛ **How many hexagonal and pentagonal rings are present in C60 molecules?**

SOLUTION

It contains 20 hexagonal and 12 pentagonal rings fused together.

ILLUSTRATION 1

Which allotrope of carbon is the hardest?

SOLUTION :

Diamond

idea box

Three different allotropes of crystalline forms of carbon are diamond, graphite and Buck minister fullerene. Mention five important uses of each form at industrial and domestic scale. You can take help of internet, television, science magazine etc.

CATENATION IN CARBON :

Carbon atom is unique amongst all the elements that are found in nature. It can form long chain molecules. The ability of carbon to form a long chain is called catenation. Carbon forms strong bond because of its small size which enables the nucleus to held on to the shared pairs of electrons strongly. The chains are formed because carbon atoms form tetravalent bonds with other carbon atoms. This structure can be repeated endlessly without disturbing the stability of the bonds and the compounds formed. The chains can form branches, and sub-branches. The carbon atoms also form rings. The rings themselves can have more rings attached to them. The list is endless. Most of the protein molecules, amino acids are long chain carbon molecules.

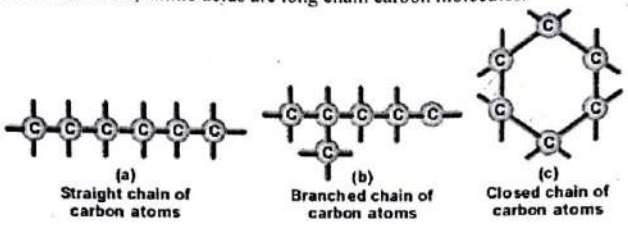


Figure : Variety of carbon chains formed when carbon atoms join together.

In addition, carbon atoms may be linked by single, double or triple bonds. Compounds of carbon, which are linked by only single bonds between the carbon atoms are called saturated compounds. Compounds of carbon having double or triple bonds between their carbon atoms are called unsaturated compounds.

Knowledge ENHANCER

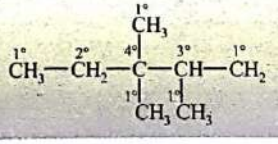
(i) A carbon atom attached to one other (or no other) carbon is called primary carbon (written) as 1° carbon).

(ii) A carbon atom attached to two other carbon atoms is called secondary carbon (written as 2° carbon).

(iii) A carbon atom attached to three other carbon atoms is called tertiary carbon (written as 3° carbon).

(iv) A carbon atom attached to four other carbon atoms is called quarternary carbon (written as 4° carbon).

Hydrogen atoms attached to 1°, 2° and 3° carbon atoms are referred to as primary, secondary and tertiary H atoms respectively. The following example clearly differentiates the various carbon atoms :



CHECK Point

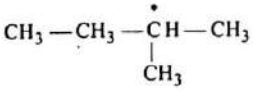
☛ **What is so special about carbon so that one entire major branch of chemistry is devoted to its compound with hydrogen?**

SOLUTION

Carbon has a unique property of catenation because of which it is capable to form long straight and branched carbon chains also rings of carbon atoms. Thus it forms large number of compounds with hydrogen called hydrocarbons. Chemistry of these hydrocarbons are studied under separate branch called organic chemistry.

ILLUSTRATION 2

In the following compound carbon atom with * mark is primary, secondary, tertiary or quaternary carbon atom?



SOLUTION:

The carbon atom with * mark is tertiary carbon atom as it is attached to three another carbon atoms.

COMPOUNDS OF CARBON:

Compounds of carbon can broadly classified into two main categories, namely inorganic and organic.

Inorganic compounds : *These include compounds of carbon with metals and non-metals* (other than hydrogen) *such as oxygen, hologens etc.* These do not have carbon-carbon bonds.

Organic compounds : *These include compounds of carbon with hydrogen* i.e., hydrocarbons and their derivatives. These contain carbon-carbon bonds.

Inorganic Compounds of Carbon :

Carbon combines with a variety of other elements like oxygen, halogens, metals, etc., to form binary compounds respectively named as oxides, halides and carbides.

Oxides of carbon :

1. **Carbon monoxide, CO**

(i) Carbon monoxide is a colourless, tasteless gas with faint odour.

(ii) Carbon monoxide is *extremely poisonous* in nature. The poisonous nature of CO is due to the fact that it combines with **haemoglobin** (a red colouring matter of blood), oxygen carrier of blood, to form a stable compound *carboxyhaemoglobin*. With the result the oxygen transporation is disturbed and thus brain and body tissues do not get necessary oxygen and ultimately death occurs. This explains why people sleeping in closed rooms with charcoal fire burning inside dies.

A victim of carbon monoxide poisoning, if unconscious, should be brought in the open place and given artificial respiration with carbogen (mixture of oxygen and 5-10% CO_2).

Uses :

(i) Carbon monoxide is used as a fuel in the form of *water gas* ($CO + H_2$) and *producer gas* ($CO + N_2$).

(ii) It is used in making metal carbonyls among which nickel tetracarbonyl is used in the extraction of nickel (*Mond's process*).

(iii) It is used in the manufacture of methanol, synthetic petrol and phosgene.

(iv) It is used as a reducing agent in metallurgy of iron and in the manufacture of hydrogen.

(v) Iron carbonyl is used in the manufacture of magnetic tapes for videos and tape recorders.

2. **Carbon dioxide, CO_2**

It is present in air to the extent of 0.03% by volume. It is found in caves, mines, etc., and it evolves from volcanoes.

(i) Carbon dioxide can be easily liquefied under pressure (50-60 atmospheres) and also solidified. Solid carbon dioxide is technically known as **dry ice** because it evaporates (sublimes) without liquefying (it sublimes at 195 K). Solid carbon dioxide is used as a refrigerant (coolant). Under the commercial name of *drikold*. When it is placed in an organic solvent like ether, it gives a constant temperature of 173 K.

(ii) Although CO_2 is not poisonous in nature, it does not support life and animals die in it for want of oxygen.

(iii) It is neither combustible nor a suporter of combustion. But certain active metals like Na, K and Mg continue burning in atmosphere of CO_2. It is due to the fact that the active metals combine with oxygen CO_2 to form the oxides. The latter combine with CO_2 to form carbonates.

$$2Mg + CO_2 \longrightarrow 2MgO + C$$

(iv) Carbon dioxide is absorbed by plants in the presence of sunlight and chlorophyll (green colouring matter) to form glucose and higher carbohydrates. This process is known as **photosynthesis**.

$$6CO_2 + 6H_2O \xrightarrow[\text{chlorophyll}]{\text{light}} C_6H_{12}O_6 + 6O_2$$

- *Solid CO_2 causes burns on skin so it should be handled carefully.*
- *Although CO_2 is used as a fire extinguisher, but it can't be used to put out metal fires.*
- *The CO_2 absorbs strongly in the infrared region, and its presence in the atmosphere decreases the loss of heat from the earth by radiation. This global warming is called the 'green house effect' (other gases, including the oxides of nitrogen from car exhausts, freons from aerosols and refrigerators and methane from bacteria in the soil and in the rumen of cows, also add to the green house effect).*

Knowledge ENHANCER

Reaction of carbon with the halogens :

Graphite reacts with fluorine, F_2, at high temperatures to make a mixture of carbon tetrafluoride, CF_4, together with some C_2F_6 and C_5F_{12}.

$$C(s) + \text{excess } F_2(g) \longrightarrow CF_4(g) + C_2F_6 + C_5F_{12}$$

At room temperature, the reaction with fluorine is complex. The result is 'graphite fluoride', a non-stoichiometric species with formula CF_x

Reaction of carbon with the transition metals :

The usual method of preparing polycrystalline transition metal carbides on the research scale involves the direct reaction of metal or metal hydride powders with carbon.

$$M + C \rightarrow MC$$

$$MH + C \rightarrow MC + H_2$$

Organic Compounds :

Origin and Defination of Organic Compounds :

According to the earlier defination *organic chemistry was defined as the chemistry of substances found in the living matter* (Berzelius in 1808). Further, Berzelius in 1815 also suggested that the organic compounds could be prepared only by living organisms under the influence of a mysterious force known as the **vital force** (*life-force*). The **vital force theory** remain unchallanged until 1828, when Friedrich Wohler, a German chemist, reported the synthesis of a typical organic compound **urea** by heating an inorganic compound, ammonium cyanate.

$$\underset{\text{Ammonium chloride}}{NH_4Cl} + \underset{\text{Potassium cyanate}}{KCNO} \longrightarrow \underset{\text{Ammonium cyanate}}{NH_4CNO} + KCl$$

$$\underset{\text{Ammonium cyanate}}{NH_4CNO} \xrightarrow{\text{Heat}} \underset{\text{Urea}}{H_2N.CO.NH_2}$$

The synthesis of urea from inorganic compounds gave a blow to the vital force theory and thus the distinction between organic and inorganic compounds based on vital force soon started fading away. The vital force theory was completely dicarded after the synthesis of acetic acid from C, H and O (Kolbe, 1845) and methane (Bertholet, 1856). Subsequently, a large number of naturally occurring organic compounds were synthesized in the laboratory and the vital force theory was decisively disproved.

However, all the organic compounds were found to contain carbon as one of the elements.Thus the emphasis for the definition of organic compounds was shifted from origin to composition. Accordingly, **organic chemistry has now been defined as the chemistry of carbon compounds.**

On the basis of this definition, the study of compounds like CO, CO_2, H_2CO_3, carbonates, cyanides, cyanates, carbides, etc., should be included in organic chemistry, but due to their greater resemblance with inorganic compounds these are studied under inorganic chemistry.

Further, since all the organic compounds could be considered as derivatives of hydrocarbons, a more appropriate definition for the **organic chemistry is the study of hydrocarbons and their derivatives.**

Chemical Properties of Carbon Compounds:

Combustion:

Combustion means the burning of a substance. It is a process that is highly exothermic i.e., produces a lot of heat.Combustion, at its most general, can mean the reaction of oxygen gas (O_2) with anything. However, we will understand combustion to mean the reaction of oxygen with a compound containing carbon and hydrogen.

it is usually shown as:

$$C_xH_y + 2O_2 \longrightarrow CO_2 + 2H_2O$$

These are some examples:

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

$$C_2H_6 + \frac{7}{2}O_2 \longrightarrow 2CO_2 + 3H_2O$$

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$$

$$C_2H_5OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$$

Notice that some compounds contain carbon, hydrogen and oxygen. However, the products are all the same, in every reaction.

- *If compound undergoing combustion contain nitrogen (burns to form NO_2) or sulfur (burns to form SO_2). Like this:*

$$C_{21}H_{24}N_2O_4 + O_2 \longrightarrow CO_2 + H_2O + NO_2$$

$$C_2H_5SH + O_2 \longrightarrow CO_2 + H_2O + SO_2$$

- *There are complexities with combustion as you get deeper into it. For example, a combustion with insufficient oxygen yields carbon monoxide rather than carbon dioxide.*

Combustion can be of two type : spontaneous and rapid combustion. When substance does not need external heat energy to start the combustion, the process is called spontaneous combustion. For example, when white phosphorus is kept in air for sometime, it will automatically catch fire. The phosphorus reacts with air to form oxide, which is an exothermic reaction. The heat generated by the formation of the oxide ignites the rest of the white phosphorus. Spontaneous combustion is also seen where dry forest or grass is present. The heat in the atmosphere may start the burning. Rapid combustion occurs when a large amount of heat is released in a very short time. Example of rapid combustion can be sighted in case of bombs. Lighting up of LPG in the kitchen, which produces heat and light immediately, is also an example of spontaneous combustion.

Carbon, in all its allotropic forms, burns in oxygen to give carbon dioxide along with the release of heat and light. Most carbon compounds also release a large amount of heat and light on burning. For example, methane burns with a blue flame in air.

$$\underset{\text{Methane}}{CH_4(g)} + \underset{\text{oxygen (from air)}}{O_2(g)} \xrightarrow{\Delta} \underset{\text{carbon dioxide}}{CO_2(g)} + \underset{\text{water}}{2H_2O\,(Vap)} + \text{Heat}$$

In a very limited supply of air methane gives carbon black.

$$\underset{\text{Air (very limited)}}{CH_4 + O_2(g)} \xrightarrow{\Delta} \underset{\text{carbon black}}{C + 2H_2O}$$

CHECK Point

☛ **Compressed Natural Gas (CNG) which is used these days in automobiles undergoes spontaneous or rapid combustion ?**

SOLUTION

CNG undergoes rapid combustion in automobile's internal combustion engine, it does not ignite by itself.

Knowledge ENHANCER

Some carbon compounds are very combustible and have an explosive reaction with air e.g., alkenes. They burn with a luminous flame to produce carbon dioxide and water vapour. Some hydrocarbon compounds undergo cracking or thermal decomposition. In this process, substances (complex hydrocarbons) are heated to high temperature (500 - 800°C) in the absence of air, and they decompose into a mixture of saturated and unsaturated hydrocarbons (lesser complex hydrocarbons) and hydrogen. Saturated hydrocarbons will generally give a clean flame while unsaturated carbon compounds will give a yellow flame with lots of black smoke,limiting the supply of air results in incomplete combustion of even saturated hydrocarbons giving a sooty flame. The gas/kerosene stove used at home has inlets for air so that a sufficiently oxygen-rich mixture is burnt to give a clean blue flame. If you observe the bottom of cooking vessels getting blackened, it means that the air holes are blocked and fuel is getting wasted. Fuels such as coal and petroleum have some amount of nitrogen and sulphur in them. Their combustion results in the formation of oxides of sulphur and nitrogen which are major pollutants in the environment.

ILLUSTRATION 3

What are the products obtained on combustion of ethylene?

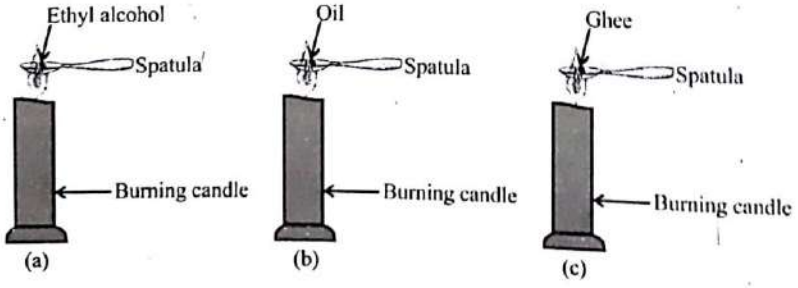
SOLUTION :

Carbon dioxide and water vapour

$$\underset{\text{ethylene}}{CH_2} = CH_2 + 3O_2 \longrightarrow 2CO_2 + 2H_2O$$

CHECK Point

☛ **Observe the following experimental setup carefully.**



Which set up will produce smoke :

SOLUTION

Setup (b) will produce smoke as the oil taken in spoon contain unsaturated hydrocarbons which burns with luminous flame and produce more smoke.

Oxidation :

Carbon undergoes oxidation by combining with oxygen at higher temperature to form oxides, viz., carbon monoxide (CO) and carbon dioxide (CO_2). Carbon monoxide is formed, when incomplete combustion of carbon or carbon containing fuels takes place.

$$C + \frac{1}{2}O_2 \longrightarrow CO(g)$$

CO is present in automobile exhausts (when there is incomplete combustion), volcanic gases, chimney gases etc.

Carbon dioxide may be prepared by the complete combustion of carbon, hydrocarbons, carbon monoxide etc.

$$C + O_2(g) \longrightarrow CO_2(g) + \text{Heat}$$

$$2CO + O_2(g) \longrightarrow 2CO_2(g) + \text{Heat}$$

$$CH_4 + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O$$

Carbon containing compounds undergo oxidation reactions when burnt in air or oxygen in the presence of catalyst or necessary condition required to give useful products other then CO_2 and H_2O. For example, when methane is mixed with oxygen and heated in presence of molybdenum oxide, it gets oxidized to methanal or formaldehyde.

$$2CH_4 + O_2(g) \xrightarrow[\text{Molybdenum oxide}]{350-500°C} HCHO + 2H_2O$$

Oxidation of carbon compounds is used as for producing other carbon compounds with different functional groups like alcohol, carboxylic acid, ethers etc. Oxidation is achieved by using an oxygen atmosphere or oxidizing agents like alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$. Methanol, an industrial alcohol, for instance, is prepared by the oxidation of methane.

$$2CH_4 + O_2(g) \xrightarrow[\text{copper tube}]{200°C} 2CH_3OH(g)$$

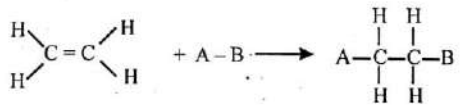
Acetic acid is manufactured by the oxidation of fermented liquors (10-15% alcohol) in air along with the presence of mycoderma aceti. A 3-7% solution of acetic acid is obtained and it is called vinegar.

$$C_2H_5OH + O_2 \longrightarrow CH_3COOH + H_2O$$

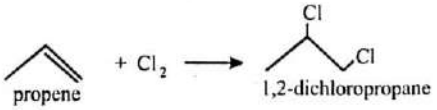
Addition Reaction :

The reactions in which two molecules react to form a single product are called addition reactions.

Consider some alkene and a second molecule A_2 or A-B (e.g. H_2 or HCl). An addition reaction occurs when the A-B bond breaks and A adds to one end of the double bond and B adds to the other end. When this happens the double bond becomes a single bond. Shown pictorially:

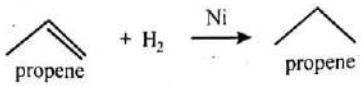


For some molecules this happens quite readily. For example, simply adding either molecular chlorine (Cl_2) or bromine (Br_2) to an alkene results in the rapid addition to form the dihalogenated product.

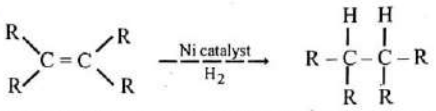


Likewise HCl, HBr, and H_2SO_4 (H-HSO_4) add rapidly across double bonds.

The addition of hydrogen (hydrogenation) to a double bond usually requires a catalyst. Nickel is usually chosen because it is both effective and inexpensive

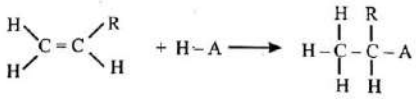


For example hydrogenation of vegetable oil using a Ni *catalyst. You have seen advertisements stating vegetable oils are 'healthy' as they contain long unsaturated carbon chains. Oils containing unsaturated fatty acids are choosen for cooking. Whereas animal fats generally contain saturated fatty acids carbon chain which are said to be harmful for health.*



The products of addition reactions are relatively simple to predict when the alkene is symmetrical (the same number of hydrogens are attached to both ends of the double bond) or when the addition product is of the A_2 type. But for A-B molecules adding to unsymmetrical double bonds some care is required. Consider the examples just given. Since in the last example we are adding two chlorine atoms across the double bond it doesn't matter which goes where because the result is the same. This is also true for Br_2. But what about H-Cl ? The reaction could place the Cl in either the 1 or 2 position. Is either preferred or is it a 50/50 mix ?

We use *Markovnikov's rule as a guide in these situations. This empirical rule based on many observations tells us that when adding a reagent of the type H-A to an unsymmetrical alkene, the hydrogen always binds to the carbon that starts off with more hydrogen atoms.*



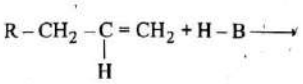
You will not encounter any exceptions to this rule in this class.

Water adds across double bonds just like the other unsymmetrical addition reagents, but unlike the others it may require forcing conditions (e.g. heat, pressure). To make an alcohol by addition of water to an alkene, the reaction requires a catalyst (usually an acid) and in many cases significant heat.

$$H_2C = CH_2 + H_2O \xrightarrow[240°C]{H^+} CH_3CH_2OH$$

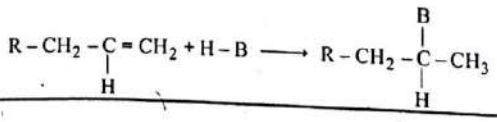
CHECK Point

Write the product of following reaction on the basis of Markovnikov's addition



SOLUTION

On the basis of Markovniv's rule reagent of type H–B to an unsymmetrical alkene, the hydrogen atom always binds to carbon atom with more hydrogen atom.



Substitution reaction :

The reactions in which an atom or group of atoms in a molecule is replaced or substituted by different atoms or group of atoms are called substitution reaction. For example,

$$CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$$

(H replaced by Cl)

$$CH_3CH_2I + KOH(aq) \longrightarrow CH_3CH_2OH$$

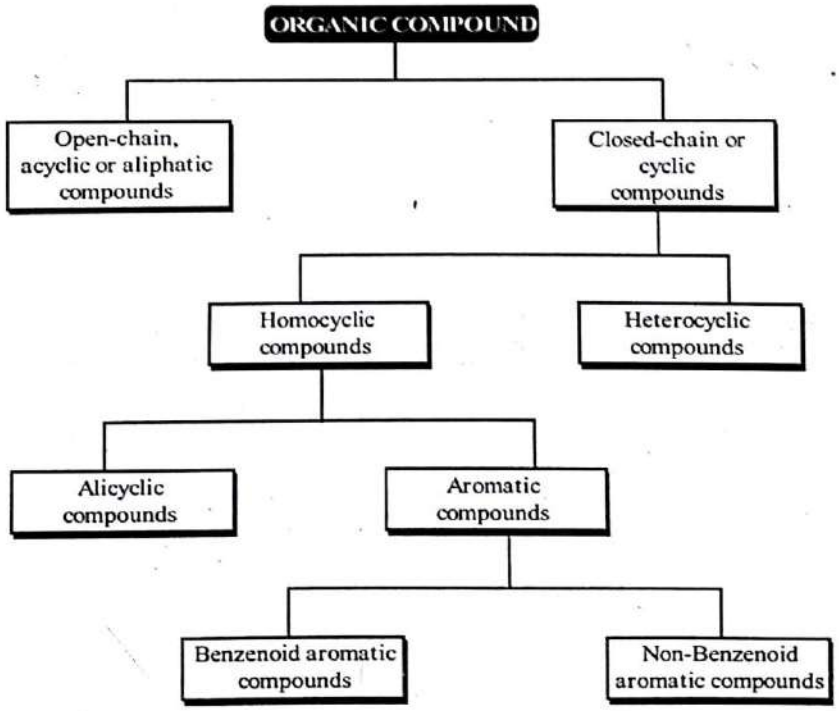
(I replaced by OH)

In substitution reactions the hydrogen of the alkane molecule is replaced by another atom or a group of atoms (like alkyl) resulting in the formation of the derivatives of that hydrocarbon. Substitution by halogen atom is generally called halogenation. This type of substitution results in chlorination, bromination or iodination.

Like methane, ethane also forms a series of substitution products in the presence of excess chlorine and sunlight.

$$\underset{\text{Ethane}}{C_2H_6} + Cl_2 \xrightarrow{\text{uv light}} \underset{\text{Chloroethane}}{C_2H_5Cl} + HCl$$

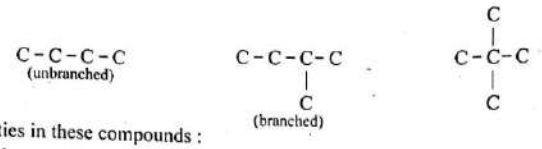
Classification of Organic Compounds :



1. Acyclic or open chain compounds :

Are those compounds in which first & last carbon atoms are not connected with each other. Branched or unbranched chains are possible in these compounds.

For example :



There are two varieties in these compounds :

2. Cyclic or closed chain compounds :

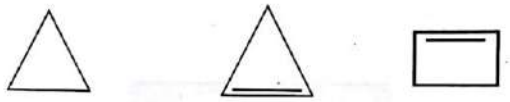
These are the compounds in which carbon atoms are linked to each other or to the atoms of other elements in such a manner that the molecule has a closed-chain or cyclic or ring structure. One or more closed chains or rings may be present in the molecule. The compounds with only one ring of atoms in the molecule are known as monocyclic but those with more than one ring of atoms are termed as polycyclic. These are further divided into two subgroups (i) Homocyclic or carbocyclic, (ii) Heterocyclic.

(i) Homocyclic compound :

These are the compounds in which the complete ring is formed by carbon atoms only. These are also of two types :

(a) Alicyclic compounds :

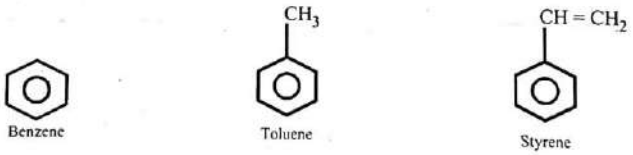
These are the compounds having the properties like aliphatic compounds. These may be saturated or unsaturated like aliphatic compounds.



(Cyclopropane) (Cyclopropene) (Cyclobutene)

(b) Aromatic compounds :

These compounds consist of at least one benzene ring i.e., a six-membered carbocyclic ring having alternate single and double bonds. These compounds have some fragrant odour and hence, named as aromatic (greek word aroma means sweet smell).



Benzene Toluene Styrene

These compounds are also called benzenoid compounds and they exhibit special property known as aromaticity

Aromatic compounds now include a much wider class of compounds, not necessarily very sweet smelling ones. A more appropriate modern day definition of aromatic compounds is "Compounds which have a sustained ring current." Infact, you will find non-benzenoid aromatics as well, the aromatic compounds without a benzene ring!

Examples of non-benzenoid aromatics include beautiful blue-colored azulene and trophylium bromide.

azulene tropylium bromide

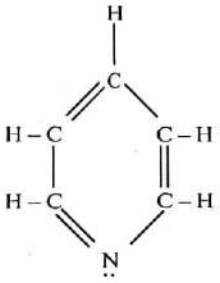
(ii) Heterocyclic compounds

These are cyclic compounds having ring or rings built up of more than one kind of atoms. Heterocyclic rings can also be aromatic.

Furan

Thiophene

Pyrrole

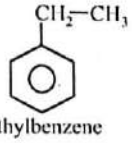

Pyridine

CHECK Point

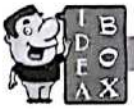
1. **Ethylbenzene comes under category of homocyclic or heterocyclic compounds?**
2. **Which atom is present in pyrole ring except carbon/atoms ?**

SOLUTION

1. Ethylbenzene is a homocyclic compound as complete ring is formed by carbon atoms only.

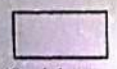

Ethylbenzene

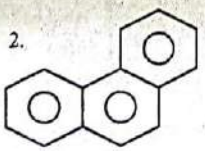
2. Nitrogen atom.

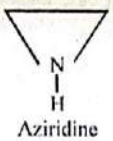


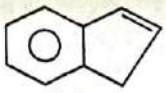
idea box

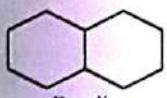
Following is the list of few compounds

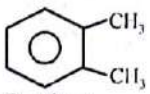
1. 
Cyclobutane

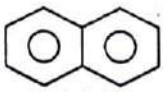
2. 
Phenonthrene

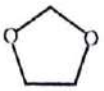
3. 
Aziridine

4. 
Indene

5. 
Decalin

6. 
O-xylene

7. 
Naphtholene

8. 
Dioxalane

9.
Cyclononone

10.
Thiane

Divide these compounds into three categories of alicyclic, aromatic and heterocyclic compounds.

The above classification of organic compounds is based upon the **basic skeleton of carbon atom**. All the above classes may further be classified into several classes depending upon the nature of the functional group.

FUNCTIONAL GROUPS :

In organic chemistry, functional groups (or moieties) are specific groups of atoms within molecules that are responsible for the characteristic chemical reactions of those molecules. The same functional group will undergo the same or similar chemical reaction(s) regardless of the size of the molecule it is a part of.

Combining the names of functional groups with the names of the parent alkanes generates a powerful systematic nomenclature for naming organic compounds.

The non-hydrogen atoms of functional groups are always associated with each other and with the rest of the molecule by covalent bonds.

The first carbon atom after the carbon that attaches to the functional group is called the alpha carbon. Functional groups are attached to the carbon backbone of organic molecules. They determine the characteristics and chemical reactivity of molecules. Functional groups are far less stable than the carbon backbone and are likely to participate in chemical reactions. Free valency or valencies of the group are shown by the single line. The functional group is attached to the carbon chain through this valency by replacing one hydrogen atom or atoms.

Important Functional Groups and the Corresponding Classes of Organic Compounds :

S.No.	Functional Group		Class of compounds
	Formula	Name	
1.	–X (–F, Cl, –Br, –I)	Halo (fluoro, chloro, bromo, iodo)	Alkyl halides or halogen compounds
2.	–OH	Hydroxy	Alcohol
3.	–OR	Alkoxy	Ethers
4.	–SH	Mercapto	Thioalcohols, mercaptans or thiols
5.	–SR		Thioethers or sulphides
6.	–CHO	Aldehydic	Aldehydes
7.	–CO–	Ketonic	Ketones
8.	–COOH	Carboxyl	Carboxylic acids
9.	–COOR	Ester	Esters
10.	–COX (X = Cl, Br or I)	Acyl halide	Acid halides or Acyl halides
11.	$-CONH_2$	Amide	Amides or acid amides
12.	–CO.O.CO–	Anhydride	Acid anhydrides
13.	$-NH_2$	Amino	Amines
14.	–NH–	Imino	Imines
15.	$-C \equiv N$	Cyano	Cyanides or Nitriles
16.	$-N \overset{-}{=} C$	Isocyano	Isocyanides or Isonitriles
17.	$-NO_2$	Nitro	Nitro compounds
18.	$-N=O$	Nitroso	Nitroso compounds
19.	$-N=N-$	Azo	Azo compounds
20.	$-SO_2-OH$	Sulphonic acid	Sulphonic acids

ILLUSTRATION 4

Why Mercapto (–SH) are also called thioalcohols

SOLUTION :

It is due to resemblance with the functional group of alcohols –OH. The only difference is the presence of sulphur atom instead of oxygen atom that's why thio is used before alcohols.

CHECK Point

☛ **What is the similarity in the following pair of organic compounds**
HCHO and CH_3CH_2CHO.

SOLUTION

Both of above compounds are aldehydes as they contain –CHO functional group.

Homologous series :

In chemistry, a homologous series is a series of organic compounds with a similar general formula, possessing similar chemical properties due to the presence of the same functional group, and shows a gradation in physical properties as a result of increase in molecular size and mass (see relative molecular mass). For example, ethane has a higher boiling point than methane since it has more Van der Waals forces(intermolecular forces) with neighbouring molecules. This is due to the increase in the number of atoms making up the molecule.

• *Organic compounds in the same homologous series vary by a CH_2. Alkanes (paraffins), alkenes (olefins), methoxyethane (ethers), and alkynes (acetylenes) form such series in which members differ in mass by 14 atomic mass units. For example, the alkane homologous series begins with methane (CH_4), ethane (C_2H_6), propane (C_3H_8), butane (C_4H_{10}), and pentane (C_5H_{12}), each member differing from the previous one by a CH_2 group (or 14 atomic mass units).*

• *A Homologation reaction is any chemical process converting one member of a homologues series to the next.*

Similarly in alcohol homologous series starts with methanol (CH_4O), ethanol (C_2H_6O), as primary alcohols, isopropanol (C_3H_8O) as a simple secondary alcohol, and a simple tertiary alcohol is tert-butanol ($C_4H_{10}O$).

Even while the general formula are the same, they have different structures that can lead the exact same compound to different properties, although they will always exhibit the same chemical properties while as a homologous compound.

Compounds in each set have the same little group of atoms called the functional group. Most chemical properties of organic compounds are due to the presence of the functional group.

Homologous series	General formula	Example	Functional group
Straight Chain Alkanes	C_nH_{2n+2} (n more than or equal to 1)	CH_4, n = 1	
Alkyl	C_nH_{2n+1} (n more than or equal to 1)	CH_3, n = 1	
Alkenes and Cyclic Alkanes	C_nH_{2n} (n more than or equal to 2)	C_2H_4, n = 2	C = C
Alkynes	C_nH_{2n-2} (n more than or equal to 2)	C_2H_2, n = 2	C = C
Alcohols	$C_nH_{2n+1}OH$ (n more than or equal to 1)	CH_4O, n = 1	– OH
Carboxylic acids	$C_nH_{2n}O_2$ (n more than or equal to 1)	CH_2O_2, n = 1	– COOH
Carbohydrates	$C_n(H_2O)_n$ (n more than or equal to 1)	$C_6H_{12}O_6$ n=6	–CHO, OH

Where n represents the number of carbon atoms present.

CHECK Point

☛ **In each homologous series member differ from previous one by which group?**

SOLUTION

In homologous series each member differ from the previous one by a CH_2 group (or 14 atomic mass units).

1 EXERCISE

FIB Fill in the Blanks :

DIRECTIONS : *Complete the following statements with an appropriate word / term to be filled in the blank space(s).*

1. The ability of carbon to form chains gives rise to a ..... series of compounds
2. Newly discovered allotrope of carbon is called .....
3. The soft crystalline form of carbon is .....
4. Next homologue of ethane is .....
5. Valency of carbon in ethylene is .....
6. Ethylene burns in air to form CO_2 and .....
7. The molecular mass of any two adjacent homologous differ by ..... amu.
8. The purest form of carbon is .....
9. The general formula of alcohols is .....
10. The functional group present in carboxylic acids is .....

T/F True / False :

DIRECTIONS : *Read the following statements and write your answer as true or false.*

1. Carbon is a versatile element.
2. Carbon forms covalent bonds with itself and other elements such as hydrogen, oxygen, sulphur, nitrogen and chlorine.
3. Carbon and its compounds are some of our major sources of fuels.
4. Graphite is a good conductor of electricity.
5. The simplest saturated hydrocarbon is methane.
6. Ethanol is the first member of the alcohol homologous series.
7. Diamond is a good conductor of electricity.
8. Graphite is used in pencils.
9. When hydrocarbons burn in air, carbon dioxide and hydrogen are produced with heat energy.
10. If a hydrocarbon has double or triple covalent bond, it is saturated.
11. Unsaturated hydrocarbons give addition reactions.
12. By hydrogenation, vegetable oils convert into vanaspati ghee

MTF Match the Following :

DIRECTIONS : *Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.*

1.

Column I		Column II
A. Combustion reaction	(p)	$C_3H_8 + Cl_2 \xrightarrow{\text{UV light}} C_3H_7Cl + HCl$
B. Oxidation reaction	(q)	$CH_2=CH_2+H_2 \xrightarrow{Ni/Pd} CH_3-CH_3$
C. Addition reaction	(r)	$2CH_4+O_2(g) \xrightarrow[\text{Molybdenum oxide}]{300-500°C} HCHO+2H_2O$
D. Substitution reaction	(s)	$C_2H_5OH+3O_2 \rightarrow 2CO_2+3H_2O$

2.

Column I	Column II
A. —CHO	(p) Azo Compounds
B. $-CONH_2$	(q) Aldehydes
C. $-NH_2$	(r) Acid amides
D. —N=N—	(s) Amines

VSAQ Very Short Answer Questions :

DIRECTIONS : *Give answer in one word or one sentence.*

1. Write the chemical equation for the reaction of ethene with hydrogen in the presence of nickel catalyst.
2. What are Fullerences ?
3. Name the following functional groups :

(a) $-\overset{\overset{O}{\|}}{C}-OH$ (b) $-\overset{\overset{O}{\|}}{C}-H$ (c) C – O – C

4. C_nH_{2n} can represent an alkane or an alkene. Explain.
5. Which of the eight classes of compounds containing hetero atoms contains a carboxyl group ? What contains a hydroxyl group ?
6. What are unsaturated hydro-carbons ?
7. Give the names of the functional groups :

(i) –CHO (ii) $-\overset{\overset{O}{\|}}{C}-$

8. Give the names of the following functional groups :
   (i) –OH (ii) –COOH
9. Name the product other than water formed on burning of ethanol in air.
10. Allotropy is a property shown by which class : Substances, elements, compounds or mixtures ? Give one example of allotropy.
11. What is Carboxyhaemoglobin ?
12. Why some hydrocarbons are subjected to cracking ?
13. Cyclodipentene belongs to which category of organic compound ?
14. What will be the formula and electron dot structure of cyclopentane?
15. The molecular formula of the consecutive members of a homologous series are C_6H_{14} and C_7H_{16}. Write the molecular formulae of members having 9 and 11 carbon atoms of this homologous series.

SQA Short Answer Questions :

DIRECTIONS : *Give answer in 2-3 sentences.*

1. Why carbon atoms cannot form ionic bonds in its compounds ?
2. What is meant by homologous series ? State any four characteristics.
3. Draw the electron dot structure of ethyne and also draw its structural formula.
4. Write the name of all amorphous form of carbon.
5. Explain the toxic effects of carbon monoxide.
6. How CO_2 resulting into global warming.
7. Give equations to show combustion of compounds containing nitrogen and sulphur.
8. Explain with example how unsaturated aromatic organic compound undergoes addition reaction.
9. What are heterocyclic compounds explain with examle.
10. Which two of the following compounds could belong to the same homologous series :
    $C_2H_6O_2, C_2H_6O, C_2H_6, CH_4O$

LAQ Long Answer Questions :

DIRECTIONS : *Give answer in four to five sentences.*

1. Write a brief note on fullerenes also by showing its structure.
2. Explain the reason for difference in properties of diamond and graphite.
3. Write a note on cracking of hydrocarbons. Also explain how we can distinguish between saturated and unsaturated hydrocarbons on the basis of nature of flame produced.
4. Why is organic chemistry studied as a separate branch of chemistry ?
5. Give one example each of :
   1. Cracking 2. Hydrogenation
   3. Dehydration 4. Substitution reaction
   5. Addition reaction
6. Differentiate between diamond and graphite.

2 EXERCISE

MCQ Multiple Choice Questions :

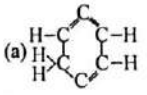
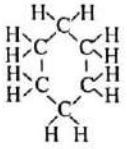
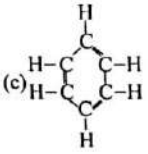
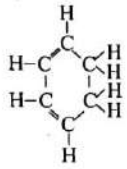
DIRECTIONS : *This section contains 25 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. The number of 4° carbon atoms in 2,2,4,4-tetramethyl pentane is –
   (a) 1 (b) 2
   (c) 3 (d) 4
2. Ethane, with the molecular formula C_2H_6 has –
   (a) 6 covalent bonds. (b) 7 covalent bonds.
   (c) 8 covalent bonds. (d) 9 covalent bonds.
3. While cooking, if the bottom of the vessel is getting blackened on the outside, it means that
   (a) the food is not cooked completely.
   (b) the fuel is not burning completely.
   (c) the fuel is wet.
   (d) the fuel is burning completely.
4. Two adjacent members of a homologous series have –
   (a) a difference of CH_2 in their structure
   (b) a different of 14 amu in molecular mass
   (c) same general method of preparation
   (d) all the above
5. Which is a general formula of alkenes –
   (a) C_nH_{2n+2} (b) C_nH_{2n}
   (c) C_nH_{2n-2} (a) None of the above
6. The functional group represent alcohol is –
   (a) –OH (b) –CHO
   (c) –COOH (d) >C=O
7. Which of the following is the purest form of carbon –
   (a) charcoal (b) coal
   (c) diamond (d) graphite
8. Organic compounds will always contain –
   (a) carbon (b) hydrogen
   (c) nitrogen (d) sulphur
9. Methane, ethane and propane are said to form a homologous series because all are –
   (a) hydrocarbons
   (b) saturated compounds
   (c) aliphatic compounds
   (d) differ from each other by a CH_2 group
10. When methane is burnt in an excess of air, the products of combustion are –
    (a) C and H_2O (b) CO and H_2O
    (c) CO_2 and H_2 (d) CO_2 and H_2O
11. Which of the following gases is called 'marsh gas' –
    (a) H_2 (b) CH_4
    (c) C_2H_4 (d) C_2H_2
12. The final product of chlorination of methane in the sun light is –
    (a) CH_3Cl (b) CH_2Cl_2
    (c) $CHCl_3$ (d) CCl_4
13. When ethane is burnt in excess of air, the products of combustion are –
    (a) C and H_2O (b) CO and H_2O
    (c) CO_2 and H_2 (d) CO_2 and H_2O
14. The number of oxygen molecules used in the combustion of 1 molecule of ethanol is –
    (a) 1 (b) 2
    (c) 3 (d) 4
15. General formula of alkyne is –
    (a) C_nH_{2n+2} (b) C_nH_{2n}
    (c) C_nH_{2n-2} (d) C_nH_n
16. When vanaspati oil reacts with hydrogen then it convert into vanaspati ghee. In this process catalyst used is :
    (a) Fe (b) Mo
    (c) V (d) Ni
17. Observe the following pairs of organic compounds :
    (I) C_4H_9OH and $C_5H_{11}OH$
    (II) $C_7H_{15}OH$ and $C_5H_{11}OH$
    (III) $C_6H_{13}OH$ and C_3H_7OH
    Which of these pair is a homologous series according to increasing order of carbon atom.
    (a) (III) only (b) (II) only
    (c) (I) only (d) All of these
18. Carbon exists in the atmosphere in the form of
    (a) carbon monoxide only
    (b) carbon monoxide in traces and carbon dioxide
    (c) carbon dioxide only
    (d) coal
19. Buckministerfullerene is an allotropic form of
    (a) phosphorus (b) sulphur
    (c) carbon (d) tin
20. Oils on treating with hydrogen in the presence of palladium or nickel catalyst form fats. This is an example of
    (a) addition reaction
    (b) substitution reaction
    (c) displacement reaction
    (d) oxidation reaction

21. Chlorine reacts with saturated hydrocarbons at room temperature in the
(a) absence of sunlight
(b) presence of sunlight
(c) presence of water
(d) presence of hydrochloric acid

22. Pentane has the molecular formula C_5H_{12}. It has
(a) 5 covalent bonds
(b) 12 covalent bonds
(c) 16 covalent bonds
(d) 17 covalent bonds

23. Structural formula of benzene is

(a) 
(b) 
(c) 
(d) 

24. Carbon forms four covalent bonds by sharing its four valence electrons with four univalent atoms, e.g. hydrogen. After the formation of four bonds, carbon attains the electronic configuration of
(a) helium (b) neon
(c) argon (d) krypton

25. Which of the following does not belong to the same homologous series?
(a) CH_4 (b) C_2H_6
(c) C_3H_8 (d) C_4H_8

MTQC More than One Correct :

DIRECTIONS : *This section contains 12 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. The acid in which –COOH group is (are) present
(a) ethanoic acid (b) picric acid
(c) lactic acid (d) palmitic acid

2. Which is not a carbon compound
(a) Wood (b) chalk
(c) paper (d) cement

3. Which of the following is (are) property of a diamond
(a) It is the hardest substance
(b) It has high refractive index
(c) Like metals, it can conduct electricity
(d) In diamond each C is at the centre of tetrahedron

4. Which of the following is(are) property of graphite
(a) It melts at 800°C
(b) It is smooth, crystallined form of carbon
(c) It is good conductor of electricity
(d) It forms a black sign on paper

5. Which of the following statements is (are) correct
(a) graphite is much less dense than diamond
(b) graphite is black and soft
(c) graphite has low melting point
(d) graphite feels smooth and slippery

6. Which of the following statements regarding diamond is (are) correct
(a) diamond does not refract light
(b) when proper cut and polished, a diamond reflects light in an array of many colours
(c) diamond is less stable than graphite
(d) Impure sample of diamond is often black and are used in abrasives

7. Which of the following is (are) incorrect
(a) graphite is a bad conductor of electricity
(b) dry graphite in a vacuum is not slippery
(c) the adsorption of substances increases the friction as layers slid pass each other
(d) the graphite does not possess metallic properties

8. Which of the following are allotropes of carbon
(a) coal (b) diamond
(c) charcoal (d) fullerene

9. Which compounds of carbon are also studied in inorganic chemistry
(a) CO_2 (b) H_2CO_3
(c) $(COOH)_2$ (d) CaC_2

10. Which of the following are combustion reactions.
(a) $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$
(b) $C_2H_5OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$
(c) $CO + O_2 \longrightarrow CO_2$
(d) $2CH_4 + O_2 \longrightarrow 2CH_3OH$

11. Which of the following is (are) heterocyclic compounds
(a) furan (b) anisole
(c) thiophene (d) chlorobenzene

12. Which of the following is are homocyclic compounds
(a) cyclopropene (b) pyrrole
(c) azulene (d) toluene

FIP Fill in the Passage :

DIRECTIONS : *Complete the following passage(s) with an appropriate word/term to be filled in the blank spaces.*

1. The phenomenon of existence of a chemical element in two or more forms differ in physical properties but of same chemical nature is known as ___(1)___ . Carbon has two major allotropes diamond, graphite and the third one is ___(2)___. Diamond has three dimensional network of

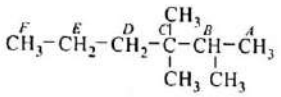
___(3)___ bonds and is hard. In graphite, the carbon atoms are arranged in flat ___(4)___ layers as regular hexagons resulting in to soft ___(5)___ nature.

2. Functional group is a group of atom within a molecule which is responsible for ___(1)___ characteristics of that molecule (compound). The first carbon atom after the carbon atom bonded to functional group is known as ___(2)___ carbon and further carbon in chain are known as Beta, Gamma progressively. ___(3)___ series is a series of organic compounds with similar formula, possessing similar chemical properties due to the ___(4)___ of the same functional group.

PBQ Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

A carbon atom attached to one, two, three and four carbon atom is called primary, secondary, tertiary and quaternary carbon respectively. Now consider following compound and answer the following questions.



1. In above compound how many carbon atom are primary?
   (a) 7 (b) 5
   (c) 6 (d) 4
2. In above compound how many carbon atoms are secondary?
   (a) 2 (b) 1
   (c) 3 (d) 0
3. In above compound which carbon atom is quaternary?
   (a) *B* (b) *D*
   (c) *F* (d) *C*

A&R Assertion & Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect.**
(d) If **Assertion** is **incorrect** but **Reason** is **correct.**

1. **Assertion :** Diamond and graphite are allotropes of carbon
   Reason : Some elements can have several different structural forms while in the same physical state. These differing forms are called allotropes.
2. **Assertion :** Carbon monoxide is extremely poisonous in nature.
   Reason : Carbon monoxide is formed by complete combustion of carbon.
3. **Assertion :** Carbon has ability to form long carbon chains.
   Reason : Carbon has a unique property of ability to form long straight and branched chains called catenation.
4. **Assertion :** Cyclopropane is heterocyclic compound.
   Reason : Cyclopropane comes into category of those compounds in which complete ring is formed by carbon atoms only.
5. **Assertion :** Alcohols have similar chemical properties.
   Reason : All alcohols contains similar hydroxy (–OH) functional group.

MMQ Multiple Matching Questions :

DIRECTIONS : *Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

Columm-I	Columm-II
(A) Cyclopropane	(p) Homocyclic compounds
(B) Benene	(q) Alicyclic compounds
(C) Azulene	(r) Aromatic compounds
(D) Furan	(s) Heterocyclic compounds

HOTS HOTS Subjective Questions :

DIRECTIONS : *Answer the following questions.*

1. Why are unsaturated hydrocarbons more reactive than saturated hydrocarbons ?
2. Why are carbon compounds not able to conduct electricity through them ?
3. How many covalent bonds are there in a molecule of cyclohexane?
4. Define covalent bond? What do you mean by the term covalency. Give an example of each containing.
   (i) single bond (ii) double bond (iii) triple bond
5. Why carbon shows the property of catenation? Can any other elements shows the property of catenation.
6. What are the two properties of carbon which lead to the huge number of carbon compounds we see around us ?
7. Why are carbon and its compounds used as fuels for most applications ?
8. Which of the following hydrocarbons undergo addition reactions : C_2H_6, C_3H_8, C_3H_6, C_2H_2 and CH_4.
9. Which phenomena is shown by zinc blende and wurtzite.
10. Which of the following C_2H_6, C_2H_5OH, CH_3COOH is a hydrocarbon?

11. (i) What is the type of reaction taking place between ethane and chlorine to form monochloroethane?
(ii) The reaction between ethene and chlorine forms only one product. Name the type of this reaction.
(iii) (a) Draw the structural formula of ethene.
(b) What is the feature of the ethene structure which allows ethene to react with chlorine in the way it does?

12. State the general formula of the homologous series to which each of the following belong :
(a) Methane (b) Butane (c) Butyne
(d) Ethyne (e) Propyne (f) Propylene
(g) Ethene

13. The molecules of alkene family are represented by a general formula C_nH_{2n}. Now answer the following :
(i) What do n and 2n signify?
(ii) What is the name of alkene when n = 4?
(iii) What is the molecular formula of alkene when n = 6?
(iv) What is the molecular formula of the alkene if there are six H atoms in it?
(v) What is the molecular formula and structural formula of the first member of the alkene family?
(vi) Write the molecular formulae of lower and higher homologues of an alkene which contains four carbon atoms.

14. Write the molecular formulae of the third and fifth members of homologous series of carbon compounds represented by the generally formula C_nH_{2n-2}.

15. Give the molecular formula of one homologues of each of the following :
(i) C_6H_{14} (ii) C_3H_6 (iii) C_4H_8

16. Give the general formula of paraffins, olefins and alkynes.

17. Ethane, Ethene, Ethanoic acid, Ethyne, Ethanol
From the list of compound given above, name :
(i) The compounds with – OH as a part of its structure.
(ii) The compounds with – COOH as a part of its structure.
(iii) Homologues and Homologous series with general formula C_nH_{2n}.

TBQ Table Based Questions :

DIRECTIOINS : *Complete the following table by filling vacant spaces.*

S. No.	Equation representing reaction	Type of reaction
1.	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	.....
2.	.....	Addition reaction
3.	$C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$	.....
4.	.....	Substitution reaction
5.	.....	Photo-synthesis reaction

SOLUTIONS

Brief Explanations of Selected Questions

Exercise 1

FILL IN THE BLANKS :

1. homologous
2. fullerene
3. Graphite
4. propane
5. 4
6. water
7. 14
8. diamond
9. $C_nH_{2n+1}OH$
10. –COOH

TRUE / FALSE

1. True
2. True
3. True
4. True
5. True
6. False
7. False
8. True
9. False
10. False
11. True
12. True

MATCH THE FOLLOWING :

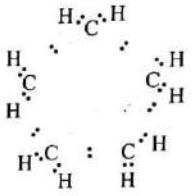
1. A – (s), B – (r), C – (q), D – (p)
2. A – (q), B – (r), C – (s), D – (p)

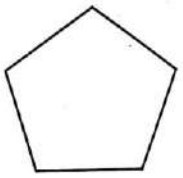
VERY SHORT ANSWER QUESTIONS :

1. $\underset{\text{ethene}}{CH_2 = CH_2} + H_2 \xrightarrow{Ni} \underset{\text{ethane}}{CH_3 - CH_3}$
2. Fullerences are special class of crystalline carbon allotropes. One of such allotropes is C-60, which is called Buck-minister Fullerene.
3. (a) carboxylic acid (b) aldehyde.
   (c) ether
4. This general formula is that of a straight-chain alkene with one double bond or of a cyclic alkane.
5. Carboxylic acids contain a carboxyl group, and alcohols contain one or more hydroxyl group.
6. Those hydro-carbon molecules, which have either double or triple covalent bonds in their molecules, are called unsaturated hydro-carbons.
7. (i) Aldehyde (ii) Ketone.
8. (i) hydroxyl group (ii) carboxyl group.
9. Carbon dioxide.
10. Elements, e.g., carbon has three allotropes graphite, diamond and fullerene.
11. It is a complex compound of carbon monoxide (CO) and haemoglobin.
12. On cracking complex hydrocarbons decomposes to give a mixture of simpler saturated and unsaturated hydrocarbons.
13. Alicyclic compounds.

Cyclodipentene

14. Formula of cyclopentane is C_5H_{10} and its electron dot structure is:





15. C_9H_{20} and $C_{11}H_{24}$

SHORT ANSWER QUESTIONS :

1. Carbon atoms have 4 electrons in their outermost shell. So if it were to gain or lose electrons :
   (i) it could gain four electrons forming C^{4-} anion. But it would be difficult for the nucleus with six protons to hold on to ten electrons.
   (ii) it could lose four electrons forming C^{4+} cation. But it would require a large amount of energy to remove four electrons from its outermost shell.
2. The series of organic compounds having same functional group is called homologous series. Characteristics :
   (i) They have same general formula.
   (ii) Each successive member is differ by CH_2.
   (iii) They have similar chemical properties.
   (iv) They have gradation in physical properties.
3. $H:C\vdots\vdots C:H$ Electron dot structure of ethyne (C_2H_2)
   $H-C\equiv C-H$ Structural formula of ethyne
4. The various amorphous form of carbon are Coal, Coke, Charcoal (or wood charcoal), Animal Charcoal (or bone black), Lamp black, Carbon black, Gas carbon and Petroleum coke.
5. The poisonous nature of CO is due to the fact that it combines with **haemoglobin** (a red colouring matter of blood), oxygen carrier of blood, to form a stable compound *carboxyhaemoglobin*. With the result the oxygen transporation is disturbed and thus brain and body tissues do not get necessary oxygen and ultimately death occurs.
7. In case a compound undergoing combustion contain nitrogen (burns to form NO_2) to the compound formula or sulfur (burns to form SO_2). Like this:

   $C_{21}H_{24}N_2O_4 + O_2 \longrightarrow CO_2 + H_2O + NO_2$

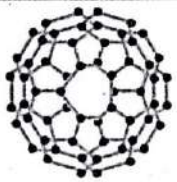
   $C_2H_5SH + O_2 \longrightarrow CO_2 + H_2O + SO_2$
9. These are cyclic compounds having ring or rings built up of more than one kind of atoms. Heterocyclic rings can also be aromatic. For e.g.

Furan

Thiophene

10. C_2H_6O (C_2H_5OH), CH_4O (CH_3OH) belong to same homologous series.

Long Answer Questions :



Structure of Buckminsterfullerene

1. Fullerenes contain 60 carbon atoms arranged in a round molecule resembling a soccer ball. C60 molecule has marvelously symmetrical structure. C60 molecule is a fused-ring of aromatic system containing 20 hexagons and 12 pentagons of C atoms. The structure bends around and closes to form a soccer ball shaped molecule. It is therefore, called buckyball also. The five numbered rings are connected to six numbered rings while the six numbered rings are fused to both five and six numbered rings. Kullerenes were first prepared by the evaporation of graphite using a loser. These can be conveniently made by heating graphite in an electric arc in the presence of helium or orgon. Fullerene looks different from diamond and graphite. It is a yellow powdery substance and are soluble in organic solvents and form coloured solutions. It turns pink on dissolution in solvents like toluene. It polymerizes on exposure to U.V. radiations. Fullerenes are fascinating because they show unusual characteristics and applications like: They are wonderful lubricants because the balls can roll between the surfaces.
2. These differences in the properties of diamond and graphite are due to the difference in their structures. In diamond, each C atom is linked to its neighbours by four single covalent bonds. This leads to a three-dimensional network of covalent bonds. In graphite, the carbon atoms are arranged in flat parallel layers as regular hexagons. Each carbon in these layers is bonded to three others by covalent bonds. Graphite thus acquires some double bond character. Each layer is bonded to adjacent layers by weak van der Waals forces. Moreover one electron is free at each carbon atom. This allows each layer to slide over the other easily. Due to this type of structure graphite is soft and slippery, and can act as a lubricant. Graphite is also a good conductor of electricity due to mobile electrons in it.
4. This is due to the following reasons :
   (i) The number of known organic compounds is very large.
   (ii) Organic compounds involve only a few elements C, H, O, N, S, P, X (Cl, Br, I)
   (iii) Organic compounds involve covalent bonds.
   (iv) The molecules of organic compounds show a wide variety of structures.
   (v) Organic compounds show isomerism.
   (vi) It is a matter of convinience to study organic chemistry as a separate branch of chemistry.
5. **1. Cracking**

$$C_8H_{18} \xrightarrow{\text{Heat}} C_6H_{14} + \underset{\text{Ethene}}{C_2H_4}$$

2. Hydrogenation

$$C_2H_4 + H_2 \xrightarrow[\text{Heat}]{\text{Ni}} \underset{\text{Ethane}}{C_2H_6}$$

3. Dehydration

$$C_2H_5OH \xrightarrow[433\ K]{\text{Conc.}H_2SO_4} C_2H_4 + H_2O$$
Ethene

4. Substitution reaction

$$CH_4 + Cl_2 \xrightarrow{\text{Diffused sunlight}} \underset{\text{Chloromethane}}{CH_3Cl} + HCl$$

5. Addition reaction

$$\underset{\text{Ethene}}{C_2H_4} + \underset{(aq)}{Cl_2} \longrightarrow \underset{\begin{matrix}|\\Cl\end{matrix}\quad\begin{matrix}|\\Cl\end{matrix}}{CH_2 - CH_2}$$
1, 2 – Dichloroethane

6. **Comparison between Diamonds and Graphite :**

Diamond	Graphite
• Diamond is the hardest material known.	• Graphite is one of the softest materials known.
• Diamond is the ultimate abrasive.	• Graphite is a very good lubricant.
• Diamond is an excellent electrical insulator.	• Graphite is a conductor of electricity.
• Diamond is the best known naturally occurring thermal conductor	• Some forms of graphite are used for thermal insulation (i.e. firebreaks and heat
• Diamond is highly transparent.	• Graphite is opaque.
• Diamond crystallizes in the cubic system.	• Graphite crystallizes in the hexagonal system.

Exercise 2

Multiple Choice Questions :

1.	(b)	2.	(b)	3.	(b)	4.	(d)
5.	(b)	6.	(a)	7.	(c)	8.	(a)
9.	(d)	10.	(d)	11.	(b)	12.	(d)
13.	(d)	14.	(c)	15.	(c)	16.	(d)
17.	(c)	18.	(b)	19.	(c)	20.	(a)
21.	(b)	22.	(c)	23.	(c)	24.	(b)
25.	(d)						

More Than One Correct :

1.	(a, c, d)	2.	(b, d)
3.	(a, b, d)	4.	(b, c, d)
5.	(b, c, d)	6.	(b, c, d)
7.	(a, c, d)	8.	(b, d)
9.	(a, b, d)	10.	(a, b, c)
11.	(a, c)	12.	(a, c, d)

Pill in the Passage :

1. (1) Allotropy (2) Buckminister fullerene
   (3) Covalent (4) parallel
   (5) Slippery
2. (1) chemical (2) alpha
   (3) Homologous (4) presence

PASSAGE BASED QUESTIONS :

1. (b) 2. (a) 3. (d)

ASSERTION & REASON :

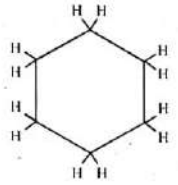
1. (a) 2. (c) 3. (a)
4. (d) 5. (a)

MULTIPLE MATCHING QUESTIONS :

1. A → (p, q), B → (p, r), C → (p, r), D → (r, s)

HOTS SUBJECTIVE QUESTIONS :

1. Unsaturated hydrocarbons are more reactive due to the presence of C = C and C ≡ C bonds. These are the reactive sites in the unsaturated hydrocarbons.
2. Carbon compounds have covalent bonds between them, these do not give ions in their solution form or in molten state. So carbon compounds do not conduct electricity through them.
3. The structure of cyclohexane is



i.e. has 18 covalent bonds

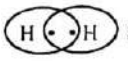
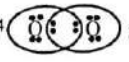
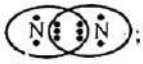
4. **Covalent Bond :** The covalent bonds are formed by the mutual sharing of electrons between the participating atoms of same or different elements.

Covalency : The number of electrons contributed by an atom for sharing is known as its covalency.

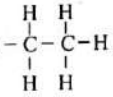
Depending upon the number of electrons shared by each atom in bond formation, covalent bond may be single, double or triple.

A single bond is represented by a single line (–), a double bond by double line (=) and a triple bond by a triple line (≡).

Example :

- $H_2 \longrightarrow H(1) = 1s^1$  ; H – H .
- $O_2 \longrightarrow O\,(8) = 1s^2\,2s^2\,2p^4$  ; O = O
- $N_2 \longrightarrow N\,(7) = 1s^2\,2s^2\,2p^3$  ; N ≡ N

5. Carbon has unique property of catenation as a result of which it is capable to form long straight and branch chains. Carbon forms strong bond with each other and atom of other elements also because of its small size which enables the nucleus to hold on to the shared pairs of electrons strongly. Sulphur elements shows property of catenation but only upto certain extent sulphur forms S_8 molecule.
6. (i) Catenation (ii) Tetravalency of carbon
7. It is because they are inflammable and have high calorific value.
8. C_3H_6 and C_2H_2 are unsaturated hydrocarbon, therefore, they will undergo addition reactions.
9. Zinc blende and wurtzite are polymorphs of zinc sulphide and the phenomena is known as polymorphism. It is a phenomena in which a compound exists in two or more forms.
10. C_2H_6, this is because it is made up of only carbon and hydrogen atoms linked together by covalent bonds.



11. (i) Substitution reaction
    (ii) Addition reaction
    (iii) (a) Structural formula of ethene is

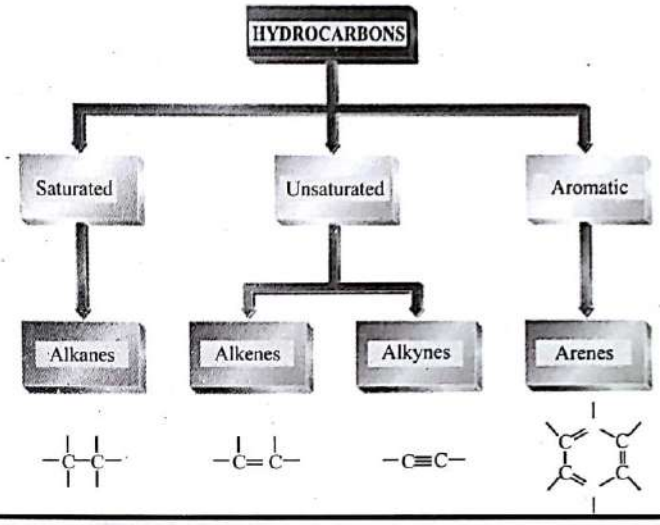
    (b) Carbon-Carbon double bond (> C = C <)
12. (a) C_nH_{2n+2} ; n = 1, 2, 3 .....
    (b) C_nH_{2n+2} ; n = 1, 2, 3 .....
    (c) C_nH_{2n-2} ; n = 2, 3, 4 .....
    (d) C_nH_{2n-2} ; n = 2, 3, 4 .....
    (e) C_nH_{2n-2} ; n = 2, 3, 4 .....
    (f) C_nH_{2n} ; n = 2, 3, 4 .....
    (g) C_nH_{2n} ; n = 2, 3, 4 .....
13. (i) *n* indicates number of carbon atoms and 2*n* indicates number of hydrogen atoms.
    (ii) Butene
    (iii) C_6H_{12}
    (iv) C_3H_6
    (v) C_2H_4
    (vi) Lower homologue – C_3H_6
    Higher homologue – C_5H_{10}
14. General formula = C_nH_{2n-2}, n = 2, 3, 4 .....
    Third member = C_4H_6
    Fifth member = C_6H_{10}
15. (i) C_5H_{12} (ii) C_2H_4 (iii) C_3H_6
16. General formula of :
    (a) Paraffins is C_nH_{2n+2} ; n = 1, 2, 3 .....
    (b) Olefins is C_nH_{2n} ; n = 2, 3, 4 .....
    (c) Alkynes is C_nH_{2n-2} ; n = 2, 3, 4 .....
17. (i) Ethanol
    (ii) Ethanoic acid
    (iii) Ethene

TABLE BASED QUESTIONS :

S. No.	Equation representing reaction	Type of reaction
1.	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	Combustion reaction
2.	$R_2C = CR_2 \xrightarrow[H_2]{\text{Nicatalyst}} R_2CH\,CH\,R_2$	Addition reaction
3.	$C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$	Oxidation reaction
4.	$CH_4 + Cl_2 \xrightarrow{UV} CH_3Cl + HCl$	Substitution reaction
5.	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$	Photo-synthesis reaction

chapter

9



GOC (NOMENCLATURE AND ISOMERISM)

Introduction

As we know that due to unique property of catenation carbon forms a large number of straight chain, branched chain and ring compounds with hydrogen called hydrocarbons. Moreover increasingly large number of organic compounds identified with a each passing day, requires that a systematic nomenclature system need to be developed. Initially common names were given to few compounds for example Methane (CH_4), Acetone (CH_3COCH_3), Toluene ($CH_3C_6H_5$), Acetylene (C_2H_2) etc. The system of nomenclature was first introduced in 1947 by International Union of pure and Applied Chemistry called IUPAC system of naming and was modified from time to time. The IUPAC system provides a unique name for each of the nearly 10 million known organic compounds, as well as the thousands of new compounds discovered or synthesized each year. The IUPAC rules are simple to learn and easy to use. With these rules you can readily write the name of any compound you might encounter or can derive the structure of any given compound from its name. Scientist adopted the IUPAC system of nomenclature for several reasons. First, scientist everywhere understand it. Second, they can readily adapt it to the indexing methods used for the chemical literature. Third, they can easily use it for computerized literature searching operations. We will learn common names (Trivial system), derived system before learning IUPAC

COMMON NAMES (TRIVIAL SYSTEM):

Initialy organic compounds are named on the basis of source from which they were obtained. For eg.

S.N.	Organic compound	Trivial name	Source
1.	CH_3OH	Wood spirit or Methyl spirit	Obtained by destructive distillation of wood.
2.	NH_2CONH_2	Urea	Obtained from urine.
3.	CH_4	Marsh gas (Fire damp)	It was produced in marsh places
4.	CH_3COOH	Vinegar	Obtained from Acetum-i.e. Vinegar
5.	$\begin{matrix}COOH\\\vert\\COOH\end{matrix}$	Oxalic acid	Obtained from oxalis plant.
6.	$HCOOH$	Formic acid	Obtained from formicus (Red ant)
7.	$CH_3-\underset{\vert}{\overset{OH}{CH}}-COOH$	Lactic acid	Obtained from sour milk
8.	$\begin{matrix}CH_2COOH\\\vert\\CH(OH)COOH\end{matrix}$	Malic acid	Obtained from apples
9.	$CH_3CH_2CH_2COOH$	Butyric acid	Obtained from butter
10.	$CH_3(CH_2)_4COOH$	Caproic acid	Obtained from goats

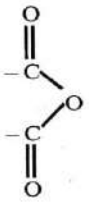
Some typical compounds in which common & trivial names are also differ.

S.N.	Compound	Trivial name	Common name
1.	CH_4	Marsh gas	Methane
2.	CH_3OH	Wood spirit	Methyl alcohol
3.	CH_3COOH	Vinegar	Acetic acid
4.	$CH_3-\overset{O}{\overset{\Vert}{C}}-CH_3$	Acetone	Dimethyl ketone
5.	$CH_2=CH-\overset{O}{\overset{\Vert}{C}}-H$	Acrolein	Acryl Aldehyde
6.	$CH_3-\underset{CH_3}{\overset{CH_3}{\underset{\vert}{\overset{\vert}{C}}}}-\overset{O}{\overset{\Vert}{C}}-H$	Pyvaldehyde	Tertiary valer aldehyde

Common Names : R is termed as alkyl

S.N.	Compound	Name	S.N.	Compound	Name
1.	$R-X$	Alkyl halide	2.	$R-OH$	Alkyl alcohol
3.	$R-SH$	Alkyl thioalcohol	4.	$R-NH_2$	Alkyl amine
5.	$R-O-R$	Dialkyl ether	6.	$R-S-R$	Dialkyl thioether
7.	$R-\overset{O}{\overset{\Vert}{C}}-R$	Dialkyl ketone	8.	$R-NH-R$	Dialkyl amine
9.	$R-\underset{R}{\overset{R}{\underset{\vert}{\overset{\vert}{N}}}}$	Trialkyl amine	10.	$R-O-R'$	Alkyl alkyl' ether
11.	$R-\overset{O}{\overset{\Vert}{C}}-R'$	Alkyl alkyl' ketone	12.	$R-S-R'$	Alkyl alkyl' thio ether
13.	$R-NH-R'$	Alkyl alkyl' amine	14.	$R-\underset{R''}{\overset{R'}{\underset{\Vert}{\overset{\vert}{N}}}}$	Alkyl alkyl' alkyl'' amine

COMMON - NAMING OF THE FUNCTIONAL GROUP HAVING CARBON :

Functional group	Suffix	Functional group	Suffix
$-\overset{\overset{O}{\|\|}}{C}-H$	-aldehyde	$-\overset{\overset{O}{\|\|}}{C}-OH$	-ic Acid
$-\overset{\overset{O}{\|\|}}{C}-X$	-yl halide	$-\overset{\overset{O}{\|\|}}{C}-NH_2$	-amide
$-C\equiv N$	-o-nitrile	$-N\equiv C$	-o-isonitrile
$-\overset{\overset{O}{\|\|}}{C}-O-R$	-ate		-ic anhydride

DERIVED SYSTEM :

According to this system any compound is given name according to the parent name of the homologous series. This system is reserved for the following nine homologous series.

S.No.	Name of Homologous series	Derived name	Structure of group
1.	Alkane	Methane	$-\overset{\|}{\underset{\|}{C}}-$
2.	Alkene	Ethylene	$>C=C<$
3.	Alkyne	Acetylene	$-C\equiv C-$
4.	Alkanol	Carbinol	$-\overset{\|}{\underset{\|}{C}}-OH$
5.	Alkanal	Acetaldehyde	$-\overset{\|}{\underset{\|}{C}}-CHO$
6.	Alkanoic acid	Acetic acid	$-\overset{\|}{\underset{\|}{C}}-COOH$
7.	Alkanoyl halide	Acetyl halide	$-\overset{\|}{\underset{\|}{C}}-COX$
8.	Alkanamide	Acetamide	$-\overset{\|}{\underset{\|}{C}}-CONH_2$
9.	Alkanone	Acetone	$-\overset{\|}{\underset{\|}{C}}-\overset{\overset{O}{\|\|}}{C}-\overset{\|}{\underset{\|}{C}}-$

IUPAC NOMENCLATURE :

In order to solve the problem of naming of organic compounds, an organisation called international chemical congress for the first time met at Geneva in 1892. They developed a certain system called Geneva system. Since then the system of naming has been improved from time to time by the International union of pure and Applied chemistry and the new system is called IUPAC system of naming. This system of nomenclature was introduced in 1947. The latest IUPAC system is based on the recommendation made in 1993.

IUPAC system of nomenclature of Aliphatic compounds

According to IUPAC system, the name of an organic compound consists of three parts :

(i) Word root

(ii) Suffix

(iii) Prefix

From C_1 to C_4 common names have been retained and from C_5 upwards Greek number roots have been used.

(i) Word root

Word root denotes the number of carbon atoms present in the principal chain which is the longest possible chain of carbon atoms.

Chain length	Word root	Chain length	Word root
C_1	Meth-	C_7	Hept-
C_2	Eth-	C_8	Oct-
C_3	Prop-	C_9	Non-
C_4	But-	C_{10}	Dec-
C_5	Pent-	C_{11}	Undec-
C_6	Hex-	C_{12}	Dodec

(ii) Suffix

Suffixes are of two types. Primary and Secondary suffixes.

(a) Primary suffix

A primary suffix indicates the type of linkage in the carbon atoms. If the carbon atoms are linked by single covalent bonds, the primary suffix is 'ane'. If these are linked by double and triple bonds, the primary suffixes 'ene' and 'yne' are respectively used to represent them. Thus,

ane : primary suffix for C – C bond

ene : primary suffix for C = C bond

yne : primary suffix for C ≡ C bond

If the parent chain of carbon atoms contains more than one double or triple bonds, numerical prefixes like di (for two), tri (for three), tetra (for four) etc. are added to primary suffix.

Example–

Hydrocarbon	Word root	Primary suffix
$CH_3 - CH_2 - CH_2 - CH_3$	But	ane
$CH_2 = CH - CH = CH_2$	Buta	diene
$CH_3 - C \equiv CH$	Prop	yne

(b) Secondary suffix

A secondary suffix is used to represent the functional group if present in organic molecule and is attached to the primary suffix while writing its IUPAC name. Secondary suffixes of some of the functional groups are listed :–

Functional group	Symbol	Suffix
Sulphonic	$-SO_3H$	sulponic acid
Carboxylic	$-COOH$	–oic acid
Ester	$R-\overset{O}{\overset{\|}{C}}-OR$	Alkyl-Alkanonate
Acid halide	$-\overset{O}{\overset{\|}{C}}-X$	-oylhalide
Acid Amide	$-\overset{O}{\overset{\|}{C}}-NH_2$	amide
Aldehyde	$-\overset{O}{\overset{\|}{C}}-H$	-al
Cynide	$-C \equiv N$	-nitrile
Isocynide	$N \equiv C$	-isocynide
Ketone	$>C=O$	-one
Hydroxy	$-OH$	-ol
Thioalcohol	$-SH$	-Thiol
Amine	$-NH_2$	-Amine
Secondary amine	$-N\begin{matrix}R\\H\end{matrix}$	N-Alkyl-amine
Tertiary amine	$-N\begin{matrix}R\\R\end{matrix}$	N-alkyl-N-alkyl-amine
Nitro	$-N^{+}\begin{matrix}R\\O^{-}\end{matrix}$	nitrite

Groups given below does not have any Suffix

Ether	$-OR$	alkoxy
Epoxide	$-O-$	epoxy
Azo	$-N=N-$	azo
Nitroso	$-NO$	nitroso
Halogen	$-X$ (F, Cl, Br, I)	halo

While adding a secondary suffix to the primary suffix, the terminal 'e' of the primary suffix (ane, ene or yne) is dropped if the secondary suffix begins with 'a', 'i' 'o' 'u' or 'y'. In case, it begins with consonant, then the terminal 'e' of the primary suffix is retained.

For example

Organic compounds	Word root	Primary suffix	Secondary suffix	IUPAC name
CH_3CH_2OH	Eth	an (e)	ol	Ethanol
CH_3CH_2CHO	prop	ane (e)	al	Propanal
$CH_3CH_2CH_2NH_2$	prop	ane (e)	amine	Propanamine
CH_3CH_2CN	prop	ane (e)	nitrile	Propanamine

CHECK Point

☛ **What is the IUPAC name of the following compound ?**

$CH_3CH_2CH_2COOH$

SOLUTION

Word root	Primary suffix	Secondary suffix	IUPAC name
But	ane(e)	oic acid	Butanoic acid

(iii) Prefix

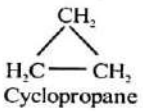
Prefix is a part of IUPAC name which appear before the word root. Prefixes are of two types.

(a) Primary prefix

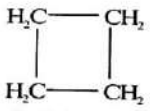
A primary prefix "cyclo" is used in order to differentiate a cyclic compound from an acyclic compound.

For example

$CH_3 - CH_2 - CH_3$
Propane


Cyclopropane

$CH_3 - CH_2 - CH_2 - CH_3$
Butane


Cyclobutane

Propane and butane are the IUPAC names of acyclic compounds while cyclopropane and cyclobutane are for cyclic compounds.

CHECK Point

☛ **What is the IUPAC name of the following compound ?**

SOLUTION

Primary prefix	word root	Primary suffix	Secondary suffix
Cyclo	hex	ane	-

Thus the IUPAC name of the compound is cyclo hexane

(b) Secondary prefix

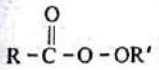
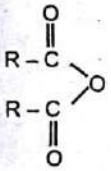
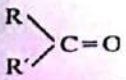
In the IUPAC system of nomenclature, certain characteristic groups are not considered as or secondary suffixes. These are regarded as substituents and are denoted by secondary prefixes. The secondary prefixes of a few substituents are given –

Substituent group	Secondary prefix	Substituent group	Secondary prefix
–F	Fluoro	$-NO_2$	Nitro
–Cl	Chloro	$-CH_3$	Methyl
–Br	Bromo	$-C_2H_5$	Ethyl
–I	Iodo	$-OCH_3$	Methoxy
–NO	Nitroso	$-OC_2H_5$	Ethoxy

Secondary prefix are added before the word root in case of acyclic compounds and before the primary prefix in case of cyclic compounds in alphabetical order.

Knowledge ENHANCER

Besides these, some other functional groups are also treated as substituents in case of poly functional organic compounds. These are listed as :

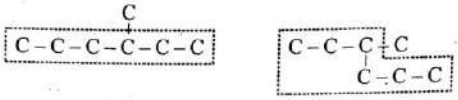
Functional group	Name	Secondary prefix
SO_3H	sulphonyl	sulpho
–COOH	carboxylic acid	carboxylic
	Ester	alkyl carboxylate or alkoxy carboyl
–COX	Acid halides	Halo formyl
$-CONH_2$	Acid amide	Carbamoyl
	Acid anhydride	Oic anhydride
–CHO	Aldehyde	Formyl
–CN	Cynide	Cyano
–N ≡ C	Isocynide	Iso cyano
	Ketone	Keto, oxo
–OH	Alcohol	Hydroxy
–SH	Thioalcohol	Mercapto
$-NH_2$	Amine	Amino
–O–	Epoxy	Epoxide

RULES FOR IUPAC NOMENCLATURE OF ORGANIC COMPOUNDS :

There are certain (following) rules for naming a complex organic molecule according to IUPAC system. These rules are illustrated below.

1. **Longest chain rule :** *The first step for naming an organic compound is to select the longest continuous chain of carbon atoms which may or may not be horizontal (straight).* This is called the *parent chain* or *main chain*, and other carbon chains attached to it are known as *side chains*. On the basis of the number of carbon atoms present in the parent chain, the parent hydrocarbon is determined. For example, if the parent chain contains six carbon atoms, the compound is considered to be a derivative of hexane.

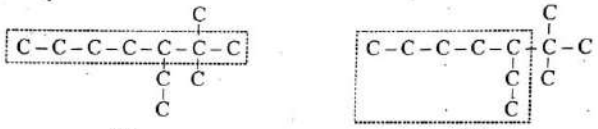
Example : The structure I has the longest chain of six carbon atoms present in a straight line; therefore it is said to be a derivative of hexane. On the other hand, in structutre II the straight chain has only four or three carbon atoms while the longest possible chain may have as many as six carbon atoms (zig-zag chain).



I (Longest chain, straight) II (Longest chain, zig-zig)

Therefore, both the structures are the derivatives of hexane.

In case a molecule contains two equally long carbon chains, the one carrying larger number of side chains is selected. For example, structures A and B have carbon chains of equal lengths (C_7), but the one carrying three substituents is selected.

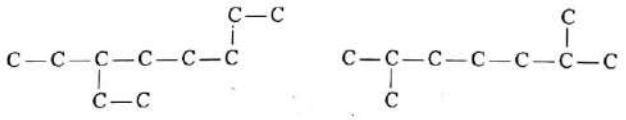


(A) Straight parent chain of C, with three side chains (correct)

(B) Zig-zag parent chain of C, with only one side chains (incorrect)

CHECK Point

☛ **Which of the following arrangement of selecting parent chain is correct**

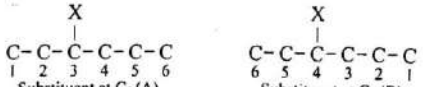


A B

SOLUTION

Arrangement B is correct with straight chain containing 7 carbons as it contains more number of side chains

2. **Lowest number or lowest sum rule :** *The carbon atom of longest carbon chain is numbered as 1, 2, 3, 4, ... etc., starting from the end that gives the lowest possible number* to the substituent. For example, in structures A and B numbering may be done in two ways;

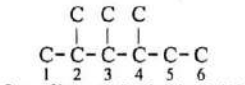


Substituent at C_3(A) (correct) Substituent at C_4(B) (incorrect)

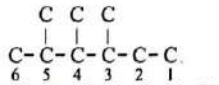
In one case (A), the substituent is assigned position 3 while in other case (B), it is assigned position 4. Hence the former, being smaller, is correct. The number that locates the position of a substituent is known as **locant**. Thus the locant for X in the above correct structure is 3.

In case, the parent chain has two or more substituents, numbering must be done in such a way that the sum of the locants on the parent chain is the lowest possible. Thus in following structures A and B numbering may be done in two ways : in one (A),

the sum of locants is 9 while in other (B), it is 12, hence the former is correct while the latter is wrong.



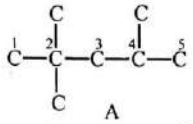
Sum of locants : 2 + 3 + 4 = 9 (A)
(correct)



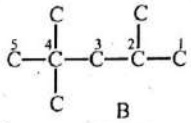
Sum of locants : 3 + 4 + 5 = 12 (B)
(incorrect)

CHECK Point

☛ **Which of the following way of numbering of carbon atoms of parent chain is correct according to lowest sum.**



A

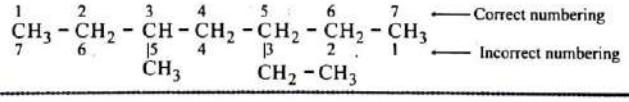


B

SOLUTION

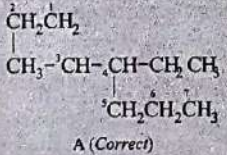
A represents correct way of numbering as sum of locants is minimum in A.
Sum of locants in A = 2 + 2 + 4 = 8
Sum of locants in B = 2 + 4 + 4 = 10

Further, if different side chains (alkyl groups) are present on the identical positions in relation to the ends of the chain, numbering must be done in a way as to give simpler side chain (i.e., containing lesser number of carbon atoms) the lower number, e.g., in the following example methyl group (simpler than the ethyl) should be given lower number, i.e., 3 in the present case.

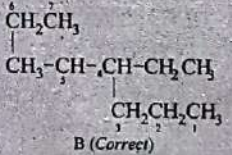


Knowledge ENHANCER

When two or more substituents are present, the carbon atoms of parent chain are numbered in such a way that the set of locants is the lowest. This is called lowest set of locants rule.
In accordance with the above rules, the carbon atoms of the alkanes given below should be numbered as shown in structures A and not as in structures B.



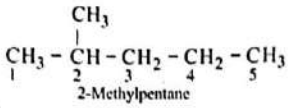
A (*Correct*)
Sum of the locants = 3 + 4 = 7
Set of locants = 3, 4



B (*Correct*)
Sum of the locants = 4 + 5 = 9
Set of locants = 4, 5

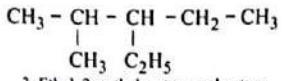
Thus, according to the lowest sum rule, the numbering of the carbon chain should be done from right to left but according to the lowest set of locants rule, the numbering should be done from left to right. To overcome such an ambiguity, the IUPAC nomenclature recommends that the lowest set of locants rule should be preferred over the lowest sum rule.

3. **Name of the complex alkane :** The name of the substituent is prefixed to the name of the parent hydrocarbon and its position on the main chain is indicated by writing the locant before the prefix. A hypen (–) is inserted between the locant and the substituent name. The final name is always written as one word. Thus the following compound is written as 2-methylpentane.



2-Methylpentane

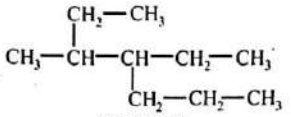
4. **Alphabetical order of side chains :** In case two or more alkyl groups (side-chains) are attached to the parent chain, these are prefixed in alphabetic order, e.g.,



3-Ethyl-2 methylpentane and not as
2-Methyl-3-ethylpentane

CHECK Point

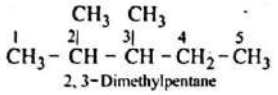
☛ **Write IUPAC name of following compound :**



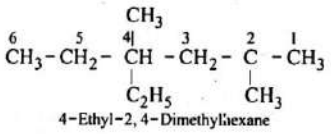
SOLUTION

4-Ethyl-3-methylheptane

5. **If a substituent is present two or more times:** *this is indicated by the prefix di-, tri-, tetra-, etc. added to the substituent* The different locants of the substituents are separated by commas. For example,

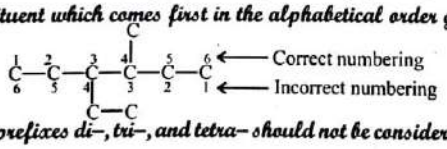


2, 3-Dimethylpentane



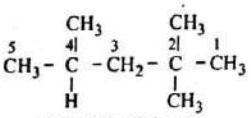
4-Ethyl-2, 4-Dimethylhexane

- ***When two different substituents are present at equivalent positions, the numbering of the parent chain is done in such a way that the substituent which comes first in the alphabetical order gets the lower number***

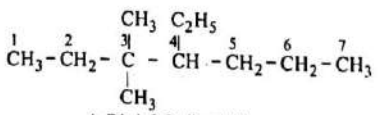


- ***The spelling of the prefixes di–, tri–, and tetra– should not be considered while arranging the substituents alphabetically,***

In case the same alkyl group occurs twice on the same carbon atom, its locant is also repeated twice. For example,



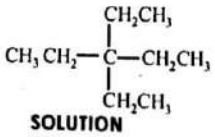
2, 2, 4-Trimethylpentane



4-Ethyl-3, 3-dimethylheptane

CHECK Point

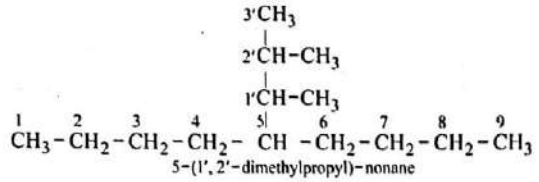
☛ **Write IUPAC name of following compound :**



SOLUTION

3, 3–Diethylpentane

6. **Naming the complex substituent :** *In case the substituent on the parent chain is complex (i.e., it has itself branched chain) then it is named as a substituted alkyl group and its carbon chain is numbered from the carbon atom attached to the main chain.* The name of this complex substituent is written in bracket to avoid confusion with the numbers of the main chain. e.g.,



5-(1', 2'-dimethylpropyl)-nonane

NOMENCLATURE OF DIFFERENT CLASSES OF ORGANIC COMPOUNDS :

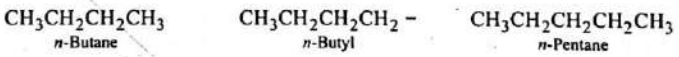
The names of various homologous series and their simple members are given below.

1. **Alkanes or Paraffins** (*Saturated hydrocarbons*) : These are compounds containing only carbon and hydrogen. The various carbon atoms present in the molecule are linked by single covalent bonds (C–C). These are represented by the general formula C_nH_{2n+2}. The IUPAC name for this homologous series is **alkanes** and the names of all alkanes will be ending with the suffix *–ane*. The common and IUPAC names for the first few members are identical.

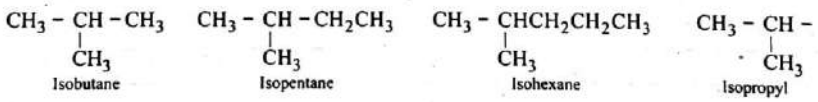
Formula	Common name	IUPAC name
CH_4	Methane	Methane
C_2H_6	Ethane	Ethane
C_3H_8	Propane	Propane
C_4H_{10}	Butane	Butane
C_5H_{12}	Pentane	Pentane

Certain prefixes commonly used in common system of nomenclature are given below.

1. **Prefix** *n* (normal) is used to indicate where the carbon chain is straight, i.e., when there is no branching. For example,



2. **Prefix** *iso* is used for those carbon chains which have a methyl group at the second carbon atom. For example,

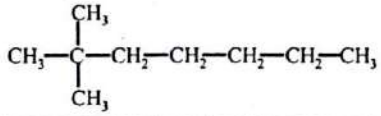


3. **Prefix** *neo* is used for those carbon chains which have two methyl groups at the second carbon atom. For example,

CHECK Point

☛ **Write the structure of compound neononane ?**

SOLUTION



2. **Alkenes or Olefins** (*Unsaturated hydrocarbons*) : These are characterised by the presence of a double bond between two carbon atoms, and contain two hydrogen atoms less than the corresponding alkanes. Their names are derived by substituting the suffix – *ane* of the corresponding alkanes by – *ylene* (common name) or –*ene* (IUPAC name). Thus IUPAC name for the series is **alkenes.**

General foumula : C_nH_{2n} *Functional group* : $-\overset{|}{C}=\overset{|}{C}-$

Alkene	*Common name*	*IUPAC name*
$CH_2 = CH_2$	Ethylene	Ethene
$CH_3 - CH = CH_2$	Propylene	Propene
$CH_3 - CH_2 - CH = CH_2$	Butylene	Butene
$CH_3CH_2CH_2CH = CH_2$	Pentylene	Pentene
$CH_3CH_2CH_2CH = CHCH_3$	β - Hexylene	Hex-2- ene
$CH_3CH_2CH = CHCH_2CH_3$	g-Hexylene	Hex-3-ene
$CH_2 = CH - CH = CH_2$	Butadience	But 1,3 diene

CHECK *Point*

☛ **Write the IUPAC name of following compound**

$$CH_3CH_2-\overset{C_2H_5}{\overset{|}{C}}=CHCH_3$$

SOLUTION

3-Ethyl pent-2-ene

3. **Alkynes or Acetylenic hydrocarbons :** These are characterised by the presence of a triple bond between two carbon atoms, and contain four Hydrogen atoms less than the corresponding alkanes. Their names are derived from the corresponding alkanes by substituting the suffix –*ane* by –*yne*.

General foumula : C_nH_{2n-2} *Functional group* : $-C\equiv C-$

Alkyne	*Common name*	*IUPAC name*
$HC \equiv CH$	Acetylene	Ethyne
$CH_3 - C \equiv CH$	Methyl acetylene	Propyne
$CH_3 - C \equiv C - CH_3$	Dimethyl acetylene	But-2-yne
$CH_3CH_2 - C \equiv C - CH_3$	Ethylmethylacetylene	Pent-2-yne
$CH_3 - CH_2 - CH_2 - C \equiv C - CH_3$	Methylpropyl acetylene	Hex-2-yne
$HC \equiv C - CH_2 - CH_2 - C \equiv CH$	Hexadiacetylene	Hex-1,5-diyne

If both double and triple bonds are present, the double bond is always given preference over the triple bond. For example,

$$\overset{5}{HC}\equiv\overset{4}{C}-\overset{3}{CH_2}-\overset{2}{CH}=\overset{1}{CH_2}$$

1 ← Correct numbering

5 ← Incorrect numbering

CHECK Point

☛ **Write the IUPAC name of following compound**

$CH_3 - CH = CH - C \equiv CH$

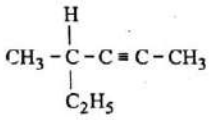
SOLUTION

Pent-3-en-1-yne

ILLUSTRATION 1

Draw the structure of compound 4-Ethylpent-2-yne.

SOLUTION:



4. **Halogen derivatives :** These are derived from alkanes by the replacement of one or more hydrogen atoms by halogen atoms (Cl, Br, or I). They are thus further classified as mono–, di–, tri–, or tetra-halogen derivatives when they contain 1, 2, 3 or 4 halogen atoms in their molecules.

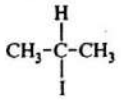
(i) ***Mono-halogen derivatives or alkyl halides*** **:** These are monohalogen derivatives of the alkanes.
General foumula : $C_nH_{2n+1}X$ or $R-X$, where R stands for an alkyl radical and X for a halogen atom.
Functional group : –X

Common names of alkyl halides are derived from the alkyl radical, while in IUPAC ***system these are named as substituted alkanes.***

Monohalogen derivative	*Common name*	*IUPAC name*
CH_3Cl	Methyl chloride	Chloromethane
C_2H_5Br	Ethyl bromide	Bromoethane
C_3H_7I	Propyl iodide	Iodopropane
$CH_3-\underset{\displaystyle Br}{\underset{\vert}{CH}}-CH_2-CH_3$	Sec-butylbromide	2-Bromobutane

CHECK Point

☛ **Give IUPAC name of following compound**

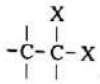


SOLUTION

2-Iodopropane

(ii) ***Di-halogen derivatives*** : These are derived from alkanes by the replacement of two hydrogen atoms by two halogen atoms.

If the two halogen atoms are attached to the same carbon atom, the derivative is known as alkylidene halide, and when the two halogen atoms are attached to two different atoms, the compound is known as alkylene halide.

(a) Alkylidene halide : 

(b) Alkylene halide :

General foumula : $C_nH_{2n}X_2$

Di-halogen derivative	Common name	IUPAC name
CH_2Br | CH_2Br	Ethylene dibromide	1, 2–Dibromoethane
CH_3 | $CHBr_2$	Ethylidine dibromide	1, 1–Dibromoethane

(iii) ***Tri-and Tetra-halogen derivatives*** : These are derivatives by the replacement of three or four hydrogen atoms from the corresponding alkanes by three or four halogen atoms respectively.

General foumula : $C_nH_{2n-1}X_3$ and $C_nH_{2n-2}X_4$, where X stands for a halogen atom.

Characteristic group : X_3 and X_4

Formula	*Common names*	*IUPAC name*
$CHCl_3$	Chloroform	Trichloromethane
CHI_3	Idoform	Triodomethane
CCl_4	Carbon tetrachloride	Tetrachloromethane

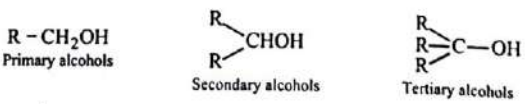
5. **Alcohols :** These are derived from alkanes by the replacement of one, two or three hydrogen atoms by hydroxy (–OH) groups, and thus these are correspondingly known as monohydric–, dihydric–, and trihydric alcohols.

(i) ***Monohydric alcohols*** : The IUPAC names of monohydric alcohols are derived from the corresponding alkane by replacing the *e* of alkane by *–ol*. Thus, in general, these are also known as **alkanols**.

The monohydric alcohols may further be classified into primary, secondary and tertiary alcohols depending on the carbon to which the alcoholic (–OH) group is attached.

General formula : $C_nH_{2n+1}OH$ or R–OH, where R stands for an alkyl radical and X for a halogen atom.

Characteristic group :



Formula	*Common name*	*IUPAC name*
R–OH	Alkyl alcohol	Alkanol
CH_3OH	Methyl alcohol	Methanol
$CH_3CH_2CH_2OH$	*n*-Propyl alcohol	1-Propanol (*pri.*)
$\begin{array}{l} CH_3 \\ \vert \\ CH—OH \\ \vert \\ CH_3 \end{array}$	*iso*-Propyl alcohol	2-Propanol (*sec.*)
$CH_3CH_2CH_2CH_2OH$	*n*-Butyl alcohol	1-Butanol (*pri.*)
$\begin{array}{l} \quad OH \\ \quad \vert \\ CH_3CH_2CHCH_3 \end{array}$	*sec.*–Butyl alcohol	2-Butanol (*sec.*)
$\begin{array}{l} \quad CH_3 \\ \quad \vert \\ H_3C—C—OH \\ \quad \vert \\ \quad CH_3 \end{array}$	*tert*-Butyl alcohol	2-Methyl-2-propanol (*ter.*)

where *pri.* = Primary; *sec.* = secondary; and *ter.* = tertiary.

ILLUSTRATION 2

Draw the structure of 2-Methyl-4-chloro pentanol

SOLUTION

$$CH_3-\underset{\displaystyle Cl}{\overset{\displaystyle H}{\underset{|}{\overset{|}{C}}}}-CH_2-\overset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-OH$$

CHECK Point

☛ **Give IUPAC name of following compound**

$$CH_3-\overset{\displaystyle CH_3}{\underset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}-CH_2-\overset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2OH$$

SOLUTION

2, 4, 4-Trimethyl pentanol

(ii) ***Dihydric alcohols*** : The alcohols having two hydroxyl groups are known as dihydric alcohols, or *glycols* or *alkanediols*.

Since compounds having two hydroxyl groups on the same carbon atom are usually unstable, the two hydroxyl groups of glycols must be attached to different carbon atoms as glycols are quite stable compounds.

General foumula	*Characteristic group*
$C_nH_{2n}(OH)_2$	$>C-OH$ $>C-OH$

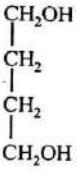
IUPAC names of the dihydric alcohols are derived by adding the suffix *–diol* to the name of the corresponding alkane and indicating the positions of the –OH groups by the numerals.

Formula	*Common name*	*IUPAC name*
$CH_2OH.CH_2OH$	Ethylene glycol	1, 2-Ethanediol
$CH_3CHOH.CH_2OH$	Propylene glycol	1, 2-Propanediol
$CH_2OH.CH_2CH_2OH$	Trimethylene glycol	1, 3-Propanediol

CHECK Point

☛ **Draw the structure of 1, 4-Butanediol**

SOLUTION



Structure of 1, 4-Butanediol

(iii) ***Trihydric alcohols*** **:** These contain three hydroxyl groups attached to different carbon atoms.

General foumula	*Characteristic group*
$C_nH_{2n-1}(OH)_3$	$-\overset{\vert}{\underset{OH}{C}}-\overset{\vert}{\underset{OH}{C}}-\overset{\vert}{\underset{OH}{C}}-$

IUPAC names are derived by adding the suffix *–triol* to the name of the corresponding alkane and indicating the positions of the –OH groups by numerals. In case the compound contains more than three hydroxyl groups, it is known as polyhydric alcohol. In IUPAC system these are named by putting a suffix polyol after the name of the parent hydrocarbon. For example,

Formula	*Common name*	*IUPAC name*
$CH_2OH.CHOH.CH_2OH$	Glycerol	Propane-1, 2, 3-triol
$CH_2OH.(CHOH)_4.CH_2OH$	Sorbitol	Hexane-1, 2, 3-hexol

6. Ethers : These are derived from alkanes by the replacement of one hydrogen atom by an alkoxy group (–OR), thus the general formula for ethers becomes R–O–R.

When the alkyl groups are same, the ether is called as simple ether; and when they are different, it is called mixed ether. Thus

General formula : $\underset{\text{Simple ether}}{R-O-R}$ $\underset{\text{Mixed ether}}{R-O-R'}$

The ether with same alkyl group are called symmetrical ether whereas the others containing different alkyl group are called unsymmetrical ether.

Characterstic group : –O–

The IUPAC (systematic) name for ethers is **alkoxyalkanes**.

Formula	*Common name*	*IUPAC name*
CH_3–O–CH_3	Dimethyl ether	Methoxymethane
CH_3–O–C_2H_5	Ethyl methyl ether	Methoxyethane
C_2H_5–O–C_2H_5	Diethyl ether	Ethoxyethane

7. **Thio-alcohols, Thiols, Mercaptans or Alkanethiols :** Thioalcohols are derived by replacing O (*oxygen*) atom of alcohol by S (sulphur) atom.

General formula R–SH *Functional group* –SH (mercapto or thiol group)

The secondary suffix used for thio alcohols or mercaptans in IUPAC system is *thiol* (in place of *–ol* used in alcohols), while the common names are obtained by adding *thio alcohol* to the name of the alkyl group.

Formula	*Common name*	*IUPAC name*
CH_3SH	Methyl mercaptan *or* Methyl thioalcohol	Methanethiol
C_2H_5SH	Ethyl thioalcohol	Ethanethiol
C_4H_9SH	Butyl thioalcohol	Butanethiol
$C_{10}H_{21}SH$	Decyl thioalcohol	Decanethiol

ILLUSTRATION 3

Write the IUPAC name of following compound.

$$CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-SH$$

SOLUTION :

2-Bromopropanethiol

8. **Thioethers or Alkyl sulphides :** These are obtained by replacing O atom of ethers by S atom. Thus

General formula R–S–R *Charactetistic group* –S–

where the two R groups may be same (simple thioether) or different (mixed thioether). The common and IUPAC names for thioethers are as below.

Formula	*Common name*	*IUPAC name*
CH_3-S-CH_3	Dimethyl thioether	Dimethyl sulphide
$CH_3-S-C_2H_5$	Methyl ethyl thioether	Ethyl methyl sulphide
$C_2H_5-S-C_2H_5$	Diethyl thioether	Diethyl sulphide

9. **Aldehyde or Alkanals :** These are derived from alkanes by the replacement of one hydrogen atom by aldehyde (–CHO) group.

General foumula : $C_nH_{2n+1}CHO$ or R–CHO

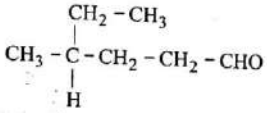
Functional group : –CHO

Formula	*Length of carbon chain*	*Common name*	*IUPAC name*
HCHO	C_1	Formaldehyde	Methanal
CH_3CHO	C_2	Acetaldehyde	Ethanal
CH_3CH_2CHO	C_3	Propionaldehyde	Propanal
C_3H_7CHO	C_4	Butyraldehyde	Butanal

While counting the carbon atoms in the parent chain, the carbon of the –CHO *group is also counted.*

CHECK Point

☛ **Write the IUPAC name of following compound ?**



SOLUTION

4-Ethylpentanol

10. **Ketones or Alkanones :** These are derived by replacing two hydrogen atoms from two molecules of alkanes by a divalent ketonic (> C = O) group.
General formula : $C_nH_{2n+1}-CO-C_nH_{2n+1}$ or $R-CO-R$
Characterstic group : $>C=O$
The two alkyl groups of a ketone may be similar (*simple ketones*) or different (*mixed ketones*).
The IUPAC names (alkanones) are derived by replacing terminal '*e*' of the alkane by '*one*'. The position of the CO group is indicated by placing the number of the carbon atom to which oxygen is attached at the end. For example,

Formula	*Common name*	*IUPAC name*
$CH_3.CO.CH_3$	Dimethyl ketone	Propanone–2
$CH_3.CO.C_2H_5$	Ethylmethyl ketone	Butanone–2
$C_2H_5.CO.C_2H_5$	Diethyl ketone	Pentanone–3

The prefix 2 or 3, indicates the position of the functional group carbon atom which must always get the lowest possible number.

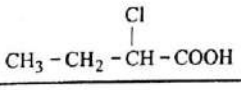
11. **Carboxylic acids :** These are derived by replacing one or two hydrogen atoms of the alkanes by carboxylic (–COOH) groups.
(i) **Monocarboxylic acids, fatty acids or alkanoic acids :** These acids contain one –COOH group. Their IUPAC names are derived from the name of the corresponding alkane by replacing the final –*e* with –*oic* acid.
General foumula : $C_nH_{2n+1}COOH$ or R–COOH
Functional group : –COOH

Formula	*Length of carbon chain*	*Common name*	*IUPAC name*
HCOOH	C_1	Formic acid	Methanoic acid
CH_3COOH	C_2	Acetic acid	Ethanoic acid
C_2H_5COOH	C_3	Propionic acid	Propanoic acid
C_3H_7COOH	C_4	Butyric acid	Butanoic acid
$C_{15}H_{31}COOH$	C_{16}	Palmitic acid	Hexadecanoic acid
$C_{17}H_{35}COOH$	C_{18}	Stearic acid	Octadecanoic acid

CHECK Point

☛ **Draw the structure of compound 2-chlorobutanoic acid**

SOLUTION



(ii) **Dicarboxylic acids or alkanedioic acids :** These compounds contain two –COOH group attached to the same or different carbon atoms.

General foumula : $C_nH_{2n}(COOH)_2$

Functional group : $-\overset{|}{C}-COOH$ $-\overset{|}{C}-COOH$ or $>C\begin{matrix}COOH\\COOH\end{matrix}$

Their IUPAC names are derived by adding the word *–dioic acid* to the name of the alkane. Thus

Formula	*Length of carbon chain*	*Common name*	*IUPAC name*
$\begin{matrix}COOH\\\vert\\COOH\end{matrix}$	C_2	Oxalic acid	Ethanedioic acid or Dicarboxylic acid
$CH_2\begin{matrix}COOH\\COOH\end{matrix}$	C_3	Malonic acid	Propanedioic acid or Methanedicarboxylic acid
$\begin{matrix}CH_2-COOH\\\vert\\CH_2-COOH\end{matrix}$	C_4	Succinic acid	Butanedioic acid or 1, 2-Ethanedicarboxylic acid
$\begin{matrix}CH.COOH\\\Vert\\CH.COOH\end{matrix}$	C_4	Maleic acid or Fumaric acid	Butene-2, 3-dioic acid

The tri– or poly– carboxylic acids are known as alkanetrioic acid or alkane-polyoic acid.

12. Acid derivatives : These are derived by the replacement of an atom or group of atoms of the carboxylic group by other atom or group of atoms. There are four important acid derivatives.

(i) **Acid halides or acyl halides :** These are obtained by the replacement of the –OH group of the –COOH group by a halogen atom. Their IUPAC (systematic) names are obtained by replacing the '*e*' of alkane '*oyl halide*'. Alternatively, they can be named by replacing the suffix *–oic acid* of the carboxylic acids by *–oyl halide*.

General foumula : $C_nH_{2n+1}COX$ or R–COX

Functional group : –COX

Formula	*Length of carbon chain*	*Common name*	*IUPAC name*
$HCOCl$	C_1	Formyl chloride	Methanoyl chloride
CH_3COBr	C_2	Acetyl bromide	Ethanoyl bromide
CH_3CH_2COCl	C_3	Propionyl chloride	Propanoyl chloride

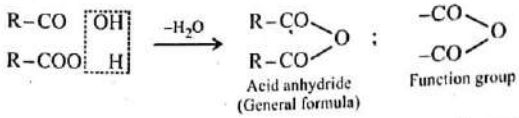
(ii) **Acid amides :** These are derived by the replacement of –OH part of the –COOH group by an amino ($–NH_2$) group. Their IUPAC names are derived by replacing the '*e*' of alkane by '*amide*' or by replacing the suffix *–oic acid* of the carboxylic acid by amide.

General formula : $C_nH_{2n+1}CONH_2$ or $R–CONH_2$

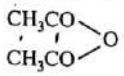
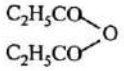
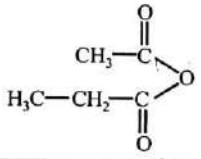
Functional group : $–CONH_2$

Formula	*Corresponding acid*	*Common name*	*IUPAC name*
$HCONH_2$	HCOOH (Formic acid or Methanoic acid)	Formamide	Methanamide
CH_3CONH_2	CH_3COOH (Acetic acid or Ethanoic acid)	Acetamide	Ethanamide
$C_3H_7CONH_2$	C_3H_7COOH (Propionic acid or Propanoic acid)	Propionamide	Propanamide

(iii) **Acid anhydrides :** These are considered to be derived from two (similar or different) carboxylic acid molecules by the elimination of a molecule of water.



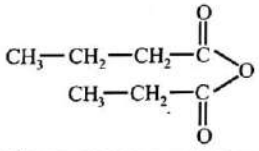
The common names of acid anhydrides are derived from the name of the parent acid or acids; while the systematic names are obtained by replacing '*acid*' from the name of the acid by 'anhydride', e.g.,

Formula	*Parent acid*	*Common name*	*IUPAC name*
	CH_3COOH, Acetic acid (Ethanoic acid)	Acetic anhydride	Ethanoic anhydride
	C_2H_5COOH, Propionic acid (Propanoic acid)	Propionic anhydride	Propanoic anhydride
	Acetic and propionic acid	Acetic propionic anhydride	Acetic propanoic anhydride

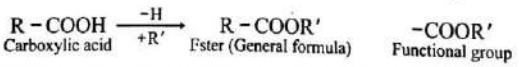
CHECK Point

Draw the structure of butanoic propanoic anhydride

SOLUTION



(iv) **Esters :** These are derived by the replacement of hydrogen atom of the carboxylic acid by an alkyl group. e.g.,



Formula	*Corresponding acid*	*Common name*	*IUPAC name*
$HCOOCH_3$	HCOOH (Formic acid)	Methyl formate	Methyl methanoate
CH_3COOCH_3	CH_3COOH (Acetic acid)	Methyl acetate	Methyl ethanoate
$C_2H_5COOCH_3$	C_2H_5COOH (Propionic acid)	Methyl propionate	Methyl propanoate
$CH_3COOC_2H_5$	CH_3COOH (Acetic acid)	Ethyl acetate	Ethyl ethanoate

13. **Amines :** Amines can be regarded as derived from ammonia by the replacement of one, two or three hydrogen atoms by the corresponding number of alkyl groups. Amines are of three types depending upon the number of alkyl groups attached to nitrogen.
   (i) **Primary amines :** These are monoalkyl derivatives of ammonia.
   (ii) **Secondary amines :** These are dialkyl derivatives of ammonia.

(iii) **Tertiary amines :** These are trialkyl derivatives of ammonia.

Types of amine	*General formula*	*Functional group*
Primary amine	$R–NH_2$	$–NH_2$
Secondary amine	R–NH–R	>NH
Tertiary amine	$R_3–N$	≥N

Common names of amines are derived according to the alkyl group (s) attached to the nitrogen atom and adding the suffix amine.

The systematic names for primary, secondary and tertiary amines are aminoalkanes, alkylaminoalkanes, and dialkylaminoalkanes respectively.

Formula	*Type of amine*	*Common name*	*IUPAC namd*
CH_3NH_2	Primary	Methyl amine	Aminomethane
$C_2H_5NH_2$	Primary	Ethyl amine	Aminoethane
$CH_3.NH.CH_3$	Secondary	Dimethyl amine (N–Methyl methyl amine)	Methylaminomethane
$CH_3.NH.C_2H_5$	Secondary	Ethyl methyl amine (N–Methyl ethyl amine)	N–Methylaminoethane
$(CH_3)_3N$	Tertiary	Trimethyl amine	N, N–Dimethylaminoethane
$(CH_3)_2N.C_2H_5$	Tertiary	Ethyl dimethyl amine	(N, N–Dimethyl ethyl amine)

14. Alkyl cyanides, or Carbonitriles : These are derived by replacing a hydrogen atom of the alkanes or a –COOH group of the acid by nitrile (–CN) group.

General foumula : $C_nH_{2n+1}CN$ or R–CN

Functional group : –CN or $-C \equiv N$

Their common names are obtained by adding suffix *cyanide* to the alkyl group or by replacing *ic acid* from the name of the corresponding acid by the suffix *onitrile*, while systematic names are derived by adding suffix *nitrile* to the parent alkane, e.g.

Formula	*Length of carbon chain*	*Common name*	*IUPAC name*
HCN	C_1	Hydrogen cyanide	*Methanenitrile*
CH_3CN	C_2	Methyl cyanide or Acetonitrile	*Ethanenitrile*
C_3H_7CN	C_4	Propyl cyanide or Propiono nitrile	*Butanenitrile*

While counting number of carbon atom in parent chain the carbon atom present in –CN group is also taken into consideration.

15. Alkyl isocyanides, Isonitriles, Carbylamines : These may be derived by replacing a hydrogen atom of alkanes or a –COOH group of carboxylic acid by the isonitrile (–NC) group. Thus

General formula : $C_nH_{2n+1}NC$ or RNC

Functional group : $-N \equiv C$ or –NC

In the trivial (common) system these are named as alkyl isocyanides or alkyl isonitriles; while in the systematic names, the word carbylamine is added to the name of the alkyl radical.

Formula	*Common name*	*IUPAC name*
CH_3NC	Methyl isocyanide	Methyl carbylamine
C_2H_5NC	Ethyl isocyanide	Ethyl carbylamine
C_4H_6NC	Butyl isocyanide	Butyl carbylamine

16. **Nitroalkanes or Nitroparaffins :** These may be regarded as derived from alkanes by the replacement of a hydrogen atom by $-NO_2$ (nitro) group.

General foumula : $R-NO_2$

Functional group : $-NO_2$ or $-N\begin{matrix}=O\\ \searrow O\end{matrix}$

The common and systematic names are generally the same.

They are named by adding the prefix *nitro* before the name of the corresponding alkane, e.g.

CH_3NO_2 Nitromethane $C_2H_5NO_2$ Nitroethane

17. **Alkyl nitrites :** These are derived from alkanes by the replacement of a hydrogen atom by nitrite ($-O-N=O$) group.

General formula : $R-O-N=O$

Functional group : $-O-N=O$

The common and IUPAC names are same. These are named by adding the suffix nitrite to the name of the alkyl group, e.g.

$CH_3-O-N=O$ Methyl nitrite

$CH_3CH_2CH_2-O-N=O$ *n*-Propyl nitrite

CHECK Point

☛ **Which of the following compound is Nitroethane?**

(a) $CH_3CH_2NO_2$

(b) CH_3CH_2ONO

SOLUTION

A represents Nitromethane as it contains nitro group whereas B is Ethylnitrite.

Knowledge ENHANCER

Rules for Nomenclature of poly functional organic compounds

1. *If the organic compound contains two or more functional groups, one of the group is selected as principal functional group containing longest possible chain of carbon atom and all the remaining functional groups are treated as substituents.*

   Order of preferance used while selecting the principal functional group

   Carboxylic acids > Sulphonic acids > anhydrides > esters > acid chlorides > acid amides > nitriles > aldehydes > cyanides > isocyanides > ketones > alcohols > phenols > thiols > amines > ethers > alkenes > alkynes.

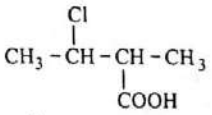
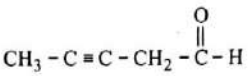
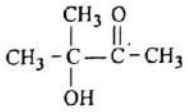
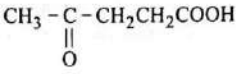
2. *Select the longest continuous chain of carbon atoms containing the principal functional group and maximum number of secondary functional groups and multiple bonds.*

3. *Number the principal chain in such a way that the principal functional group gets the lowest locant followed by double bond, triple bond and the substituents.*

   Note : How you have to write secondary functional group while writing name of a particular compound is already discussed in previous knowledge enhancer as secondary prefix.

CHECK Point

☛ **Write IUPAC names of following compound :**

1. 
2. 
3. 
4.
5. 

SOLUTION

1. 3-Chloro-2-methylbutan-1-oic acid
2. Pent-3-yn-1-al
3. 3-Hydroxy-3-methylbutan-2-one
4. 4-Aminopentan-2-ol
5. 4-Oxopentanoic acid

idea box

Tell your friends to go through to go through the concept of nomenclature of organic compounds throughly. Then divide your friends among two groups and you also become a part of one group. Play a quiz sort of game like one group can show the structure of any compound and can ask for its correct IUPAC name or can give IUPAC name and ask to draw structure of that compound. Then other group can pose a question. Keep 10 marks for right answer and there is no negative marking as such. lets see which group win the quiz.

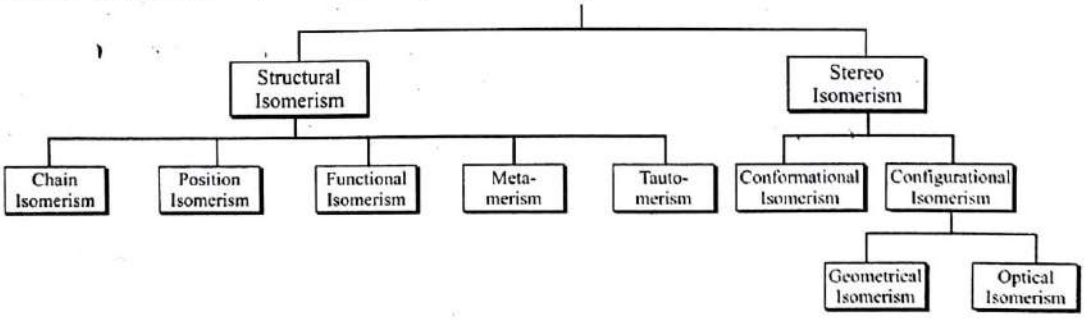
ISOMERISM :

In the study of organic chemistry we come across many cases when two or more compounds are made of equal number of like atoms. A molecular formula does not tell the nature of organic compound, sometimes several organic compounds may have same molecular formula. These compounds possess the same molecular formula but differ from each other in physical or chemical properties, are called isomers and the phenomenon is termed isomerism proposed by Berzelius (Greek, isos = equal ; meros = parts). Since isomers have the same molecular formula, the difference in their properties must be due to different modes of the combination of arrangement of atoms within the molecule. Broadly speaking, isomerism is of two types.

(i) Structural Isomerism (ii) Stereoisomerism

Types of Isomerism

The following figure shows the pictorial representation of different types of isomerism



Structural isomerism (constitutional isomerism)

Structural isomers possess the same molecular formula but different arrangement of atoms or groups within the molecules. The phenomenon is called structural isomerism.

The term constitutional isomerism is a more modern term of structural isomerism. It arises because of the difference in the sequence of covalently bonded atoms in the molecule without reference to space.

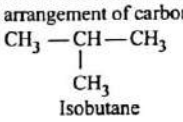
e.g., The oxygen in ethanol is bonded to a carbon and to a hydrogen while the oxygen in dimethyl ether is bonded to two carbon CH_3-CH_2-O-H, CH_3-O-CH_3 though these compounds have same molecular formula C_2H_6O.

Types of structural isomerism :

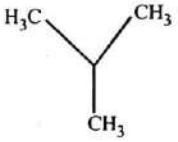
1. Chain/Nuclear/Skeleton isomerism

This type of isomerism arises from the difference in the structure of carbon chain which forms the nucleus of the molecule. This type of compounds have same molecular formula but different arrangement of carbon chain.

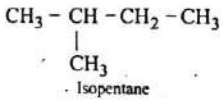
Ex. C_4H_{10} : $CH_3-CH_2-CH_2-CH_3$ (n-Butane) and



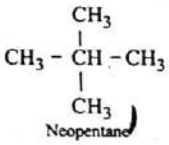
Isobutane



C_5H_{12} : $CH_3-CH_2-CH_2-CH_3$ (n-Pentane)

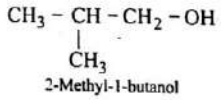


Isopentane



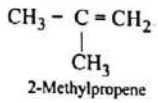
Neopentane

$C_4H_{10}O$: $CH_3-CH_2-CH_2-CH_2-OH$ (1-Butanol) and



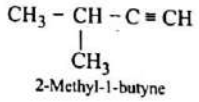
2-Methyl-1-butanol

C_4H_8 : $CH_3-CH_2-CH=CH_2$ (1-Butene)



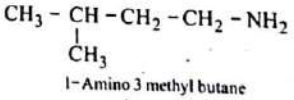
2-Methylpropene

C_5H_8 : $CH_3-CH_2-CH_2-C\equiv CH$ (1-Pentyne)

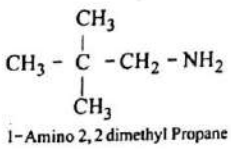


2-Methyl-1-butyne

$C_5H_{13}N$: $CH_3-CH_2-CH_2-CH_2-CH_2-NH_2$ (1- Amino Pentane)



1-Amino 3 methyl butane



1-Amino 2, 2 dimethyl Propane

Chain isomers have somewhat different physical and chemical properties, n-butane boiling at –0.5°C and isobutane at –10.2°C.

- *Chain isomerism is observed when the number of carbon atoms is four or more than four in an organic compounds. Chain isomers differ in the nature of carbon chain, i.e., in the length of the carbon chain.*
- *The isomers showing chain isomerism belong to the same homologous series, i.e., functional group, class of the compound (cyclic or open) remains unchanged.*

2. Position Isomerism

It is the type of isomerism in which the compounds possessing same molecular formula differ in their properties due to the difference in the position of either the functional group or the multiple bond or the branched chain attached to the main carbon chain. For example:

C_4H_8 : $CH_3-CH_2-CH=CH_2$ (1-Butene) ; $CH_3-CH=CH-CH_3$ (2-Butene)

C_3H_6 : $CH_3-CH_2-C\equiv CH$ (1-Butyne) ; $CH_3-C\equiv C-CH_3$ (2-Butyne)

C_3H_7Cl : $CH_3-CH_2-CH_2-Cl$ (1-Chloropropane) ; $CH_3-\underset{Cl}{\underset{|}{CH}}-CH_3$ (2-Chloropropane)

$C_2H_4Cl_2$: $CH_3-CH\begin{matrix}Cl\\Cl\end{matrix}$ (1,1-Dichloroethane) ; $\begin{matrix}CH_2-CH_2\\|\quad\quad|\\Cl\quad Cl\end{matrix}$ (1,2-Dichloroethane)

C_3H_8O : $CH_3-CH_2-CH_2-OH$ (1-Propanol) ; $CH_3-\underset{OH}{\underset{|}{CH}}-CH_3$ (2-Propanol)

$C_5H_{10}O$: $CH_3-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$ (2-Pentanone) ; $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_2-CH_3$ (3-Pentanone)

- *Aldehydes, carboxylic acids (and their derivatives) and cyanides do not exhibit positional isomerism.*
- *Chain and positional isomerism cannot be possible together between two isomeric compounds. If two compounds are chain isomers then these two will not be positional isomers.*

(C) Functional Isomerism : If the molecules have the same molecular formula but differ in the type of the functional group, then it is known as functional group isomerism.

The following pairs of families show this isomerism.

(i) Monohydric alcohol and ether (ii) Aldehyde and ketone (iii) Acid and ester
(iv) Cyanide and isocyanide (v) nitroalkane and alkyl nitrite
(vi) Oxime, amide and many more (vii) Alkene and Cycloalkane
(viii) Alkyne, Alkadiene and Cycloalkene

Ex. (i) Molecular formula : C_2H_6O

CH_3-CH_2-OH and CH_3-O-CH_3 : Functional isomers.

Ethyl alcohol (Alcohol) — Dimethyl ether (Ether)

(ii) Molecular formula : C_3H_6O

$CH_3-CH_2-\overset{O}{\overset{\|}{C}}-H$ and $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$: Functional isomers

Propanal (Aldehyde) — Propanone (Ketone)

(iii) Molecular formula : $C_3H_6O_2$

CH_3-CH_2-COOH and $CH_3-\overset{O}{\overset{\|}{C}}-O-CH_3$: Functional isomers

Propanoic acid (Acid) — Methyl acetate (Ester)

(iv) Molecular formula : CH_3NO_2

$CH_3-\overset{O}{\overset{\|}{N}}\rightarrow O$ $CH_3-O-N=O$: Functional isomers

Nitromethane (Nitro) — Methyl nitrite (Nitrite)

CHECK Point

1. Which type of isomerism following pair of compound represents?

$CH_3-\overset{O}{\overset{\|}{C}}-OH$ (Acetic acid) **and** $H-\overset{O}{\overset{\|}{C}}-OCH_3$ (Methyl formate)

SOLUTION

Above pair of compounds represents functional isomerism.

ILLUSTRATION 4

Which type of isomerism is exhibit by following compound

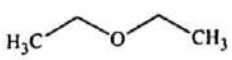
$CH_3-C\equiv N$ (A) $CH_3-N\ddot{=}C$ (B)

SOLUTION :

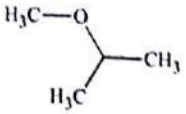
A and B exhibit functional isomerism

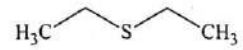
(iv) Metamerism : The compounds having same molecular formula but different number of carbon atoms (or alkyl groups) on either side of functional group, are called metamers.

e.g. ethers, thioethers, secondary amines, ketones, esters etc.

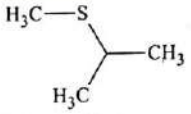
(a)  diethyl ether

1-methoxypropane

 2-methoxypropane

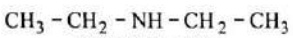
(b) 
(ethylthio)ethane

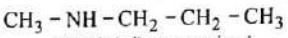
1-(methylthio)propane


2-(methylthio)propane

CHECK Point

☛ **Which type of isomerism is exhibited by following pair of compounds ?**


N-Ethylethanamine


1-(N-Methyl) propanamine

SOLUTION

Above pair of compounds represents metamerism.

(v) **Tautomerism :** This is a special type of functional isomerism in which the isomers differ in the arrangement of atoms but they exist in eynamic equilibrium with each other. For example, acetaldehde and vinyl alcohol are tautomers.

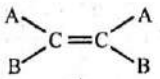
Acetal dehyde ⇌ Vinyl alcohol

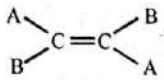
Knowledge ENHANCER

Stereoisomerism was exhibit by compounds which have same structural formula and sequence of bonds but differ in the relative positions of atoms or groups of atoms in space. It is majorly of two types

1. Geometrical isomerism 2. Optical ismorism

Geometrical isomerism also called cis trans isomerism is exhibited by alkenes because of the presence of double bond. This is due to the restricted rotation around carbon-carbon double bond. As a result, the position of the groups attached to these carbons is fixed in space. Here cis-isomer have identical group on either side whereas in trans-isomer identical groups are at opposite side of C = C.


cis


trans

Optical isomerism is shown by substances which can rotate the plane of polarised light

For example this type of isomerism is shown by amino acid alanine.

Note : You will study about stereoisomerism in detail in higher classes.

idea box

Make a list of pair of different type of compounds exhibiting various types of isomerism. Now play with your classmates by showing them the pair of compound and ask which type of isomerism they exhibit and see whether your friends can make out correct answer or not.

1 EXERCISE

FIB Fill in the Blanks :

DIRECTIONS : *Complete the following statements with an appropriate word / term to be filled in the blank space(s).*

1. Trivial name of methanol is _____.
2. The first system developed for naming of _____ compounds was called Geneva system
3. Suffix in IUPAC nomenclature are of two types _____ and secondary
4. A primary suffix indicates the type of _____ in the carbon atoms.
5. _____ is the primary suffix for C ≡ C bond.
6. In IUPAC nomenclature, certain groups are regarded as _____ and are denoted by secondary prefixes.
7. The _____ step in naming of an organic compound is to select longest continuous chain of carbon atom which may or may not be straight.
8. While numbering carbon atoms of long carbon chain numbering should be done is such a manner that substituent gets _____ possible number.
9. Alkenes are characterised by presence of a _____ bond between two carbon atoms
10. Alcohols containing two hydroxyl groups are known as glycols or _____
11. The _____ name for ethers is alkoxy alkanes.
12. Secondary amines are _____ derivatives of ammonia.
13. Structural isomers possess the same _____ formula but different arrangement of atoms or groups with in the molecules.
14. 1-Butene and 2-Butene exhibits _____ isomerism.
15. In _____ the isomers are in dynamic equilibrium with each other.
16. n-butane and isobutane exhibit .....
17. Alkynes exhibit three types of isomerism
    i. ..... ii. ..... iii. .....
18. Metamers include ..... class of compounds.
19. Glucose and fructose are ..... isomers.
20. Isobutyl group is a ..... alkyl group.
21. Among two substituents ..... is assigned to that substituent which comes first in alphabetic order.

T/F True / False :

DIRECTIONS : *Read the following statements and write your answer as true or false.*

1. Lactic acid is called so because it is obtained from sour milk.
2. Prefix is a part of IUPAC name which appear after the word root.
3. Secondary prefix for Ketone is Keto or oxo.
4. A primary prefix "cyclo" is used in order to differentiate a cyclic compound from an acyclic compound.
5. When two or more substituents are attached to the parent chain, these are prefixed in anti alphabetic order.
6. $C_nH_{2n+1}X$ is a general formula for monohalogen derivatives of alkane.
7. $C_nH_{2n+1}(OH)_3$ is a general formula for trihydric alcohols.
8. IUPAC name of compound C_4H_9SH is butanethiol
9. $CH_3 - CH_2 - CH_2 - CH_3$ and $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-CH_3$ are chain isomers.
10. Geometrical isomerism is also known as Cis-trans isomerism
11. A functional group is a reactive part of molecule.
12. Homologues can be isomers.
13. When two or more substituents are present on the parent chain. Such groups are named in alphabetical order irrespective of their position number
14. If an organic compound contains a functional group, multiple bonds and a side chain then the order of preference will
    Double bond > Triple bond > Side chain > Functional group
15. Aldehyde and Ketone show metamerism
16. 1 chloro propane and 2 chloro propane and position isomers.
17. Ketones show metamerism
18. $CH_3-\underset{\displaystyle Cl}{\overset{\displaystyle |}{CH}}-CH_2-CH_2-CH_3$ and $CH_3-CH_2-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{\overset{\displaystyle Cl}{\overset{\displaystyle |}{C}}}}-CH_3$ show both positional and chain isomerism
19. The correct IUPAC name of $CH_3CH_2CH_2CH_3$ is n-butane not butane

MTF Match the Following :

DIRECTIONS : *Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.*

1. Column II give prefix or suffix for functional group mentioned in column I match them correctly.

Column I	Column II
(A) Aldehyde	(p) Chloro
(B) Ketone	(q) ol
(C) Alcohol	(r) one
(D) Halogen	(s) al

2. Column II give formula for compounds given in column I, match them correctly.

Column I	Column II
(A) Propane	(p) C_2H_5OH
(B) ethyl alcohol	(q) C_3H_8
(C) carboxylic acid	(r) CH_3COOH
(D) ethyl ethanoate	(s) $CH_3COOC_2H_5$

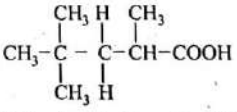
3.

Column-I	Column-II
(A) CH_3COOH & $HCOOCH_3$	(p) Functional isomers
(B) 1 butene & 2-butene	(q) Metamers
(C) diethyl ether & methyl propyl ether	(r) Position isomers
(D) dimethyl ether and ethanol	(s) Chain isomers and ethanol

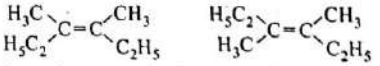
VSAQ Very Short Answer Questions :

DIRECTIONS : *Give answer in one word or one sentence.*

1. Name the simplest ketone.
2. Show a structural formula for the straight-chain isomer of C_5H_{12}. What is the name of the alkyl group formed by removing a hydrogen atom from one of the terminal carbon atoms ?
3. Write the formulation for the functional groups of alcohols and carboxylic acids.
4. Write the (i) name and (ii) formula of the functional group present in the compound, CH_3COOH.
5. How are the molecules of aldehydes and ketones structurally different ?
6. Write the structural formula of any isomer of *n*-heptane (C_7H_{16}).
7. Write the structural formula of one isomer of *n*-hexane (C_6H_{14}).
8. Write molecular formula of (i) ethanol and (ii) propanoic acid.
9. Write the names of functional groups present in (a) ethanol (b) ethanoic acid.
10. Write the IUPAC names of the following :
    (i) CH_3OH (ii) CH_3COOH
11. Write names of two alkanes, one having three carbon atoms and the other having four carbon atoms.
12. Write names of alkenes having two and three carbon atoms respectively.
13. Name the classes of organic compounds represented by the following formulae :
    (i) $C_2H_5NH_2$ (ii) $H_3C-\overset{\overset{O}{\|}}{C}-CH_3$
14. Write the IUPAC name of following compound



15. Write chemical formula of chloral.
16. What type of isomerism is exhibited by following pair of compounds.

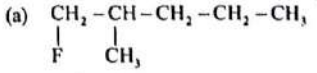


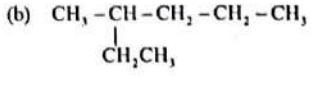
17. Draw the structure of compound
    3, 3 - Dimethyl butan-2-one
18. Give one example of functional isomerism
19. Write the metamer of diethyl ether. What is its IUPAC name?

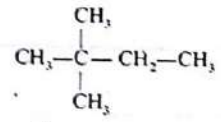
SQA Short Answer Questions :

DIRECTIONS : *Give answer in 2-3 sentences.*

1. Name the following compounds :

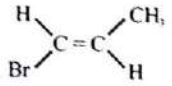
(a) 

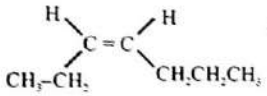
(b) 

(c) 

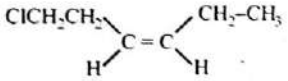
(d)

2. Name the following unsaturated compounds :

(a) 

(b) 

(c) $CH_3C \equiv CCH_2CH_2F$

(d) 

3. Which of the following are formulas of open-chain alkanes
   (a) C_4H_{10} (b) C_3H_7
   (c) C_8H_{14} (d) $C_{10}H_{22}$
   (e) C_7H_{15} (f) C_7H_{14}
   (g) $C_{18}H_{38}$ (h) C_9H_{15}
4. To which class does each of the following compounds belong –
   (a) 3-heptanone (b) 3-nonene
   (c) 2-methylpentane (d) 2-ethylhexanal
   (e) ethylbenzene (f) propanol
5. Identify the following as an alkane, alkene, or alkyne. Assume that the compounds are not cyclic.
   (a) C_8H_{16} (b) C_5H_{12}
   (c) C_4H_8 (d) $C_{20}H_{38}$
   (e) CH_4 (f) $C_{10}H_{20}$
   (g) $C_{18}H_{38}$
6. What category of compounds will be formed when one hydrogen of alkane molecule is replaced by a hydroxyl group? Name the functional groups present in following compounds :

   (i) $R-\overset{\overset{O}{\|}}{C}-OR$ (ii) $R-\overset{\overset{O}{\|}}{C}-OH$
7. Write the formula for the given compounds and name the functional groups present in each of them :
   (i) Ethanoic acid (ii) Propanone (iii) Nitromethane
8. Give IUPAC name of the compound :
   $C_2H_5-CH(CH_3)-CH_2-CH(CH_3)_2$
9. Give IUPAC name of following compound :
   $CH_3-CH=CH-CH_2-C\equiv C-CH_2-CH_3$
10. How do two structural isomers differ from each other?
11. Draw all the structural isomers for hydrocarbons having the molecular formula C_4H_{10}.
12. What primary and secondary suffixes are as applied to IUPAC nomenclature?
13. A compound is formed by the substitution of two chlorine atoms for two hydrogen atoms in propane. What is the number of structural isomers possible ?
14. (a) What are hydrocarbons?
    (b) State two main groups of aliphatic hydrocarbons with examples.
15. Give the names of the following functional groups :- CHO, CO, –OH, –COOH
16. Write the formula of the following :
    (i) Methane (ii) Ethane
    (iii) Methyl alcohol (iv) Ethyl alcohol
    (v) Methyl chloride (vi) Ethyl chloride
17. List three characteristics of isomers.
18. Write the structural formula of isomers of pentane other than n-pentane.

Long Answer Questions :

DIRECTIONS : *Give answer in four to five sentences.*

1. Tell how the following differ in structure.
   (a) alcohols and ethers
   (b) aldehydes and ketones
   (c) amines and amides
   (d) carboxylic acids and esters
2. Two compounds 'A' and 'B' have the same molecular formula $C_4H_8O_2$. Compound 'A' is an acid and compound 'B' has a fruity smell. Suggest (i) chemical formulae and (ii) the structural formulae of compounds A and B. Name the functional group of compound B. What name would you give to the relationship between the compounds A and B?
3. Draw all the structural isomers for hydrocarbons having the molecular formula C_6H_{14}.
4. Draw the relevant structural formula of :
   (a) ethane (b) Marsh gas
   (c) ethene (d) acetylene
   (e) 2-Butyne (f) propylene
   (g) 1-Butyne (h) 1-Butene
   (i) Propane (j) n-Butane
5. Define isomerism. How many isomers are possible for a alkane having molecular formula C_4H_{10}?
6. Write the structural formula for each of the following alkanes:
   (a) 2-methylhexane
   (b) 3,3-dichloroheptane
   (c) 2-methyl-3-phenyloctane
   (d) 1,1-diethylcyclohexane

2 EXERCISE

MCQ Multiple Choice Questions :

DIRECTIONS : *This section contains 42 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. The functional group present in $CH_3COOC_2H_5$ is –
(a) ketonic (b) aldehydic
(c) ester (d) carboxylic

2. The IUPAC name of the compound having the formula $(CH_3)_3CCH=CH_2$ is –
(a) 3,3,3-trimethyl-1-propane (b) 1,1,1-trimethyl-1-butene
(c) 3,3-dimethyl-1-butene (d) 1,1-dimethyl-1,3-butene

3. The IUPAC name of the following compound is :
$CH_2=CH-CH(CH_3)_2$
(a) 1, 1dimethyl-2-propene (b) 3-methyl-1-butene
(c) 2-vinyl propane (d) 1-isopropyl ethylene

4. The IUPAC name of :
$CH_3-C(CH_3)(OH)CH_2-CH(CH_3)CH_3$ is –
(a) 2,4-dimethyl pentane-2-ol
(b) 2,4-dimethyl pentane-4-ol
(c) 2,2-dimethyl butane
(d) butanol-2-on

5. Butanone is a four-carbon compound with the functional group –
(a) carboxylic acid (b) aldehyde.
(c) ketone (d) alcohol.

6. Which of the following is incorrectly matched –
(a) vinegar → carboxylic acid
(b) C_2H_6 → alkane
(c) ethanol → alcohol
(d) methanol → ketone

7. The number of structural isomer for an alkane with a molecular weight 72 is –
(a) 2 (b) 3
(c) 4 (d) 5

8. In C_6H_{14}, the number of possible isomers is –
(a) 3 (b) 6
(c) 4 (d) 5

9. If a hydrocarbon has any double bond, it is –
(a) alkyne (b) alkane
(c) alkene (d) all the above

10. The functional group present in organic, acid is –
(a) –OH (b) –CHO
(c) –COOH (d) >C=O

11. The IUPAC name of $CH_3\underset{\displaystyle CH_2CH_3}{\underset{|}{C}}HCH_2CH_3$ is –
(a) 1,1-methylethylpropane
(b) 2-ethylbutane
(c) 1-methyl-1-ethylpropane
(d) 3-methylpentane

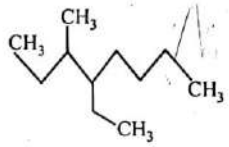
12. Alkynes are characterized by –
(a) C – C bonds (b) C = C bonds
(c) C ≡ C bonds (d) cyclic structure

13. The fruity smell is of a/an –
(a) aldehyde (b) ketone
(c) alcohol (d) ester

14. How many different isomers are possible for a hydrocarbon with the molecular formula C_4H_{10} –
(a) 1 (b) 2
(c) 3 (d) 5

15. Which of these contains the carbonyl group?
(a) ketones (b) aldehydes
(c) esters (d) all of these

16. Name of the given compound -



(a) 2, 3-diethyl heptane (b) 5-ethyl-6-methyl octane
(c) 4-ethyl-3-methyl octane (d) 3-methyl-4-ethyl octane

17. What is the IUPAC Name of t-butyl alcohol.
(a) Butanol–2 (b) 2–Methyl–propan–2-ol
(c) Butanol–1 (d) Propanol–2

18. The IUPAC name of $CH_3COCH(CH_3)_2$ is -
(a) isopropyl methyl ketone
(b) 2-methyl-3-butanone
(c) 4-methylisopropyl ketone
(d) 3-methyl-2-butanone

19. The general formula $C_nH_{2n}O_2$ could be for open chain -
(a) diketones (b) carboxylic acids
(c) diols (d) dialdehydes

20. $CH_3CH_2-\underset{\displaystyle CH_3}{\underset{|}{C}}H-\underset{\displaystyle CHO}{\underset{|}{C}}H-CH_2CH_3$ has the IUPAC name –
(a) 2–sec Butylbutanal
(b) 2, 3–Diethylbutanal
(c) 2–Ethyl–3–methylpentanal
(d) 3–Methyl–2–ethylpentanal

21. The name of $ClH_2C-\underset{\displaystyle Br}{\underset{|}{C}}=\underset{\displaystyle Br}{\underset{|}{C}}-CH_2Cl$ according to IUPAC system is –
(a) 2,3–Dibromo–1,4–dichlorobutene–2
(b) 1,4–Dichloro–2,3–dibromobutene–2

(c) Dichlorodibromobutene
(d) Dichlorodibromobutane

22. Compound $CH_3CH_2CH(CH_3)CH_2\overset{\overset{O}{\|}}{C}-Cl$ has the IUPAC name –
(a) 3–Methylpentanoyl chloride
(b) 1–Chloroformyl1–2–methylbutane
(c) 1–Chloroformyl–1–sec butylmethane
(d) None

23. The incorrect IUPAC name is –
(a) $CH_3-\overset{\overset{O}{\|}}{C}-CH(CH_3)-CH_3$ 2–Methyl–3–butanone
(b) $CH_3-\underset{CH_3}{\overset{|}{CH}}-\underset{CH_2CH_3}{\overset{|}{CH}}-CH_3$ 2, 3–Dimethylpentane
(c) $CH_3-C\equiv CCH(CH_3)_2$ 4–Methyl–2–pentyne
(d) $CH_3-\underset{Cl}{\overset{|}{CH}}-\underset{Br}{\overset{|}{CH}}-CH_3$ 2–Bromo–3–chlorobutane

24. The IUPAC name of $CH_3-CH_2-\underset{CH_3CH_3}{\overset{H\quad C_4H_9}{C-C}}-CH_3$: –
(a) 3, 4, 4–Trimethyl octane
(b) 3, 4, 4–Trimethyl heptane
(c) 2–Ethyl, 3,3–dimethyl heptane
(d) 2–Butyl, 2 methyl,3–ethyl butane

25. The IUPAC name of the compound
$CH_3-\underset{CH_3}{\overset{|}{CH}}-CH_2-CH_2-Cl$ is –
(a) 1–Chloro–3–methylbutane
(b) 2–Methyl–4–chlorobutane
(c) 2–Methyl–1–chlorobutane
(d) 1–Chloropentane

26. The IUPAC name of crotonaldehyde is –
(a) Prop–2–ene–1–al (b) Propenal
(c) But–2–ene–1–al (d) Butenal

27. IUPAC name of the following compound will be
$CH_3-CH=\underset{CH_2-CH_2-CH_3}{\overset{|}{C}}-CH_2-CH_3$
(a) 3–Ethyl–2–hexene (b) 3–Propyl–2–hexene
(c) 3–Propyl–3–hexene (d) 4–Ethyl–4–hexene

28. The IUPAC name of $CH_3COOC_2H_5$ will be –
(a) Ethyl acetate (b) Ethyl ethanoate
(c) Methyl propanoate (d) None of these

29. The IUPAC name of the compound having the formula $Cl_3C.CH_2CHO$ is –
(a) 3, 3, 3–Trichloropropanal
(b) 1, 1, 1–Trichloropropanal
(c) 2, 2, 2–Trichloropropanal
(d) Chloral

30. The IUPAC name for the formula $CH_3-\underset{CH_3}{\overset{|}{C}}=\overset{H}{\overset{|}{C}}-COOH$
(a) 2–Methyl–2–butenoic acid
(b) 3–Methyl–3–butenoic acid
(c) 3–Methyl–2–butenoic acid
(d) 2–Methyl–3–butenoic acid

31. IUPAC name for the compound
$H_3C(Cl)C=C(CH_2CH_3)I$
(a) Trans–3–iodo–4–chloro–3–pentene
(b) Cis–3–chloro–3–iodo–2–pentene
(c) Trans–2–chloro–3–iodo–2–pentene
(d) Cis–3–iodo–4–chloro–3–pentene

32. IUPAC name of the compound
$CH_3-\underset{CH_2}{\overset{|}{CH}}-CH_2-CH(OH)-CH_3$ is –
(a) 4–Ethyl–2–pentanol (b) 4–Methyl–2–hexanol
(c) 2–Ethyl–2–pentanol (d) 3–Methyl–2–hexanol

33. IUPAC name of the compound is $CH_3-CH=\underset{CH_2-CH_3}{\overset{|}{C}}-CH_3$
(a) 2–Ethyl–2–butene (b) 3–Ethyl–2–butene
(c) 3–Methyl–3–pentene (d) 3–Methyl–2–pentene

34. The IUPAC name for $CH_3CHOHCH_2-\underset{CH_3}{\overset{CH_3}{\overset{|}{\underset{|}{C}}}}-OH$ is –
(a) 1, 1–Dimethyl–1,3–butanediol
(b) 2–Methyl–2,4–pentanediol
(c) 4–Methyl–2,4–pentanediol
(d) 1, 3, 3–Trimethyl–1, 3–propanediol

35. The IUPAC name of $CH_3-\underset{Cl}{\overset{|}{C}}=\underset{CH_3}{\overset{|}{C}}-\underset{C_2H_5}{\overset{|}{CH}}-CH_2-C\equiv CH$ is–
(a) 6–Chloro–4–ethyl–5–methyl–hept–5–en–1–yne
(b) 6–Chloro–4–ethyl–5–methyl–hept–1–yn–5–ene
(c) 2–Chloro–4–ethyl–3–methyl–hept–2–en–6–yne
(d) 2–Chloro–4–ethyl–3–methyl–hept–6–yn–2–ene

36. The IUPAC name of the following structure is
$CH_3-\overset{\overset{O}{\|}}{C}-CH_2-COOH$
(a) 3–Ketobutanoic acid (b) 2–Ketobutanoic acid
(c) 4–Ketobutanoic acid (d) 3–Oxopropanoic acid

37. Which is correct IUPAC name of the following compound
$CH_3-\underset{CH_2-CH_3}{\overset{CH_3}{\overset{|}{\underset{|}{CH}}}}-\overset{CH_3}{\overset{|}{CH}}-CH_3$
(a) 3–Isopropyl–2–methylpentane
(b) 3–Ethyl–2,4–dimethylpentane

(c) 2,4–Dimethyl–3–ethylpentane
(d) 3–Isopropyl–4–methylpentane

38. IUPAC name of $CH_3-CH=CH-C\equiv CH$ is –
(a) Pent–2–en–4–yne (b) Pent–3–en–1–yne
(c) Pent–3–yne–1–en (d) Pent–2–yne–1–en

39. IUPAC name of $(CH_3)_2CH-CH_2-CH_2Br$ is –
(a) 1–Bromopentane
(b) 2–Methyl–4–bromobutane
(c) 1–Bromo–3–methylbutane
(d) 2–Methyl–3–bromopropane

40. The IUPAC name of CH_3CH_2COCl is –
(a) Propanoyl chloride (b) Ethanoyl chloride
(c) Acetyl chloride (d) Chloroethane

41. Which of the following statements is false for isopentane–
(a) It has three CH_3 groups
(b) It has one CH_2 group
(c) It has one CH group
(d) It has a carbon which is not bonded to hydrogen

42. General formula of alkenes and alkyl radicals are respectively:
(a) C_nH_{2n} and C_nH_{2n+1} (b) C_nH_{2n} and C_nH_{2n+2}
(c) C_nH_{2n-1} and C_nH_{2n} (d) C_nH_{2n+1} and C_nH_{2n+2}

MTQC More than One Correct

DIRECTIONS : *This section contains 17 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. Which of the following are dihydric alcohols
(a) ethylene glycol (b) trimethylene glycol
(c) glycerol (d) 1, 2-propanediol

2. Which of the following is(are) secondary suffix
(a) al (b) oxo
(c) amino (d) amine

3. Which of the following is(are) secondary prefix?
(a) sulpho (b) carbamoyl
(c) cyano (d) isocyanide

4. Which of the following is(are) alcohol?
(a) $(CH_3)_3C-OH$ (b) $CH_3-O-C_2H_5$
(c) C_3H_7CHO (d) $CH_2OH(CHOH)_4CH_2OH$

5. Which of the following pair shows functional isomerism
(a) CH_3CH_2OH and CH_3OCH_3
(b) CH_3CH_2COOH and $CH_3-\overset{O}{\overset{\|}{C}}-OCH_3$
(c) CH_3COOH and $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$
(d) $CH_3CH_2-\overset{O}{\overset{\|}{N}}\rightarrow O$ and $CH_3CH_2-O-N=O$:

6. Which of the following statements about tautomerism is (are) correct
(a) it is a type of functional isomerism
(b) it is shown by alcohols only
(c) isomers exists in dynamic equilibrium with each other
(d) tautoisomers possess different functional group

7. Which of the following statements about geometrical isomerism is (are) correct
(a) it is also called cis-trans isomerism
(b) it is exhibited by alkenes only
(c) alkynes can also show geometrical isomerism
(d) in cis-isomer identical group are on same side

8. In order of preference while selecting the principal functional group which compounds are preferable to ketones
(a) alcohols (b) anhydrides
(c) aldehydes (d) alkenes

9. Which of the following pairs is (are) isomers
(a) ethanol and dimethyl ether
(b) ethanol and ethoxy ethane
(c) diethyl ether and 1-methoxy propane
(d) isopentane and neopentane

10. Which of the following statements are true for isopentane.
(a) it has three CH_3 groups
(b) it has one CH_2 group
(c) it has one CH group
(d) it has a carbon which is not bonded to hydrogen

11. A hydrocarbon of molecular formula C_5H_{10} could be a
(a) monosubstituted alkene
(b) disubstituted alkene
(c) trisubstituted alkene
(d) tetrasubstituted alkene

12. The compound with molecular formula $C_4H_{10}O$ can exhibit
(a) metamerism (b) functional isomerism
(c) position isomerism (d) optical isomerism

13. Which of the following statements about Ethylmethylamine is true
(a) it is a dialkyl amine
(b) it is a tertiary amine
(c) its IUPAC name is N–methylaminoethane
(d) its IUPAC name is N–ethylaminomethane

14. Which of the following statements about IUPAC system of nomenclature is (are) correct
(a) this system of nomenclature was introduced in 1947
(b) on the basis of this system, the name of an organic compound consists of word root, suffix and prefix
(c) this system was first introduced in 1892
(d) the latest IUPAC system is based on recommendation made in 1993

15. Which of the following are trivial name of organic compounds
(a) vinegar (b) acryl aldehyde
(c) acetic acid (d) acrolein

16. Which of the following compound is (are) metamers
(a) diethyl ether (b) 2-methoxy propane
(c) 1-methoxy propane (d) butanal

17. Which of the following molecules are alkanes –
(a) C_8H_{16} (b) C_4H_9F
(c) C_6H_{14} (d) C_9H_{20}

FIB Fill in the Passage :

DIRECTIONS : *Complete the following passage(s) with an appropriate word/term to be filled in the blank spaces.*

I The functional group in an alcohol is known as a .....(1)..... group. In ethers, the oxygen is attached to two .....(2)..... groups. The functional group in aldehydes and ketones is a .....(3)..... group. In aldehydes, the functional group is attached to at least one .....(4)..... atom. The hetero atom in an amine is a ....(5)..... atom. The functional group of carboxylic acid is a .....(6)..... group. In esters, a hydrocarbon group replaces a .....(7)..... in the carboxylic acid, whereas in amides an amine group replaces the .....(8)..... group of the acid.

II Identify the class of compound from the following functional groups.

		when $R' = CH_3$	when $R' = H$
1.	$R-\overset{\overset{O}{\|}}{C}-O-R'$	(a)	(b) .....
2.	$R-O-R'$	(c)	(d) .....
3.	$R-\overset{\overset{O}{\|}}{C}-R'$	(e)	(f) .....
4.	$R-\overset{\overset{O}{\|}}{C}-NH_2$	(g)	
5.	$R-NH-R'$	(h)	

FBQ Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

PASSAGE – I

Compounds having the same molecular formula but having different properties are said to be isomers. If compounds have same molecular formula but different in the sequence of bonded atoms they are said to be- structural isomers. Structural isomers may be of several types like chain isomers, ring chain isomers, positional isomers, functional isomers, metamers and tautomers

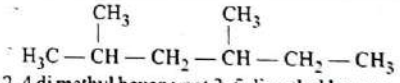
1. How many structural isomers are possible for organic compound having formula $C_4H_{10}O$?
   (a) 4 (b) 5
   (c) 6 (d) 7
2. How many structural isomers are possible for organic compound having molecular formula C_4H_8 ?
   (a) 4 (b) 5
   (c) 6 (d) 7
3. How many structural isomers are possible for organic compound having molecular formula $C_4H_8I_2$?
   (a) 7 (b) 8
   (c) 6 (d) 5

A&R Assertion & Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect.**
(d) If **Assertion** is **incorrect** but **Reason** is **correct.**

1. **Assertion :** A primary suffix indicates the type of linkage in the carbon atom.
   Reason : CN is a Prmary suffix
2. **Assertion :** The general formula for a dihydric alcohol is $C_nH_{2n}(OH)_2$
   Reason : Ethylene glycol is a dihydric alcohol.
3. **Assertion :** The correct IUPAC name for the compound.



   is 2, 4 di methyl hexane not 3, 5 dimethyl hexane
   Reason : In case the parent chain has two or more substitutents numbering must be done in such a way that the sum of the locants on the parent chain is the lowest possible
4. **Assertion :** Chain isomerism is observed in compounds containining four or more than four carbon atoms
   Reason : Only alkanes show chain isomerism
5. **Assertion :** But-1-ene and 2-methylprop-1-ene are position isomers.
   Reason : Position isomers have same molecular formula but differ in position of functional group or C = C.
6. **Assertion :** Ethers are regarded as anhydrides of alcohols.
   Reason : Ethers may be obtained by elimination of water molecule from two alcohol molecules.

MMQ Multiple Matching Questions :

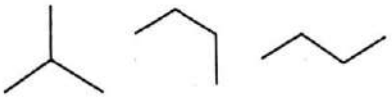
DIRECTIONS : *Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

Column-I		Column-II	
(A)	$C_5H_{10}O$	(p)	Functional isomers
(B)	$C_4H_{10}O$	(q)	Metamerism
(C)	C_4H_8	(r)	Chain isomerism
(D)	$C_5H_{10}O_2$	(s)	Position isomerism

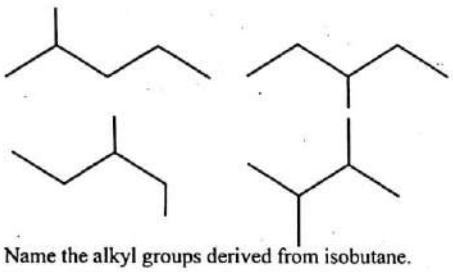
HOTS Subjective Questions :

DIRECTIONS : *Answer the following questions.*

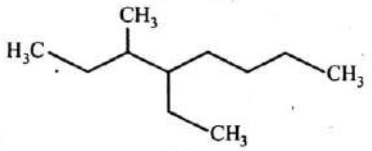
1. How many structural isomers are shown here?



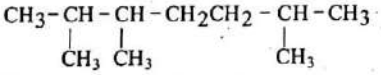
2. Which two of these four structures are of the same structural isomer?



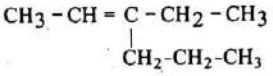
3. Name the alkyl groups derived from isobutane.
4. Name of the compound given below is



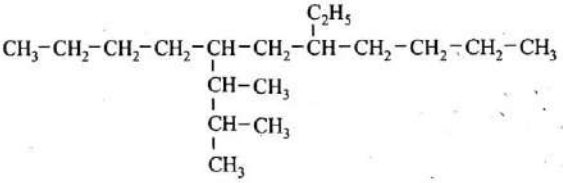
5. Write the IUPAC name of following compound



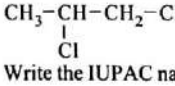
6. IUPAC name of the following compound will be



7. The number of possible open chain (acyclic) isomeric compounds for molecular formula C_5H_{10} would be
8. The total number of isomers for C_4H_8 is
9. What is the main difference is the structural formula of the compounds n-pentane, isopentane and neopentane?
10. (a) What are saturated and unsaturated aliphatic hydrocarbons
    (b) State which of the following are saturated or unsaturated aliphatic hydrocarbons CH_4, C_2H_4, C_3H_6, C_2H_6, C_2H_{-2}, C_3H_4.
11. Give the molecular formula, structural formula, electronic formula and condensed formula of butane.
12. **Fill in the blanks :**
    (a) Organic compounds of carbon and hydrogen containing a double covalent bond betwen the carbon atoms are called .....
    (b) Organic compounds of carbon and hydrogen containing one triple covalent bond between the carbon atoms are called .....
    (c) Organic compound of carbon and hydrogen containing single bonds between the carbon atoms are called .....
13. Write the IUPAC name of following compound



14. Draw the structure of 2, 3 – Dimethyl – 6 – (2– methylpropyl) decane
15. Write the IUPAC name of compound

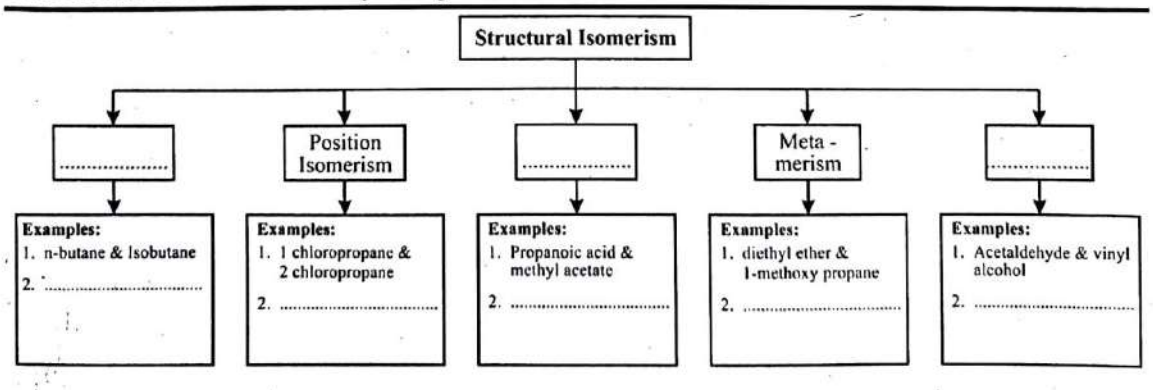


16. Write the IUPAC name of following compound

17. Draw structure of 1–Bromo–3–methylpentan–2–one

PBQ Chart Based Questions :

DIRECTIONS : *Fill in the blanks in the following chart.*



SOLUTIONS

Brief Explanations of Selected Questions

Exercise 1

FILL IN THE BLANKS :

1. Wood spirit
2. Organic
3. primary
4. linkage
5. Yne
6. substituents
7. first
8. lowest
9. double
10. Alkanediols
11. IUPAC
12. Dialkyl
13. molecular
14. position
15. Tautomerism
16. Chain isomerism
17. Chain, position, functional
18. Same
19. Functional
20. Primary
21. Lower locane

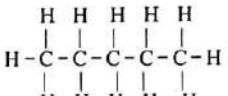
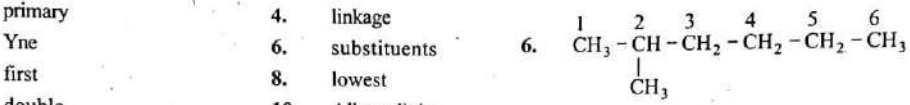
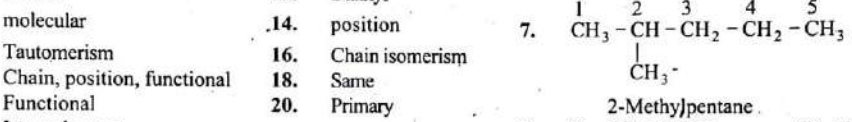
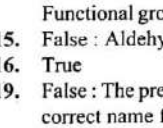
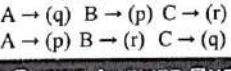
RUE / FALSE

1. True
2. False
3. True
4. True
5. False
6. True
7. False
8. True
9. True
10. True
11. True
12. False
13. True
14. False : The correct order is
    Functional group > Double bond > Triple bond > Side chain
15. False : Aldehydes and Ketone show functional isomerism
16. True
17. True
18. True
19. False : The prefix n- is not part of IUPAC system. Hence the correct name for $CH_3(CH_2)_2CH_3$ is butane, n-butane is its common name, although now-a-days, it too is also acceptable to IUPAC system.

MATCH THE FOLLOWING :

1. A → (s) B → (r) C → (q) D → (p)
2. A → (q) B → (p) C → (r) D → (s)
3. A → (p) B → (r) C → (q) D → (p)

VERY SHORT ANSWER QUESTIONS :

1. The simplest ketone is acetone, CH_3COCH_3.
2. 
   alkyl group = n-pentyl
3. (i) –OH (ii) –COOH
4. (i) Carboxyl (ii) –COOH
5. In aldehyde, $\rangle C = O$ group is attached to 'H' and alkyl group whereas in ketone, $\rangle C = O$ group is attached with two alkyl groups.
6. 
   2-Methylhexane
7. 
   2-Methylpentane
8. (i) CH_3CH_2OH (ii) CH_3CH_2COOH
9. (a) –OH (alcohol) (b) –COOH (carboxyl).
10. (i) Methanol (ii) Ethanoic acid.
11. Propane (C_3H_8) and Butane (C_4H_{10}).
12. Ethene (C_2H_4) and Propene (C_3H_6).
13. (i) Amines (ii) Ketones.
14. 2, 3, 3, – Trimethylpentanoic acid.
15. CCl_3CHO
16. Geometrical or cis-trans isomerism
17. 
18. CH_3CH_2OH and CH_3OCH_3
19. 1-methoxypropane, $CH_3OCH_2CH_2CH_3$ or
    2-methoxypropane 

SHORT ANSWER QUESTIONS :

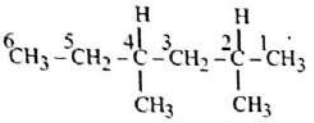
1. (a) 1-fluoro-2-methylpentane
   (b) 3-methylhexane
   (c) 2,2-dimethylbutane
   (d) cyclopropane
2. (a) trans-1-bromopropene
   (b) cis-3-heptene
   (c) 5-fluoro-2-pentyne
   (d) cis – 1 – chloro–3–hexene
3. (a) C_4H_{10} (d) $C_{10}H_{22}$
   (g) $C_{18}H_{38}$
4. (a) ketone (b) alkene
   (c) alkane (d) aldehyde
   (e) aromatic (f) alcohol

5. (a) alkene (b) alkane
(c) alkene (d) alkyne
(e) alkane (f) alkene
(g) alkane

6. Alcohol (i) Ester (ii) Carboxylic acid

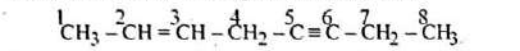
7. (i) CH_3COOH (carboxyl)
(ii) CH_3COCH_3 (ketone)
(iii) CH_3NO_2 (nitro)

8. The compound, $C_2H_5-CH(CH_3)-CH_2-CH(CH_3)_2$ may be written as



The main carbon chain consists of six carbon atoms and hence the parent alkane is hexane. The numbering is done from the right end of the chain. The two methyl groups are attached to the main chain at carbon number 2 and 4. Thus, the name of the compound is, 2, 4 dimethylhexane.

9. The longest chain contains 8 carbons. So, the parent alkane is octane. Numbering of the chain is done from the end nearer to the carbon having double bond. Thus,

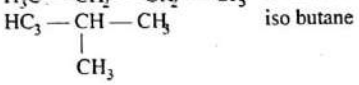


The compound contains one double and one triple bond. So, the suffix would contain both the suffixes ene and yne. Indicating their positions in the chain, the suffix is 2-en-5-yne. So, the molecule is named as, oct-2-en-5-yne.

10. Structural isomers have some molecular formula but differ in arrangements of atoms or groups for example. C_5H_{12} has following isomers

n-pentane iso pentane neo pentane

11. C_4H_{10} have following structural isomer



$H_3C-CH_2-CH_2-CH_3$ n–butane
$HC_3-CH(CH_3)-CH_3$ iso butane

12. The primary suffix indicates whether the carbon chain is saturated or unsaturated while the secondary suffix indicates the functional group present in the molecule.

13. Four: 1, 1–dichloropropane ($CH_3CH_2CHCl_2$), 1, 2–dichloropropane ($CH_3CHClCH_2Cl$), 2, 2–dichloropropane ($CH_3CCl_2CH_3$) and 1, 3–dichloropropane ($ClCH_2CH_2CH_2Cl$).

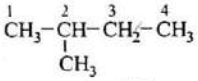
14. (a) **Hydrocarbons :** The covalent compounds of carbon and hydrogen are called hydrocarbons.
(b) Aliphatic hydrocarbons are mainly of two types.
1. Saturated aliphatic hydrocarbons
e.g., Methane, CH_4
Ethane, C_2H_6
2. Unsaturated aliphatic hydrocarbons
e.g., Ethene, C_2H_4
Ethyne, C_2H_2

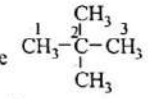
15.

S. No.	Formula of the functional group	Name of the functional group
(i)	$-CHO$	Aldehydic
(ii)	$>C=O$	Ketonic
(iii)	$-OH$	Alcoholic
(iv)	$-COOH$	Carboxyl

16. (i) Methane, CH_4
(ii) Ethane, C_2H_6
(iii) Methyl alcohol, CH_3-OH
(iv) Ethyl alcohol, CH_3-CH_2-OH
(v) Methyl chloride, CH_3-Cl
(vi) Ethyl chloride, CH_3-CH_2-Cl

17. (i) They have same molecular formula
(ii) They have different physical & chemical properties.
(iii) They have same molecular mass

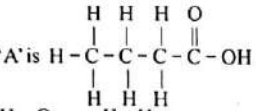
18. Isopentane 
2-methylbutane

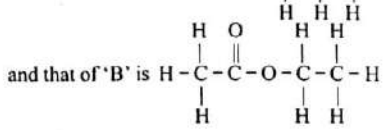
Neopentane 
2, 2-dimethyl-propane

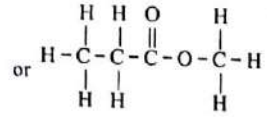
Long Answer Questions :

1. (a) In an ether, the oxygen is between two carbon atoms where as, in an alcohol the oxygen is between a carbon and a hydrogen.
(b) In an aldehyde, a carbonyl group (i.e., C=O) is between a carbon and a hydrogen, whereas in a ketone the carbonyl group is present between two carbon atoms.
(c) In an amine, the NH_2 group is attached to a alkyl group, in an amide, the NH_2 group is attached to a carbonyl group.
(d) Carboxylic acids have a hydrogen attached to an oxygen, in esters the hydrogen is replaced by a alkyl group.

2. (i) 'A' is $CH_3CH_2CH_2COOH$ 'B' is $CH_3COOCH_2CH_3$ or $C_2H_5COOCH_3$

(ii) Structural formula of 'A' is 

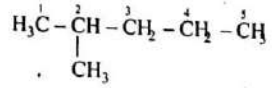
and that of 'B' is 

or 

Ester group is present in 'B'.
'A' and 'B' are functional isomers.

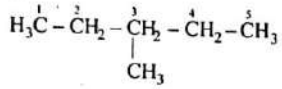
3. C_6H_{14}

$H_3C-CH_2-CH_2-CH_2-CH_2-CH_3$

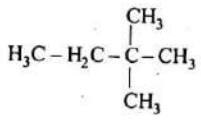
 n–hexane

iso hexane (branching at 2nd position)

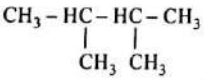
or



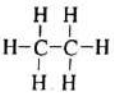
iso hexane (branching at 3rd position)

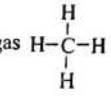
 neo hexane

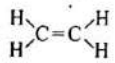
(branching at 2nd and 2nd position)

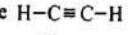


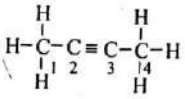
(branching at 2nd and 3rd position)

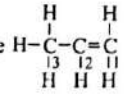
4. (a) Ethane 

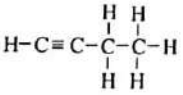
(b) Marsh gas 

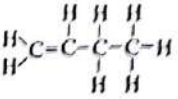
(c) Ethene 

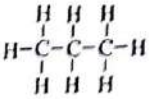
(d) Acetylene 

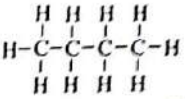
(e) 2-Butyne 

(f) Propylene 

(g) 1-Butyne 

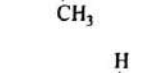
(h) 1-Butene 

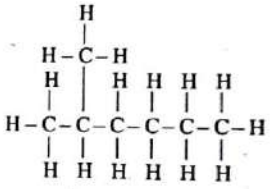
(i) Propane 

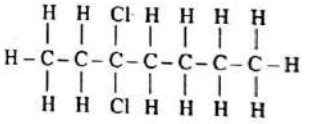
(j) n-Butane 

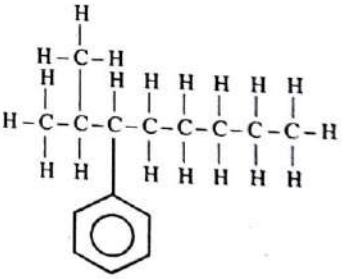
5. **Isomerism :** Two or more compounds having same molecular formula but different properties are called isomers and this phenomenon is known as isomerism.
e.g. C_4H_{10} represents two chain isomers.

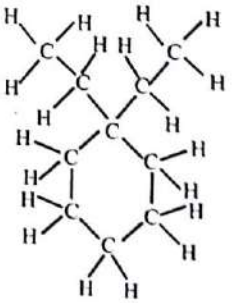
$H_3C-CH_2-CH_2-CH_3$ *n*-Butane

 Isobutane

6. (a) 

(b) 

(c) 

(d) 

Exercise 2

Multiple Choice Questions :

1.	(c)	2.	(c)	3.	(b)	4.	(a)
5.	(c)	6.	(d)	7.	(b)	8.	(d)
9.	(c)	10.	(c)	11.	(d)	12.	(c)
13.	(d)	14.	(b)	15.	(d)	16.	(c)
17.	(b)	18.	(d)	19.	(b)	20.	(c)
21.	(a)	22.	(a)	23.	(a)	24.	(a)
25.	(a)	26.	(c)	27.	(a)	28.	(b)
29.	(a)	30.	(c)	31.	(c)	32.	(b)
33.	(d)	34.	(b)	35.	(a)	36.	(a)
37.	(b)	38.	(a)	39.	(c)	40.	(a)
41.	(d)	42.	(a)				

More Than One Correct :

1.	(a, b, d)	2.	(a, d)	3.	(a, b, c)	4.	(a, d)
5.	(a, b, d)	6.	(a, c, d)	7.	(a, b, d)	8.	(b, c)
9.	(a, c, d)	10.	(a, b, c)	11.	(a, b, c)	12.	(a, b, c)
13.	(a, c)	14.	(a, b, d)	15.	(a, d)	16.	(a, b, c)
17.	(c, d)						

Fill in the Passage :

I (1) hydroxyl (2) alkyl (3) carbonyl (4) hydrogen (5) nitrogen (6) – COOH (7) hydrogen atom (8) – OH

II
(1)	ester	(2)	carboxylic acid
(3)	ether	(4)	alcohol
(5)	ketone	(6)	aldehyde
(7)	amide	(8)	methylalkyl amine

Passage Based Questions :

1. (d) 2. (b) 3. (c)

Assertion & Reason :

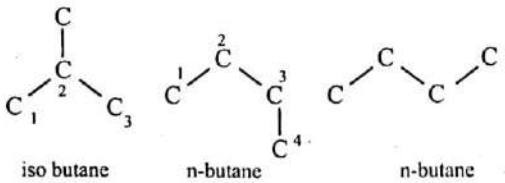
1. (c) – CN is a secondary suffix. 2. (b)
3. (a) 4. (c) 5. (d)
6. (a) $R-O\boxed{H+HO}-R \longrightarrow R-O-R+H_2O$

Multiple Matching Questions :

A → (p, q, r, s) B → (r, s) C → (q, s) D → (p, r, s)

HOTS Subjective Questions :

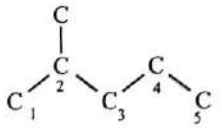
1.



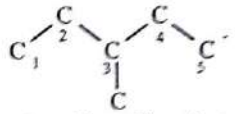
iso butane (1) n-butane (2) n-butane (3)

(2) & (3) are same straight chain hydrocarbon with 4 carbon atoms. Therefore there are only 2 structural isomers

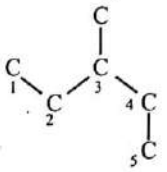
2.



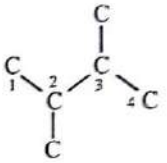
(branching at 2nd position) (I)



(branching at 3rd position) (II)



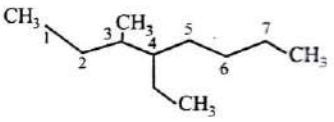
(branching at 2nd position) (III)



(branching at 2nd & 3rd position) (IV)

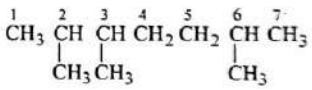
3. (i) $(CH_3)_2CHCH_2-$ (isobutyl) and $(CH_3)_3C-$ (t-butyl).

4.



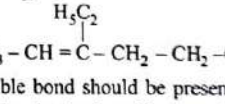
4 ethyl - 3 methyl octane

5.



2, 3, 6 – Trimethylheptane

6.



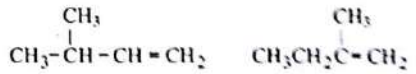
Double bond should be present on the minimum possible number in the lengthiest possible carbon chain. Thus the IUPAC name of above compound will be 3-ethylhexan-2-ene

7. C_5H_{10} has 1° degree of unsaturation since the isomers are acyclic, all of these are alkenes. For writing the isomers, first introduce the double bond at different possible positions, and then consider the possibility of branching in the alkyl group.

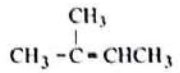
$CH_3CH_2CH_2CH=CH_2$ $CH_3CH_2CH=CHCH_3$

1-pentene (*i*) 2- pentene, (*cis*, - *trans*) (*ii*)

(*iii*)



3-methyl-1-butene, (*iv*) 2-methyl-1-butene, (*v*)



2-methyl-2-butene, (*vi*)

Thus the total number of isomers will be six.

8. C_4H_8 has 1° of unsaturation which may be in the form of either one double bond or a ring having 3 or 4 carbon atoms in a ring.

$CH_3CH_2CH=CH_2$ 1-butene (*i*) $CH_3CH=CHCH_3$ 2- butene (*ii*), (*cis*,- *trans*) $(CH_3)_2C=CH_2$ (*iii*) 2-methylpropene

cyclobutane (*iv*)

methylcyclopropane (*v*)

9. n-pentane $CH_3-CH_2-CH_2-CH_2-CH_3$

Isopentane $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$

Neopentane $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$

The difference in the three isomers in that they have different lengths of chain of C-atoms.

10. (a) Saturated aliphatic hydrocarbons :

There are the organic compounds which contain carbon-carbon single bonds (C – C) in their molecules are called saturated aliphatic hydrocarbons.

Unsaturated aliphatic hydrocarbons :

These are the organic compounds containing C = C or –C = C – in their molecules.

(b) CH_4 – Saturated aliphatic hydrocarbons

C_2H_4 – Unsaturated aliphatic hydrocarbons

C_3H_6 – Unsaturated aliphatic hydrocarbons

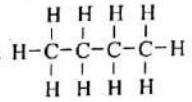
C_2H_6 – Saturated aliphatic hydrocarbons

C_2H_2 – Unsaturated aliphatic hydrocarbons

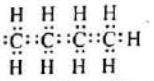
C_3H_4 – Unsaturated aliphatic hydrocarbons

11. (a) Molecular formula C_4H_{10}

(b) Structural formula



(c) Electronic formula

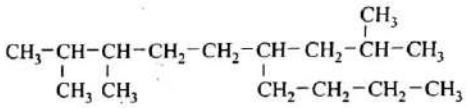


(d) Condensed formula $CH_3-CH_2-CH_2-CH_3$

12. (a) Alkenes (b) Alkynes (c) Alkanes

13. 5 – (1, 2 – Dimethylpropyl) –7– ethylundecane

14.



15. 4 – chloropentanamide

16. Hexa – 1, 3, 5 – triene

17.

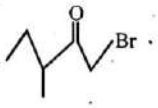
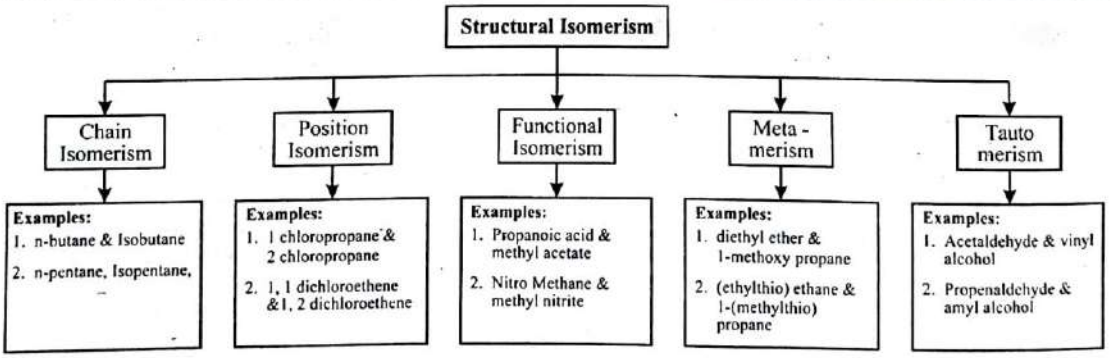
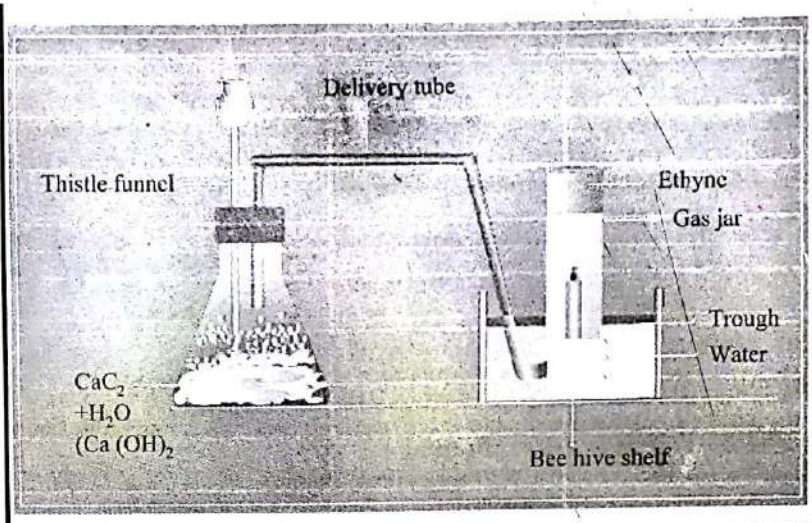


Chart Based Questions :



chapter

10



ORGANIC COMPOUNDS

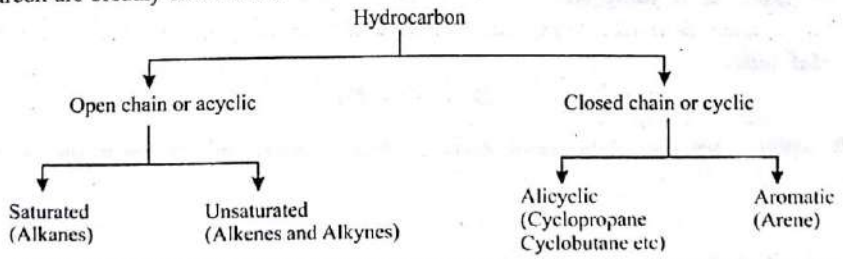
Introduction

As you have studied in previous chapters about large number of organic compounds and their nomenclature. These organic compounds has wide application in our routine life. Organic compounds are used in manufacturing of substances like drugs, plastics, rubber, clothes, dyes, cosmetics, cooking oil, ghee etc.

It was previously believed that organic compound can be derived from living organism thus organic compounds play a vital role in the proper functioning of body mechanism of all living organism also.

The simplest organic compounds are hydrocarbons. Hydrocarbons are compounds formed by the combination of carbon and hydrogen only. Petrol, diesel, kerosene, liquid petroleum gas (LPG) and condensed natural gas (CNG) etc. are all hydrocarbons or are their mixtures. All organic compounds can be considered as the derivatives of hydrocarbons, obtained by substituting hydrogen with an appropriate functional group.

Hydrocarbon are broadly classified as



There are many other classes of organic compounds. Compounds containing oxygen in its functional group like alcohols, ethers, ketones, aldehydes and carboxylic acid. Compounds containing halogen in his functional group like alkyl halide, chloroform, carbon tetrachloride. All compounds with these functional groups also have wide applications. In this chapter we will going to discuss the general chemistry of all these compounds.

ALKANES:

Alkanes or saturated hydrocarbons are those in which all the four valancies of carbon atom(s) are satisfied by four atoms or groups i.e. the carbon valencies are fully satisfied. These compound contain only C–C and C–H types of covalent bonds. Since they are relatively inert towards most of the chemical reagent under ordinary conditions, they are called Paraffins (latin-parum, little; affinis-affinity). The IUPAC name for these compounds is alkanes. Their general formula is C_nH_{2n+2}. Alkanes are quite stable compound and gives substitution reaction while alkenes and alkynes are more reactive and give addition reactions.

General Methods of Preparation of Alkanes :

(i) From Carboxylic acids :

(a) By de-carboxylation of sodium salt of fatty acid :–

$$\underset{\text{Sodium carboxylate}}{RCOONa} + \underset{\text{soda lime}}{NaOH/CaO} \xrightarrow{\Delta} \underset{\text{Alkane}}{R-H} + Na_2CO_3$$

e.g. $CH_3COONa + NaOH\,(CaO) \longrightarrow CH_4 + Na_2CO_3$

$CH_3CH_2COONa + NaOH(CaO) \longrightarrow \underset{\text{main product}}{C_2H_6} + Na_2CO_3$

Since the – COOH present in the original acid is removed in the process (in the form of carbonate), the reaction is commonly known as decarboxylation.

The reaction is employed for stepping down a homologous series.

(b) Kolbe's electrolytic synthesis :

$$2RCOONa + 2H_2O \xrightarrow{\text{Electrolysis}} R-R + 2CO_2 + H_2 + 2NaOH$$

Methane cannot be prepared by this method.

The side products are olefins, alcohols (particularly in alkaline solution) and esters.

Presence of alkyl groups in α-position decreases the yield of alkane.

Kolbe's electrolysis method can be used to prepare alkanes only with even number of carbon atoms and not alkanes with odd number of carbon atoms.

(ii) Reduction of Alkyl Halides :

$$\underset{\text{Alkyl halide}}{R-X} \xrightarrow{\text{Reductant}} \underset{\text{Alkane}}{R-H} + HX$$

Reductants : Zn–Cu couple/EtOH, Na–EtOH, Zn–HCl, Pt or Pd or Ni/H_2, Al–Hg/EtOH, $LiAlH_4$ etc.

e.g. : $CH_3I + Zn - Cu/\text{alcohol} \rightarrow CH_4 + H - I$

$$RBr + 2HI \xrightarrow[\Delta]{\text{Red P}} R-H + HBr + I_2$$

The purpose of red phosphorous is to remove iodine. It combines with iodine forming phosphorus triiodide (PI_3). Therefore, it makes the reaction to proceed in the forward direction. If I_2 is not removed, it will convert back alkane into alkyl halide.

$$2P + I_2 \longrightarrow 2PI_3$$

(iii) Wurtz Reaction : When an alkyl halide (usually bromide or iodide) is treated with sodium in dry ether, a symmetrical alkane containing twice the number of carbon atoms of alkyl halide is obtained.

$$R-Br + 2Na + Br-R \xrightarrow[\text{Dry Ether}]{\Delta} \underset{\text{Alkane}}{R-R} + 2NaBr$$

e.g. $\underset{\text{Methyl iodide}}{CH_3-I} + 2Na + I-CH_3 \xrightarrow[\text{Ether}]{\Delta,\ \text{Dry}} \underset{\text{Ethane}}{CH_3-CH_3} + 2NaI$

Methane cannot be prepared by this method.

When the two reacting alkyl halides are different, a mixture of three different alkanes is obtained. This is because the two different alkyl halides not only react with each other but they react amongst themselves also the wurtz reaction is not useful for preparing unsymmetrical alkanes. For e.g.

$$\underset{\text{Methyl bromide}}{CH_3Br} + 2Na + \underset{\text{Ethyl bromide}}{BrCH_2CH_3} \xrightarrow[\text{ether}]{\text{Dry}} \underset{\text{Propane}}{CH_3CH_2CH_3} + 2NaBr$$

$$\underset{\text{Ethyl bromide}}{CH_3CH_2Br} + 2Na + \underset{\text{Ethyl bromide}}{BrCH_2CH_3} \xrightarrow[\text{ether}]{\text{Dry}} \underset{\text{Butane}}{CH_3CH_2CH_2CH_3} + 2NaBr$$

$$\underset{\text{Methyl bromide}}{CH_3Br} + 2Na + \underset{\text{Methyl bromide}}{BrCH_3} \xrightarrow[\text{ether}]{\text{Dry}} \underset{\text{Ethane}}{CH_3CH_3} + NaBr$$

Wurtz reaction can not be used to prepare pure alkanes with odd number of carbon atoms because it is difficult to separate the mixture obtained

- *Wurtz reaction generally fails with tertiary alkyl halides.*

$$3^\circ \text{ halide} + \text{base} \xrightarrow{\text{elimination}} \text{alkene}$$

- *In presence of moist ether the product is alcohol*

$$Na + H_2O \longrightarrow NaOH \xrightarrow{R-X} \underset{\text{alcohol}}{ROH}$$

ILLUSTRATION 1

What is the product we obtained when propyl iodide is heated in presence of dry ether?

SOLUTION :

Hexane is obtained as a major product by wurtz reaction.

$$\underset{\text{Propyl Iodide}}{CH_3CH_2CH_2I} + 2Na + \underset{\text{Propyl Iodide}}{ICH_2CH_2CH_3} \xrightarrow[\text{ether}]{\text{Dry}} \underset{\text{hexane}}{CH_3CH_2CH_2CH_2CH_2CH_3} + 2NaI$$

(iv) From Alkenes and Alkynes : (Sabatier Senderen's Reaction) :

$$\underset{\text{Olefin}}{R-CH=CH-R'} + H_2 \xrightarrow[Ni/300^\circ]{H_2} \underset{\text{Alkane}}{R-CH_2-CH_2-R'}$$

$$R-C\equiv C-R' + 2H_2 \xrightarrow[Ni/300^\circ]{H_2} R-CH_2-CH_2-R'$$

When the catalyst are Pt or Pd, the hydrogenation proceeds smoothly at ordinary temperature and pressure.

With Nickel catalyst, higher temperature (250° – 300°C) and pressure are needed. The hydrogenation reaction of unsaturated hydrocarbons using nickel at a temperature of 523-573 K is commonly known as sabatier and senderen's reaction or reduction.

With Raney Nickel, the reaction takes place at room temperature.

$Ni/Al + NaOH \longrightarrow$ mixture of Ni (Raney Nickel)

- *These metals (Ni, Pt or Pd) adsorb dihydrogen gas on their surface and activate the hydrogen-hydrogen bond.*
- *In this reaction following alkanes can not be prepared from unsaturated hydrocarbon*

Alkene / Alkyne ⟶ Methane

Alkene / Alkyne ⟶ Neopantane

Alkyne ⟶ Isobutane

CHECK Point

☛ **What is the product of following reaction**

$$CH_3 - C \equiv CH + 2H_2 \xrightarrow[\text{Normal temperature}]{\text{Pd or Pt}}$$

SOLUTION

Propane by following reaction

$$CH_3 - C \equiv CH + 2H_2 \xrightarrow[\text{Normal temperature}]{\text{Pd or Pt}} \underset{\text{propane}}{CH_3CH_2CH_3}$$

Method of preparation of (CH_4):

(i) $Al_4C_3 \xrightarrow{+12H_2O} 3CH_4 + 4Al(OH)_3$

$Be_2C \xrightarrow{+4H_2O} CH_4 + 2Be(OH)_2$

(ii) **Sabatier senderens :–**

$CO_2 + 4H_2 \longrightarrow CH_4 + 2H_2O$

$\underset{\text{(water gas)}}{CO + 3H_2} \longrightarrow CH_4 + H_2O$

Physical Properties of Alkane :

(i) Alkanes from $C_1 - C_4$ are colourless, odourless gases. $C_5 - C_{17}$ are colourless liquids, C_{18} – onwards are waxy, white solids.

(ii) **Boiling points :–** Boiling points of alkanes are the lowest of all the groups of organic compounds. The forces of attraction among the alkane molecules are the weakest vander Waal's forces of attraction. With the increase in the number of carbon chain length (molecular size), the magnitude of vander waals forces also increases.

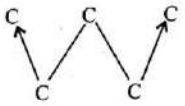
Among isomeric alkanes, the boiling point decreases with increasing branching.

Example n-Pentane > iso-Pentane > neo-Pentane

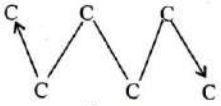
It is because with branching surface area of molecule decreases consequently forces of attraction which depends upon surface area decreases resulting into decrease of boiling point.

(iii) **Melting point :–** The melting point of alkanes do not show regular variation with increase in molecular size. Unbranched alkanes containing 2, 4, 6, 8 etc. carbon atoms have higher melting points than the unbranched alkanes containing 3, 5, 7, 9 etc. carbon atoms.

Example



n-pentane



n-hexane

(both methyl group are directed on same side) (both methyl group are adversely directed)

M.P. even no. of carbon atoms > odd no. of carbon atoms

M.P. ∝ mol.wt. ∝ packing efficiency.

This is because the alkanes with even number of carbon atoms have more symmetrical structures and result in closer packing in the crystal structure as compared to alkanes with odd number of carbon atoms.

CHECK Point

☛ **Out of C_3H_8 and C_4H_{10} which one has high melting point?**

SOLUTION

C_4H_{10} as we know that alkanes with even number of carbon atoms has higher boiling point.

(iv) Fuel capacity : Decreasing order of fuel capacity is

more branched > less branched > unbranched

lower mol.wt. > higher mol. wt.

(v) Solubility : Alkanes are lighter than water. These are insoluble in water and soluble in organic solvents.

Chemical Properties of Alkanes

1. **Substitution reactions :** *Reactions in which an atom or group in a compound is replaced by another atom or group is known as* **substitution reaction** and the product formed is known as **substituted product**. The common substitution reactions found in aliphatic compounds are halogenation, nitration and sulphonation.

   (i) Halogenation (*Replacement of hydrogen atom by halogen atom*). Alkanes react with halogens (Cl_2, Br_2) in presence of sunlight or UV light or in dark at high temperature (523-673 K) to form the corresponding substituted product. For example,

$$\underset{\text{Methane}}{CH_4} + Cl_2 \xrightarrow[520-670\ K]{\text{light}} \underset{\text{Methyl chloride}}{CH_3Cl} + HCl$$

$$\underset{\substack{\text{Chloro methane}\\(\text{Methyl chloride})}}{CH_3Cl} + Cl_2 \xrightarrow[520-670\ K]{\text{light}} \underset{\substack{(\text{Dichloromethane})\\(\text{Methylene chloride})}}{CH_2Cl_2} + HCl$$

$$\underset{\text{Dichloromethane}}{CH_2Cl_2} + Cl_2 \xrightarrow[520-670\ K]{\text{light}} \underset{\text{Trichloromethane}}{CHCl_3} + HCl$$

$$\underset{\substack{\text{Trichloromethane}\\(\text{Chloroform})}}{CHCl_3} + Cl_2 \xrightarrow[520-670\ K]{\text{light}} \underset{\substack{\text{Tetrachloromethane}\\(\text{Carbon tetrachloride})}}{CCl_4} + HCl$$

Knowledge ENHANCER

Bromination *occurs similarly but less rapidly.*

Iodination *is carried out by heating alkane with iodine in the presence of some oxidising agent like iodic acid (HIO_3) or nitric acid (HNO_3) which oxidises HI formed during the reaction. If HI is not oxidised, being a strong reducing agent, it will make the reaction reversible in nature.*

$$\underset{\text{Methane}}{CH_4} + I_2 \xrightarrow{\text{Heat}} \underset{\substack{\text{Iodomethane}\\\text{Methyl iodide}}}{CH_3I} + HI$$

$$5HI + HIO_3 \xrightarrow{\text{Heat}} 3H_2O + 3I_2$$

or $$2HI + 2HNO_3 \longrightarrow 2H_2O + 2NO_2 + I_2$$

Fluorination *of alkanes takes place explosively resulting even in the rupture of C—C bond in higher alkanes. The reactivity of halogenation is, therefore,*

fluorine > chlorine > bromine > iodine

(ii) Nitration : This involves the replacement of a hydrogen atom of alkane with $-NO_2$ group.

$$\underset{\text{Hexane}}{C_6H_{14}} + HNO_3 \longrightarrow \underset{\text{Nitrohexane}}{C_6H_{13}NO_2} + H_2O$$

(iii) Sulphonation : This involves the replacement of a hydrogen atom of alkane by $-SO_3H$ group.

$$\underset{\text{Hexane}}{C_6H_{14}} + H_2SO_4 \xrightarrow{\Delta} \underset{\text{Hexane Sulphonic acid}}{C_6H_{13}SO_3H} + H_2O$$

Lower alkanes particularly methane, ethane, etc. do not give this Sulphonation substitution reaction.

2. **Thermal decomposition, cracking or pyrolysis :** *The decomposition of a compound by heat is known as pyrolysis. Thermal decomposition when applied to alkanes is known as cracking* and leads to the formation of lower alkanes, alkenes and hydrogen, e.g.

$$\underset{\text{Ethene}}{CH_2 = CH_2} + \underset{\text{Methane}}{CH_4} \xleftarrow{675-875\ K} \underset{\text{Propane}}{CH_3.CH_2CH_3} \xrightarrow{675-875\ K} \underset{\text{Propene}}{CH_3CH = CH_2} + H_2$$

Note : Cracking has been used for the manufacture of petrol, petrol gas, oil gas, etc.

3. **Isomerisation :** *The process of conversion of one isomer of a compound to another isomer is known as isomerisation.* This is achieved by heating the normal alkane with anhydrous $AlCl_3$ and HCl or anhydrous $AlBr_3$ and HBr under pressure of 35 atmospheres.

$$\underset{n\text{-Butane}}{CH_3 - CH_2 - CH_2 - CH_3} \xrightarrow[473\ K/35\ atm.]{AlCl_3/HCl} \underset{\text{Isobutane}}{CH_3 - \overset{\overset{CH_3}{|}}{CH} - CH_3}$$

Isomerisation of alkanes is of great importance in petroleum industry.

CHECK Point

☛ **Complete the following reaction**

$$\underset{\text{n-hexane}}{CH_3(CH_2)_4\,CH_3} \xrightarrow[\Delta]{\text{anhy } AlCl_3,\ HCl}$$

SOLUTION

$$\underset{\text{n-hexane}}{CH_3(CH_2)_4\,CH_3} \xrightarrow[\Delta]{\text{anhy } AlCl_3,\ HCl} \underset{\text{2-Methyl pentane}}{CH_3 - \overset{\overset{CH_3}{|}}{CH} - CH_2CH_2 - CH_3}$$

5. **Complete oxidation or combustion :** Alkanes, when burnt in excess of air or oxygen, form carbon dioxide and water, e.g.

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + 212.8\ \text{kcal/mole}$$

$$2C_2H_6 + 7O_2 \longrightarrow 4CO_2 + 6H_2O + 745.6\ \text{kcal/mole}$$

In general, $$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \longrightarrow nCO_2 + (n+1)H_2O + \text{heat}$$

QUESTION 2

Why hydrocarbons are used as a fuel?

SOLUTION

It is because during their combustion, a large amount of heat is evolved for example cooking gas, which is often called L.P.G. is a mixture of propane and butane

ALKENES:

Alkenes are those unsaturated hydrocarbons which contain at least one carbon-carbon double bond in their molecules. Their general formula is C_nH_{2n} and hence they contain two hydrogen atoms less than the corresponding alkanes and hence designated as unsaturated hydrocarbon. Their general formula is C_nH_{2n+2}. These are also known as olefins or olifines (olifiant = oil forming) Their IUPAC name is alkenes.

The carbon-carbon double bond (C = C), commonly known as ethylenic or olifinic bond or linkage is the characteristic group of alkenes.

METHODS OF PREPARATION OF ALKENES:

(i) From Alkane:

(a) Pyrolysis:

$$CH_3-CH_2-CH_3 \xrightarrow{400°-600°C} \begin{cases} CH_3-CH=CH_2+H_2 \\ CH_2=CH_2+CH_4 \end{cases}$$

(b) Dehydrogenation:

e.g. $CH_3-CH_2-CH_3 \xrightarrow[600°C]{Cr_2O_3/Al_2O_3} CH_3-CH=CH_2+H_2$

Propane Propene

(ii) From alkyne (Partial hydrogenation): By partial reduction of alkynes in the presence of $Pd/CaCO_3$.

$$R-C\equiv CH+H_2 \xrightarrow{Pd/CaCO_3} RCH=CH_2$$

(iii) By dehydrohalogenation of alkyl halides: When mono halide react with alcoholic KOH or NaOH then respective alkenes are formed along with the elimination of HX molecule this reaction is called dehydrohalogenation.

$$R-\underset{H}{\overset{H}{C}}-\underset{H}{\overset{H}{C}}-X + Alc.KOH \xrightarrow[-HX]{} R-CH=CH_2$$

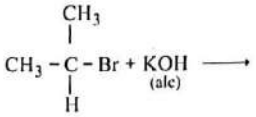
The reactivity of alkyl halide towards dehydrohalogenation follows the order $3° > 2° > 1°$ halide

Further the ease of dehydrohalogenation for different halogens is in the order Iodide > Bromide > Chloride

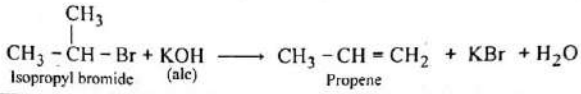
In this reaction, the hydrogen atom is eliminated from β carbon atom (carbon atom next to the carbon to which halogen is attached). Therefore, the reaction is also called β-elimination reaction.

CHECK *Point*

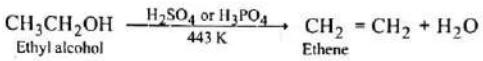
☛ **Complete the following reaction**



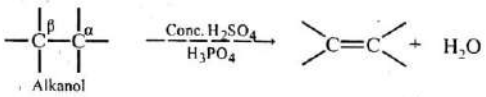
SOLUTION



(iv) From alcohols : Alkenes are prepared from alcohols by heating with protonic acids such as sulphuric acid or phosphoric acid at about 443 K. This reaction is called dehydration of alcohols.

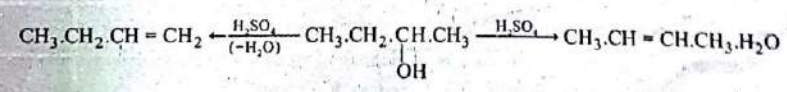


This reaction is also an example of β-elimination reaction because –OH group takes out one hydrogen atom from the β-carbon atom.



Knowledge ENHANCER

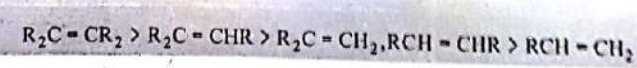
Dehydration of unsymmetrical secondary or tertiary alcohols leads to the formation of two different alkenes. For example, dehydration of butanol-2 gives 2- burene and 1-butene.



1–Butene (20%) *2-Butene (80%)*

In such cases, dehydration is governed by Saytzeff's rule according to which hydrogen is preferentially eliminated from the carbon atom with fewer number of hydrogen atoms, i.e., "poor becomes poorer". Thus 2-butene is the main product in the above example.

In other words, Saytzeff rule states that greater the number of alkyl groups attached to the doubly bonded carbon atoms, the more stable is the alkene; and hence more is its ease of formation. Thus the ease of formation of alkenes is :



Physical Properties of Alkene :

(i) From $C_2 - C_4$ they are colourless, odourless gases, from $C_5 - C_{17}$ they are colourless liquids, C_{18} onwards alkenes are solids. Ethylene C_2H_4 is a sweet smelling gas. It produces anaesthesia on smelling in large amounts.

(ii) Alkenes are practically insoluble in water. They dissolve freely in organic solvents like benzene, chloroform, CCl_4, petroleum ether., etc.

(iii) The boiling and melting points of alkenes are slightly higher than the corresponding alkanes. The branched chain alkenes, however, have lower boiling points than the corresponding straight chain alkenes. Alkenes are therefore, lesser volatile than the corresponding alkanes.

Their boiling points, melting points and specific gravities rise with the increase of molecular weight.

The boiling points of alkenes show a regular gradation with the increase in number of carbon atom like alkanes. In general, for each added $-CH_2$ group the boiling point rises by 20-30°C.

The increase in branching in carbon chain decreases the boiling point among isomeric alkenes.

$$\textbf{MP. and BP.} \propto \text{mol. wt.} \propto \frac{1}{\text{branching in alkenes}}$$

CHECK Point

☛ **Which out of two ethylene and butylene have higher boiling point ?**

SOLUTION

Butylene because in case of alkenes B.P. increases with increase in length of carbon chain.

Chemical Properties of Alkene :

1. **Addition reactions :** Reactions in which an atoms or group of atom are added to a molecule (i.e., there is simply a net gain of the reagent atoms in product molecule) are called **addition reactions**. The important examples of addition reactions are summarised below.

(i) **Addition of hydrogen** (*Catalytic hydrogenation, hydrogenation in presence of catalyst*). Hydrogen adds to unsaturated hydrocarbons (alkenes, and alkynes) in presence of metallic catalyst, such as finely divided platinum, palladium or nickel and forms the corresponding alkane. For example,

$$\underset{\text{Ethene}}{H_2C = CH_2} + H_2 \xrightarrow[473-573\,K]{Ni} \underset{\text{Ethane}}{H_3C - CH_3}$$

(ii) **Addition of halogens :** When an alkene is treated at room temperature with chlorine or bromine in an inert solvent (usually chloroform or carbon tetrachloride), the halogen adds rapidly to the double bond of the alkene and produces a vicinal dihalide.

$$\underset{\text{Alkene}}{R - CH = CH_2} + Br_2 \xrightarrow{CCl_4} \underset{\text{Vicinal dibromide}}{R - \underset{Br}{\underset{|}{CH}} - \underset{Br}{\underset{|}{CH_2}}}$$

The reaction with fluorine is explosive whereas iodine does not react under normal conditions.

This reaction can be used as a test for unsaturation of alkenes because the brown colour of bromine will get decolourised by alkenes.

$$\underset{\text{Alkene}}{-\overset{|}{C} = \overset{|}{C}-} + \underset{\text{Brown colour}}{Br_2} \longrightarrow \underset{\text{Colourless}}{-\overset{|}{\underset{Br}{C}} - \overset{|}{\underset{Br}{C}}-}$$

This reaction is also given by alkynes.

idea box

In your school lab take two beakers containing bromine solution and label them A and B. Then add vanaspati ghee in beakers A and vegetable oil in B respectively. In beaker A color of bromine solution remains same but in beaker B color of solution gets faded up or discharged. What is the reason for this observation try to find out. Consult your friends and teacher in case of any problem.

(iii) **Addition of hydrogen halides** (*hydrohalogenation*) : Alkenes add hydrogen halides to form alkyl halides, e.g.

$$CH_2 = CH_2 + HX \longrightarrow CH_3CH_2X$$

If the alkene is unsymmetrical. Then the negative part of attacking molecule joins with the carbon atom which carries lesser number of hydrogen atoms as per Markovnikov's rule

$$\underset{\text{Propene}}{CH_3 - CH = CH_2} + HBr \longrightarrow \underset{\text{2-Bromopropane}}{CH_3 - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_3}$$

In the presence of organic peroxides (R–O–O–R) *e.g. benzoyl peroxide the reaction takes place opposite to the Markovnikov's rule. This is known as an anti Markovnikov's rule or Kharasch effect*

$$\underset{\text{Propene}}{CH_3 - CH = CH_2} + HBr \xrightarrow{(R-O-O-R)} \underset{\text{1-Bromopropane}}{CH_3 - CH_2 - CH_2Br}$$

Only HBr *shows this peroxide effect not* HF, HCl *and* HI *their addition occurs according to Markovnikov's rule even in presence of peroxides.*

(iv) **Addition of hypohalous acids :** Alkenes react with hypochlorous and hypobromous acids readily to form the corresponding halohydrin.

$$\underset{\text{Ethylene}}{CH_2 = CH_2} + HOCl \longrightarrow \underset{\text{Ethylene chlorohydrin}}{\overset{OH}{\overset{|}{CH_2}} - \overset{Cl}{\overset{|}{CH_2}}}$$

ILLUSTRATION 3

Complete the following reaction

$$CH_3 - CH = CH_2 + HOBr \longrightarrow$$

SOLUTION :

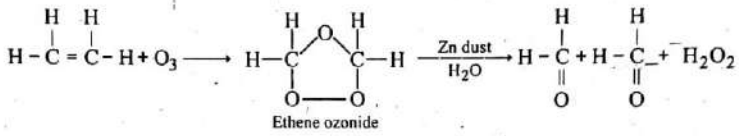
Addition of HOBr also occur according to Markovnikov's rule. Thus reaction will be

$$\underset{\text{Propene}}{CH_3 - CH = CH_2} + HOBr \longrightarrow \underset{\text{1-Bromopropan-2-ol}}{CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_2Br}$$

2. **Polymerisation :** *The process in which simple molecules combine together to form large molecules is called* **polymerisation**; the starting simple molecule is called **monomer** while the larger (final product) is called a **polymer**. Polymers are used as plastics, rubber and several other important products. Lower alkenes and alkynes undergo polymerisation to form products of great importance.
Lower alkenes (e.g. ethene, propene etc.) readily undergo polymerisation at high temperature and pressure and in the presence of a suitable catalyst like $AlCl_3$, BF_3, etc.

$$\underset{\text{Ethylene}}{n CH_2 = CH_2} \xrightarrow[\text{high pressure, } O_2]{473-673\ K} \underset{\text{Polyethylene}}{(-CH_2 - CH_2-)_n}$$

3. **Ozonolysis :** On the addition of ozone to an unsaturated hydrocarbon, it forms an intermediate ozonide which on further hydrolysis in the presence of zinc dust gives corresponding carbonyl compound (aldehydes and ketones). This reaction is used to identify and locate the multiple bonds in an unsaturated hydrocarbon.
When a molecule of ozone is added to ethene at room temperature it forms a compound called ethene ozonide which on hydrolysis in the presence of zinc dust gives two molecules of formaldehyde.

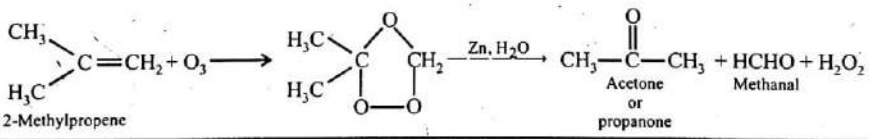


CHECK Point

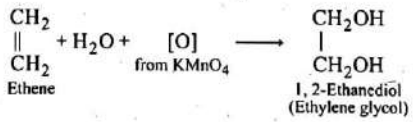
☛ **What are the products obtained by ozonolysis of 2-Methylpropene ?**

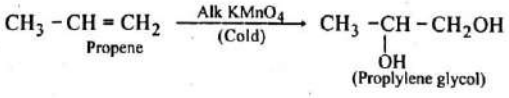
SOLUTION

Acetone and Methanal

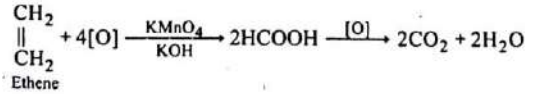


4. **Oxidation with oxidizing agents :**
   (i) **Oxidation with cold, neutral or alkaline $KMnO_4$:**
   Alkenes are readily oxidised by cold dilute (1%) neutral or alkaline $KMnO_4$ solution (Baeyer's reagent) to give **glycols** (1, 2-diols or *vic*-diols); $KMnO_4$ itself is reduced to MnO_2.





   (ii) **Oxidation with hot alkaline $KMnO_4$ solution :**
   Alkenes when treated with hot $KMnO_4$ solution give carboxylic acids, ketones and carbon dioxide depending upon the nature of the alkene.
   $= CH_2$ (terminal alkene) group is oxidised to formic acid which on further oxidation gives CO_2 and H_2O. For example,



$$\underset{\text{Propene}}{CH_3-CH=CH_2} \xrightarrow[\text{Hot } KMnO_4]{[O]} \underset{\text{Acetic acid}}{CH_3COOH} + CO_2 + H_2O$$

5. **Oxidation with oxygen :**
Alkenes, when heated with oxygen in presence of silver catalyst at 575 K, are oxidised to epoxides. For example,

$$\begin{matrix}CH_2\\ \|\\ CH_2\end{matrix} + \frac{1}{2}O_2 \xrightarrow[575\ K]{Ag} \underset{\text{Epoxyethane or (Ethylene oxide)}}{\begin{matrix}CH_2\diagdown\\ |\\ CH_2\diagup\end{matrix}O}$$

Bromine water test and Baeyer's test are used for detecting the presence of double bond while ozonolysis helps in locating the position of double bond.

ALKYNES:

Alkynes are characterised by the presence of a triple bond between two carbon atoms i.e. $-C\equiv C-$. They contain four hydrogen atoms less than the corresponding alkane and hence correspond to the general formula C_nH_{2n-2}. The first and the most important member of this series is acetylene $CH\equiv CH$ and hence these compounds are also called as acetylenes and the triple bond is usually referred to as the acetylenic linkage or acetylenic bond.

Methods of Preparation of Alkyne :

(i) **Action of water on calcium carbide.** Acetylene is prepared in the laboratory as well as on industrial scale by the action of water on calcium carbide.

$$\underset{\text{Calcium carbide}}{CaC_2} + 2H_2O \longrightarrow \underset{\text{Acetylene}}{HC\equiv CH} + Ca(OH)_2$$

(ii) **By dehydrohalogenation of vicinal dihalides**

$$\underset{\text{vic-dihalide}}{R-\overset{X}{\overset{|}{C}}H-\overset{X}{\overset{|}{C}}H-R'} \xrightarrow{\text{alc KOH}} \underset{\text{Alkyne}}{RC\equiv CR'}$$

$$\underset{\begin{matrix}Br\ \ Br\end{matrix}}{CH_3-\overset{Br}{\overset{|}{C}}H-\overset{Br}{\overset{|}{C}}H_2} + 2KOH \xrightarrow[(alc)]{} \underset{\text{propyne}}{CH_3-C\equiv CH} + 2KBr + 2H_2O$$

(iii) Higher alkyne can be prepared from lower alkynes by treating its sodium salt with alkyl halide

$$\underset{\text{Acetylene}}{HC\equiv CH} + \underset{\text{Sodamide}}{NaNH_2} \xrightarrow[-NH_3]{O(C_2H_5)_2} \underset{\text{Sod. acetylide}}{HC\equiv CNa} + R-X \xrightarrow[-NaX]{} HC\equiv C-R$$

(iv) From Vinyl Halide :

$$R-CH=CH-Cl \xrightarrow{NaNH_2} R-C\equiv C-H+NaCl+NH_3$$

CHECK Point

☛ **How can we prepare pent-2-yne from prop-1-yne ?**

SOLUTION

$$\underset{\text{prop-1-yne}}{CH_3-C\equiv CH} \xrightarrow[O(C_2H_5)_2]{NaNH_2} CH_3-C\equiv CNa \xrightarrow[-NaI]{CH_3CH_2I} \underset{\text{pent-2-yne}}{CH_3-C\equiv C-CH_2-CH_3}$$

PHYSICAL PROPERTIES OF ALKYNE :

(i) Alkynes are colourless, odourless and tasteless.

(ii) Lower alkynes are partially soluble in H_2O. (It is due to its polarizability)

(iii) Higher alkynes are insoluble in water due to more % of covalent character.

(iv) Completely soluble in organic solvents.

(v) Melting point and boiling point are directly proportional to molecular mass and inversely proportional to number of branches.
(vi) Upto C_4 alkynes are gaseous. $C_5 - C_{11}$ liquid, C_{12} & above are solids.
(vii) Acetylene & 1 – alkyne are acidic in nature. It is due to presence of active H.

Order of Solubility, density, B.P., M.P. *and Acidic nature :*
Alkyne > Alkene > Alkane
All terminal alkynes are acidic in nature.

Chemical Properties of Alkynes

1. **Addition reactions :** Reactions in which an atoms or group of atoms are added to a molecule (i.e., there is simply a net gain of the reagent atoms in product molecule) are called **addition reactions**. The important examples of addition reactions are summarised below.

(i) **Addition of hydrogen** (*Catalytic hydrogenation, hydrogenation in presence of catalyst*). Hydrogen adds to unsaturated hydrocarbons (alkenes and alkynes) in presence of metallic catalyst, such as finely divided platinum, palladium or nickel and forms the corresponding alkane. For example,

$$\underset{\text{Acetylene}}{CH \equiv CH} + H_2 \xrightarrow{Ni} \underset{\text{Ethylene}}{CH_2 = CH_2} \xrightarrow{H_2} \underset{\text{Ethane}}{CH_3 - CH_3}$$

It may be noted that the hydrogenation can be controlled at the alkene stage only. This is possible by using a Lindlar's catalyst which is a mixture of palladium and barium sulphate poisoned by quinoline.

$$HC \equiv CH \xrightarrow[\substack{\text{Quinoline} \\ \text{(Lindlar's catalyst)}}]{Pd, BaSO_4} \underset{\text{Ethene}}{CH_2 = CH_2}$$

Alkynes add one or two molecules of halogens to form *di-* and *tetra*-halides respectively.

$$\underset{\text{Acetylene}}{\overset{CH}{\overset{\|\|}{CH}}} + Cl_2 \longrightarrow \underset{\substack{\text{(1, 2-Dichloroethane)} \\ \text{(Acetylene dichloride)}}}{\overset{CHCl}{\overset{\|}{CHCl}}} \xrightarrow{Cl_2} \underset{\substack{\text{1, 1, 2, 2-Tetrachloroethane} \\ \text{(Acetylene tetrachloride, Westron)}}}{\overset{CHCl_2}{\overset{|}{CHCl_2}}}$$

(iii) **Addition of hydrogen halides** (*hydrohalogenation*) **:**
Alkynes add two molecules of hydrogen halides. For example,

$$\underset{\text{Acetylene}}{\overset{CH}{\overset{\|\|}{CH}}} + HCl \longrightarrow \underset{\text{Vinyl chloride}}{\overset{CH_2}{\overset{\|}{CHCl}}} \xrightarrow{HCl} \underset{\text{Ethylidene chloride}}{\overset{CH_3}{\overset{|}{CHCl_2}}}$$

In case of unsymmetrical alkynes addition of attacking reagent take place according to Morkovnikov's rule

(iv) **Ozonolysis :** Addition of a molecule of ozone to a molecule of ethyne at room temperature gives an addition compound, ethyne ozonide which on further hydrolysis with zinc dust gives glyoxal.

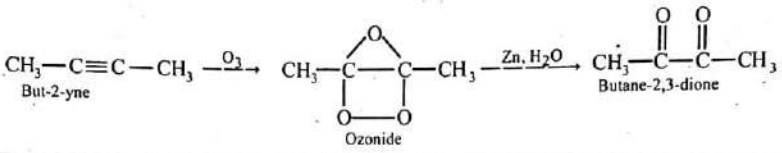
$$HC \equiv CH + O_3 \rightarrow \underset{\text{Acetylene ozonide}}{\overset{O}{\overset{\diagup\diagdown}{HC - CH}}\underset{O - O}{|\quad|}} + H_2O \xrightarrow[\text{dust}]{Zn} \underset{\text{Acetylene ozonide}}{\overset{CHO}{\overset{|}{CHO}}} + HCOOH$$

ILLUSTRATION 4

Which product is obtained when But-2-yne is first reacted with O_3 molecule and product obtained is then hydrolysed in the presence of zinc metal ?

SOLUTION :

The product obtained is Butane-2, 3-dione as per following reaction



(v) **Formation of sodium acetylides :** Acetylene and monoalkylacetylene react with sodium in liquid ammonia or sodamide to form sodium acetylides.

$$H-C\equiv C-H+Na \xrightarrow[453\ K]{liq.NH_3} \underset{\text{Monosodium acetylide}}{H-C\equiv C-Na} \xrightarrow[483\ K]{Na/liq.NH_3} \underset{\text{Disodium acetylide}}{NaC\equiv CNa}$$

This reaction shows that alkynes are acidic in nature

(vi) **Formation of copper and silver acetylides :** Copper and silver alkynides are obtained by passing alkynes in the ammonical solution of cuprous chloride and silver nitrate respectively.

$$HC\equiv CH + Cu_2Cl_2 + 2NH_4OH \longrightarrow \underset{\text{Copper acetylide (red ppt.)}}{CuC\equiv CCu\downarrow} + 2NH_4Cl + 2H_2O$$

$$HC\equiv CH + 2AgNO_3 + 2NH_4OH \longrightarrow \underset{\text{Silver acetylide (white)}}{AgC\equiv CAg\downarrow} + 2NH_4NO_3 + 2H_2O$$

CHECK Point

What will happen when 1-Butyne and 2-Butyne are reacted with ammonical solution of silver nitrate

SOLUTION

1-Butyne will give white ppt but 2-Butyne gives no reaction as this reaction is given by terminal alkynes (having hydrogen at first carbon only).

$$\underset{\text{1-Butyne}}{CH_3CH_2C\equiv CH} + 2\left[Ag(NH_3)_2\right]NO_3 \longrightarrow \underset{\text{White ppt}}{CH_3CH_2C\equiv CAg}$$

$$\underset{\text{2-Butyne}}{CH_3-C\equiv C-CH_3} + 2\left[Ag(NH_3)_2\right]NO_3 \longrightarrow \text{No reaction}$$

(vii) Oxidation with alkaline $KMnO_4$: They give diketones under normal conditions

$$\underset{\text{But-2-yne}}{CH_3-C\equiv C-CH_3} + \underset{\text{Baeyer's reagent}}{KMnO_4 + H_2O} \xrightarrow{298K-303K} \underset{\text{Butane-2, 3-dione}}{CH_3-C(=O)-C(=O)-CH_3}$$

$[3HC\equiv CH \xrightarrow{Fe/\Delta} C_6H_6]$

idea box

Make a project report on fuel value comparison of alkane, alkene and alkynes. Mentioning which out of three class of hydro carbons have maximum and minimum fuel value. Also provide reasons for that why a particular class have highest fuel value while other have lowest. Provide some statistical data to support your answer. You can consult your teacher and can take help of internet.

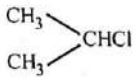
SOME IMPORTANT ORGANIC COMPOUNDS :

1. **Alkyl halides :** Halogen derivatives of alkanes, alkenes and alkynes are known as ***alkyl halides*** (haloalkenes), ***alkenyl halides*** (haloalkanes) and ***alkynyl halides*** (haloalkynes) respectively.

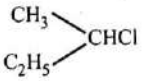
On the basis of the number of halogen atom, they are further classified as mono-, di-, tri- poly- and per-halohydrocarbons. The word perhalohydrocarbon means all the hydrogen atoms of the compound are replaced by corresponding number of halogen atoms.

Besides the nature and number of halogen atoms, alkyl halides may be classified into ***primary*** (1°), ***secondary*** (2°) and ***tertiary*** (3°) according to the nature of the carbon atom bearing halogen.

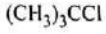
Isobutyl chloride (1°)
(–Cl present on 1°C)



Isopropyl chloride (2°)
(–Cl on 2°C)

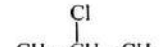


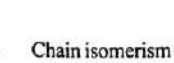
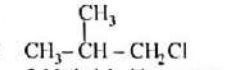
sec. Butyl chloride (2°)
(–Cl on 2°C)



tert. Butyl chloride (3°)
(–Cl on 3°C)

ISOMERISM IN ALKYL HALIDES :

1. Position isomerism example 1-chloropropane  2-chloropropane

2. Chain isomerism example  1-chlorobutane  2-Methyl-1-chloropropane

Methods of preparation :

(i) From alochols with HX, PX_3 (X = I, Br, Cl) or $SOCl_2$

$$R-OH + HX \longrightarrow RX + H_2O$$

$$R-OH + PCl_5 \longrightarrow R-Cl + POCl_3 + HCl$$

$$R-OH + SOCl_2 \xrightarrow{\text{Pyridine}} R-Cl + SO_2 + HCl$$

$$R-OH + PX_3 \longrightarrow R-X + H_3PO_3$$

(ii) Addition of hydrogen halides to alkenes.

$$CH_2 = CH_2 + HBr \longrightarrow CH_3-CH_2Br$$

CHECK *Point*

☛ **Complete the following reaction**

$$\underset{\text{1-Butylene}}{CH_3-CH_2-CH=CH_2} + HBr \xrightarrow[\text{peroxides}]{}$$

SOLUTION

$$\underset{\text{1-Butylene}}{CH_3-CH_2-CH=CH_2} + HBr \xrightarrow{\text{peroxides}} \underset{\text{1-Bromobutane}}{CH_3-CH_2-CH_2-CH_2Br}$$

(iii) From alkanes

$$R-H + X_2 \xrightarrow{\Delta \text{ or } h\nu} R-X + HX\,(X = Cl, Br \text{ or } I)$$

(iv) From ethers

$$R-O-R + 2HX \xrightarrow{373\ K} 2R-X + H_2O\,(X = Br \text{ or } I)$$

$R-Cl + NaI \xrightarrow{\text{Acetone or Methanol}} R-I + NaCl$

This reaction is called Finkelstein reaction. Only iodoalkanes can be obtained by this method by displacing Cl *or* Br.

Physical Properties :

(i) Alkyl halides are insoluble in water.

(iv) **Boiling points :** Greater the molecular mass, stronger the Vander Waals forces of attraction and hence higher is the boiling point.

(a) RI > RBr > RCl > RF (For same alkyl group)

(b) Boiling point decreases as the size of alkyl group decreases

$$CH_3(CH_2)_2CH_2X > CH_3CH_2CH_2X > CH_3CH_2X\,[\text{For the same halogen atom}]$$

(v) **Dipole moment :** Except fluoride, dipole moment decreases with the decrease in electronegativity from Cl to I. Thus the order for dipole moment is $CH_3Cl > CH_3F > CH_3Br > CH_3I$.

Fluorides, however, have lower dipole moment than chlorides due to very small size of fluorine which out weighs the effect of greater electronegativity.

(vi) **Stability order :** Stability decreases as the strength of C–X bond decreases

$$R-F > R-Cl > R-Br > R-I$$

Thus, iodides on standing become violet or brown due to their decomposition in the presence of sunlight.

$$2R-I \xrightarrow{\text{sunlight}} R-R+I_2$$

The I_2 thus liberated dissolves in alkyl iodide to impart it a dark colour.

CHEMICAL PROPERTIES:

1. Reaction with metals :

(i) $R-X + 2Li \xrightarrow{\text{dry ether}} R-Li + LiX$

(ii) $R-X + Mg \xrightarrow{\text{dry ether}} R-MgX$

(Grignard reagent)

2. Reduction :

(a) $R-X + Zn + HX \longrightarrow R-H + ZnX_2$

In this reaction, zinc atoms transfer electrons to the carbon atom of the alkyl halide. Zinc is a good reducing agent because it has two electrons in an orbital far from the nucleus, which are readily donated to an electron acceptor.

(b) $R-X \xrightarrow{LiAlH_4} R-H$

(c) $R-X + H_2 \xrightarrow{Ni} R-H + HX$

(d) $R-X \xrightarrow{Mg} R-MgX \xrightarrow[\text{or } D_2O]{H_2O} RH \text{ (or RD)}$

(e) $R-I + HI \xrightarrow[423\ K]{\text{red P}} R-H + I_2$

CHECK *Point*

☛ **How ethane can be prepared by reduction method.**

SOLUTION

It can be prepared by reduction of ethyliodide with hydrogen iodide in presence of red phosphorus.

$$\underset{\text{Ethyliodide}}{CH_3CH_2I} + HI \xrightarrow[423\ K]{\text{red P}} \underset{\text{Ethane}}{CH_3-CH_3} + I_2$$

3. Halogenation :

$$CH_3Cl \xrightarrow{Cl_2,\ h\nu} CH_2Cl_2 \xrightarrow{Cl_2,\ h\nu} CHCl_3 \xrightarrow{Cl_2,\ h\nu} CCl_4$$

4. Isomerisation :

$$\underset{\text{1-Bromopropane}}{CH_3CH_2CH_2Br} \xrightarrow[\text{anhy. } AlCl_3]{573\ K} \underset{\text{2-Bromopropane}}{CH_3CHBrCH_3}$$

5. $R-X + \text{aq KOH} \longrightarrow \underset{\text{Alcohol}}{R-OH} + KX \text{ (Hydrolysis)}$

6. $R-X + R'ONa \xrightarrow{\Delta} \underset{\text{Ether}}{R-O-R'} + NaX$ (Williamson's synthesis)

CHECK Point

☛ **How you can prepare methyl ethyl ether ?**

SOLUTION

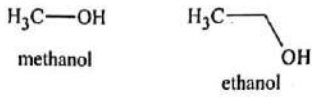
By reaction of methylchloride with sodium ethoxide.

$$\underset{\text{Methyl Chloride}}{CH_3-Cl} + \underset{\text{Sol.ethoxide}}{C_2H_5ONa} \xrightarrow{\Delta} \underset{\text{Methylethylether}}{CH_3-O-CH_2-CH_3} + NaCl$$

2. ALCOHOLS :

Alcohols, generally indicated by R–OH, can be regarded as derivative of alkanes where one or more of the hydrogens have been replaced by hydroxyl group (–OH)

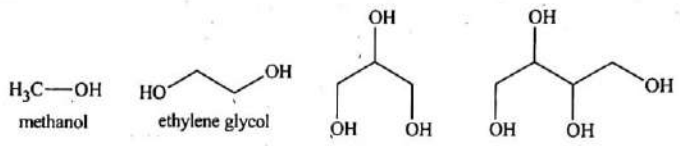
$$\underset{\text{Alkane}}{R-H} \xrightarrow[+OH]{-H} \underset{\text{Alcohol}}{R-OH}$$



methanol ethanol

Classification of Alcohols :

Alcohols are classified according to no. of hydroxyl groups present in their molecule. Alcohols with one, two and three hydroxyl groups are called monohydric, dihydric & trihydric alcohols respectively. Alcohols which contains four or more hydroxyl groups are called polyhydric alcohols.

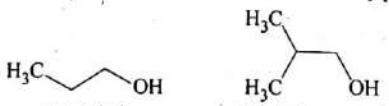


methanol ethylene glycol glycerol butane-1,2,3,4-tetrol

When two or more hydroxyl groups attached to the same carbon atom the compound is generally unstable. The compound loses a water molecule and is converted into a stable compound.

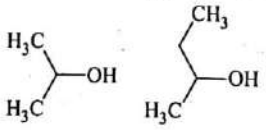
Monohydric alcohols are further divided into three classes:

(i) **Primary alcohol:** The hydroxyl group is attached with primary carbon atom. They possess a characteristic group —CH_2—OH

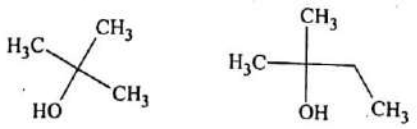


propan-1-ol 2-methylpropan-1-ol

(ii) **Secondary alcohol:** The hydroxyl group attached to secondary carbon atom. They possess >CH—OH group.



(iii) **Tertiary alcohol:** The hydroxyl group attached to tertiary carbon atom. They possess a characteristic group $-\overset{|}{\underset{|}{C}}-OH$.

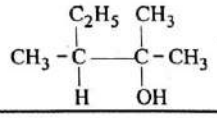


CHECK *Point*

☛ **3 Ethyl-2-methyl butan-2-ol is primary, secondary or tertiary alcohol?**

SOLUTION

It is a tertiary alcohol as –OH functional group is attach to tertiary carbon atom.



Isomerism :

The alcohols exhibit

(i) **Position isomerism:** eg. $CH_3-CH_2-CH_2-OH$ (1 – Propanol); $CH_3-\underset{OH}{\underset{|}{CH}}-CH_3$ (2 – Propanol)

(ii) **Chain isomerism:** eg. $CH_3-CH_2-CH_2-CH_2OH$ (1 – Butanol); $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2OH$ (2 – Methyl – 1 – propanol)

Ethyl Alcohol :

Physical properties :

Ethanol is a versatile solvent, miscible in all proportions with water and many organic solvents, including acetic acid, acetone, benzene, carbon tetrachloride, chloroform. Ethanol's miscibility with water is in contrast to longer chain alcohols (five or more carbons), whose water solubility decreases rapidly as the number of carbons increases. The polar nature of the hydroxyl group causes ethanol to dissolve in many ionic compounds.

Ethanol and mixtures with water greater than about 50% ethanol are flammable and easily ignited.

Reactions of Ethanol:

1. **Halogenation**

Ethanol reacts with hydrogen halides to produce ethyl halides such as ethyl chloride and ethyl bromide:

$$CH_3CH_2OH + HCl \longrightarrow CH_3CH_2Cl + H_2O$$

This reaction requires a catalyst such as zinc chloride. Hydrogen chloride in the presence of zinc chloride is known as Lucas reagent.

$$CH_3CH_2OH + HBr \longrightarrow CH_3CH_2Br + H_2O$$

Ethyl halides can also be produced by reacting ethanol with more specialized halogenating agents, such as thionyl chloride for preparing ethyl chloride, or phosphorus tribromide for preparing ethyl bromide.

2. **Ester formation**

Under acid-catalyzed conditions, ethanol reacts with carboxylic acids to produce ethyl esters and water:

$$RCOOH + HOCH_2CH_3 \xrightarrow{H_3O^+} RCOOCH_2CH_3 + H_2O$$

For this reaction to produce useful yields it is necessary to remove water from the reaction mixture as it is formed.

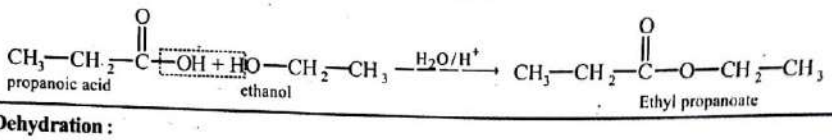
Ethanol can also form esters with inorganic acids. Diethyl sulfate and triethyl phosphate, prepared by reacting ethanol with sulfuric and phosphoric acid respectively, are both useful ethylating agents in organic synthesis. Ethyl nitrite, prepared from the reaction of ethanol with sodium nitrite and sulfuric acid, was formerly a widely-used diuretic.

CHECK Point

☛ **How you can prepare ethylpropanoate from ethanol?**

SOLUTION

By reaction of ethanol with propanoic acid in acidic media.



3. Dehydration :

Strong acids such as sulfuric acid cause ethanol's dehydration to form ethylene:

$$CH_3CH_2OH \xrightarrow[453K]{H^+} H_2C=CH_2+H_2O$$

Dehydration of alcohols to ethers or a alkenes can also be brought about by passing the vapour of alcohols over heated alumina catalyst under different conditions

$$2CH_3CH_2OH \xrightarrow[513-523\,K]{Al_2O_3} \underset{\text{Diethyl ether}}{CH_3CH_2OCH_2CH_3}+H_2O$$

$$CH_3CH_2OH \xrightarrow[633\,K]{Al_2O_3} \underset{\text{Ethene}}{CH_2=CH_2}+H_2O$$

4. Oxidation :

Ethanol can be oxidized to acetaldehyde (also known as ethanal, is an organic chemical compound with the formula CH_3CHO, occurs naturally in ripe fruit, coffee, and fresh bread), and further oxidized to acetic acid. This oxidation is brought about by common oxidising agent used in organic chemistry like alkaline $KMnO_4$, acidified $K_2Cr_2O_7$, chromic acid etc.

$$CH_3CH_2OH \xrightarrow{\text{alk } KMnO_4} CH_3CHO \xrightarrow{\text{alk } KMnO_4} CH_3COOH$$

5. Chlorination :

When exposed to chlorine, ethanol is both oxidized and its carbon chlorinated to form the compound, chloral.

$$4Cl_2+C_2H_5OH \longrightarrow \underset{\text{Chloral}}{CCl_3CHO}+5HCl$$

6. Combustion :

Combustion of ethanol forms carbon dioxide and water:

$$C_2H_5OH+3\,O_2 \longrightarrow 2CO_2+3H_2O$$

Manufacture of Ethanol : (Industrial Scale)

Ethanol is made by fermentation of sugar or by hydration of ethene.

Ethanol preparation from fermentation of sugar

Fermentation is a process whereby an organic compound is broken into smaller molecules by enzymes such as yeast. Ethanol is prepared by fermentation of molasses. Molasses, a by product of sugar industry, has traces of sugar, which gets broken down into ethanol. The fermentation is initiated by yeast, which secretes two enzymes called invertase and zymase. These enzymes act as catalyst for converting sugar into ethanol. The reaction is exothermic and carbon dioxide is released in the process.

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}}+H_2O \xrightarrow{\text{Invertase}} \underset{\text{Glucose}}{C_6H_{12}O_6}+\underset{\text{Fructose}}{C_6H_{12}O_6}$$

$$\underset{\text{Glucose or fructose}}{C_6H_{12}O_6} \xrightarrow{\text{Zymase}} \underset{\text{Ethanol}}{2C_2H_5OH}+2CO_2$$

Molasses is heated to about 20 to 30°C in an airtight container. Yeast is mixed in the molasses. The airtight container has outlets for removal of carbon dioxide released during the fermentation process. Ethanol and water vapours that are formed are collected and separated by fractional distillation.

- *Presence of air will tend to oxidize the mixture whose resultant is vinegar (acetic acid or ethanoic acid)* CH_3COOH. *This is undesirable.*
- *Ethanol can also be made by fermentation of starchy materials such as barley, potatoes and maize. The starch is first converted to sugar by fermentation and then to ethanol.*

Ethanol preparation by hydration of ethene

The molecular formula of ethene is C_2H_4. An addition of a water molecule will make ethene into ethanol (C_2H_5OH).

Addition of water to a molecule and its assimilation into the molecular formula is known as hydration.

Ethene obtained from cracking of petroleum along with water is pressurized to 60 atmosphere and passed over heated (300°C) phosphoric acid (H_3PO_4). Ethanol is directly produced in this reaction. The phosphoric acid is a catalyst. The reaction is shown below.

$$\underset{\text{Ethene}}{CH_2 = CH_2} + \underset{\text{Water}}{H - OH} \xrightarrow[300°C,60atm]{H_3PO_4} \underset{\substack{\text{Ethanol}\\(\text{Ethyl alcohol})}}{CH_3 - CH_2 - OH}$$

ILLUSTRATION 5

Preparation of ethylalcohol from ethene is which type of reaction ?

SOLUTION :

It is an example of addition reaction in which addition of water molecule to ethene takes place.

Knowledge ENHANCER

How do alcohols affect living beings?

Superficially, ethanol evokes a distinctive heat-like sensation in the mouth and a stinging sensation on the skin. In the body it is metabolized to other substances, affecting the central nervous system in particular. The effect varies between individuals, and can be worse when applied in addition to certain drugs. When applied to open wounds (as for disinfection) it produces a strong stinging sensation. Pure or highly concentrated ethanol may permanently damage living tissue on contact. When large quantities of ethanol are consumed, it tends to slow metabolic processes and to depress the central nervous system. This results in lack of coordination, mental confusion, drowsiness, lowering of the normal inhibitions, and finally stupour.

The individual ethanol consuming may feel relaxed but does not realise that his sense of judgement, sense of timing, and muscular coordination have been seriously impaired. Ethanol within the human body is converted into acetaldehyde by alcohol dehydrogenase and then into acetic acid by acetaldehyde dehydrogenase. The product of the first step of this breakdown, acetaldehyde, is more toxic than ethanol. It has been shown to increase the risk of developing cirrhosis of the liver, multiple forms of cancer, and alcoholism. Death from ethyl alcohol consumption is possible when blood alcohol level reaches 0.4%. A blood level of 0.5% or more is commonly fatal.

Uses of ethanol :

- Ethanol is used for manufacturing of paints, dyes, varnishes.
- Ethanol is used in medicines especially for disinfecting area on the skin before giving an injection. It is used for sterilization of syringes in hospitals.
- Ethanol is used for preparation of compounds such as chloroform and ether.
- Ethanol is used for making thermometer that is used for measuring low temperatures.
- Ethanol is used in spirit lamps.
- Ethanol blended fuel called rectified spirit is used as a substitute for fuels in vehicles that are environmental pollutant. They give off low emissions of carbon monoxide gas that is harmful to the environment. Ethanol is the main component of alcoholic beverages such as rum, whisky and beer.

Ethanol is used as an organic solvent. Ethanol is used as an antifreeze a mixture for radiators in cars. Ethanol mixed with water freezes at a lower temperature than water. Such an antifreeze mixture is used for radiators in cold countries.

idea box

Make a project report on the use of ethylalcohol as a fuel. Mention which type of ethyl alcohol blended fuels are currently or in near by future sold in market. Also mention the composition of these fuels. Make into account the merits and demerits of using ethyl alcohol as fuel. You can consult your teacher and can take help of internet, television, science magazine etc.

3. ETHERS:

Ethers are a class of compounds which contains an oxygen atom bonded to two alkyl groups (same or different). When both the alkyl groups are same they are designated as simple or symmetrical ethers and when both are different they are called mixed or unsymmetrical ethers.

R—O—R	R—O—R'
CH_3—O—CH_3	CH_3—O—C_2H_5
C_2H_5—O—C_2H_5	CH_3—O—C_3H_7
Simple ethers	Mixed ethers.

Ethers form a homologous series with the general formula $C_nH_{2n+2}O$ where the value of 'n' is always more than 1; the functional group of ether is (—O—). Ethers may be regarded as anhydrides of alcohol as they may be obtained by elimination of water molecule from two alcohol molecules.

$$\underset{\text{Alchol}}{R-O\boxed{H}} + \underset{\text{Alchol}}{\boxed{H}O-R} \longrightarrow \underset{\text{Ether}}{R-O-R} + H_2O$$

Just as alcohol are considered as monoalkyl derivatives of water ethers may be considered as dialkyl derivatives of water.

Isomerism in ethers: Ethers exhibit two types of isomerism

(i) ***Functional Isomerism:*** They are isomeric with monohydric alcohols eg.

CH_3-O-CH_3 $CH_3.CH_2OH$

methoxy methane ethyl alcohol

(ii) ***Metamerism:*** eg. $C_2H_5-O-C_2H_5$

ethoxy ethane

$CH_3-O-\overset{1}{C}H_2-\overset{2}{C}H_2-\overset{3}{C}H_3$ $CH_3-O-\overset{2}{C}H(\overset{1}{C}H_3)(\overset{3}{C}H_3)$

1-methoxy propane 2-methoxy propane

Preparation of Ethers :

1. **By dehydrating excess of alcohols :** Simple ethers can be prepared by heating an excess of primary alcohols with conc. H_2SO_4 at 413 K. Alcohol should be taken in excess so as to avoid its dehydration to alkenes.

$$\underset{\text{Ethanol (2 molecules)}}{C_2H_5-OH + HO-C_2H_5} \xrightarrow[413K]{\text{Conc. } H_2SO_4} \underset{\text{Diethyl ether}}{C_2H_5-O-C_2H_5} + H_2O$$

Dehydration may also be done by passing alcohol vapours over heated catalyst like alumina under high pressure and temperature.

2. **By heating alkyl halide with dry silver oxide :** (Only for simple ethers).

$$C_2H_5I + Ag_2O + IC_2H_5 \longrightarrow C_2H_5-O-C_2H_5 + 2AgI$$

3. **By heating alkyl halide with sod. or pot. alkoxides** (Williamson synthesis). This is the most important industrial and laboratory method and may be used for preparing simple as well as mixed ethers. For example,

$$C_2H_5ONa + ICH_3 \longrightarrow C_2H_5-O-CH_3 + NaI$$

CHECK *Point*

☛ **How you can prepare methyl butyl ether by methyliodide ?**

SOLUTION

By williamson's synthesis by reaction of methyliodide with sodium butoxide

$$CH_3I + CH_3-CH_2-CH_2-CH_2ONa \longrightarrow CH_3-CH_2-CH_2-CH_2-O-CH_3 + NaI$$

Physical Properties :

1. The first two members, i.e., dimethyl ether and methylethyl ether are gases; while all others are colourless, volatile pleasant smelling liquids. Anisole and phenetole are liquids; diphenyl ether is solid at room temperature.
2. Ethers are highly inflammable.
3. They are sparingly soluble in water, but readily soluble in organic solvents such as benzene, chloroform, etc.
4. Ethers are slightly polar and have a small net dipole moment (e.g., 1.18 D for diethyl ether)

Chemical Properties :

(A) Properties due to alkyl groups :

1. **Halogenation :** When ethers are treated with chlorine or bromine in the dark, substitution occurs at the α-carbon atom. The extent of substitution depends upon the reaction conditions.

$$CH_3-CH_2-O-CH_2-CH_3 \xrightarrow[\text{in dark}]{Cl_2} \underset{\text{1-Chlorodiethyl ether}}{CH_3-CH(Cl)-O-CH_2-CH_3} \xrightarrow[\text{in dark}]{Cl_2} \underset{\text{1,1'-Dichlorodiethylether}}{CH_3-CH(Cl)-O-CH(Cl)-CH_3}$$

2. **Combustion :** Since ethers are volatile and highly inflammable they burn in air to form CO_2 and water.

$$C_2H_5-O-C_2H_5 + 6O_2 \longrightarrow 4CO_2 + 5H_2O$$

(B) Properties due to ethereal Oxygen :

1. **Chemical inertness :** Since ethers do not have an active group, in their molecules, these do not react with active metals like Na, strong bases like NaOH, reducing or oxidising agents.

2. **Basic nature :** Owing to the presence of unshared electron pairs on oxygen, ethers behave as Lewis bases. Hence they dissolve in strong acids (e.g., conc. HCl, conc. H_2SO_4) at low temperature to form oxonium salts.

$$\underset{\text{Diethyl ether}}{(C_2H_5)_2O} + H_2SO_4 \longrightarrow \underset{\text{Diethyloxonium hydrogen sulphate}}{[(C_2H_5)_2OH]^+HSO_4^-}$$

On account of this property, ether is removed from ethyl bromide by shaking with conc. H_2SO_4. The oxonium salts are stable only at low temperature and in a strongly acidic medium. On dilution, they decompose to give back the original ether and acid.

Being Lewis bases, ethers also form coordination complexes with Lewis acids like BF_3, $AlCl_3$, RMgX, *etc.*

$$R_2\ddot{O}: + BF_3 \longrightarrow R_2\ddot{O} \longrightarrow BF_3$$

(C) Properties due to carbon-oxygen bond :

1. **Hydrolysis :** Ethers, when heated with dilute sulphuric acid under pressure, are hydrolysed to the corresponding alcohols.

$$C_2H_5 - O - C_2H_5 + H_2O \xrightarrow{H_2SO_4} \underset{\text{Ethyl alcohol}}{2C_2H_5OH}$$

The hydrolysis may also be effected by boiling the ether with water or by treating it with steam.

2. **Action of conc. sulphuric acid :** On heating with concentrated sulphuric acid, ethers are converted to alkyl hydrogen sulphates.

$$C_2H_5 - O - C_2H_5 + H_2SO_4(\text{conc.}) \longrightarrow \underset{\text{Ethyl alcohol}}{C_2H_5OH} + \underset{\text{Ethyl hydrogen sulphate}}{C_2H_5HSO_4}$$

$$C_2H_5OH + H_2SO_4 \longrightarrow C_2H_5HSO_4 + H_2O$$

3. **Action of hydroiodic or hydrobromic acid :** In this reaction, the ethereal linkage is broken; the product depends upon the temperature of the reaction.

(i) **In cold**, ethers react with HI or HBr to give the corresponding alkyl halide and alcohol.

$$\underset{\text{Ethyl ether}}{C_2H_5 - O - C_2H_5} + HI \longrightarrow \underset{\text{Ethyl iodide}}{C_2H_5I} + \underset{\text{Ethyl alcohol}}{C_2H_5OH}$$

In case of mixed ethers, the halogen atom attaches itself to the smaller alkyl group.

$$\underset{\text{Ethylmethyl ether}}{CH_3 - O - C_2H_5} + HI \longrightarrow \underset{\text{Mehyl iodide}}{CH_3I} + \underset{\text{Ethyl alcohol}}{C_2H_5OH}$$

(ii) Ethers **when heated** with conc. halogen acids form two molecules of alkyl halides.

$$C_2H_5 - O - C_2H_5 + 2HI \xrightarrow{\Delta} 2C_2H_5I + H_2O$$

$$CH_3 - O - C_2H_5 + 2HI \xrightarrow{\Delta} CH_3I + C_2H_5I + H_2O$$

4. **Action of phosphorus pentachloride :** Ethers react with phosphorus pentachloride on heating to form alkyl halides.

$$\underset{\text{Diethyl ether}}{C_2H_5 - O - C_2H_5} + PCl_5 \longrightarrow \underset{\text{Ethyl chloride}}{C_2H_5Cl} + C_2H_5Cl + POCl_3$$

CHECK Point

☛ **Complete the following reaction**

$$\underset{\text{Methylethylether}}{CH_3 - O - C_2H_5} + PCl_5 \longrightarrow$$

SOLUTION

$$\underset{\text{Methylethylether}}{CH_3-O-C_2H_5} + PCl_5 \longrightarrow \underset{\text{Methylchloride}}{CH_3Cl} + \underset{\text{Ethyl chloride}}{C_2H_5Cl} + POCl_3$$

5. **Dehydration :** Ether vapours when passed over heated alumina undergoes dehydration to form alkenes.

$$C_2H_5-O-C_2H_5 \xrightarrow[633K]{Al_2O_3} \underset{\text{Ethene}}{2CH_2=CH_2} + H_2O$$

Uses of ethers :

1. The most important use of ethers is as in laboratory as well as industrial solvent (for oils, resins, gums, etc.)
2. Ether is used as a reaction medium for preparing Gringnard reagents and for reductions with lithium aluminium hydride.
3. Ether is used as an anaesthic in surgery.
4. Ether may be used as a referigerant.

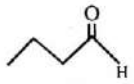
4. ALDEHYDES AND KETONES :

Aldehydes and Ketones are characterised by the presence of carbonyl group >C = O in their molecules. Aldehydes contain group $\overset{H}{\underset{R}{>}}C=O$ and ketones the $\overset{R}{\underset{R}{>}}C=O$. If the groups attached to carbonyl carbon are the same, the ketone is symmetrical and if they are different the ketone is unsymmetrical.

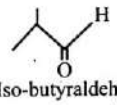
Isomerism in Aldehydes and Ketones :

isomerism in aldehydes :

(i) Chain isomerism : Example

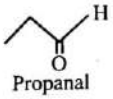


n-butyraldehyde



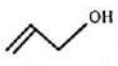
Iso-butyraldehyde

(ii) Functional isomerism :



Propanal

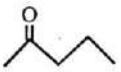
Acetone



Allylalcohol

isomerism in ketones : They exhibit three types of isomersim

(i) Chain isomerism :

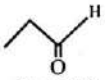


2-pentanone

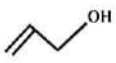
3-methyl-2-butanone

(ii) Functional isomersim :

Acetone

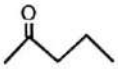


Propanal

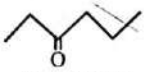


Allylalcohol

(iii) Position isomerism :



2-pentanone



3-pentanone

1. FORMALDEHYDE OR METHANAL (HCHO)

Formaldehyde is the first member of the aldehyde series. It is present in green leaves of plants where its presence is supposed to be due to the reaction of CO_2 with water in presence of sunlight and chlorophyll.

$$CO_2 + H_2O \longrightarrow HCHO + O_2$$

Traces of formaldehyde are formed when incomplete combustion of wood, sugar, coal, etc., occurs.

Preparation : It can be prepared by the following methods :

(i) **By oxidation of methyl alcohol :** The oxidation is done either by oxygen in presence of a catalyst like platinised asbestos or copper at 300-400°C or by acidified potassium dichromate.

$$2CH_3OH + O_2 \xrightarrow[300\text{-}400°C]{\text{Copper}} 2HCHO + O_2$$

$$CH_3OH + [O] \xrightarrow[H_2SO_4]{K_2Cr_2O_7} HCHO + H_2O$$

(ii) **By dehydrogenation of methyl alcohol :** Vapours of methyl alcohol are passed over copper or silver at 300-400°C when hydrogen is eliminated with the formation of formaldehyde.

$$CH_3OH \xrightarrow[300-400°C]{\text{Cu or Ag}} HCHO + H_2$$

CHECK Point

What are the products obtained when ethylalcohol is dehydrogenated?

SOLUTION

Acetaldehyde and hydrogen is obtained

$$\underset{\text{ethylalcohol}}{CH_3CH_2OH} \xrightarrow[300-400°C]{\text{Cu or Ag}} \underset{\text{Acetaldehyde}}{CH_3CHO} + \underset{\text{hydrogen}}{H_2}$$

(iii) **By heating calcium formate :** Dry calcium formate is strongly heated to obtain formaldehyde.

$$\underset{\text{Calcium formate}}{Ca(HCOO)_2} \xrightarrow{\text{Heat}} CaCO_3 + \underset{\text{Formaldehyde}}{HCHO}$$

Above method is, however, not very suitable for the preparation of other aldehydes because the yield is very low

(iv) **By ozonolysis of ethylene :** Ethylene is dissolved in an organic solvent such as chloroform or carbon tetrachloride and a stream of ozonised oxygen is passed. The ozonide formed is decomposed by passing hydrogen in presence of palladium as catalyst.

$$CH_2 = CH_2 + O_3 \longrightarrow \underset{\text{Ozonide}}{\text{(ozonide ring)}} \xrightarrow[Pd]{H_2} 2HCHO + H_2O$$

In above reaction methanal can also be obtained by hydrolysis of ozonide in presence of zinc

Manufacture : It is also manufactured by controlled oxidation of methane by air in presence of various metallic oxides as catalyst.

$$\underset{\text{Methane}}{CH_4} + O_2 \xrightarrow[\text{Catalyst}]{\text{Mo-oxide}} \underset{\text{Formaldehyde}}{HCHO} + H_2O$$

It is also prepared by passing water gas at low pressure through an electric discharge of low intensity.

$$\underbrace{CO + H_2}_{\text{Water gas}} \xrightarrow{\text{Elec. discharge}} \underset{\text{Formaldehyde}}{HCHO}$$

Physical Properties :

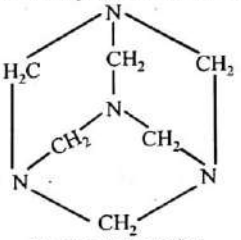
(i) It is a colourless, pungent smelling gas.
(ii) It is extremely soluble in water.
(iii) It can easily be condensed into liquid. The liquid formaldehyde boils at –21°C.
(iv) It causes irritation to skin, eyes, nose and throat.
(v) Its solution acts as antiseptic and disinfectant.

Chemical Properties :

Formaldehyde is structurally different from other aldehydes as it contains no alkyl group in the molecule. Through it shows general properties of aldehydes, it differs in certain respects. The abnormal properties of formaldehyde are given below :

(i) **Reaction with ammonia :** Like other aldehydes, formaldehyde does not form addition products but a crystalline compound, hexamethylene tetramine, with ammonia.

$$\underset{\text{Formaldehyde}}{6HCHO} + 4NH_3 \longrightarrow \underset{\substack{\text{Urotropine}\\\text{(Hexamethylene tetramine)}}}{(CH_2)_6N_4} + 6H_2O$$



Structure of Urotropine

Urotropine is used as a medicine to treat urinary infections.

(ii) **Reaction with sodium hydroxide (Cannizzaro's reaction) :** It does not form resin with sodium hydroxide like acetaldehyde but when treated with a concentrated solution of sodium hydroxide, two molecules of formaldehyde undergo mutual oxidation and reduction forming formic acid salt and methyl alcohol (Disproportionation).

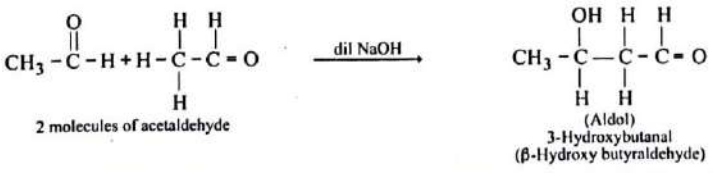
$$\underset{\text{Formaldehyde}}{2HCHO} + \underset{\text{(Conc)}}{NaOH} \longrightarrow \underset{\text{Sod. formate}}{HCOONa} + \underset{\text{Methyl alcohol}}{CH_3OH}$$

The above reaction is also called cannizaro's reaction and is only given by those aldehdes which do not have α-hydrogen atom (hydrogen atoms present at carbon atom bonded to functional group of organic compounds)

(iii) **Aldol condensation :** Formaldehyde in presence of a weak base undergo repeated aldol condensation to give formose (α-acrose).

$$\underset{\text{Formaldehyde}}{6HCHO} \xrightarrow{Ca(OH)_2} \underset{\text{Formose (hexose)}}{C_6H_{12}O_6}$$

Aldol condensation : Aldehydes and Ketones containing α-hydrogen (H-atoms attached to the C-atom adjacent to the carbonyl group) undergo self condensation in the presence of a dilute alkali to form β-hydroxyaldehyde or β-hydroxyketone.



2 molecules of acetaldehyde

(Aldol)
3-Hydroxybutanal
(β-Hydroxy butyraldehyde)

Uses :

(i) The 40% solution of formaldehyde (formalin) is used as disinfectant, germicide and antiseptic. It is used for the preservation of biological specimens.
(ii) It is used in the preparation of hexamethylene tetramine (urotropine) which is used as an antiseptic and germicide.
(iii) It is used in silvering of mirror.
(iv) It is employed in the manufacture of synthetic dyes such as *para*-rosaniline, indigo, etc.
(v) It is used in the manufacture of **formamint** (by mixing formaldehyde with lactose) – a throat lozenges.
(vi) It is used for making synthetic plastics like bakelite, urea-formaldehyde resin, etc.

As a methylating agent for primary and secondary amines, e.g.,

$$\underset{\text{Ethylamine}}{C_2H_5NH_2} + 2HCHO \longrightarrow \underset{\text{Ethyl methylamine}}{C_2H_5NH - CH_3} + HCOOH$$

2. ACETONE OR PROPANONE, CH_3COCH_3 OR DIMETHYL KETONE

It is a symmetrical (simple) ketone and is the first memeber of the homologous series of ketones. In traces, it is present in blood and urine.

Laboratory preparation : Acetone is prepared in laboratory by heating anhydrous calcium acetate.

$$\underset{\text{Calcium acetate}}{(CH_3COO)_2Ca} \xrightarrow{\Delta} CaCO_3 + \underset{\text{Acetone}}{CH_3COCH_3}$$

Manufacture : Acetone is manufactured by following methods :

By air-oxidation of isopropyl alcohol : The air oxidation occurs at 500°C.

$$\underset{\text{Isopropyl alcohol}}{CH_3CHOHCH_3} + O_2 \xrightarrow{500°C} \underset{\text{Acetone}}{CH_3COCH_3} + 2H_2O$$

By dehydrogenation of isopropyl alcohol

$$\underset{\text{2-Propanol}}{CH_3 - \overset{\overset{OH}{|}}{CH} - CH_3} \xrightarrow{Cu,\ 573\ K} \underset{\text{Acetone}}{CH_3COCH_3} + H_2$$

Physical Properties :

(i) It is a colourless liquid with characteristic pleasant odour.
(ii) It is highly flammable liquid. It boils at 56°C.
(iii) It is highly miscible with water, alcohol and ether.

Chemical Properties :

it shows general properties of ketones. Some special properties of acetone are described below :

(i) **Haloform reaction :** When heated with iodine and sodium hydroxide, it forms yellow crystals of iodoform.

$$\underset{\text{Acrtone}}{CH_3COCH_3} + 3I_2 \longrightarrow \underset{\text{Triiodo acetone}}{CI_3COCH_3} + 3HI$$

$$CI_3COCH_3 + NaOH \longrightarrow \underset{\text{Iodoform}}{CHI_3} + \underset{\text{Sodium acetate}}{CH_3COONa}$$

(ii) Acetone forms chloroform when heated with bleaching powder in water.

$$CaOCl_2 + H_2O \longrightarrow Ca(OH)_2 + Cl_2$$

$$CH_3COCH_3 + 3Cl_2 \longrightarrow CCl_3COCH_3 + 3HCl$$

$$2CCl_3COCH_3 + Ca(OH)_2 \longrightarrow \underset{\text{Chloroform}}{2CHCl_3} + \underset{\text{Calcium acetate}}{Ca(CH_3COO)_2}$$

Uses :

(i) As a solvent for cellulose acetate, cellulose nitrate, celluloid, lacquers, resins, etc.
(ii) For storing acetylene.
(iii) In the manufacture of cordite – a smokeless powder explosive.
(iv) In the preparation of chloroform, iodoform, sulphonal and chloretone.
(v) As a nailpolish remover.
(vi) In the preparation of an artificial scent (ionone), plexiglass (unbreakable glass) and synthetic rubber.

CHECK Point

☛ **For which alkyne acetone is used as a storage material ?**

SOLUTION

Acetylene

5. CARBOXYLIC ACID :

The compounds containing the carboxyl functional group $-\overset{\overset{O}{\|}}{C}-OH$ are called **Carboxylic** acids. The word carboxyl is a combination of two words Carbonyl (>C = O) and hydroxyl (–OH).

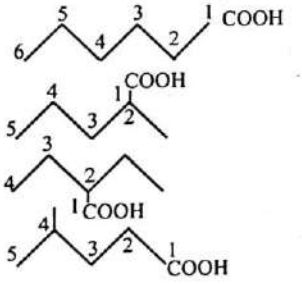
Classification :

Depending upon the number of –COOH groups they are classified as (i) Manocarboxylic acids: containing one - COOH group (ii) dicarboxylic acids: containing two -COOH groups and so on

Fatty acids: Aliphatic monocarboxylic acids are commonly called fatty acids because higher members are obtained by the hydrolysis of oils and fats.

Isomerism: They exhibit following type of isomerism

(i) **Position and chain Isomerism** eg. $C_6H_{12}O_2$ represent

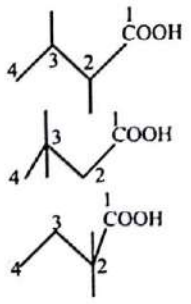


1-Hexanoic acid

2-methyl pentanoic acid } Position isomers

2-ethyl butanoic acid

4-methyl pentanoic acid



2, 3 - dimethyl butanoic acid

3, 3 - dimethyl butanoic acid } All the three are chain Isomers

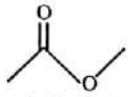
2, 2 - dimethyl butanoic acid

(ii) Functional Isomerism

Formula Functional Isomers

$C_3H_6O_2$

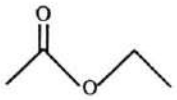
Propanoic acid



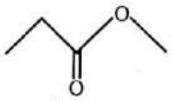
methylacetate

$C_4H_8O_2$

butanoic acid



ethyl ethanoate



methyl propanoate

ACETIC ACID (ETHANOIC ACID):

Acetic acid, also known as ethanoic acid, is an organic chemical compound best recognized for giving vinegar its sour taste and pungent smell. Its structural formula is represented as CH_3COOH. The trivial name acetic acid is the most commonly used and officially preferred name by the IUPAC. Glacial acetic acid is a trivial name for water-free acetic acid, the name comes from the ice-like crystals that form slightly below room temperature at 16.7 °C (about 62 °F).

It is a weak acid because at standard temperature and pressure the dissociated acid exists in equilibrium with the undissociated form in aqueous solutions, in contrast to strong acids, which are fully dissociated.

Physical Properties :

Pure, water-free acetic acid (glacial acetic acid) is a colourless liquid that attracts water from the environment (hygroscopy), and freezes below 16.7°C (62°F) to a colourless crystalline solid. Acetic acid is one of the simplest carboxylic acids (the second-simplest, next to formic acid). Acetic acid is corrosive, and its vapour causes irritation to the eyes, a dry and burning nose, sore throat and congestion to the lungs.

Chemical Properties :

The hydrogen (H) atom in the carboxyl group (-COOH) in carboxylic acids such as acetic acid can be given off as an H^+ ion (proton), giving them their acidic character. Acetic acid is a weak, effectively monoprotic acid in aqueous solution, Acetic acid is corrosive to metals including iron, magnesium, and zinc, forming hydrogen gas and metal salts called acetates.

Reactions of ethanoic acid:

1. **Esterification reaction :**

Esterification is the reaction between an acid and an alkanol producing ester(sweet-smelling substances). An acid, containing the -COOH functional group, can react with an alkanol, containing the-OH functional group, to produce an ester and water.

$$\underset{\text{alkanol}}{\text{R-OH}} + \underset{\text{acid}}{\text{HOOC-R'}} \rightleftharpoons \underset{\text{ester}}{\text{R-OOC-R'}} + \underset{\text{water}}{H_2O}$$

Ester are used in making perfumes and as flavouring agents.

Esters react in the presence of an acid or a base to give back the alcohol and carboxylic acid. This reaction is known as saponification because it is used in the preparation of soap.

2. **Reaction with a base:** Like mineral acids, ethanoic acid reacts with a base such as sodium hydroxide to give a salt (sodium ethanoate or commonly called sodium acetate) and water:

$$NaOH + CH_3COOH \rightarrow CH_3COONa + H_2O$$

3. **Reaction with carbonates and hydrogencarbonates:** Ethanoic acid reacts with carbonates and hydrogencarbonates to give rise to a salt, carbon dioxide and water. The salt produced is commonly called sodium acetate.

$$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$$
$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$$

CHECK Point

☛ **What are the products obtained on reaction of acetic acid with magnesium carbonate?**

SOLUTION

Magnesium acetate, water and carbondioxide

$$\underset{\text{Acetic acid}}{2CH_3COOH} + \underset{\text{Magnesium carbonate}}{MgCO_3} \longrightarrow \underset{\text{Magnesium acetate}}{(CH_3COO)_2Mg} + H_2O + CO_2\uparrow$$

4. **Reaction with (PCl_5, PCl_3 and $SOCl_2$)**

$$\underset{\text{Acetic acid}}{CH_3COOH} + PCl_5 \longrightarrow \underset{\text{Acetylchloride}}{3CH_3COCl} + POCl_3 + HCl$$

$$\underset{\text{Acetic acid}}{3CH_3COOH} + PCl_3 \longrightarrow \underset{\text{Acetylchloride}}{3CH_3COCl} + H_3PO_3$$

$$\underset{\text{Acetic acid}}{CH_3COOH} + SOCl_2 \longrightarrow \underset{\text{Acetylchloride}}{CH_3COCl} + SO_2 + HCl$$

5. **Decarboxylation :** Sodium or potasium salts of carboxylic acids on heating with soda lime (NaOH + CaO) give alkanes which contain one carbon atom less than the parent acid

$$CH_3COONa + NaOH \xrightarrow{CaO,\ Heat} CH_4 + Na_2CO_3$$

Uses :

It is an important chemical reagent and industrial chemical that is used in the production of polyethylene terephthalate mainly used in soft drink bottles; cellulose acetate, mainly for photographic film; and polyvinyl acetate for wood glue, as well as synthetic fibres and fabrics. In households diluted acetic acid is often used in descaling agents. In the food industry acetic acid is used under the food additive code E260 as an acidity regulator.

SOAPS AND DETERGENTS :

Soaps are sodium or potassium fatty acids salts, produced from the alkaline hydrolysis of fats (higher fatty acids) in a chemical reaction called saponification. They are of biological origin. Each soap molecule has a long hydrocarbon chain, sometimes called its 'tail', with a carboxylate 'head'. In water, the sodium or potassium ions float free, leaving a negatively-charged head.

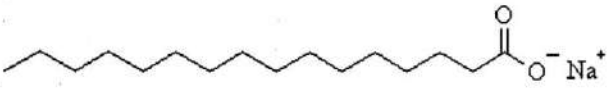
$$\underset{\text{Glyceride}}{\begin{matrix}CH_2-O-COC_{17}H_{35}\\ |\\ CH-O-COC_{17}H_{35}\\ |\\ CH_2-O-COC_{17}H_{35}\end{matrix}} + 3NaOH \longrightarrow \underset{\text{Glycerol}}{\begin{matrix}CH_2OH\\ |\\ CHOH\\ |\\ CH_2OH\end{matrix}} + \underset{\text{Soap}}{3C_{17}H_{35}COO^-\ Na^+}$$

Soap is an excellent cleanser because of its ability to act as an emulsifying agent. An emulsifier is capable of dispersing one liquid into another immiscible liquid. This means that while oil (which attracts dirt) doesn't naturally mix with water, soap can suspend oil/dirt in such a way that it can be removed.

Detergents(syndets) are emulsifying agents that are very similar to soaps but they do not have a biological origin and they do not have a carboxylate group instead of it they have sulphonate group as they are sodium salt of long chain sulphonic acid. Synthetic detergents usually have a long hydrocarbon chain with between 12 and 20 carbon atoms. At the head of this chain there is a polar group that is a strong hydrogen bonder to water but does not form strong ionic bonds with the ions of hard water.

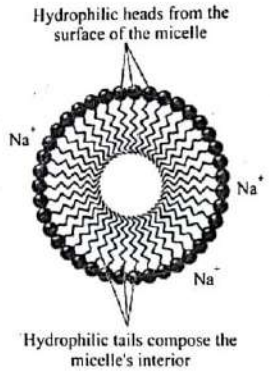
Before dealing with cleaning action of soap let us understand few points:

- The salts of fatty acids have an ionic polar head group, the carboxylate and a long non-polar hydrocarbon chain.



non-polar hydrocarbon polar head

- The polar group is hydrophilic (i.e. water-loving), the non-polar portion is hydrophobic (i.e. water-hating) or lipophilic.
- As a result aqueous solutions of this type of compound tend to form ordered aggregates such as micelles and bilayers.
- Here the polar groups are exposed to the water and the hydrocarbon portions are "buried" within the core, away from the water.



How Soap Cleans :

The things we call "dirt" are actually grease, oils, and fats which are insoluble in water. When cleaning, soap micelles come into contact with the "dirt" and because like dissolves like, a minute amount of "dirt" dissolves within the central region of the soap micelle. Nonpolar dirts are emulsified and washed away in polar wash water. This is why soaps are said to be good emulsifying agents.

How Does Detergent Remove Grease From Clothing ?

Detergent molecules contain hydrophobic tails (hydrocarbon chain) which are water hating and will attach themselves to any substance which is non-polar (grease). The tails surround the grease with the polar heads in the aqueous phase, forming the structure shown . With agitation the grease is pulled from the clothing allowing the grease to be completely surrounded and moved into the aqueous phase. This is then washed away leaving only clean clothes behind.

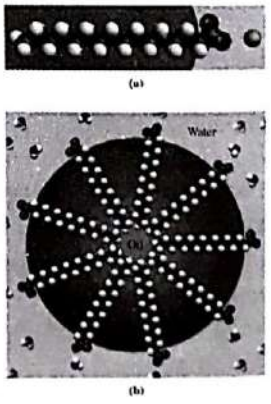


Figure : Structure of a detergent molecule and it's cleaning action

Knowledge ENHANCER

Surface tension/surfactants :

Water, the liquid commonly used for cleaning, has a property called surface tension. In the body of the water, each molecule is surrounded and attracted by other water molecules. However, at the surface, those molecules are surrounded by other water molecules only on the water side. A tension is created as the water molecules at the surface are pulled into the body of the water. This tension causes water to bead up on surfaces (glass, fabric), which slows wetting of the surface and inhibits the cleaning process. You can see surface tension at work by placing a drop of water onto a counter top. The drop will hold its shape and will not spread. In the cleaning process, surface tension must be reduced so water can spread and wet surfaces. Chemicals that are able to do this effectively are called surface active agents, or surfactants. Surfactants perform other important functions in cleaning, such as loosening, emulsifying (dispersing in water) and holding soil in suspension until it can be rinsed away. Surfactants can also provide alkalinity, which is useful in removing acidic soils. Surfactants are classified by their ionic (electrical charge) properties in water: anionic (negative+ charge), nonionic (no charge), cationic (positive charge) and amphoteric (either positive or negative charge). Soap is an anionic surfactant. Carboxymethycellulose (CMC) is a compound which is added to synthetic detergent to keep the dust particles suspended in water, and helps in cleansing process.

When the alkali is sodium hydroxide, a sodium soap is formed. Sodium soaps are "hard" soaps. When the alkali is potassium hydroxide, a potassium soap is formed. Potassium soaps are softer and are found in some liquid hand soaps and shaving creams.

1 EXERCISE

FIB Fill in the Blanks :

DIRECTIONS : *Complete the following statements with an appropriate word / term to be filled in the blank space(s).*

1. _____ are also called olefins.
2. _____ are also called paraffins.
3. The melting point of alkenes increases with _____ in molecular weight.
4. _____, _____ or _____ is used as a catalyst in the hydrogenation reactions.
5. Wurtz reaction is good for preparation of only _____ alkanes.
6. The difference in melting point of alkane with even and odd number of carbon atom is due to difference in _____.
7. Alkenes can be prepared by _____ hydrogenation of alkynes.
8. Reaction of bromine to $>C=C<$ or $-C\equiv C-$ can be used as a test for _____.
9. Alkenes on ozonolysis gives _____ compounds.
10. Lindlar's catalyst is _____ along with quinoline.
11. In presence of _____ addition across double bond occurs by kharasch effect.
12. In case of alkyl halide with same alkyl group the _____ decreases with decrease in electronegativity from Cl to I.
13. Most convenient way to prepare ethers are by _____ synthesis.
14. Dehydrohalogenation of alkyl halide with alcoholic KOH is governed by _____
15. Alcohols show _____ and _____ isomerism.
16. Chlorination of _____ gives chloral
17. In preparation of mixed ethers, the halogen atom attaches itself to _____ alkyl group.
18. Detergents causes _____ pollution.

T/F True / False :

DIRECTIONS : *Read the following statements and write your answer as true or false.*

1. Alkanes can be prepared by reduction of alkyl halide
2. Kolbe's electrolysis is the only method to prepare methane
3. Boiling point of alkanes increases with increase in length of carbon-carbon chain.
4. Isomerisation of straight chain alkane is of great importance in petroleum industry.
5. Ethene can be prepared by from ethanol by dehydration with H_2SO_4.
6. In case of alkene, for each added $-CH_2$ group the boiling point rises by 20 – 30°C
7. In kharasch effect addition of HBr across $>C=C<$ takes place according to markovnikov's rule.
8. $HC\equiv CH + Cu_2Cl_2 + 2NH_4OH \longrightarrow$

   $CuC\equiv CCu + 2NH_4Cl + 2H_2O$ (red ppt)

   Above reaction shows that alkynes are acidic in nature.
9. Finkelstein reaction can be used to prepare iodoalkanes only.
10. When two or more hydroxyl groups are attached to the same carbon atom the compound is generally unstable.
11. Invertase and amylase are two enzymes involved in fermentation of ethanol from sugar.
12. Ethers are used as an anaesthetic in surgery.
13. Position isomerism is only shown by ketones not by aldehydes.
14. Methanal can be prepared by oxidation of ethyl alcohol.
15. Molecular formula of urotropine is $(CH_2)_6N_4$
16. Acetone when heated with iodine and sodium hydroxide forms $CHCl_3$.
17. Glacial acetic acid contains appreciable amount of dissolved water.
18. The salts of fatty acids have non-polar head group, the carboxylate and a long polar hydrocarbon chain.

MTF Match the Following :

DIRECTIONS : *Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.*

1.	**Column-I**		**Column-II**
(A)	Alkanes	(p)	Isobutene
(B)	Alkenes	(q)	Dehydrohalogenation
(C)	Alkynes	(r)	Wurtz reaction
(D)	Alkylhalide	(s)	Partially soluble in water
2.	**Column-I**		**Column-II**
(A)	Ethyl alcohol	(p)	Willamson synthesis
(B)	Symmetrical ethers	(q)	Cannizaro's reaction
(C)	Methanal	(r)	Fermentation
(D)	Glacial acetic acid	(s)	Water free

VSAQ Very Short Answer Questions :

DIRECTIONS : *Give answer in one word or one sentence.*

1. Why is the conversion of ethanol to ethanoic acid an oxidation reaction ?
2. What are oxidising agents ?

3. Would you be able to check if water is hard by using a detergent ?
4. Which gas is evolved during the process of fermentation?
5. Name the organic compound, which can be produced by fermentation of sugar and is a constituent of beer.
6. Complete the following reaction equation :

$$CH_3CH_2OH + Na \longrightarrow$$

7. Write the chemical equation for the combustion of methane.
8. Complete the following chemical equation. Also name this equation.

$$CH_3COOH + C_2H_5OH \xrightarrow{Conc.H_2SO_4}$$

9. What is the need to denaturate alcohol?
10. What are the constituents of an antifreeze?
11. Which chemical is used as nail polish remover?
12. What is formalin?
13. Write chemical equation of decarboxylation of ethanoic acid?
14. Give the name of the by product of soap industry? How is it formed?
15. Name the type of organic compounds which give fruity smell.
16. Write down the equation for the preparation of ethyne from calcium carbide.
17. What happens when ethene is polymerised? Write the chemical equation representing this reaction write any one use of the product formed?

SQA Short Answer Questions

DIRECTIONS : *Give answer in 2-3 sentences.*

1. Why does miscelle formation take place when soap is added to water ? Will micelle be formed in other solvents such as ethanol also ?
2. Give four uses of ethyl alcohol.
3. What is a detergent ? Name one detergent.
4. What is meant by 'fermentation' ? Write chemical equations for the two steps involved in preparing ethanol by the fermentation of molasses.
5. Complete the following reaction equations :

(i) $CH_3CH_2OH \xrightarrow[CH_3COOH]{CrO_3 \text{ in}}$

(ii) $CH_3COOH + C_2H_5OH \xrightarrow{H_2SO_4}$

6. (a) Name the organic acid present in vinegar. Write its chemical formula also.
   (b) Name the products formed when
   (i) ethanol burns in air.
   (ii) sodium ethanoate is heated with soda lime.
7. What are esters ? How are they formed ? Where do they occur in nature ? Give one example.
8. Write two reaction to demonstrate that acetic acid is acidic in nature?
9. Give two differences between soap and synthetic detergents?
10. What are the harmful effects of drinking alcohol?
11. What are enzymes? Name the enzymes required for fermentation of sugar cane to ethanol?
12. What is meant by denatured alcohol? How is it prepared?
13. How is ethanol converted to ethene using a solid as dehydrating agent and not concentrated H_2SO_4? Give balanced equation also.
14. Write the following properties of ethylene :
   (a) Physical state
   (b) Odour
   (c) Density as compared to air
   (d) Solubility
15. How would you convert :
   (i) Ethylene to 1, 2 – dibromoethane
   (ii) Ethylene to ethyl bromide?
16. What is the molecular formula of alcohol, which is derived from pentane?
17. Give any three uses of ethanoic acid.
18. Give balanced equations for the following reactions. Also name the products formed in each reaction.
   (i) Butane is heated in an exces of air.
   (ii) Propane is heated in insufficient supply of air
   (iii) n-Hexane is subjected to strong heat in the absence of air.

LAQ Long Answer Questions

DIRECTIONS : *Give answer in four to five sentences.*

1. What happens when :
   (i) ethanoic acid reacts with sodium metal.
   (ii) ethanoic acid reacts with sodium carbonate
   (iii) ethanoic acid reacts with sodium hydroxide
   (iv) ethanoic acid reacts with an alcohol in the presence of a little of concentrated sulphuric acid ?
2. Explain the cleansing action of soaps.
3. (a) Which compound should be heated with soda lime to obtain ethane gas in the laboratory?
   (b) Write the equation for the reaction in (1) above.
   (c) Write balanced equation for the complete combustion of ethane.
   (d) Name a solid which can be used instead of concentrated sulphuric acid to prepare ethylene by the dehydration of ethanol.
   (e) Name a reagent which can be used to distinguish between ethane and ethene.
   (f) Ethylene forms an addition product with chlorine. Name this addition product and write its structural formula.

4. What happens when (give balanced equations) :
   (a) Ethylene is burnt in an excess of oxygen.
   (b) Ethylene reacts with chlorine.
   (c) Ethylene combines with hydrogen chloride.
   (d) A mixture of ethylene and hydrogen is passed over nickel at 150°C.
   (e) Ethylene is heated at 400°C under very high pressure in the presence of traces of O_2.
5. Write the name and formula of the product formed in each case below :
   (i) $CH_4 + Cl_2 \xrightarrow{\text{Sunlight}}$ ..... (Final product)
   (ii) $C_2H_5Br + KOH\,(alc) \xrightarrow{\Delta}$ .....
   (iii) $H_2C=CH_2 \xrightarrow[\text{Alk. } KMnO_4]{\text{Sunlight}}$ .....
   (iv) $H_2C=CH_2 + HBr \longrightarrow$ .....
   (v) $H_2C=CH_2 + O_3 \longrightarrow$ .....
6. Arrange n-pentane, isopentane and neopentane in the decreasing order of boiling points and justify.
7. Give balanced equations for the following conversions :
   1. 1, 2 dibromoethane $\xrightarrow{A}$ Acetylene $\xrightarrow{B}$ Silver acetylide
   2. Ethanol $\xrightarrow{C}$ Ethene $\xleftarrow{D}$ Ethyl iodide
   3. Bromoethane $\xrightarrow{E}$ Ethane $\xleftarrow{F}$ Sodium propanoate
   4. Calcium carbide $\xrightarrow{G}$ Acotylene $\xrightarrow{H}$ Acetylone tetrahloride
8. Explain the following terms :
   (i) Esterification, (ii) Saponification, (iii) Decarboxylation and (iv) Polymerisation.

EXERCISE

MCQ Multiple Choice Questions :

DIRECTIONS : *This section contains 51 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Methane and ethane both can be obtained in single step from –
   (a) CH_3I (b) C_2H_5I
   (c) CH_3OH (d) C_2H_5OH
2. Petroleum consists mainly of –
   (a) aliphatic hydrocarbons
   (b) aromatic hydrocarbons
   (c) aliphatic alcohols
   (d) none of these
3. Cyclohexane, a hydrocarbon floats on water because –
   (a) it is immiscible with water
   (b) its density is low as compared to water
   (c) it is non-polar substance
   (d) it is immiscible and lighter than water
4. LPG is a mixture of –
   (a) $C_6H_{12} + C_6H_6$ (b) $C_4H_{10} + C_3H_8$
   (c) $C_2H_4 + C_2H_2$ (d) $C_2H_4 + CH_4$
5. Which of the following has maximum boiling point –
   (a) iso-octane
   (b) n-octane
   (c) 2,2,3,3-tetramethyl butane
   (d) n-butane
6. Water gas is –
   (a) $CO + CO_2$ (b) $CO + N_2$
   (c) $CO + H_2$ (d) $CO + N_2 + H_2$
7. As the number of carbon atoms in a chain increases the boiling point of alkanes –
   (a) increases
   (b) decreases
   (c) remains same
   (d) may increase or decrease
8. What is the chief product obtained when n-butane is treated with bromine in the presence of light at 130°C –
   (a) $CH_3-CH_2-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-Br$
   (b) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-CH_2-Br$
   (c) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{\displaystyle |}{\underset{\displaystyle |}{C}}}}-Br$
   (d) $CH_3-CH_2-CH_2-CH_2-Br$
9. A liquid hydrocarbon can be converted to gaseous hydrocarbon by –
   (a) cracking

(b) hydrolysis
(c) oxidation
(d) distillation under reduced pressure

10. Which one gives only one monosubstitution product on chlorination –
(a) n-pentane (b) Neopentane
(c) Isopentane (d) n-butane

11. B.P. of branched chain alkanes as compared to straight chain alkanes are –
(a) lower
(b) equal
(c) higher
(d) independent of the chain length

12. The reaction $CH_4 + Cl_2 \xrightarrow{\text{uv light}} CH_3Cl + HCl$ is an example of –
(a) addition reactions (b) substitution reaction
(c) elimination reaction (d) rearrangement reaction

13. Aluminium carbide on reacting with water gives –
(a) methane (b) ethane
(c) ethene (d) ethyne

14. Maximum carbon-carbon bond distance is found in –
(a) ethyne (b) ethene
(c) ethane (d) benzene

15. Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds is –
(a) n-hexane (b) 2,3-dimethylbutane
(c) 2,2-dimethylbutane (d) 2-dimethylpentane

16. Alkenes usually show which type of reaction –
(a) addition (b) substitution
(c) elimination (d) superposition

17. In a reaction, it half of the double bond is broken and two new bonds are formed, this is a case of –
(a) elimination (b) addition
(c) displacement (d) rearrangement

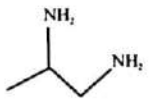
18. Ethyl bromide gives ethylene when reacted with –
(a) ethyl alcohol (b) dilute H_2SO_4
(c) aqueous KOH (d) alcoholic KOH

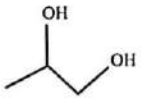
19. Which is the most reactive hydrocarbon in the following–
(a) ethane (b) ethyne
(c) ethene (d) methane

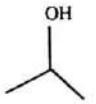
20. Which of the following gases is used for welding –
(a) methane (b) ethane
(c) acetylene (d) ethene

21. Identify Z in the series

$CH_3 - CH_2 - CH_2 - OH \xrightarrow{\text{Conc. } H_2SO_4} X \xrightarrow{Br_2} Y \xrightarrow{\text{alc. KOH}} Z$

(a) 

(b) 

(c) 

(d)

22. Which of the following compound is oxidised to prepare methyl ethyl ketone –
(a) 2-propanol (b) 1-butanol
(c) 2-butanol (d) 1-butyl alcohol

23. In CH_3CH_2OH the bond that undergoes heterolytic cleavage most readily is –
(a) C–C (b) C–O
(c) C–H (d) O–H

24. Which of the following has the highest boiling point –
(a) $CH_3CH_2CH_2OH$ (b) CH_3OH
(c) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-OH$ (d) CH_3-CH_2-OH

25. Power alcohol contains –
(a) 50% petrol and 50% ethanol
(b) 75% petrol and 25% ethanol
(c) 25% petrol and 75% ethanol
(d) 70% petrol and 30% ethanol

26. The enzyme involved in the oxidation of ethanol to form vinegar is –
(a) zymase (b) oxidase
(c) acetobacter (d) invertase

27. Glacial acetic acid is –
(a) 100% acetic acid free of water
(b) solidified acetic acid
(c) gaseous acetic acid
(d) frozen acetic acid

28. When ethanoic acid is heated with $NaHCO_3$ the gas evolved is –
(a) H_2 (b) CO_2
(c) CH_4 (d) CO

29. During decarboxylation of ethanoic acid with sodalime (NaOH + CaO), CO_2 is removed as –
(a) CO_2 (b) CO
(c) Na_2CO_3 (d) $CaCO_3$

30. The highest boiling point is of –
(a) ethane (b) ethanol
(c) methanol (d) ethanoic acid

31. When ethanoic acid reacts with ethanol a sweet smelling product is formed. The functional group in the product is–
(a) aldehyde (b) ketone
(c) alcohol (d) ester

32. To prepare methane, we heat sodalime with –
(a) calcium acetate (b) sodium acetate
(c) sodium propionate (d) sodium formate

33. Detergents can lather well in –
(a) soft water (b) hard water
(c) river water (d) any one of the above

34. Which of the following gases produced when cold water is added on calcium carbide –
(a) C_2H_2 (b) C_2H_4
(c) C_2H_6 (d) C_3H_8

35. When acetylene is passed over heated iron tube, the product obtained is –
(a) C_2H_2 (b) C_4H_4
(c) C_6H_6 (d) C_8H_8

36. 'Drinking alcohol' is very harmful and it ruins the health. 'Drinking alcohol' stands for –
(a) drinking methyl alcohol
(b) drinking ethyl alcohol
(c) drinking propyl alcohol
(d) drinking isopropyl alcohol

37. The treatment of acetic acid with lithium aluminium hydride produces –
(a) methanol (b) ethanol
(c) ethanal (d) methanal

38. The fermentation reactions are carried out in temperature range of –
(a) 20-30°C (b) 30-40°C
(c) 40-50°C (d) 50-60°C

39. Soaps are sodium salts of fatty acids. Which of the following fatty acids does not form soap –
(a) butyric acid (b) oleic acid
(c) palmitic acid (d) stearic acid

40. The OH group of an alcohol or the —COOH group of a carboxylic acid can be replaced by —Cl using :–
(a) phosphorus pentachloride
(b) hypochlorous acid
(c) chlorine
(d) hydrochloric acid

41. Which compound represent the circa -
(a) HCOOH (b) CH_3CHO
(c) HCHO (d) CH_3COOH

42. A & B both compounds give H_2 gas with sodium. If A & B react in presence of acid catalyst then they form ethyl acetate. Thus, A & B would be -
(a) CH_3COOH, CH_3OH (b) $HCOOH, CH_3COOH$
(c) CH_3COOH, C_2H_5OH (d) C_3H_7COOH, C_3H_7OH

43. Reaction of HBr with propene in the presence of peroxide gives -
(a) isopropyl bromide (b) 3-bromo propane
(c) allyl bromide (d) n-propyl bromide

44. Correct position of double bond in alkene is identified with-
(a) hydrogenation (b) ozonolysis
(c) baeyer's reagent (d) dehydration

45. Alkene, which on ozonolysis yields acetone -
(a) $CH_2=CH-CH_2-CH_3$
(b) $CH_3-CH=CH_2$
(c) $CH_3-CH=CH_2$
(d) $(CH_3)_2C=C(CH_3)_2$

46. A reagent used to test unsaturation in alkene is -
(a) ammonical Cu_2Cl_2 (b) ammonical $AgNO_3$
(c) solution of Br_2 in CCl_4 (d) conc. H_2SO_4

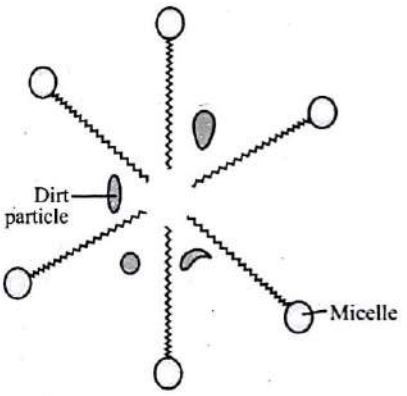
47. Acidic hydrogen is present in -
(a) ethyne (b) ethene
(c) ethane (d) benzene

48. Which of the following hydrocarbon has the maximum boiling point -
(a) CH_4 (b) C_2H_6
(c) C_3H_8 (d) C_4H_{10}

49. By which reaction ethene is obtained from ethyne -
(a) oxidation (b) polymerisation
(c) hydrogenation (d) dehydrogenation

50. Which of the following has highest percentage of hydrogen
(a) CH_4 (b) C_2H_4
(c) C_6H_6 (d) C_2H_2

51. During the cleansing action of soap dirt is surrounded by soap molecules as in the given figure.



Soap molecule is like a tadpole which has a head and tail. These head and tail are :
(a) hydrophobic and hydrophilic
(b) hydrophobic and hydrophobic
(c) hydrophilic and hydrophilic
(d) hydrophilic and hydrophobic

MTQC More than One Correct :

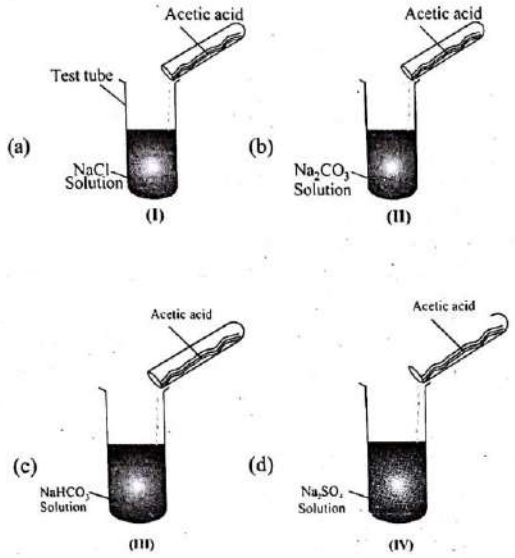
DIRECTIONS : *This section contains 17 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. Which of the following statements is not true for ethane
(a) It can be chlorinated with chlorine
(b) Ethane undergoes addition reactions
(c) When oxidised produces CO_2 and H_2O
(d) It can be catalytically hydrogenated

2. Which gives CH_4 when treated with water
(a) Aluminium carbide (b) Iron carbide
(c) Beryllium carbide (d) Calcium carbide

3. When ethyl iodide and propyl iodide reacts with Na in the presence of ether, they form
(a) pentane (b) butane
(c) hexane (d) propane

4. In which test tube effervescence will occur ?



5. Soaps are
   (a) Sodium salts of fatty acids containing carbon atom 16 to 18.
   (b) Sodium salts of trihydroxy alcohols
   (c) Week cleansing agent in hard water
   (d) Biodegradable.

6. Which of the following compound decolorises alkaline $KMnO_4$ solution -
   (a) CH_4 (b) Butanol
   (c) Butan-2-ol (d) C_2H_2

7. Alkynes have in their molecule
   (a) two hydrogen atom less than in a molecule of corresponding alkene
   (b) two hydrogen atom less than in a molecule of corresponding alkane.
   (c) $-C\equiv C-$ linkage
   (d) $>C=C<$ linkage

8. Which of the following is true for soaps
   (a) Soaps are bio-degradable
   (b) Soaps cannot be used in acidic solutions
   (c) Soaps form a white curdy precipitate with hard water
   (d) Soaps are relatively stronger in their cleansing action than synthetic detergents

9. Detergents differ from soaps in that
   (a) their ironically charged heads are derivatives of sulfur
   (b) they can not be broken down by micro organism
   (c) detergents are human-made while soaps are produced naturally
   (d) detergents have weak cleansing properties in hard water.

10. Which of the following is (are) detergents
    (a) Sodium alkyl sulphonate
    (b) Sodium chloride
    (c) Sodium sulphonate
    (d) Sodium stearate

11. In sabatier senderen's reaction alkanes can be prepared from
    (a) alkenes (b) alkyl halides
    (c) alcohols (d) alkynes

12. Which of the following statements regarding ozonolysis is (are) correct
    (a) It involves addition of ozone to unsaturated hydrocarbon
    (b) Ozonide formed is hydrolysed by H_2O in presence of Zn.
    (c) Ozonide formed is subjected to alkaline hydrolysis
    (d) Ozonolysis results in the formation of carbonyl compounds (containing $-\overset{O}{\overset{\|}{C}}-$ group)

13. In which of the following reactions addition takes place across double bond according to Markovnikov's rule

    (a) $CH_3-CH=CH_2 + HBr \longrightarrow CH_3-\underset{Br}{\underset{|}{CH}}-CH_3$
    2 – Bromopropane

    (b) $CH_3-CH=CH_2 + HBr \xrightarrow{(R-O-O-R)}$
    Propene
    $CH_3-CH_2-CH_2Br$
    1 – Bromopropane

    (c) $CH_3-CH=CH_2 + HOBr \longrightarrow CH_3-\underset{OH}{\underset{|}{CH}}-CH_2Br$
    Propene
    1 – Bromopropan – 2 – ol

    (d) $CH_3-CH=CH_2 + H_2O \xrightarrow[300°C,\,60\,atm]{H_3PO_4}$
    Propene
    $CH_3-\underset{OH}{\underset{|}{CH}}-CH_3$
    propan – 2 – ol

14. Which of the following enzymes is (are) involved in fermentation of sugar to manufacture ethanol.
    (a) Invertase (b) Zymase
    (c) Amylase (d) Proteinase

15. Which of the following are used as dehydrating agent
    (a) conc. H_2SO_4 (b) Al_2O_3
    (c) alkaline $KMnO_4$ (d) Pd or pt/H_2

16. Which of the following aldehyde undergoes cannizzaro's reaction.
    (a) HCHO (b) CH_3CHO
    (c) C_6H_5CHO (d) $(CH_3)_3C-CHO$

17. Which of the following statements about esters are correct
(a) prepared by reaction between alcohol and carboxylic acid
(b) esters are used in making perfumes and as flavouring agents
(c) ester react with H_2O in presence of an acid or a base to give back the alcohol and carboxylic acid.
(d) esters are one of reactant in williamson's synthesis

FIP Fill in the Passage :

DIRECTIONS : *Complete the following passage(s) with an appropriate word/term to be filled in the blank spaces.*

I. The conversion of ethanol to ethene is an example of .....(1)..... converting ethanol to ethene requires the use of .....(2)..... the conversion of ethene to ethane is an example of .....(3)..... the catalyst used in the conversion of ethene to ethane is commonly .....(4).....

II. Alkenes are the .....(1)..... series of .....(2)..... hydrocarbons. They differ from alkanes due to the presence of .....(3)..... bonds. Alkenes mainly undergo .....(4)..... reactions.

PBQ Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

Reactions in which an atom or a group of atoms is replaced by some other atom or another group of atoms without causing any change in the structure of the remaining part of the molecule, are called substitution reactions.
All organic compounds contain double or triple bonds give addition reactions, i.e., alkenes, alkynes and aromatic hydrocarbons give addition reactions.
Reactions in which the products get formed by rearrangement of atoms in the reacting molecule, are termed as rearrangement reactions.

1. The reaction [$CH_3CH_2Br + HS^- \longrightarrow$ product] is
(a) substitution reaction
(b) addition reaction
(c) rearrangement reaction
(d) elimination reaction

2. The reaction [$(CH_3)_2C = CH_2 + HCl \longrightarrow$ product] is
(a) substitution reaction
(b) addition reaction
(c) rearrangement reaction
(d) elimination reaction

3. The reaction [$(CH_3)_3C - CH_2OH + HBr \longrightarrow$ product] is
(a) substitution reaction (b) rearrangement reaction
(c) addition reaction (d) elimination reaction

A&R Assertion & Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion is correct** but **Reason is incorrect**.
(d) If **Assertion is incorrect** but **Reason is correct**.

1. **Assertion :** Hydrogenation is the process of converting an oil into a fat, called vegetable ghee.
Reason : Hydrogenation is carried out in presence of a catalyst usually finely divided nickel.

2. **Assertion :** Iodination of alkane is slow and reversible in nature
Reason : Iodination of alkane is performed in the presence of strong oxidising agent like HIO_3, which consumes the by-product HI and recycles into reactants I_2.

3. **Assertion :** When n-butane is heated in the presence of $AlCl_3/HCl$ it will be converted into propane
Reason : In the presence of $AlCl_3/HCl$ if any alkane having more than four carbon is heated than isomerisation is held to give isomer of reactant alkane.

4. **Assertion :** Branched alkanes have lower boiling point than their unbranched isomers.
Reason : Branched alkanes has relatively small surface area, so less Vander Wall forces of attraction act among molecules.

5. **Assertion :** Alkanes float on the surface of water.
Reason : Density of alkanes is in the range of 0.6 – 0.9 g/ml, which is lower than water.

6. **Assertion :** 1-Butene on reaction with HBr in the presence of a peroxide produces 1-bromo-butane.
Reason : It involves the free radical mechanism.

MMQ Multiple Matching Questions :

DIRECTIONS : *Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

Column I		Column II
(A)	$2CH_3COONa + 2H_2O \xrightarrow{\text{Electrolysis}}$	(p) CH_3CHO
(B)	$CH_3OH \xrightarrow[300°-400°C]{\text{Cu or Ag}}$	(q) $CH_3 - CH_3$
(C)	$CH_3 - CH = CH_2 \xrightarrow[\text{(ii) Zn / } H_2O]{\text{(i) } O_3}$	(r) HCHO
(D)	$CH_3CH_2OH \xrightarrow[CrO_3]{CH_3COOH}$	(s) H_2

HOTS Subjective Questions

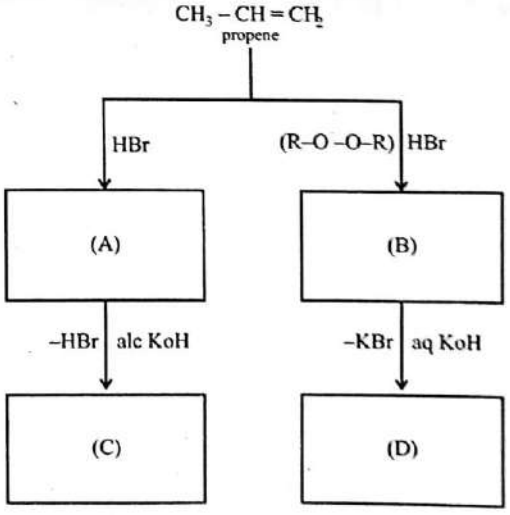
DIRECTIONS : *Answer the following questions.*

1. People use a variety of methods to wash clothes. Usually after adding the soap, they 'beat' the clothes on a stone, or beat it with a paddle, scrub with a brush or the mixture is agitated in a washing machine. Why is agitation necessary to get clean clothes?
2. How can ethanol and ethanoic acid be differentiated on the basis of their physical and chemical properties ?
3. Explain the formation of scum when hard water is treated with soap.
4. A mixture of oxygen and ethyne is burnt for welding. Explain.
5. What happens when soap solution in a test tube is shaken with (i) soft water (ii) hard water ?
6. An organic compound 'A' is a constituent of antifreeze. The compound on heating with oxygen forms another compound B which has a molecular formula $C_2H_4O_2$. Identify the compound 'A' and 'B'. Write the chemical equation of the reaction that takes place to form the compound 'B'.
7. Compound 'X' is a symmetrical gaseous hydrocarbon. Its empirical formula is CH_2. 'X' decolourises Bromine water. It forms monochloro addition product 'Y' with gaseous HCl. Molecular weight of 'Y' is 64.5. Identify the compounds 'X' and 'Y'. Write the chemical reaction of 'X' with Bromine water and HCl gas.
8. An organic compound has been obtained by reacting an unsaturated hydrocarbon with water in presence of phosphoric acid. The compound was found to burn with a blue flame. Its mixture with water has a freezing point much below 0°C. Identify the compound. Write the equation of the reaction by which it was formed.
9. An organic compound 'X' which is sometimes used as an anti-freeze has the molecular formula C_2H_6O. 'X' on oxidation gives a compound 'Y' which gives effervescence with a baking soda solution. What can 'X' and 'Y' be ? Write their structural formulae.
10. What are detergents chemically ? Why are they more effective than soaps in cleansing action ? How can detergent molecules be altered to make them biodegradable?
11. An organic compound A of molecular formula C_2H_6O on oxidation gives an acid B with the same number of carbon atoms in the molecule as A. Compound A is often used for sterilization of skin by doctors. Name the compounds A and B. Write the chemical equation involved in the formation of B from A.
12. Two compounds 'A' and 'B' have the same molecular formula $C_4H_8O_2$. Compound 'A' is an acid and compound 'B' has a fruity smell. Suggest (i) chemical formulae and (ii) the structural formulae of compounds A and B. Name the functional group of compound B. What name would you give to the relationship between the compounds A and B?
13. Give balanced equations for the reactions of acetylene with each of the following :
    (i) Oxygen
    (ii) Chloride
    (iii) Bromine water
    (iv) Liquid bromine
14. Giving equations, write under what conditions do ethane get converted to (i) Ethyl alcohol (ii) Acetaldehyde (iii) Acetic acid.
15. What are the final products when acetylene (ethyne) adds to
    (i) Iodine (ii) Chlorine
    (iii) Broinine (iv) Hydrogen ?

CBQ Chart Based Questions

DIRECTIONS : *Complete the following flow chart by identifying A, B, C and D.*

1.



SOLUTIONS

Brief Explanations of Selected Questions

Exercise 1

FILL IN THE BLANKS :

1. Alkenes
2. Alkanes
3. increase
4. Nickel, platinum, palladium
5. Symmetrical
6. Packing efficiency
7. Partial
8. Unsaturation
9. Carbonyl
10. Pd, $BaSO_4$
11. Organic peroxides
12. Dipole moment
13. Williamson's
14. Saytzeff rule
15. position, chain
16. Ethanol
17. smaller
18. water

TRUE / FALSE

1. True
2. False
3. True
4. True
5. True
6. True
7. False
8. False
9. True
10. True
11. False
12. True
13. True
14. False
15. True
16. False
17. False
18. False

MATCH THE FOLLOWING :

1. A → (r); B → (p); C → (s); D → (q)
2. A → (r); B → (p); C → (q); D → (s)

VERY SHORT ANSWER QUESTIONS :

1. It involves addition of oxygen and removal of hydrogen therefore, it is an oxidation process.
2. Those compounds which add oxygen to other materials or remove hydrogen from other materials are called oxidising agents, e.g., alkaline solution of $KMnO_4$.
3. No, we cannot determine whether the water is hard or not by using a detergent.
4. Carbon dioxide is evolved during the process of fermentation.
5. Ethanol.
6. $2CH_3CH_2OH + 2Na \longrightarrow 2CH_3CH_2ONa + H_2(g)$
7. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
8. $CH_3COOH + C_2H_5OH \xrightarrow{Conc.H_2SO_4} CH_3COOC_2H_5 + H_2O$
   (Ethyl acetate)
   This equation is known as esterification.
9. To prevent its misuse for drinking purpose.
10. Ethylene glycol and ethanol.
11. Acetone.
12. Aqueous solution of methanal.
13. $CH_3COONa + NaOH \longrightarrow CH_4 + Na_2CO_3$.
14. Glycerol.

$$\begin{array}{l} CH_2-O-COC_{17}H_{35} \\ | \\ CH-O-COC_{17}H_{35} \\ | \\ CH_2-O-COC_{17}H_{35} \end{array} + 3NaOH \longrightarrow \begin{array}{l} CH_2OH \\ | \\ CHOH \\ | \\ CH_2OH \end{array} + \underset{\text{Soap}}{3C_{17}H_{35}COONa}$$

Glyceride Glycerol.

15. Esters give fruity smell.
16. $CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$
    Calcium carbide — Ethyne
17. Polyethene is formed.

$$nCH_2 = CH_2 \xrightarrow{Catalyst} \text{-}(CH_2-CH_2)\text{-}_n$$

Polythene

Use : In making toys, containers, pipes, bags.

SHORT ANSWER QUESTIONS :

1. When soap is added to water, micelle formation takes place. This is because the hydrocarbon chains of soap molecules are hydrophobic which are insoluble in water, but the ionic ends of soap molecules are hydrophilic and hence soluble in water. Since soap is soluble in ethanol, so micelle formation does not occur.

2. Uses of ethyl alcohol :
(i) It is used in the manufacture of paints, medicines, dyes, soaps, etc.
(ii) It is used in the preparation of organic compounds like ether, chloroform and iodoform.
(iii) It is used as a fuel in internal combustion engines.
(iv) It is used in low temperature thermometers.

3. Detergents are sodium salt of long chain sulphonic acids. They are also called soap-less soaps because although they act like a soap in having the cleaning properties, they do not contain the usual 'soaps' like sodium stearate, etc.

4. It is a process in which organic compounds are changed to simple substances in presence of biological catalysts called enzymes.

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\text{Invertase}} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{Zymase}} \underset{\text{Ethanol}}{2C_2H_5OH} + 2CO_2$$

5. (i) $CH_3CH_2OH \xrightarrow[\text{in } CH_3COOH]{CrO_3} CH_3-\overset{\overset{O}{\|}}{C}-H + H_2O$

(ii) $CH_3COOH + C_2H_5OH \xrightarrow{H_2SO_4} CH_3COOC_2H_5 + H_2O$

6. (a) Acetic acid, CH_3COOH

(b) (i) $C_2H_5OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$

(ii) $CH_3COONa + NaOH(CaO) \xrightarrow{heat} CH_4 + Na_2CO_3$

7. Esters are pleasant fruity smelling compounds containing $-COOR$ as functional group where 'R' is an alkyl group. They are formed by the reaction of carboxylic acid with alcohol in presence of conc. H_2SO_4. They occur in fruits e.g., banana contains isoamyl acetate.

8. (i) $2CH_3COOH + 2Na \longrightarrow 2CH_3COONa + H_2$.

(ii) $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$.

9. 1. Soaps are sodium or potassium salts of fatty acids while detergents are sodium salts of sulphonic acid.
2. Soaps do not work well with hard water, acidic & saline water while detergents work well.

10. If a person drinks alcohol regularly even in small doses he/she becomes dependent on it. This person becomes an addict of alcohol and is called an alcoholic. Medically, alcohol is an intoxicant. The person loses all senses of discrimination under its influence, as the intake of alcohol increases, the body loses its control gradually. The consumption of large quantities may even cause death by damaging the liver.

11. (a) The catalysts which bring about biochemical changes are known as enzymes.
(b) Invertase and Zymase.

12. Denatured alcohol is one which is unfit for drinking purposes. It is also known as methylated spirit. It is prepared by mixing ethanol with some poisonous substances such as methanol, pyridine, copper sulphate etc.

13. $\underset{\text{(Ethanol)}}{C_2H_5OH} \xrightarrow[300°C]{Al_2O_3} \underset{\text{Ethene}}{C_2H_4} + H_2O$

Dehydrating agent used is alumina Al_2O_3.

14. Physical properties of ethylene :
(a) **Physical state :** It is colourless gas.
(b) **Odour :** It has a peculiar odour (first sweetish smell).
(c) **Density as compared to air :** It is slightly lighter as compared to air (V.D. of $C_2H_4 = 14$, V.D. of air 14.4)
(d) **Solubility :** It is sparingly soluble in water but soluble in organic solvents like alcohol, ether, chloroform etc.

15. (i) Ethylene to 1, 2 – dibromoethane :

$$\underset{\text{Ethylene}}{CH_2=CH_2} + Br-Br(aq) \rightarrow \underset{\text{1, 2 – Dibromoethane}}{\overset{Br}{\overset{|}{CH_2}}-\overset{Br}{\overset{|}{CH_2}}}$$

(ii) Ethylene to ethyl bromide

$$CH_2=CH_2 + H-Br \rightarrow \underset{\text{Ethyl bromide}}{CH_3-\overset{Br}{\overset{|}{CH_2}}}$$

16. $\underset{\text{n-pentane}}{CH_3-CH_2-CH_2-CH_2-CH_3} \xrightarrow[+OH]{-H}$

(Molecular formula = C_5H_{12})

$\underset{\text{Pentanol}}{CH_3-CH_2-CH_2-CH_2-CH_2OH}$

(Molecular formula = $C_5H_{11}OH$)

Thus, molecular formula of alcohol which is derived from pentane is → $C_5H_{11}OH$. It is called pentanol.

17. (i) Ethanoic acid is used as a reagent in chemistry laboratory.
(ii) Ethanoic acid is used for making white lead $[2PbCO_3.Pb(OH)_2]$.
(iii) Ethanoic acid is used for coagulating rubber from latex and casein from milk.

18. (i) $\underset{\text{Butane}}{2C_4H_{10}} + 13O_2 \rightarrow \underset{\text{Carbon dioxide}}{8CO_2} + \underset{\text{Water}}{10H_2O} + \text{Heat energy}$

(ii) $2C_3H_8 + 7O_2 \rightarrow \underset{\text{Carbon monoxide}}{6CO} + \underset{\text{Water}}{8H_2O}$ + Heat energy
Propane

(iii) $C_6H_{14} \xrightarrow{\text{Heat}} C_4H_{10} + C_2H_4$
n-Hexane Butane Ethene

Long Answer Questions :

1. (i) Ethanoic acid reacts with sodium metal to form sodium ethanoate and hydrogen gas

$$2CH_3COOH + 2Na \longrightarrow 2CH_3COONa + H_2$$

(ii) Ethanoic acid reacts with sodium carbonate to form sodium ethanoate and carbon dioxide gas

$$2CH_3COOH + Na_2CO_3 \longrightarrow 2CH_3COONa + CO_2 + H_2O$$

(iii) Ethanoic acid reacts with sodium hydroxide to form a salt called sodium ethanoate and water.

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

This reaction shows that the hydrogen atom present in – COOH group is acidic in nature.

(iv) Ethanoic acid reacts with alcohols in the presence of a little of concentrated sulphuric acid to form esters.

$$CH_3COOH + C_2H_5OH \xrightarrow{\text{Conc. } H_2SO_4} CH_3COOC_2H_5 + H_2O$$

2. Most of the dirt is oily in nature, and the oil does not dissolve in water . The soap molecules are sodium or potassium salts of long chain carboxylic acids. The acid end of soap dissolves in water while its carbon chain dissolves in the oil. In this way soap molecules form rounded special type of structures called micelles in which one end of the soap molecule is towards the oil droplet while the acid end is towards the water. This forms an emulsion in water. In this way soap micelles help in removing the dirt particles from the surface and this dirt dissolves in water and the surface gets cleaned and washed.

3. (a) Sodium propionate

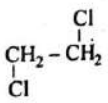
(b) $C_2H_5COONa + NaOH \xrightarrow[\text{Heat}]{CaO} \underset{\text{Ethane}}{C_2H_6} + Na_2CO_3$

(c) $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$
Ethane

(d) Alumina or Aluminium oxide, Al_2O_3.

(e) Alkaline potassium permanganate solution

(f) Ethylene dichloride ($C_2H_4Cl_2$)
Structural formula of ethylene dichloride is :



4. (a) $\underset{\text{Ethylene}}{C_2H_4} + 3O_2 \rightarrow 2CO_2 + 2H_2O$

(b) $\underset{\text{Ethylene}}{CH_2 = CH_2} + Cl - Cl \rightarrow \underset{Cl}{\overset{Cl}{CH_2 - CH_2}}$
1, 2 – Dichloroethane

(c) $\underset{\text{Ethylene}}{CH_2 = CH_2} + HCl \rightarrow \underset{Cl}{CH_3 - CH_2}$
Ethyl chloride (chloroethane)

(d) $CH_2 = CH_2 + H_2 \xrightarrow[150°C]{Ni} \underset{\text{(Ethane)}}{CH_3 - CH_3}$

(e) $\underset{\text{Ethylene}}{nC_2H_4} \xrightarrow[\text{High pressure}]{\text{High temp.}} \underset{\text{Polyethylene or Polythene}}{(-CH_2 - CH_2-)_n}$

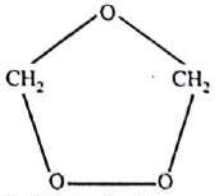
5. (i) Tetrachloromethane, CCl_4 or $Cl-\overset{Cl}{\underset{Cl}{C}}-Cl$

(ii) Ethylene, C_2H_4

(iii) Ethylene glycol $\overset{CH_2 - CH_2}{OH \quad OH}$

(iv) Ethyl bromide $\underset{Br}{CH_3-CH_2}$

(v) Ethylene ozonide



6. The decreasing order of boiling points is
n-pentane > Isopentane > Neopentane
or

$$H_3C - CH_2 - CH_2 - CH_2 - CH_3 >$$

$$CH_3 - \underset{CH_3}{\overset{CH_3}{CH}} - CH_2 - CH_3 > H_3C - \underset{CH_3}{\overset{CH_3}{C}} - CH_3$$

This is because with the increase in branching, the surface area decreases. As a result of it the magnitude of Vander Waals forces of attraction decreases, therefore boiling point goes on decreasing.

7. 1. (a)

$$\underset{\text{1, 2 Dibromoethane}}{H_2C(Br)-CH_2(Br)} + 2KOH \xrightarrow[\text{(alc.)}]{\Delta} \underset{\text{(Acetylene)}}{CH\equiv CH} + 2KBr + (2H_2O)$$

(b) $CH \equiv CH + 2AgNO_3 + 2NH_4OH \longrightarrow$

$$\underset{\text{(white)}}{AgC \equiv CAg} + 2NH_4NO_3 + 2H_2O$$

2. (c) $\underset{\text{Ethanol}}{C_2H_5OH} \xrightarrow[160-170°C]{\text{Conc. } H_2SO_4} \underset{\text{(Ethene)}}{C_2H_4} + H_2O$

(d) $\underset{\text{Ethyl iodide}}{C_2H_5I} + \underset{\text{(alc.)}}{KOH} \xrightarrow{\Delta} \underset{\text{Ethene}}{C_2H_4} + KI$

3. (e) $C_2H_5Br + 2[H] \xrightarrow[\text{Alcohol}]{Zn/Cu} C_2H_6 + HBr$

(f) $\underset{\text{Sodium propionate}}{C_2H_5COONa} + NaOH \xrightarrow[\Delta]{CaO} \underset{\text{(Ethane)}}{C_2H_6} + Na_2CO_3$

(g) $\underset{\text{Calcium carbide}}{CaC_2} + 2H_2O \longrightarrow Ca(OH)_2 + \underset{\text{Acetylene}}{C_2H_2}$

(h) $\underset{\text{Acetylene}}{C_2H_2} + 2Cl_2 \xrightarrow{CCl_4} Cl-\overset{H}{\overset{|}{C}}(Cl)-\overset{H}{\overset{|}{C}}(Cl)-Cl$

(Acetylene tetrachloride)

8. (i) **Esterification :** Reaction between an alcohol and carboxylic acid in presence of H_2SO_4 to form esters is known as **esterification**.

(ii) **Saponification :** When an oil or a fat (glyceride) is treated with sodium hydroxide solution, it gets converted to sodium salt of the acid (soap) and glycerol. The reaction is known as **saponification**.

(iii) **Decarboxylation :** When carboxylic acids are heated strongly with sodalime, it losses CO_2 and the reaction is known as **decarboxylation**.

(iv) **Polymerisation :** The process by which simple molecules (Called monomers) are converted into big molecules (called polymers) with or with out elimination of simpler molecules like water, NH_3, HCl etc., is called **polymerisation**.

Exercise 2

Multiple Choice Questions :

1. (a). $CH_3I + 2H \xrightarrow{Zn/HCl} CH_4 + HI$

$$CH_3I + 2Na + ICH_3 \xrightarrow{\text{Dry ether}} CH_3 - CH_3 + 2NaI$$

2. (a).
3. (d). Cyclohexane, is immiscible and lighter than water. Hence, floats on the surface of water.
4. (b). Liquified petroleum gas is a mixture of ethane, propane and butane. The main component is butane.
5. (b). n-octane.
   Boiling point firstly depend on molecular mass. Greater the molecular mass higher will be the boiling point. Boiling point also depends on the structure. If two compounds have same molecular mass then straight chain or linear compound has higher boiling point because of more surface area
6. (c).
7. (a). Boiling point of alkanes increases with the number of carbon atoms because surface area increases which increases the Vander Waal forces of attraction.
8. (a).

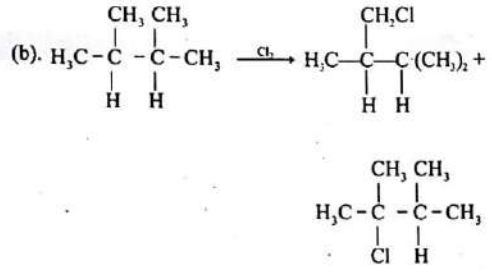
$$CH_3-CH_2-CH_2-CH_3 + Br_2 \xrightarrow[130°C]{light} \underset{\text{2-Bromobutane (main product)}}{CH_3-\overset{Br}{\overset{|}{C}}H-CH_2-CH_3} + \underset{\text{1-Bromobutane (minor)}}{CH_3-CH_2-CH_2-CH_2-Br}$$

9. (a). $\underset{\text{Hexane liquid}}{C_6H_{14}} \xrightarrow{\Delta} \underset{\text{Butane}}{C_4H_{10}} + \underset{\text{Ethene gas}}{C_2H_2}$

10. (b). $CH_3-\overset{CH_3}{\overset{|}{C}}(CH_3)-CH_3$

Replaceable hydrogen atoms are present only on 4 primary carbon atoms. Hence, it gives only one monochlorosubstituted product.

11. (a). As the number of branches increases, surface area decreases, due to which Vander Waal forces of attraction decreases. Hence, boiling point also decreases.
12. (b).
13. (a). $Al_4C_3 + 6H_2O \longrightarrow \underset{\text{Methane}}{3CH_4} + 2Al_2O_3$
14. (c). In C_2H_6, C – C bond length is 1.54 Å.

15. (b). 

16. (a). Alkenes are unsaturated hydrocarbon having double bond so generally gives addition reaction.

17. (b). e.g., $CH_2 = CH_2 + Br_2 \longrightarrow \underset{Br}{\overset{}{CH_2}} - \underset{Br}{\overset{}{CH_2}}$

Half of the double bond is broken. It means π bond is broken while sigma bond is retained also two new C – Br bonds are formed.

18. (d).

$$CH_3 - CH_2 - Br + KOH \xrightarrow[(alc)]{} CH_2 = CH_2 + KBr + H_2O$$

19. (b). Alkyne > Alkene > Alkane

20. (c).

$$2CH \equiv CH + 5O_2 \longrightarrow 4CO_2 + 2H_2O, \quad \Delta H = -1300\ KJ$$

The combustion of acetylene is highly exothermic and the heat produced during the combustion can be used for welding purposes in the form of oxy acetylene flame.

21. (d)	22. (c)	23. (d)
24. (a)	25. (b)	26. (c)
27. (a)	28. (b)	29. (c)
30. (d)	31. (d)	32. (b)
33. (d)	34. (a)	35. (c)
36. (b)	37. (b)	38. (a)
39. (a)	40. (a)	41. (d)
42. (c)	43. (d)	44. (b)
45. (d)	46. (c)	47. (a)
48. (d)	49. (c)	50. (a)
51. (d)		

MORE THAN ONE CORRECT :

1. (b, d)
2. (a, c)

$$\underset{\text{Aluminium carbide}}{Al_4C_3} + 12H_2O \longrightarrow \underset{\text{Methane}}{3CH_4} + 4Al(OH)_3$$

$$Be_2C + 4H_2O \longrightarrow \underset{\text{Methane}}{CH_4} + 2Be(OH)_3$$

3. (a, b, c)

$$C_2H_5I + 2Na + C_3H_7I \xrightarrow{\text{Dry ether}} \underset{\text{(Pentane)}}{CH_3 - (CH_2)_3 - CH_3} + 2NaI$$

$$C_2H_5I + 2Na + C_2H_5I \xrightarrow{\text{Dry ether}} \underset{\text{(Butane)}}{CH_3 - (CH_2)_2 - CH_3} + 2NaI$$

$$C_3H_7I + 2Na + C_3H_7I \xrightarrow{\text{Dry ether}} \underset{\text{(Hexane)}}{CH_3 - (CH_2)_4 - CH_3} + 2NaI$$

4. (b, c)

$$2CH_3COOH + Na_2CO_3 \longrightarrow 2CH_3COONa + H_2O + CO_2\uparrow$$

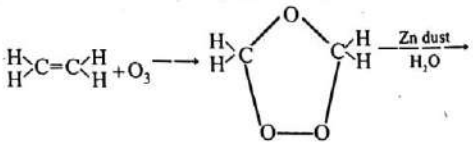
$$CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2\uparrow$$

5. (a, c, d) Soap not forms lather in hard water and are biodegradable in comparison to detergents.
6. (b, c, d) As these organic compound will get oxidised by alkaline $KMnO_4$ resulting in colour change.
7. (a, c)
8. (a, b, c)
9. (a, b, c) Detergents are sodium salt of sulphonic acid. Detergents are non biodegradable and are prepared from hydrocarbons of petroleum
10. (a, c)
11. (a, d)

$$CH_2 = CH_2 + H_2 \xrightarrow{Ni/300°C} CH_3 - CH_3$$

$$CH \equiv CH + 2H_2 \xrightarrow{Ni/300°C} CH_3 - CH_3$$

12. (a, b, d) when ethene undergoes ozolysis



Ethene ozonide

$$2H - \overset{O}{\overset{\|}{C}} - H + H_2O_2$$

13. (a, c, d) In presence of peroxides addition of unsymmetrical molecule takes place according to antimarkovnikov rule.
14. (a, b)
15. (a,b) Sulphuric acid and alumina eliminate a water molecule from reactant molecule.
16. (a, c, d) Aldehydes which do not contain any α - hydrogen atom undergoes canizzaro's reaction.
17. (a, b,c)

Fill in the Passage :

I 1. dehydration; 2. conc. H_2SO_4; 3. hydrogenation; 4. nickel.

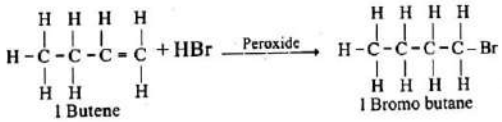
II 1. homologous; 2. unsaturated; 3. double; 4. addition.

Passage Based Questions :

1. (a) $CH_3CH_2Br + HS^- \longrightarrow CH_3CH_2SH + Br^-$
2. (b) $(CH_3)_2C=CH_2 + HCl \longrightarrow (CH_3)_2ClC-CH_3$
3. (c) $(CH_3)_3C-CH_2OH + HBr \longrightarrow (CH_3)_2CBrCH_2CH_3$
   Reaction A is substitution reaction (nucleophilic),
   (b) is addition reaction and (c) is rearrangement reaction.

Assertion & Reason :

1. (b) Hydrogenation or hardening of oil is a process in which various unsaturated fatty glycerides are converted into more highly or completely saturated glycerides by the addition of hydrogen in the presence of a catalyst, usually finely divided nickel.
2. (b)
3. (d). When n-butane is heated in the presence of $AlCl_3$/HCl it will be converted into isobutane. In the presence of $AlCl_3$/HCl if any alkane having more than four carbon is heated than isomerisation is held to give isomer of reactant alkane.
4. (a)
5. (a)
6. (a). This reaction is followed by anti Markovnikov rule



1 Butene — 1 Bromo butane

In this reaction anti Markovnikov's addition is explained on the basis of the fact that it is in the presence of peroxide the addition takes place via a free radical mechanism.

Multiple Matching Questions :

1. A → (q, s); B → (r, s); C → (p, r); D → (p)

Hots Subjective Questions :

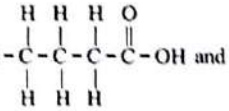
1. It is because dirt and grease are strongly attracted towards the clothes, therefore, beating or scrubbing with a brush or agitation in machine is necessary to overcome the force of attraction between fibre and dirt. Secondly it will produce heat, which helps in removing dirt and grease from fibres.
2. Ethanol has specific smell whereas ethaonic acid has vinegar like smell.
   Ethanol does not react with sodium hydrogen carbonate whereas ethanoic acid liberates CO_2 on treatment with $NaHCO_3$.
   $CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2$
3. Soap is sodium or potassium salts of fatty acids. Hard water contains Ca^{2+} and Mg^{2+} ions, which react with soap to form calcium and magnesium salts of fatty acids which are insoluble and are called scum.

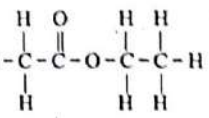
   $2C_{17}H_{35}COONa + Ca^{2+} \longrightarrow 2Na^{+} + Ca(C_{17}H_{35}COO)_2$
   (Sodium stearate) (Calcium ion) (Sodium ion) (Calcium stearate)
4. If a mixture of oxygen and ethyne is burnt, then ethyne burns completely producing a blue flame. The oxygen-ethyne flame is extremely hot and produces a very high temperature which is used for welding metals.
5. (i) A lot of lather is formed.
   (ii) Little lather is formed.
6. $CH_3CH_2OH + O_2 \longrightarrow CH_3COOH + H_2O$
   Ethanol 'A' — Ethanoic acid 'B'
7. $CH_2=CH_2 + HCl(g) \longrightarrow CH_3CH_2Cl$
   'X' — 'Y'
   'Y' has molecular weight 64.5

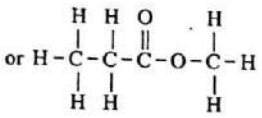
   $CH_2=CH_2 + Br_2(aq) \longrightarrow CH_2Br-CH_2Br$
8. $CH_2=CH_2 + H_2O \xrightarrow{H_3PO_4} CH_3CH_2OH$
   The compound is CH_3CH_2OH (Ethanol)
9. $C_2H_5OH + O_2 \longrightarrow CH_3COOH + H_2O$
   'X' ('Y')

   $CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2$
   'Y' (Baking soda)
10. They are sodium salts of sulphonic acids. They do not react with Ca^{2+} and Mg^{2+} ions present in hard water to form insoluble compounds, therefore, they are more effective than soaps. Detergent molecules should have minimum branching so as to make them biodegradable.
11. 'A' is CH_3CH_2OH 'B' is CH_3COOH

    $CH_3CH_2OH + 2[O] \xrightarrow[KOH]{KMnO_4} CH_3COOH + H_2O$
    'A' — 'B'
    Ethanol — Acetic acid
12. (i) 'A' is $CH_3CH_2CH_2COOH$ 'B' is $CH_3COOCH_2CH_3$ or $C_2H_5COOCH_3$

    (ii) Structural formula of 'A' is  and

    that of 'B' is 



Ester group is present in 'B'.
'A' and 'B' are functional isomers.

13. Reactions of acetylene with

(i) Oxygen

$$2C_2H_2 + 5O_2 \longrightarrow 4CO_2 + 2H_2O + \text{Heat energy}$$

(ii) Chloride

$$CH \equiv CH + 2Cl_2 \longrightarrow \underset{\text{(Acetylene tetrachloride)}}{\overset{CHCl_2}{\underset{CHCl_2}{|}}}$$

(iii) Bromine water

$$C_2H_2 + \underset{(aq)}{Br_2} \longrightarrow \underset{\text{(1,2 dibromo ethene)}}{\overset{CHBr}{\underset{CHBr}{\|}}}$$

(iv) Liquid bromine

$$C_2H_2 + Br_{2(l)} \longrightarrow \underset{\text{(Acetylene dibromide)}}{\overset{CHBr_2}{\underset{CHBr_2}{|}}}$$

14. (i) Ethane into ethyl alcohol

$$2C_2H_6 + O_2 \xrightarrow[\text{Cu tubes, 475K}]{\text{120 atm}} \underset{\text{Ethyl alcohol}}{2C_2H_5OH}$$

(ii) Ethane into acetaldehyde

$$\underset{\text{Ethane}}{C_2H_6} + O_2 \xrightarrow[\Delta]{MoO} \underset{\text{(Acetaldehyde)}}{CH_3CHO} + H_2O$$

(iii) Ethane into acetic acid

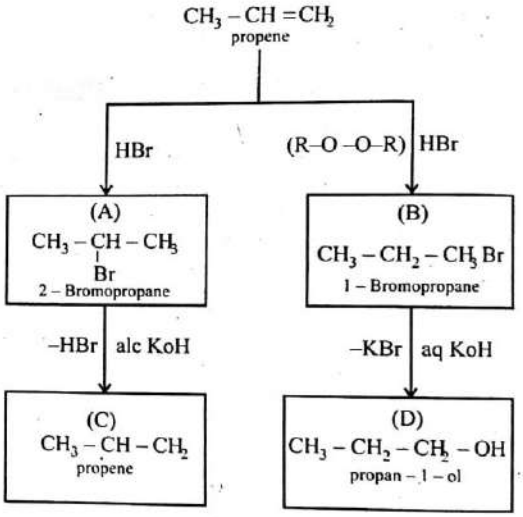
$$\underset{\text{Ethane}}{C_2H_6} + O_2 \xrightarrow[\Delta]{MoO} \underset{\text{Acetaldehyde}}{CH_3CHO} + H_2O$$

$$CH_3CHO + (O) \xrightarrow{K_2Cr_2O_7/H^+} \underset{\text{Acetic acid}}{CH_3COOH}$$

15. (i) 1, 1, 2, 2 – tetraiodoethane $C_2H_2I_4$
(ii) 1, 1, 2, 2 – tetraiodoethane $C_2H_2Cl_4$
(iii) 1, 1, 2, 2 – tetraiodoethane $C_2H_2Br_4$
(iv) Ethane, C_2H_6

Chart Based Questions :

1.



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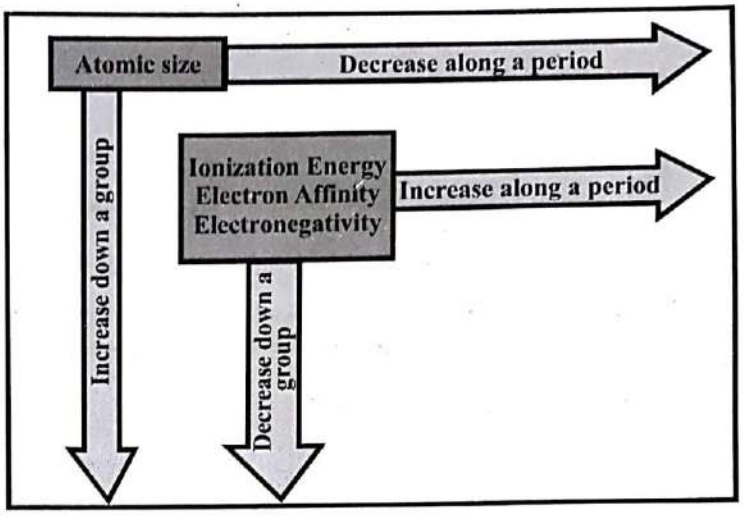


PERIODIC CLASSIFICATION OF ELEMENTS

Introduction

Matter around us is present in the form of elements, compounds and mixtures and the elements contain atoms of only one type. At present, 114 elements are known to us. Around the year 1800, only 30 elements were known. All these had seemingly different properties. As different elements were being discovered, scientists gathered more and more information about the properties of these elements. They found it difficult to organise all that was known about the elements. They started looking for some pattern in their properties, on the basis of which they could study such a large number of elements with ease.

You must have visited a library. There are thousands of books in a large library. In spite of this if you ask for a particular book, the library staff can locate it easily. How is it possible? In library the books are classified into various categories and sub-categories. They are arranged on shelves accordingly. Therefore location of books becomes easy. On similar basis scientists made periodic table to arrange elements in proper manner. The periodic table of the chemical elements is a tabular display of the chemical elements.

EARLY ATTEMPTS AT THE CLASSIFICATION OF ELEMENTS:

It wasn't until 1649, however, until the first element was discovered through scientific inquiry by Hennig Brand . That element was phosphorous (P). By 1869, 63 elements had been discovered.

Between 1817-1829, *Johann Dobereiner began to group elements with similar properties in to groups of three or triads.*

This began in 1817 when Johann Dobereiner noticed that the atomic weights of strontium, Sr, was halfway between the weights of calcium and barium. These elements possessed similar chemical properties. By 1829, he had discovered that a halogen triad made up of chlorine, bromine, and iodine and a alkali metal triad of lithium, sodium and potassium.

Element	Atomic mass
Lithium, Li	7
Sodium, Na	23
Potassium, K	39

He postulated that nature contained triads of elements in which the middle element had properties that were an average of the other two elements.

Mean of the atomic masses of the first (Li) and the third (K) elements(7+39)/2 u The atomic mass of the middle element, sodium, Na is equal to 23 u. Two more examples of Dobereneir's triads

I		II	
Element	Atomic mass	Element	Atomic mass
Calcium, Ca	48	Clorine, Cl	35.5
Strontium, Sr	88	Bromine, Br	80
Barium, Ba	137	Iodine, I	127

Mean of the atomic masses of the first and third elements (I) = (40 +137)/2 = 88.5 u

Mean of the first atomic masses of the first and third elements (II) = (35.5 +127)/2 = 81.5 u

Actual atomic mass of the second element (I) = 88 u

Actual atomic mass of the second element (II) = 80 u

Dobereneir's idea of classification of elements into triads did not receive wide acceptance as he could arrange only a few elements in this manner.

Later, other scientists found other triads and recognized that elements could be grouped into set large than three. The poor accuracy of measurements such as that of atomic weights hindered grouping more elements.

In 1862, **A.E.Beguyer de Chancourtois** was the first person to make use of atomic weights to reveal that the elements were arranged according to their atomic weights with similar elements occurring at regular intervals. He drew the elements as a continuous spiral around a cylinder divided into 16 parts. A list of elements was wrapped around a cylinder so that several sets of similar elements lined up, creating the first geometric representation of the periodic law.

CHECK Point

☛ **Who discovered first element and which element it was ?**

SOLUTION

Phosphorus was the first element discovered by Hennig Brand.

Newland's law of Octaves :

In 1864 John Alexander Newland, an English chemist noticed that "when elements are arranged in the increasing order of their atomic masses every eighth element had properties similar to the first element." Newland called it the Law of Octaves.

It was due to its similarity with musical notes where, in every octave, after seven different notes the eighth note is repetition of the first one as shown below.

1	2	3	4	5	6	7	8
सा	रे	गा	म	प	धा	नी	सा

Look carefully at the Newland's arrangement of elements shown below:

Li	Be	B	C	N	O	F
(6.9)	(9.0)	(10.8)	(12.0)	(14.0)	(16.0)	(19.0)
Na	Mg	Al	Si	P	S	Cl
(23.0)	(24.3)	(27.0)	(28.1)	(31.0)	(32.1)	(35.5)
K	Ca					
(39.1)	(40.1)					

With the help of the arrangement given above, can you tell starting from lithium eighth element is the Sodium. And starting from sodium? It is potassium. Properties of all three are similar. Similarly, aluminium is the eighth element from boron it shows properties similar to it.

However, Newland could arrange elements in this manner only up to calcium out of a total of over sixty elements known at his time. Because of this shortcoming his work was not received well by the scientific community. The next break through in classification of elements came in the form of Mendeleev's work.

MENDELEEV'S LAW :

A periodic function is the one which repeats itself after a certain interval. Thus, according to the periodic law (given by Mendeleev) the chemical and physical properties of elements repeat themselves after certain intervals when they are arranged in the increasing order of their atomic mass.

The chemical and physical properties of elements are a periodic function of their atomic masses.

A tabular arrangement of the elements based on the periodic law is called periodic table. Mendeleev believed that atomic mass of elements was the most fundamental property and arranged them in its increasing order in horizontal rows till he encountered an element which had properties similar to the first element. He placed this element below the first element and thus started the second row of elements. Proceeding in this manner he could arrange all the known elements according to their properties and thus created the first periodic table.

Dmitri Mendeleev (1834-1907) periodic table :

Series	Groups of Elements								
	0	I	II	III	IV	V	VI	VII	VIII
1.		Hydrogen H 1.008							
2.	Helium He 4.0	Lithium Li 7.03	Beryllium Be 9.1	Boron B 11.0	Carbon C 12.0	Nitrogen N 14.04	Oxygen O 16.00	Fluorine F 19.0	
3.	Neon Ne 19.9	Sodium Na 23.5	Magnesium Mg 24.3	Aluminium Al 27.0	Silicon Si 28.4	Phosphorus P 31.0	Sulphur S 32.06	Chlorine Cl 35.45	
4.	Argon Ar 38	Potassium K 39.1	Calcium Ca 40.1	Scandium Sc 44.1	Titanium Ti 48.1	Vanadium V 51.4	Chromium Cr 52.1	Manganese Mn 55.0	Iron Fe 55.9 Cobalt Co 59 Nickel Ni 59 (Cu)
5.		Copper Cu 63.6	Zinc Zn 65.4	Gallium Ga 70.0	Germanium Ge 72.3	Arsenic As 75	Selenium Se 79	Bromine Br 79.95	
6.	Krypton Kr 81.8	Rubidium Rb 85.4	Strontium Sr 87.6	Yttrium Y 89.0	Zirconium Zr 90.6	Niobium Nb 94.0	Molybedenum Mo 96.0		Ruthenium Ru 101.7 Rhodium Rh 103.0 Palladium Pd 106.5 (Ag)
7.		Silver Ag 107.9	Cadmium Cd 112.4	Indium In 114.0	Tin Sn 119.0	Antimony Sb 120.0	Tellurium Te 127.6	Iodine I 126.9	
8.	Xenon Xe 128	Caesium Cs 132.9	Barium Ba 137.4	Lanthanum La 139	Cerium Ce 140				
9.	–	–	–	–	–	–	–	–	–
10.				Ytterbium Yb 173		Tantalum Ta 183	Tungsten W 184		Osmium Os 191 Iridium Ir 193 Plantinum Pt 194.9 (Au)
11.		Gold Au 197.2	Mercury Hg 200	Thalium Ti 204.1	Lead Pb 206.9	Bismuth Bi 208			
12.			Radium Ra 224		Thorium Th 232		Uranium U 240		
	R	R_2O	RO	R_2O_3	Higher Saline Oxides RO_2 R_2O_5 RO_3 R_2O_7 Higher Gaseous Hydrogen Compounds RH_4 RH_3 RH_2 RH				RO_4

Mendeleev predicted the existence and properties of unknown elements which he called eka-aluminum, eka-boron, and eka-silicon. The elements gallium, scandium and germanium were found later to fit his predictions quite well.

Main features of Mendeleev's periodic table :

In Mendeleev's periodic table elements are arranged in tabular form in rows and columns. Now let us learn more about these rows and columns and the elements present in them.

1. The horizontal rows present in the periodic table are called series. These series are further divided into horizontal columns called periods. You can see that there are seven periods in the periodic table.
2. Properties of elements in a particular period show regular gradation (i.e. increase or decrease) from left to right.
3. The vertical columns present in it are called groups. You must have noticed that these are nine in number and are numbered from I to VIII and Zero (Roman numerals).

4. Groups I to VII are subdivided into A and B subgroups. Groups Zero and VIII don't have any subgroups.
5. All the elements in a particular group are chemically similar in nature. They show regular gradation in their physical properties and chemical reactivities.

After learning about the main features we shall now learn about the main merits of Mendeleev's periodic table.

Merits of Mendeleev's periodic classification :

1. Classification of all elements : Mendeleev's was the first classification which successfully included all the elements.
2. Prediction of new elements : Mendeleev's periodic table had some blank spaces in it. These vacant spaces were for elements that were yet to be discovered. For example, he proposed the existence of an unknown element that he called eka aluminium. The element gallium was discovered four years later and its properties matched very closely with the predicted properties of eka aluminium.

CHECK Point

☛ **Why Meendeleev left some places empty in his periodic table?**

SOLUTION

For the elements which were not discovered at that time. The elements which supposed to be discovered later

Defects in Mendeleev's periodic classification :

In spite of being a historic achievement Mendeleev's periodic table had some defects in it. The following were the main defects in it:

1. **Position of hydrogen :** Hydrogen resembles alkali metals (forms H^+ ion just like Na^+ ions) as well as halogens (forms H^- ion similar to Cl^- ion).Therefore, it could neither be placed with alkali metals (group I) nor with halogens (group VII).
2. **Position of isotopes :** Different isotopes of same elements have different atomic masses, therefore, each one of them should be given a different position in the periodic table. On the other hand, because they are chemically similar, they had to be given same position.
3. **Anomalous pairs of elements :** At certain places, an element of higher atomic mass has been placed before an element of lower atomic mass. For example, Argon (39.91) is placed before potassium (39.1)

MODERN CLASSIFICATION :

Henry Moseley, an English physicist discovered in the year 1913 that atomic number, is the most fundamental property of an element and not its atomic mass. Atomic number, (Z), of an element is the number of protons in the nucleus of its atom. The number of electrons in the neutral atom is also equal to its atomic number. This discovery changed the whole perspective about elements and their properties to such an extent that a need was felt to change the periodic law also.

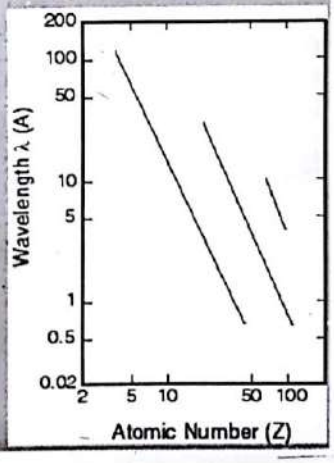
Knowledge ENHANCER

Moseley's Law

Henry Moseley discovered that the wavelength (energy) of an X-ray depended upon the nuclear charge of an atom. In 1913, working at the University of Manchester, he photographed the X-ray spectrum of 10 elements that occupied consecutive places in the periodic table. He concluded that there was "a fundamental quantity which increases by regular steps as we pass from one element to the next." In 1920, Rutherford at Cambridge identified this quantity as the atomic number.

Moseley's Law describes the relationship between atomic number and wavelength of a spectral line : $\lambda = \frac{K}{(Z-\sigma)^2}$, *where K and s are constants for a given spectral line.*

Moseley's determination of this relationship provided a simple test of the order of the elements according to Z. It showed where elements were missing from the periodic table and led to the discovery of some of these elements. For example, hafnium (Hf), which is chemically almost identical to zirconium (Zr), was identified by D.Coster and G. von Hevesy in 1923 from its X-ray spectrum.



MODERN PERIODIC LAW :

After discovery of atomic number the periodic law was modified and the new law was based upon atomic numbers in place of atomic masses of elements. After the change in the periodic law many changes were suggested in the periodic table. When the elements are arranged in the increasing order of their atomic number, most of the defects of Mendeleev classification get rectified.

Modern periodic law : It states that "the properties of the elements are periodic functions of their atomic numbers.

Both the lanthanide and actinide series of elements were placed under the rest of the periodic table. These elements technically should be placed between the alkaline earth metals and the transition metals, however, since this would make the periodic table too wide, they were placed below the rest of the elements.

CHECK *Point*

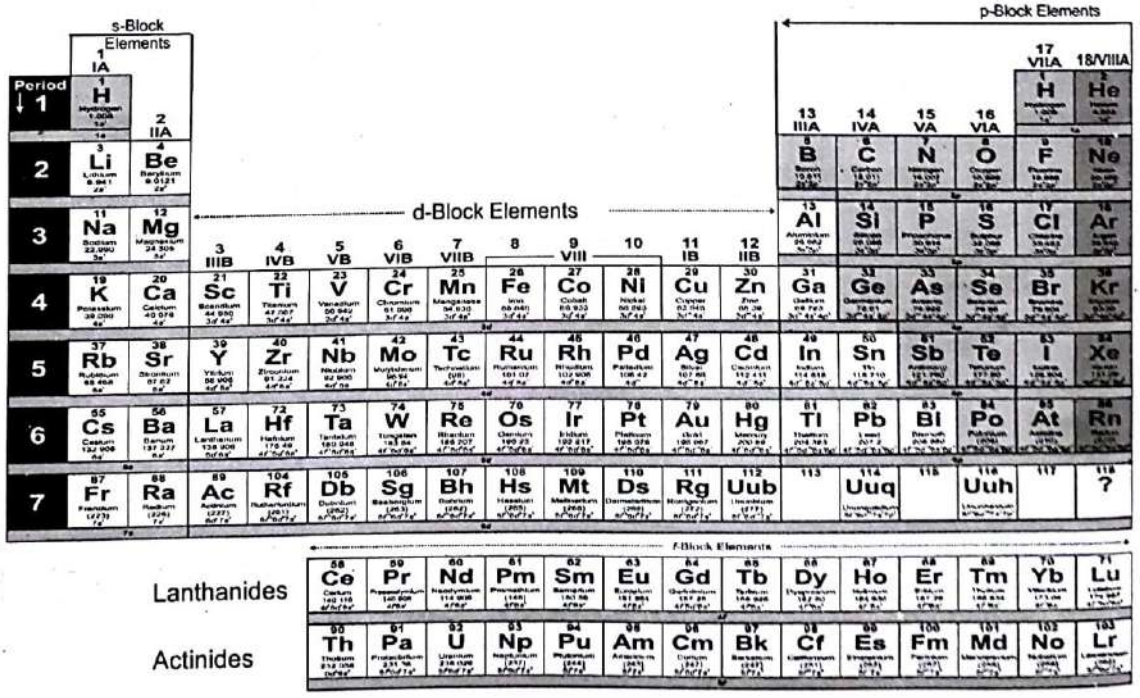
☛ **Modern periodic law is based on which property of elements ?**

SOLUTION

It is based on atomic number of elements.

The modern periodic table is also known as the long form of the periodic table or the extended form of the periodic table.

Modern periodic table :



The main features of this periodic table:

Groups:

There are 18 vertical columns in the periodic table. Each column is called a group. The groups have been numbered from 1 to 18 (in Arabic numerals) from left to right. Group 1 on extreme left position contains alkali metals (Li, Na, K, Rb, Cs and Fr) and group 18 on extreme right side position contains noble gases (He, Ne, Ar, Kr, Xe and Rn).

All elements present in a group have similar electronic configurations and have same number of valence electrons.

Group 1		Group 17	
Element	Electronic configuration	Element	Electronic configuration
Li	2, 1	F	2, 7
Na	2, 8, 1	Cl	2, 8, 7
K	2, 8, 8, 1	Br	2, 8, 8, 7
Rb	2, 8, 8, 8, 1	I	2, 8, 18, 8, 7

Elements present in groups 1 and 2 on left side and groups 13 to 17 on the right side of the periodic table are called normal elements or representative elements. Their outermost shells are incomplete. They are also called typical or main group elements. Elements present in groups 3 to 12 in the middle of the periodic table are called transition elements. However, it should be noted here that more and more electrons are added to valence shell only in case of normal elements. In transitions elements, the electrons are added to incomplete inner shells.

Group 12 elements (Zn, Cd, Hg) *are not regarded as a transition metal because according to modern,* IUPAC *definition "an element whose atom has an incomplete d sub-shell, or which can give rise to cations with an incomplete d sub-shell are considered as transition elements. Elements of these group does not satisfy this condition but they resemble transition metal in their properties.*

Group 18 on extreme right side of the periodic table contains noble gases. Their outermost shells contain 8 electrons and are completely filled that's why they are inert. Inner transition elements:14 elements with atomic numbers 58 to 71 (Ce to Lu) are called lanthanides and they are placed along with the element lanthanum (La), atomic number 57 in the same position (group 3 in period 6) because of very close resemblance between them. However, for convenience sake they are shown separately below the main periodic table. 14 elements with atomic numbers 90 to103 (Th to Lr) are called actinides and they are placed along with the element actinium (Ac), atomic number 89 in the same position (group 3 in period 7) because of very close resemblance between them. They are shown also separately below the main periodic table along with lanthanides.

Periods :

There are seven rows in the periodic table. Each row is called a period. The periods have been numbered from 1 to 7 (Arabic numerals). In each period a new shell starts filling up. The period number is also the number of shell which starts filling up in it. For example, in elements of 3rd period, the third shell (M shell) starts filling up as we move from left to right. The first element of this period sodium Na (2,8,1) has only one electron in its valence shell (third shell) while the last element of this period, argon Ar (2,8,8) has eight electrons in its valence shell.

The gradual filing of the third shell can be seen below.

Element	Na	Mg	Al	Si	P	S	Cl	Ar
Electronic configuration	2,8,1	2,8,2	2,8,3	2,8,4	2,8,5	2,8,6	2,8,7	2,8,8

The first period is the shortest period of all and contains only 2 elements, H and He.

The second and third periods are called short periods and contain 8 elements each.

Fourth and fifth periods are long periods and contain 18 elements each.

Sixth and seventh periods are very long periods containing 32 elements each(Including elements up to atomic number 118. Elements 114, 116 and 118 have been reported only recently.)

The period of an element signifies the highest energy level an electron in that element occupies in an unexcited state.

The elements in the Periodic Table are classified as :

Metalloids , Alkali metals, Alkaline Earth Metals, Transition Metals, Other Metals, Non-metals, Halogens, Noble Gases, Rare Earth Elements

Elements classified as Metalloids :

The 7 elements classified as "Metalloids" are located in Groups 13, 14, 15, 16 and 17 elements of the Periodic Table. Elements classified as Metalloids have properties of both metals and non-metals. Some are semi-conductors and can carry an electrical charge making them useful for electronic appliances like calculators and computers.

The Metalloids in the Periodic Table are : Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium, Polonium,

Elements classified as Alkali Metals :

The 6 elements classified as "Alkali Metals" are located in Group 1 elements of the Periodic Table. Elements classified as Alkali Metals are very reactive metals that do not occur freely in nature. Alkali metals are soft, malleable, ductile, and are good conductors of heat and electricity. The Alkali Metals are: Lithium, Sodium, Potassium, Rubidium, Cesium, Francium

Elements classified as Alkaline Earth Metals :

The 6 elements classified as "Alkaline Earth Metals" are located in Group 2 elements of the Periodic Table. Elements classified as Alkaline Earth Metals are all found in the Earth's crust, but not in the elemental form as they are so reactive (Inspite of being so reactive there reactivity is less than alkali metals). Instead, they are widely distributed in rock structures. The Alkaline Earth Metals on the Periodic Table are : Beryllium, Magnesium, Calcium, Strontium, Barium, Radium.

Elements of group 1 and group 2 are called alkali and alkaline earth metal respectively. As they all are basic in nature and form alkaline oxides and hydroxides.

Elements classified as Transition Metals :

The elements classified as "Transition Metals" are located in Groups 3 - 12 of the Periodic Table. Elements classified as Transition Metals are ductile, malleable, and conduct electricity and heat. The Transition Metals on the Periodic Table are:

Scandium, Titanium, Vanadium, Chromium, Manganese, Iron, Cobalt, Nickel, Copper, Zinc, Yttrium, Zirconium, Platinum, Gold, Mercury, Rutherfordium, Dubnium, Seaborgium, Bohrium, Hassium, Meitnerium, Ununbium, Niobium, Iridium, Darmstadtium, Molybdenum, Technetium, Ruthenium, Rhodium, Palladium, Silver, Cadmium, Hafnium, Tantalum, Tungsten, Rhenium, Osmium.

They are called 'transition elements because their position in the periodic table is between the s-block and p-block elements. Their properties are transitional between highly reactive metallic elements of the s-block, which typically form ionic compounds, and the elements of the p-block, which majorly forms covalent compounds.

Elements classified as Other Metals :

The 7 elements classified as "other metals" are located in groups 13, 14, and 15 of the Periodic Table. All of these elements are solid, have a relatively high density and are opaque. The "Other Metals" on the Periodic Table are: Aluminum, Gallium, Indium, Tin, Thallium, Lead, Bismuth

Elements classified as Non-metals

The 7 elements classified as "Non-metals" are located in Groups 14,15 and 16 of the Periodic Table. Non-metals are not easily able to conduct electricity or heat and do not reflect light . Non-metallic elements are very brittle, and cannot be rolled into wires or

pounded into sheets. Non-metallic elements exist, at room temperature, in two of the three states of matter : gases (such as oxygen) and solids (such as carbon). The Non-Metal elements on the Periodic Table are : Carbon, Nitrogen, Oxygen, Phosphorus, Sulfur, Selenium etc.

These elements are called Non-metals because of their properties approximately opposite to those of metals.

CHECK Point

☛ **Phosphorus does not conduct electricity whereas Lithium does why ?**

SOLUTION

Phosphorus is present in group 15. Thus it is a non-metal which are poor conducter of electricity. Lithium on the other hand is a alkali metal which are good conductors of electricity.

Elements classified as Halogens :

The 5 elements classified as "halogens" are located in Group 17 of the Periodic Table. Term halogen is derived from Greek words Hals = Sea salt, gennao = producer i.e. sea salt producer

The term "halogen" means "salt-former" and compounds containing halogens are called "salts". The Halogen elements on the Periodic Table are: Fluorine, Chlorine, Bromine, Iodine, Astatine.

The halogens exist, at room temperature, in all three states of matter - Gases such as Fluorine & Chlorine, Solids such as Iodine and Astatine and Liquid as in Bromine.

Elements classified as Noble Gases

The 6 elements classified as "Noble Gases" are located in Group 18 of the Periodic Table. Six Noble Gases on the Periodic Table are: Helium, Neon, Argon, Krypton, Xenon, Radon

These gases are said to be noble or inert because they have completely filled outermost shell. Thus they are chemically inert

CHECK Point

☛ **Neon and fluorine both are gases in which fluorine are extremely reactive whereas neon will not react at all ?**

SOLUTION

Fluorine gas belongs to halogen family which have one electron short in outermost shell. Thus it has tendency to get completely filled that shell. Thus it is highly reactive whereas neon has completely filled outermost shell thus it is inert in nature.

Elements classified as Rare Earth Elements

The elements classified as "Rare Earth Elements" are located in Group 3 of the Periodic Table and in the 6th and 7th periods. The Rare Earth Elements consists of the Lanthanide and Actinide series. Most of the elements in the Actinide series are synthetic or man-made. The Lanthanide and Actinide series of Rare Earth Elements on the Periodic Table are:

Lanthanide Elements	Actinide Elements
Lanthanum	Actinium
Cerium	Thorium
Praseodymium	Protactinium
Neodymium	Uranium
Promethium	Neptunium
Samarium	Plutonium
Europium	Americium
Gadolinium	Curium
Terbium	Berkelium
Dysprosium	Californium
Holmium	Einsteinium
Erbium	Fermium
Thulium	Mendelevium
Ytterbium	Nobelium
Lutetium	Lawrencium

Classification based on differentiating electron :

This classification divides the elements into four types. i.e., s-, p-, d- and f-block elements depending on the type of the atomic shell into which the last electron enters.

The last electron is called deferentiating electron.

s-block elements : Those elements of the periodic table in which the last electron enters in s–orbital, are called s–block elements. s-orbital can accommodate a maximum of two electrons. Their general formulae are ns^1 and ns^2 respectively, where n = (1 to 7). I A group elements are known as alkali metals because they react with water to form alkali. II A group elements are known as alkaline earth metals because their oxides react with water to form alkali and these are found in the soil or earth. s-block element are soft and have low melting and boiling points. They mostly form ionic compounds. The total number of s block elements are 14. Fr^{87} and Ra^{88} are radioactive elements while H and He are gaseous elements. Cs and Fr are liquid elements belonging to s-block.

p-block elements : Those elements of the periodic table in which the last electron gets filled up in the p-orbital, called p-block elements. A p-orbital can accommodate a maximum of six electrons. Therefore, p-block elements are divided into six groups which are III A, IV A, V A, VI A, VII A and VIII A groups. They include both metals and non-metals. They form mostly covalent compounds. The general formula of p block elements is ns^2np^{1-6} (where n = 2 to 6)

The VIIIA group elements having general formula ns^2np^6 are inert, because their energy levels are fully filled. The total number of p block elements in the periodic table is 30 (excluding He). There are nine gaseous elements (Ne, Ar, Kr, Xe, Rn, F_2, Cl_2, O_2 and N_2) belonging to p-block. Gallium (Ga) and bromine (Br) are liquids.

d-block elements : Those elements of the periodic table in which the last electron gets filled up in the d orbital, called d block elements. The d block elements are placed in groups named IIIB, IV B, V B, VI B, VII B, VIII, I B and II B. In d block elements the electron gets filled up in the d orbital of the penultimate shell. Though the total number of d block elements is 39 in the periodic table but there are only 36 transition elements. Because only those elements are transition in which d orbital is partially filled. The general formula of these elements is $(n-1)d^{1-10}\ ns^{1-2}$ where n = 4 to 7. They are metals having high melting and boiling points. Most of them form coloured compounds and exhibit several oxidation states.

All of these elements are metals. Out of all the d block elements mercury is the only liquid element.

f-block elements : Those elements of the periodic table in which the last electron gets filled up in the f orbital, called f block elements. There are 28 f block elements in the periodic table. The elements from atomic number 58 to 71 are called lanthanides because they come after lanthanum (57). The elements from 90 to 103 are called actinides because they come after actinium (89). They are heavy

metals with high melting and boiling points. They form coloured ions. All the actinide elements are radioactive. All the elements after atomic number 92 (i.e. U^{92}) are transuranic elements. The elements which do not occur in nature and are produced in the laboratories artificially are called transuranic or synthetic elements. The general formula of these elements is $(n-2)\ f^{(1-14)}\ (n-1)\ d^{0-1}\ ns^2$ where $n = 6$ & 7.

The lanthanides occur in nature in low abundance and therefore, these are called rare earth elements.

CHECK Point

☛ **Write general electronic configuration of d-block elements.**

SOLUTION

$(n-1)\ d^{1-10}\ ns^{1-2}$

Merits of modern periodic table over Mendeleev's periodic table :

The modern periodic table is based on atomic number which is more fundamental property of an atom than atomic mass. The long form of modern periodic table is therefore free of main defects of Mendeleev's periodic table.

1. **Position of isotopes :** All isotopes of the same elements have different atomic masses but same atomic number. Therefore, they occupy the same position in the modern periodic table which they should have because all of them are chemically similar.
2. **Anomalous pairs of elements :** When elements are arranged in the periodic table according to their atomic numbers the anomaly regarding certain pairs of elements in Mendeleev's periodic table disappears. For example, atomic numbers of argon and potassium are 18 and 19 respectively. Therefore, argon with smaller atomic number comes before potassium although its atomic mass is greater.
3. It explains the periodicity of the properties of the elements and relates them to their electronic configurations.
4. The table is simple, synthetic and easy way for remembering the properties of various elements and moreover lanthanides and actinides are placed separately.

CHECK Point

☛ **Hydrogen has three isotopes ^1H, ^2H and ^3H respectively. On what basis these elements were placed in modern periodic table ?**

SOLUTION

Modern periodic table is based on more fundamental property atomic number. All isotopes of the same element have same atomic number. Thus all three of them are placed on same position in modern periodic table.

TRENDS IN THE MODERN PERIODIC TABLE :

In a period the number of valence electrons and the nuclear charge increases from left to right. It increases the force of attraction between them. In a group the number of filled shells increases and valence electrons are present in higher shells. This decreases the force of attraction between valence electrons and the nucleus of the atom. These changes affect various properties of elements and they show gradual variation in a group and in a period and they repeat themselves after a certain interval of atomic number. Such properties are called periodic properties.

Valency :

The combining capacity of an atom or radical is known as its valency. It is measured in terms of hydrogen, chlorine and oxygen atom. Valency of an element is defined as the number of hydrogen, chlorine and double the number of oxygen atom with which atom of an element can combine.

For eg. Valency of oxygen in water (H_2O) is 2

Valency of carbon in methane (CH_4) is 4.

(a) **Valency in a period :** You have already learnt in the previous section that the number of valence electrons increases in a period. In normal elements it increases from 1 to 8 in a period from left to right. It reaches 8 in group 18 elements (noble gases) which show practically no chemical activity under ordinary conditions and their valency is taken as zero. Carefully look at the table given below. Valency of normal elements with respect to oxygen increases from 1 to 7 as shown below for elements of third

period. This valency is equal to the number of valence electrons or group number for groups 1 and 2, or (group number-10) for groups 13 to 17.

Group	1	2	13	14	15	16	17
Element	Na	Mg	Al	Si	P	S	Cl
No. of valence electrons	1	2	3	4	5	6	7
Valency with respect to oxygen	1	2	3	4	5	6	7
Formula of oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3	Cl_2O_7

In the following table for elements of second period you will observe that valency of elements with respect to hydrogen and chlorine increases from 1 to 4 and then decreases to 1 again.

Group	1	2	13	14	15	16	17
Element	Li	Be	B	C	N	O	F
No. of valence electrons	1	2	3	4	5	6	7
Valency with respect to hydrogen and chlorine	1	2	3	4	3	2	1
Formula of hydride	LiH	BeH_2	BH_3	CH_4	NH_3	H_2O	HF
Formula of chloride	LiCl	$BeCl_2$	BCl_3	CCl_4	NCl_3	Cl_2O	ClF

(b) Valency in a group : All the elements of a group have the same number of valence electrons. Therefore, they all have the same valency. Thus valency of all group 1 elements, alkali metals, is 1. Similarly valency of all group 17 elements, halogens, is 1 with respect to hydrogen and 7 with respect to oxygen.

group 1 elements	Formulae of oxides	Valency	Formulae of Hydrides	Valency
Li	Li_2O	1	LiH	1
Na	Na_2O	1	NaH	1
K	K_2O	1	KH	1
Rb	Rb_2O	1	RbH	1
Cs	Cs_2O	1		

CHECK Point

1. **Valency remains constant for elements in group or in period ?**
2. **Observe the following periodic table :**

H 1							He 2
Li 2, 1	Be 2, 2	B 2, 3	C 2, 4	Y 2, 5	O 2, 6	F 2, 7	Ne 2, 8
Na 2, 8, 1	Mg 2, 8, 2	Al 2, 8, 3	Z 2, 8, 4	P 2, 8, 5	S 2, 8, 6	Cl 2, 8, 7	Ar 2, 8, 8
K 2, 8, 8, 1	X 2, 8, 8, 2						

Arrange the following elements X, Y, Z in increasing order of their valencies :

SOLUTION

1. It remain constant for elements in group.
2. Z > Y > X

Atomic Radii

A number of physical properties like density and melting and boiling points are related to the sizes of atoms. Atomic size is difficult to define. Atomic radius determines the size of an atom. For an isolated atom atomic radius may be taken as the distance between the centre of nucleus of atom and the outermost shell of electrons. Practically, measurement of size of an isolated atom is difficult; therefore, it is measured when an atom is in company of another atom of same element. It is defined as one-half the distance between the nuclei of two atoms when they are linked to each other by a single covalent bond.

Variation of atomic radii in a period :

Atomic radii (in picometer) of 2nd and 3rd period elements are given in the table given below. In a period, atomic radius generally decreases from left to right with increase in atomic number

2nd period	Li	Be	B	C	N	O	F
	152	111	88	77	74	66	42
3rd period	Na	Mg	Al	Si	P	S	Cl
	190	145	118	111	98	88	79

In a period there is a gradual increase in the nuclear charge with increase in atomic number. Since valence electrons are added in the same shell since, the electrons in the same shell do not screen each other from the nucleus, the increase in nuclear charge is not neutralised by the extra valence electron. As a result effective nuclear charge increases therefore valence electrons are more and more strongly attracted towards nucleus. This gradually decreases atomic radii.

Variation of atomic radii in a group :

Atomic radii increases in a group from top to bottom. This can be seen from the data of atomic radii in picometers given for groups 1 and 17 elements below.

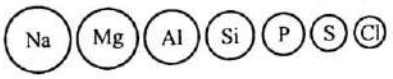
Element (group 1)	Atomic radius	Element (group 2)	Atomic radius
Li	155	F	72
Na	190	Cl	99
K	235	Br	114
Rb	248	I	133

In moving down the group the nuclear charge increases with increase in atomic number. However, while going down in a group from one atom to another the number of inner shells also increases, although the number of electrons in the outermost shell remains the same. The effect of increase in the size of the electron cloud (due to increase in number of shells) is more pronounced than the effect of increased nuclear charge. Thus the distance of outermost electron from the nucleus increases as we move down a group. For example, in lithium the valence electron is present in 2nd shell while in sodium it is present in 3rd shell. Also, the number of filled shells between valence electrons and nucleus increases. Thus in group 1 Li (2,1) has one filled shell between its nucleus and valence electron while Na (2,8,1) has two filled shells between them. Both the factors decrease the force of attraction between nucleus and valence electron. Therefore, atomic size increases on moving down a group.

Note : Isoelectronic species are those which have same number of electrons for e.g. K^+ and Ca^{2+}.

CHECK Point

1. **Why Iodine atom is larger in size in comparison to chlorine atom ?**
2. **Observe the following figure :**



In above series, why the size of atom decreases from Na to Cl

SOLUTION

1. This is because iodine is below chlorine in group 17th, we know that atomic size increases as we move down the group.
2. The size of atom decreases due to increase in nuclear charge as we know nuclear charge increases while going from left to right in a period.

Knowledge ENHANCER

Ionic radii :

Ionic radius is the radius of an ion. On converting into an ion the size of a neutral atom changes. Anion is bigger than the neutral atom. This is because addition of one or more electrons increases repulsions among electrons and they move away from each other. On the other hand a cation is smaller than the neutral atom. When one or more electrons are removed, the repulsive force between the remaining electrons decreases and they come a little closer.

Variation of Ionic radii in periods and groups :

Ionic radii show variations similar to those of atomic radii. Thus, ionic radii increase in a group. You can see such increases in groups 1 and 16 elements from the data given below.

Group 1		Group 16	
Element	**Electron radius**	**Element**	**Ionic radius**
Li^+	60	O^{2-}	140
Na^+	95	S^{2-}	184
K^+	133	Se^{2-}	198
Rb^+	148	Te^{2-}	221

Ionic radii decrease in a period. It can be seen from the data of ionic radii in picometer for 2nd period elements given below.

Element	Li^+	Be^{2+}	B	C	N^{3-}	O^{2-}	F^-
radii	60	31	–	–	171	140	136

In the data given above, the positions of boron and carbon have been left vacant as they do not form ions. The trend in radii of cations is seen in Li^+ and Be^{2+} and in radii of anions is seen in N^{3-}, O^{2-} and F^-.

Ionization Energy :

Negatively charged electrons in an atom are attracted by the positively charged nucleus. For removing an electron this attractive force must be overcome by spending some energy. The minimum amount of energy required to remove an electron from a gaseous atom in its ground state to form a gaseous ion is called ionization energy. It is measured in unit of kJ mol^{-1}.

It is a measure of the force of attraction between the nucleus and the outermost electron. Stronger the force of attraction, greater is the value of ionization energy.

It corresponds to the following process : If only one electron is removed, the ionization energy is known as the first ionization energy. If second electron is removed the ionization energy is called the second ionization energy same goes for third ionization energy and so on. Now we shall study the variation of ionization energy in the periodic table.

Variation of ionization energy in a group :

We have already seen earlier, that the force of attraction between valence electrons and nucleus decreases in a group from top to bottom because of increase in atomic size due to addition of inner shells. Moreover there is increase in shielding effect on outermost (valence) electrons due to increase in the number of inner electrons. As a result, the electron becomes less and less firmly held to nucleus as we move down the group. Ionization energy decreases in a group from top to bottom. This can be seen from ionization energy values (in kJ mol^{-1}) of groups 1 and 17 elements given below.

Group 1		Group 17	
Element	**Ionization Energy**	**Element**	**Ionization Energy**
Li	520	F	1680
Na	496	Cl	1251
K	419	Br	1143
Rb	403	I	1009

Variation of ionization energy in a period :

We know that the force of attraction betweens valence electron and nucleus increases in a period from left to right due to increase in nuclear charge. As a consequence of this, the ionization energy generally increases in a period from left to right. This trend can be seen in ionization energies (in kJ mol^{-1}) of elements belonging to 2nd and 3rd periods.

2nd Period Elements :	Element	Li	Be	B	C	N	O	F	Ne
	Ionization Energy	520	899	801	1086	1400	1314	1680	2080
3rd Period Elements :	Element	Na	Mg	Al	Si	P	S	Cl	Ar
	Ionization Energy	496	738	578	786	1011	1000	1251	1521

However some irregularities in the general trend have been noticed. These are due to extra stability of half filled and completely-filled configurations. For example as shown in data above nitrogen have more ionization energy in comparison to oxygen which is due electronic configuration of nitrogen atom ($1s^2, 2s^2, 2p^3$). Here 2p-orbital is exactly half filled, is more stable than that of oxygen ($1s^2, 2s^2, 2p^4$).

CHECK Point

1. **Why ionisation energy of beryllium is more in comparison to Boron?**
2. **Examine the following elements of the third period of Modern periodic table :**
   Na, Mg, Al, Si, P, S, Cl
   How will tendency to lose electrons vary in given series :

SOLUTION

1. Beryllium has more ionisation energy in comparision to Boron, because of the electronic configuration of Be atom ($1s^2 2s^2$). Here 2s orbital is fully filled which is more stable than that of B ($1s^2 2s^2 2p^1$)
2. Tendency to lose electrons decreases in given series because we know that atomic size decreases as we move in a period which results in to increase in ionisation energy along a period. Thus removal of electrons become difficult along a period.

Electron Affinity :

Another important property that determines the chemical properties of an element is the tendency to gain an additional electron. This ability is measured by electron affinity. It is the energy change when an electron is accepted by an atom in the gaseous state. It corresponds to the process : $X(g) + e^- \longrightarrow X^-(g) + E$

Here, X is an atom of an element. The energy change is measured in the unit kJ mol^{-1}. By convention, electron affinity is assigned a positive value when energy is released during the process. Greater the value of electron affinity, more energy is released during the process and greater is the tendency of the atom to gain electron.

Electron affinity depend on factors like nuclear charge, size of atom and electronic configuration.

Variation of electron affinity in a group :

In a group, the electron affinity decreases on moving from top to bottom, that is, less and less amount of energy is released. On moving down a group, the size and nuclear charge increases. The effect of increase in atomic size is much more pronounced than the effect of nuclear charge therefore additional electron feels less attraction by nucleus. Hence lower is electron affinity. Such trends in its values (in kJ mol-1) for group 1 and group 17 elements are given below.

Group 1		Group 17	
Element	**Electron affinity**	**Element**	**Electron affinity**
Li	58	F	333
Na	53	Cl	348
K	48	Br	324
Rb	45	I	295

Variation of electron affinity along a period

On moving across a period, the size of atom decreases and nuclear charge increases. Both of these factors result into greater attraction for incoming electron. Thus electron affinity increases in a period from left to right. However certain irregularities are due to stable electronic configurations of certain atoms.

- *Halogens have the highest electron affinity.*
- *Electron affinity values of noble gases are positive while those of* Be, Mg, N *and* P *are almost zero.*

CHECK Point

☛ **Why electron affinity value of bromine atom is lower than of chlorine atom ?**

SOLUTION

This is because the value of electron affinity decreases as we move down the group.

Electronegativity :

You have learnt in the previous section that electron affinity of an element is a measure of an isolated atom to attract electrons towards itself. We normally do not deal with isolated atoms. Mostly we come across atoms which are bonded to other atoms. There is another property which deals with the power of bonded atoms to attract electrons. This property is known as electronegativity. Electronegativity is relative tendency of a bonded atom to attract the bond-electrons towards itself. Electronegativity is a dimensionless quantity and does not have any units. It just compares the tendency of various elements to attract the bond-electrons towards themselves. The most widely used scale of electronegativity was devised by Linus Pauling. Now let us learn about its variation in groups 1 and 17.

Group 1		Group 17	
Element	**Electronegativity**	**Element**	**Electronegativity**
Li	1.0	F	4.0
Na	0.9	Cl	3.0
K	0.8	Br	2.8
Rb	0.8	I	2.5

Fluorine atom has the greatest power of attracting electrons and is the most electronegative element.

Electronegativity generally decreases in a group from top to bottom. As we move in a group from top to bottom the atomic size increases as a result bonding electrons become away from the nucleus. Now let us see its variation in 2nd and 3rd period elements.

Electronegativity generally increases in a period from left to right. This is due to decrease in atomic size and increase in effective nuclear charge. Now as a result of increase in effective nuclear charge, the attraction for the outermost electrons increases.

2nd Period Elements :							
Element	Li	Be	B	C	N	O	F
Electronegativity	1.0	1.5	2.0	2.5	3.0	3.5	4.0
3rd Period Elements :							
Element	Na	Mg	Al	Si	P	S	Cl
Electronegativity	0.9	1.2	1.5	1.8	2.1	2.5	3.0

- *Halogens have the highest electronegativities.*
- *Non-metals have high electronegativity values in comparison to metals.*

Metallic and Non-Metallic Character

You know what are characteristic properties of a metal. They are electropositive in nature (the tendency to lose electrons), have luster, ductility, malleability and electrical conductance.

Metallic character of an element largely depends upon its ionization energy. Smaller the value of ionization energy, more electropositive and hence more metallic the element would be.

Variation of metallic character in a group :

You know about the variation of ionization energy in a group. Metallic character of elements increases from top to bottom. As we move down in a group atomic size increases therefore distance between valence electrons and nucleus also increase. Thus electrostatic force of attraction on valence electrons decreases and they can be easily removed. This can best be seen in elements of group 14. Its first element, carbon is a typical non metal, next two elements Si and Ge are metalloids and the remaining elements Sn and Pb, are typical metals as shown below.

Group 14					
Element :	C	Si	Ge	Sn	Pb
Nature :	Non-metal	Metalloid	Metalloid	Metal	Metal

Variation of metallic character in a period :

Metallic character of elements decreases in a period from left to right as shown below for 3rd period elements. As we move from left to right in a period atomic size decreases. Thus electrostatic force of attraction increases for valence electrons thereby decreasing electropositive character.

Element :	Na	Mg	Al	Si	P	S	Cl
Character :	Metal	Metal	Metal	Metalloid	Non-metal	Non-metal	Non-metal

CHECK Point

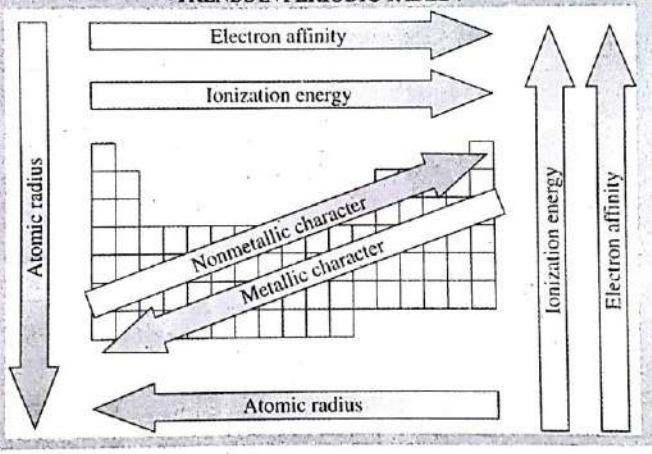
Even being in same group lead is a metal while silicon is a metalloid why ?

SOLUTION

It is because the metallic character increases as we move down a group. Thus lead being the last member of group 14 possess metallic character.

Knowledge ENHANCER

TRENDS IN PERIODIC TABLE :



idea box

Make a complete periodic table on chart paper and paste it on a thin card board. Then cut square piece of every element separately in such a manner that you will get flash cord for each and every element separately. Now tell your friends to go through the concept of variation of periodic properties in periodic table. Now divide your friends in two groups and you also become a part of one group. Play a game in which one group will name a property and a group or period while other group members have to arrange the elements of that particular group or period according to variation of that asked property. Now the other group will perform in the same manner. Ten points for right arrangement and deduction of five points for wrong arrangement at the end let's see which group wins.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : *Complete the following statements with an appropriate word / term to be filled in the blank space(s).*

1. The law of triads was given by .....
2. According to Modern periodic law, the elements are arranged in the periodic table in the order of their increasing .....
3. Elements with eight electrons in their outermost energy shell are called .....
4. If two elements have the same number of valence electrons, then they belong to the same ..... of the periodic table.
5. The elements in groups 1, 2 and 13 to 18 are known as ..... elements.
6. The valency of a atom is equal to its .....
7. The atomic size in a period .....
8. Dobereiner grouped the elements into triads and Newlands gave the .....
9. Mendeleev arranged the elements in increasing order of their ..... and according to their ..... properties.
10. Mendeleev predicted the existence of some yet to be discovered elements on the basis of ..... in his Periodic Table.
11. Elements in the Modern Periodic Table are arranged in ..... vertical columns called ..... and ..... horizontal rows called .....

True / False :

DIRECTIONS : *Read the following statements and write your answer as true or false.*

1. As nuclear charge increases, atomic orbitals become smaller and more stable.
2. As number of shells increases, atomic orbitals become larger and less stable.
3. Atomic radii decrease from left to right across a row of the periodic table.
4. Atomic radii increase from top to bottom down a column of the periodic table.
5. Fluorine has highest electron affinity in the periodic table.
6. Noble gases are placed extremely left in the periodic table.
7. Magnesium is more metallic in nature than sodium.
8. The number of shells increases in a given period from left to right in the periodic table.
9. The elements silicon, germanium and arsenic are called metalloids.
10. Elements are classified on the basis of similarities in their properties.
11. Rows in the periodic table are called periods.
12. The columns of the periodic table are called groups.
13. You will find metals on the extreme right side of the periodic table.
14. Although the order of elements is based on atomic number, vertical families share similar chemical properties.

Match the Following :

DIRECTIONS : *Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.*

1. Column II give group to which element in column I belong match them correctly.

Column I	Column II
(A) Nitrogen	(p) 15
(B) Aluminium	(q) 16
(C) Chlorine	(r) 17
(D) oxygen	(s) 13

2. Column II give period to which element in column I match them correctly.

Column I	Column II
(A) Hydrogen	(p) 3
(B) Sodium	(q) 4
(C) Calcium	(r) 6
(D) Barium	(s) 1

3. Match the column –

Column I	Column II
(A) Element with largest size in second period	(p) boron
(B) Element with smallest size in group 13	(q) fluorine
(C) Element with maximum non-metallic character.	(r) bromine
(D) Element with smallest size in fourth period	(s) lithium

4. Match the column –

Column I	Column II
(A) 's' block elements	(p) Cr
(B) 'p' block elements	(q) Na
(C) 'd' block elements	(r) Ce
(D) 'f' block elements	(s) Si

VSAQ Very Short Answer Questions :

DIRECTIONS : *Give answer in one word or one sentence.*

1. Did Dobereiner's triads also exist in the columns of Newlands' Octaves ? Compare and find out.
2. What were the limitations of Dobereiner's classification?
3. Besides gallium, which other elements have since been discovered that were left by Mendeleev in his Periodic Table?
4. What were the criteria used by Mendeleev in creating his Periodic Table?
5. Why do you think the noble gases are placed in a separate group?
6. How could the Modern Periodic Table remove various anomalies of Mendeleev's Periodic Table?
7. Name two elements you would expect to show chemical reactions similar to magnesium. What is the basis for your choice?
8. Name (a) three elements that have a single electron in their outermost shells. (b) two elements that have two electrons in their outermost shells. (c) three elements with filled outermost shells.
9. In the Modern Periodic Table, which are the metals among the first ten elements ?
10. By considering their position in the Periodic Table, which one of the following elements would you expect to have maximum metallic characteristic? Ga Ge As Se Be.
11. What is the basis of the modern periodic table ?
12. In how many blocks has the modern periodic table been divided ?
13. Why elements in any given group have similar properties?
14. Why some gaps were left in Mendeleev's periodic table ?
15. Why are the group 2 elements called alkaline earth metals?
16. What was Dobereiner's basis of classifying elements ?
17. Out of Li, Ge and N, which forms the most basic oxide and which forms the most acidic oxide ?
18. Define group.
19. Name the scientist who proposed Modern Periodic table.
20. Predict the location in the periodic table (row and column) of element with atomic number 111.
21. What element immediately follows xenon in the periodic table?
22. Name two metals that react with bromine to give compounds with the chemical formula MBr.
23. Write the names and symbols of all elements that occupy the same row of the periodic table as nitrogen.
24. Which is the smallest atom in group VIIA ?
25. Use the second period of the periodic table as an example to show that the size of atoms decreases as we move from left to right. Explain the trend.
26. What is the atomic number of the element that would occupy, the position in row 7, column 17 of the periodic table

SQA Short Answer Questions :

DIRECTIONS : *Give answer in 2-3 sentences.*

1. Arrange these atoms and ions in order of decreasing size: Mg^{2+}, Ca^{2+} and Ca.
2. What were the limitations of Newlands' Law of Octaves?
3. (a) Lithium, sodium, potassium are all metals that react with water to liberate hydrogen gas. Is there any similarity in the atoms of these elements ?
   (b) Helium is an unreactive gas and neon is a gas of extremely low reactivity. What, if anything, do their atoms have in common ?
4. What do you understand by the term periodicity ? Are the properties of the elements placed in a group, same ? Illustrate.
5. Why does the size of the atom increases down the group ?
6. Define and explain Medeleev's Periodic law ?
7. (a) Which elements or ions from among Ar, S^{2-}, Si^{-} and Cl^{3+} are isoelectronic with P^{+} ?
   (b) Which ions from among Fe^{3+}, Ni^{3+}, and Co^{3+} are isoelectronic with Mn^{2+}?
8. Explain why the first ionization energy of lithium is less than that for beryllium, but the second ionization energy of beryllium is less than that for lithium.
9. What is the general electron configuration for the valence electrons in
   (a) group IA (b) group IV A (c) group VIIA?
10. Write the symbol for a cation with a 1+ charge that has the electron configuration
    (a) $1s^2 2s^2 2p^6 3s^2 3p^1$ (b) $1s^2 2s^2 2p^6 3s^1$
    (c) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^6$
11. Which species in each of the following pairs is larger ? Give an explanation for your answer.
    (a) Na or Na^{+} (b) O^{2-} or F^{-} (c) Ni^{2+} or Ni^{3+}
12. Using only a periodic table as a guide, arrange each of the following series of atoms in order of increasing size.
    (a) B, O, Li (b) C, N, Si (c) S, As, Sn
13. Using only a periodic table as a guide, arrange each of the following series of species in order of increasing size.
    (a) Li, Be^{2+}, Be (b) Cl, S, S^{2-} (c) N, C, Si
14. Indicate which species in each pair has the higher ionization energy. Explain the reason for your answer.
    (a) Na and Rb (b) O^{2-} and F^{-}
15. Which will be greater, the second ionization energy of boron or that of beryllium ? Explain your answer.
16. What are the names and chemical symbols of the elements that are vertical and horizontal neighbors of sulfur in the periodic table? Which of these have chemical properties similar to those of sulfur?
17. Why does the metallic character increase down the group?
18. Arrange the following atoms in order of decreasing atomic radius : Na, Al, P, Cl, Mg.

19. Arrange the following atoms in order of increasing radius: P, Si, N.

LAQ Long Answer Questions :

DIRECTIONS : *Give answer in four to five sentences.*

1. Identify the larger species of each of the following pairs:
   (a) K or K^+ (b) S^{2-} or Cl^- (c) Co^{2+} or Co^{3+}
2. Predict which species in each of the following pairs has the higher ionization energy:
   (a) Mg or P (b) B or Cl (c) K^+ or Ca^{2+}.
   (d) Indicate which species in each pair has the higher ionization energy.
   (I) Ge and Cl (II) B and F
   (III) Al^{3+} and Na^+
3. (a) Which atom in Group VIA should have a smaller first ionization energy : oxygen or sulfur?
   (b) Which atom in the second period should have a higher second ionization energy : lithium or beryllium?
   (c) Specify the group of the periods table in which each of the following elements is found :
   (I) [Ne] $3s^1$, (II) [Ne] $3s^2\,3p^3$, (III) [Ar] $4s^2\,3d^8$
4. An atom has electronic configuration 2, 8, 7.
   (a) What is the atomic number of this element ?
   (b) To which of the following elements would it be chemically similar ? (Atomic numbers are given in parentheses)
   N(7) F(9) P(15) Ar(18)

2 EXERCISE

MCQ Multiple Choice Questions

DIRECTIONS : *This section contains 71 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.*

1. Cl, Br, I, if this is Dobereiner's triad and the atomic masses of Cl and I are 35.5 and 127 respectively the atomic mass of Br is –
(a) 162.5 (b) 91.5
(c) 81.25 (d) 45.625

2. Newlands could classify elements only upto –
(a) copper (b) chlorine
(c) calcium (d) chromium

3. Mendeleev classified elements in –
(a) increasing order of atomic groups
(b) eight periods and eight groups
(c) seven periods and nine groups
(d) eight periods and seven groups

4. Noble gases were included in Mendeleev's periodic table in the –
(a) 1st group (b) 7th group
(c) 8th group (d) none of these

5. The long form of periodic table consists of –
(a) seven periods and eight groups
(b) seven periods and eighteen groups
(c) eight periods and eighteen groups
(d) eighteen periods and eight groups

6. In the modern periodic table one of the following does not have appropriate position –
(a) transition elements
(b) inert gases
(c) inner transition elements
(d) galogens

7. An element M has an atomic number 9 and atomic mass 17. Its ion will be represented by –
(a) M (b) M^{+2}
(c) M^{-} (d) M^{-2}

8. The correct order of first IE of C, N, O, F is –
(a) F>O>N>C (b) C>N>O>F
(c) O>N>F>C (d) F>N>O>C

9. Elements belonging to the same group have similar properties because –
(a) they have similar electronic configuration of the outermost shell
(b) their atomic numbers go on increasing as we move down the group
(c) all of them are metallic elements.
(d) none of the above

10. The atoms of elements belonging to the same group of periodic table have the same –
(a) number of protons
(b) number of electrons
(c) number of neutrons
(d) number of electrons in the outermost shell

11. Which of the following is the correct order of relative size
(a) $I^{-}>I^{+}>I$ (b) $I^{-}>I>I^{+}$
(c) $I>I^{+}>I^{-}$ (d) $I^{+}>I^{-}>I$

12. The element with the smallest size in the group 13 is –
(a) beryllium (b) carbon
(c) aluminium (d) boron

13. Which of the following hydroxides is most basic –
(a) $Be(OH)_2$ (b) $Ba(OH)_2$
(c) $Ca(OH)_2$ (d) $Mg(OH)_2$

14. The element with smallest size in the 4th period is –
(a) chlorine (b) iodine
(c) fluorine (d) bromine

15. The most metallic element in the fifth period is –
(a) silver (b) rubidium
(c) gold (d) rhodium

16. If the two members of a Dobereiner triad are chlorine and iodine, the third member of this triad is –
(a) fluorine (b) bromine
(c) sodium (d) calcium

17. If the two members of a Dobereiner triad are phosphorus and antimony, the third member of this triad is –
(a) arsenic (b) sulphur
(c) iodine (d) calcium

18. According to Mendeleef periodic law, the properties of elements are periodic function of their –
(a) atomic masses (b) atomic numbers
(c) atomic volumes (d) densities

19. The elements with atomic numbers 2, 10, 18, 36, 54 and 86 are all –
(a) halogens (b) noble gases
(c) noble metals (d) light metals

20. How many periods are there in the long form of the periodic table –
(a) 6 (b) 7
(c) 8 (d) 9

21. The elements with atomic numbers 3, 11, 19, 37 and 55 belong to
(a) alkali metals (b) alkaline earth metals
(c) halogens (d) noble gases

22. The elements with atomic numbers 9, 17, 35, 53 and 85 belong to
(a) alkali metals (b) alkaline earth metals
(c) halogens (d) noble gases

23. Each transition series contains a total of –
(a) 2 elements (b) 8 elements
(c) 10 elements (d) 18 elements

24. The number of elements in each of the inner transition series is –
(a) 2 (b) 8
(c) 10 (d) 14

25. The number of elements in the third period of the periodic table is –
(a) 2 (b) 8
(c) 18 (d) 32

26. The total number of elements in VII group of the periodic table is –
(a) 3 (b) 5
(c) 7 (d) 9

27. The total number of elements in the group IB is –
(a) 3 (b) 5
(c) 7 (d) 9

28. Which of the following elements has the least nonmetallic character –
(a) fluorine (b) chlorine
(c) bromine (d) iodine

29. Element X forms a chloride with the formula XCl_2, which is a solid with a high melting point. X would most likely be in the same group of the Periodic Table as –
(a) Na (b) Mg
(c) Al (d) Si

30. About how many known elements are there –
(a) 10 (b) 50
(c) 100 (d) 200

31. Elements in the periodic table are arranged by –
(a) atomic number (b) atomic weight
(c) number of neutrons (d) chemical reactivity

32. Which of these things you will not find in the periodic table on the wall –
(a) element Name and Symbol
(b) atomic Weight
(c) atomic Orbital Radius
(d) atomic Number

33. Which scientist came up with the concept of a periodic table that included all of the known elements?
(a) Joseph Priestly (b) Dmitri Mendeleev
(c) Antoine Lavoisier (d) Albert Einstein

34. The alkali metals are in which group of the periodic table?
(a) Group 1 (b) Group 2
(c) Group 3 (d) Group 4

35. As you go down the group, the alkali metals become –
(a) brighter (b) hotter
(c) more reactive (d) less reactive

36. Where are the transition metals in the periodic table –
(a) In group 0
(b) In group 1
(c) In group 2
(d) In a central block with no group number

37. The noble gases are unreactive because
(a) they react with sodium
(b) they have a full outer shell of electrons
(c) they have a half outer shell of neutrons
(d) they are too thin

38. Which of the following element has most ionisation energy
(a) Al (b) In
(c) Ga (d) B

39. Which of the following element is not in the liquid state :-
(a) Hg (b) Li
(c) Ga (d) Br

40. Which sequence of ionisation potential is correct –
(a) B < Be (b) Be < B
(c) Be = B (d) None

41. In which of the following process highest energy is required
(a) $Cu \rightarrow Cu^+$ (b) $Al \rightarrow Al^+$
(c) $Zn \rightarrow Zn^+$ (d) $Li \rightarrow Li^+$

42. Arrange F, Cl, O, N in the decreasing order of electronegativity –
(a) O > F > N > Cl (b) F > N > Cl > O
(c) Cl > F > N > O (d) F > O > N = Cl

43. Which of the following is most electronegative –
(a) carbon (b) silicon
(c) lead (d) tin

44. The electron affinity for the inert gases is –
(a) zero (b) high
(c) begative (d) positive

45. The long form of periodic table has –
(a) eight horizontal rows and seven vertical columns
(b) seven horizontal rows and eighteen vertical columns
(c) seven horizontal rows and seven vertical columns
(d) eight horizontal rows and eight vertical columns

46. Which element has the highest electronegativity –
(a) C (b) Mg
(c) O (d) S

47. In the following, the element with the highest electropositivity is –
(a) copper (b) caesium
(c) barium (d) chromium

48. Which of the following elements are analogous to the lanthanides –
(a) actinides (b) borides
(c) carbides (d) hydrides

49. Arrange the following in increasing order of their atomic radius : Na, K, Mg, Rb –
(a) Mg < K < Na < Rb (b) Mg < Na < K < Rb
(c) Mg < Na < Rb < K (d) Na < K < Rb < Mg

50. Which is metalloid –
(a) Pb (b) Sb
(c) Si (d) Zn

51. Which shows variable valency –
(a) s–block elements (b) p–block elements
(c) d–block elements (d) Radioactive elements

52. Dobereiner traids is –
(a) Na, K, Rb (b) Mg, S, As
(c) Cl, Br, I (d) P, S, As

53. Elements in which 4f orbitals are progressively filled are called as –
(a) transition elements (b) lanthanides
(c) actinides (d) inert gases

54. To which block is related an element having electronic configuration $1s^2\ 2s^2 2p^6\ 3s^2 3p^6 3d^{10}\ 4s^1$ in the periodic table–
(a) s–block (b) p–block
(c) d–block (d) f–block

55. Which of the following elements is a lanthanide (Rare–earth element) –
(a) cadmium (b) californium
(c) cerium (d) cesium

56. If the valene shell electronic conjiguration for an element is ns^2np^5, this element will belong to the group of –
(a) alkali metals (b) inert metals
(c) noble gases (d) halogens

57. If an atom has electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$, it will be placed in –
(a) second group (b) third group
(c) fifth group (d) sixth group

58. On moving from left to right across a period in the table the metallic character –
(a) increases
(b) decreases
(c) remains constant
(d) first increases and then decreases

59. Which of the following is the atomic number of a metal –
(a) 32 (b) 34
(c) 36 (d) 38

60. All the elements in a group in the periodic table have the same –
(a) atomic number
(b) electronic configuration
(c) atomic weight
(d) number of electrons in the outermost shell or number of electrons for bonding

61. Which has the maximum atomic radius –
(a) Al (b) Si
(c) P (d) Mg

62. Which one of the following ions has the highest value of ionic radius –
(a) O^{2-} (b) B^{3+}
(c) Li^+ (d) F^-

63. Which one of the following is the smallest in size –
(a) N^{3-} (b) O^{2-}
(c) F^- (d) Na^+

64. The size of the following species increases in the order –
(a) $Mg^{2+} < Na^+ < F^- < Al$ (b) $F^- < Al < Na^+ > Mg^{2+}$
(c) $Al < Mg < F^- < Na^+$ (d) $Na^+ < Al < F^- < Mg^{2+}$

65. Elements of which group form anions most readily –
(a) oxygen family (b) nitrogen group
(c) halogens (d) alkali metals

66. The correct order of radii is –
(a) N < Be < B (b) $F^- < O^{2-} < N^{3-}$
(c) Na < Li < K (d) $Fe^{3+} < Fe^{2+} < Fe^{4+}$

67. Which of the following element has maximum, first ionisation potential –
(a) V (b) Ti
(c) Cr (d) Mn

68. Which of the following order is wrong –
(a) $NH_3 < PH_3 < AsH_3$-acidic nature
(b) $Li^+ < Na^+ < K^+ < Cs^+$ - ionic radius
(c) $Al_2O_3 < MgO < Na_2O < K_2O$ - basic
(d) Li < Be < B < C - I^{st} ionisation potential

69. Which one of the following arrangements represents the correct order of electron gain enthalpy (with negative sign) of the given atomic species –
(a) Cl < F < S < O (b) O < S < F < Cl
(c) S < O < Cl < F (d) F < Cl < O < S

70. Arrange S, O and Se in ascending order of electron affinity
(a) Se < S < O (b) Se < O < S
(c) S < O < Se (d) S < Se < O

71. Which of the following is correct regarding ionic radii
(a) $Ti^{4+} < Mn^{+7}$ (b) $^{35}Cl^- < {}^{37}Cl^-$
(c) $K^+ > Cl^-$ (d) $P^{3+} > P^{5+}$

MTQs More than One Correct :

DIRECTIONS : *This section contains 19 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.*

1. Which of the following are representative element –
(a) Fe (b) K
(c) Ba (d) N

2. Which one of the following are electropositive element –
(a) sodium (b) calcium
(c) oxygen (d) chlorine

3. Which of the following pair of element has same property
(a) 10, 12 (b) 11, 20
(c) 20, 38 (d) 13, 31

4. All the members in a group of a long form of periodic table have the same –
(a) valency
(b) number of valence electrons
(c) chemical properties
(d) Physical characteristics

5. Metals are included in the long form of periodic table in the
(a) s-block only (b) p-block only
(c) d-block only (d) f-block only

6. Important merits of modern periodic table is –
(a) it explains why element in the same group have the same chemical properties
(b) hydrogen has been placed accurately
(c) isotopes have been placed of same position
(d) it is based on classifying elements according to their atomic number

7. The difference between ions and atoms is of –
(a) relative size (b) configuration
(c) presence of charge (d) mass of nucleus

8. I.E. increases with –
(a) decrease in atomic size
(b) increase in nuclear charge
(c) increase in penetration effect of electrons
(d) decrease in nuclear charge

9. While moving in a period left to right –
   (a) atomic size decrease (b) nuclear charge increase
   (c) I.E. increases (d) I.E. decreases
10. Which of the following properties generally decrease along a period –
   (a) atomic size (b) non-metallic character
   (c) metallic character (d) ionic size
11. Which of the following elements will form acidic oxide –
   (a) Na (b) Si
   (c) Mg (d) P
12. In the periodic table, the metallic character of elements –
   (a) increases, (i) from left to right across a period and (ii) on descending a group
   (b) decreases, (i) from left to right across a period and (ii) on moving up group from bottom to top
   (c) increases from left to right across a period and decreases on descending a group
   (d) increases from right to left across a period and increases on descending a group
13. Which of the following statements is (are) correct statement about the trends when going from left to right across the periods of periodic Table.
   (a) the elements become less metallic in nature.
   (b) the number of valence electrons increases.
   (c) the atoms lose their electrons more easily.
   (d) the oxides become more acidic.
14. Which of these choices is not a family of elements?
   (a) halogen (b) metal
   (c) inert Gas (d) fire extinguishers
15. Transition metals can –
   (a) show variable oxidation states
   (b) form coloured compounds
   (c) float in air
   (d) create oxygen
16. Which one of the following is not an s–block element –
   (a) aluminium (b) chromium
   (c) niobium (d) potassium
17. The statement that is true for the long form of the periodic table is –
   (a) if reflects the sequence of filling the electrons in the order of sub-energy levels s, p, d and f.
   (b) it helps to predict the stable valency states of the elements
   (c) if reflects trends in physical and chemical properties of the elements.
   (d) it helps to predict the relative atomicity of the bonds between any two elements.
18. The correct statement among the following is –
   (a) the first ionisation potential of Al is less than the first ionisation potential of Mg
   (b) the second ionisation potential of Mg is greater than the second ionisation potential of Na.
   (c) the first ionisation potential of Na is less than the first ionisation potential of Mg
   (d) the third ionisation potential of Mg is greater than the third ionisation potential of Al.
19. The statement that is correct for the period classification of elements is –
   (a) the properties of elements are the periodic functions of their atomic numbers.
   (b) non-metallic elements are lesser in number than metallic elements
   (c) the first ionisation energies along a period do not vary in a regular manner with increase in atomic number
   (d) for transition elements the d-sub-shells are filled with electrons monotonically with increases in atomic number

FIP Fill in the Passage :

DIRECTIONS : *Complete the following passage(s) with an appropriate word/term to be filled in the blank spaces.*

I. In the periodic table, elements are ordered in increasing .....(1)..... so that .....(2)..... of elements, fall into vertical columns. Of the four general categories of elements, calcium is a .....(3)..... element, nickel is a .....(4)..... ..... and xenon is .....(5)..... An element in Group IIA, such as calcium, is also known as an .....(6)..... metal.

II. Metallic character increases from ....(1)..... to .....(2)..... and from .....(3)..... to .....(4)..... with respect to position of elements in the periodic table. Nonmetallic character increases from .....(5)..... to ....(6)..... and from .....(7)..... to .....(8)..... in the periodic table.

FBQ Passage Based Questions :

DIRECTIONS : *Study the given paragraph(s) and answer the following questions.*

The atomic radii of first group elements are given below:
Table-1

Group I elements :	Na	Li	Rb	Cs	K
Atomic radius (pm)	86	152	244	262	231

Atomic radii of the elements of the second period are given below

Period II elements	B	Be	O	N	Li	C
Atomic radius (pm)	88	111	66	74	152	77

1. Table 1 give atomic radii variation of group 1 elements. What is the common name of elements of first group.
   (a) alkali metals (b) alkaline earth metals
   (c) halogens (d) transition metals
2. In Table 1 which element has the largest atomic radius.
   (a) Na (b) K
   (c) Cs (d) Li
3. In Table 2 which element has the smallest atomic radii.
   (a) B (b) C
   (c) O (d) N

4. In table 2 which element has the largest atomic radii
   (a) Be (b) Li
   (c) N (d) B

A&R Assertion & Reason :

DIRECTIONS : *Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.*

(a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Group I (1s) elements are known as the alkali elements.
   Reason : s-orbital can accommodate only two electrons.
2. **Assertion :** Nitrogen has higher ionization energy than that of oxygen.
   Reason : Nitrogen gas smaller atomic size than that of oxygen.
3. **Assertion :** According to Mendeleev, periodic properties of elements is a function of their atomic number.
   Reason : Atomic number is equal to the number of protons.
4. **Assertion :** Elements in the same vertical column have similar properties.
   Reason : Elements have periodic dependence upon the atomic number.

MMQ Multiple Matching Questions :

DIRECTIONS : *Following question has four statements (A, B, C and D) given in Column I and four statements (p, q, r and s) in Column II. Any given statement in Column I can have correct matching with one or more statement(s) given in Column II. Match the entries in column I with entries in column II.*

1.

Column I	Column II
(A) s-block elements	(p) Alkali metals
(B) p-block elements	(q) Alkaline earth metals
(C) Representative elements	(r) Halogens
(D) High ionisation energy	(s) Noble gases

HOTS HOTS Subjective Questions :

DIRECTIONS : *Answer the following questions.*

1. Referring to a periodic table, arrange the following atoms in order of increasing first ionization energy : Ne, Na, P, Ar, K.
2. For each of the following pairs, indicate which one of the two species is larger:
   (a) N^{3-} or F^-; (b) Mg^{2+} or Ca^{2+}; (c) Fe^{2+} or Fe^{3+}
3. Using only a periodic table as a guide, arrange each of the following series of species in order of increasing first ionization energy.
   (a) O, O^{2-}, F (b) C, Si, N (c) Te, Ru, Sr
4. Predict which element of the following pairs will have the higher density and support your prediction using periodic trends :
   (a) Cr or W; (b) W or Os; (c) Pd or Ag; and (d) Y or Nb.
5. Group the following electron configurations in pairs that would represent similar chemical properties of their atoms:
   (a) $1s^2 2s^2 2p^5$
   (b) $1s^2 2s^1$
   (c) $1s^2 2s^2 2p^6$
   (d) $1s^2 2s^2 2p^6 3s^2 3p^5$
   (e) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
   (f) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$
6. Why are the elements of group 18 called zero valent ?
7. List the following series of elements in order of increasing atomic radius : (a) Be, C, Mg (b) Rb, I, Br.
8. Arrange the ions S^{2-}, Cl^-, K^+ and Ca^{2+} in order of decreasing order.
9. List the following ions in order of increasing ionic radius : N^{3-}, Na^+, F^-, Mg^{2+}, O^{2-}.
10. Why are the electron affinities of the alkaline earth metals, either negative or small positive values?

CBQ Chart Based Questions :

DIRECTIONS : *Complete the following table by writing Symbol of elements, atomic no. and group name in vacant spaces.*

Name and Symbol of element	Atomic Number	Category of element
Sodium (Na)	.....	s'-block element
.....	5	.....
.....	.....	d' block element
Chlorine (Cl)	17	.....
.....	20	s' block element

SOLUTIONS

Brief Explanations of Selected Questions

Exercise 1

Fill in the Blanks :

1. Dobereiner
2. Atomic number
3. Noble gases
4. Group
5. Main group
6. Combining capacity
7. Decreases
8. Law of Octaves.
9. Atomic masses, Chemical
10. Gaps
11. 18, groups, 7, periods.

True / False

1. True
2. True
3. True
4. True
5. True
6. False
   Noble gases are placed extremely right in the periodic table.
7. False
8. False.
   The number of shells remain same in a given period.
9. True
10. True
11. True
    Rows in the periodic table are called periods. The columns of the periodic table are called groups.
12. True
13. False
    Inert gases are found on the far right of the periodic table. Halogens are in the second group form the right. Metals of all types are found around the left and middle side of the periodic table. There may be alkalitransition, or alkaline earth metals across the table.
14. True

Match the Following :

1. A → (p); B → (s); C → (r); D → (q)
2. A → (s); B → (p); C → (q); D → (r)
3. A → (s); B → (p); C → (q); D → (r)
4. A → (q); B → (s); C → (p); D → (r)

Very Short Answer Questions :

1. Yes, they also exist in the columns of Newlands' Octaves.
2. Dobereiner could identify only three triads from the elements known at that time. Thus he could arrange only a few elements in this manner and his classification not receive wide acceptance.
3. Germanium, Scandium
4. Mendeleev's periodic table was based on atomic weight.
5. It is because noble gases resemble each other in properties and do not resemble with other group of elements.
6. Modern periodic table is based on atomic number, therefore problem of placing Ar, K, Co, Ni was easily removed. Secondly, problem of isotopes was also solve because have same atomic number.
7. Calcium and barium. It is because they belong to same group and have same number of valence electrons.
8. (a) Na, K, Li (b) Be, Mg (c) He, Ne, Ar.
9. Li and Be are metals
10. Gallium has maximum metallic character.
11. The basis of the modern periodic table is Mosley periodic law that is, the properties of the elements are periodic functions of their atomic numbers.
12. The periodic table is divided into four blocks (s-block, p-block, d-block and f-block) on the basis of their electronic configuration.
13. The chemical properties of an atom are largely determined by its valance electrons. In a given group, the number of valance electrons are same, hence they have the same properties.
14. Mendeleev left some gaps in his periodic table for the elements yet to be discovered. He even predicted the properties of these elements by studying the properties of the neighbouring elements.
15. These elements are called alkaline earth metals because they form basic hydroxides which are less soluble in water than group I hydroxides.

16. According to the Doberenier, elements having similar properties can be arranged in triads in which atomic mass of the middle element is the mean of the atomic mass of the other two elements.
17. Lithium forms the most basic oxide as it is a metal and nitrogen forms the most acidic oxide as it is a non-metal.
18. The vertical columns in Mendeleev's, as well as in Modern periodic table, are called groups.
19. Henry Mosely, a scientist, proposed the Modern periodic table based on modern periodic law.
20. Row = 7, column = Group 11 (directly below gold)
21. Cesium
22. Metals from group 1 : Li, Na, K, Rb, Cs.
23. Lithium, (Li); beryllium, (Be); boron, (B); carbon, (C); oxygen, (O); fluorine, (F); and neon, (Ne).
24. F
25. The effective nuclear charge on the outermost electrons increases across the period.
26. 117

SHORT ANSWER QUESTIONS :

1. Cations are smaller than their parent atoms, so Ca^{2+} is smaller than the Ca atom. Because Ca is below Mg in group IIA of the periodic table, Ca^{2+} is larger than Mg^{2+}. Consequently, $Ca > Ca^{2+} > Mg^{2+}$.
2. (i) It was found that law of octaves was applicable only up to calcium because after Ca, every eighth element did not possess properties similar to first.
   (ii) New elements discovered could not fit into the law of octaves on the basis of their properties.
   (iii) Newland adjusted two elements in same slot e.g., Co and Ni which differ entirely from halogens. Iron resembles with Co and Ni, has been placed far away from these elements.
3. (a) All of them are highly reactive and all of them have one electron in their outer most shell. i.e., they lose electrons easily.
   (b) Helium and Neon both have completely filled outermost shell.
4. When the elements are arranged in order of increasing atomic numbers, elements with similar chemical properties appear at definite intervals known as periodicity. Yes, this periodicity is due to the periodicity in the number of electrons in the outermost shell of the atoms of the elements. If elements having the same number of valence electrons are grouped together, the elements falling within each group are similar in chemical properties.
5. In moving down a group, the charge on the nucleus increases with increase in atomic number, but at the same time, there is an increase in the energy levels. The number of electrons in the outermost shell, however, remains the same. Since the effect of additional energy levels outweigh the effect of increased atomic or nuclear charge, the distance of the outermost electron from the nucleus increase on going down the group.
6. **Mendleev's periodic law :** It states that, the properties of elements are the periodic functions of their atomic masses. It means the properties of the elements depend on their atomic masses and the elements are positioned in the periodic table on the basis of their increasing atomic masses.
7. (a) The P^+ cation has 14 electrons in the configuration $1s^2 2s^2 2p^6 3s^2 3p^2$. Ar and S^{2-} have 18 electrons, Si^- has 15 electrons, and Cl^{3+} has 14 electrons. Only Cl^{3+} is isoelectronic with P^+.
   (b) The Mn^{2+} cation has 23 electrons and the electron configuration $[Ar]3d^5$ (remember to remove first the valence electrons with the highest n value, 4s in this case). The Fe^{3+} ion also has 23 electrons and the same electron configuration as Mn^{2+}; they are isoelectronic. The Ni^{3+} ion has 25 electrons ($[Ar]3d^7$) and Co^{3+} has 24 electrons ($[Ar]3d^6$). Neither is isoelectronic with Mn^{2+}.
8. For both lithium and beryllium, the first electron is ionized from the 2s sublevel. As we know that fully or exactly half filled configurations are extra stable. Thus while removing first electron is case of Beryllium 2s sublevel is completely filled therefore being more stable in comparison to partially filled 2s subshell of lithium. Thus first ionisation energy is more for beryllium. While removing second electron in case of lithium it has to be removed from 1s subshell which acquire extra stability due to completely filled configuration. Thus require more energy in comparison to 1s subshell of beryllium.
9. (a) ns^1 (b) ns^2np^2 (c) ns^2np^5
10. (a) Si^+ (b) Mg^+ (c) Fe^+
11. (a) Na. A cation always has a smaller size that a neutral atom.
    (b) O^{2-}. The size of anion increase as the charge increases.
    (c) Ni^{2+}. The size of cation decreases as the charge increases.
12. (a) O < B < Li (b) N < C < Si (c) S < As < Sn
13. (a) Be^{2+} < Be < Li (b) Cl < S < S^{2-} (c) N < C < Si
14. (a) In a group, the ionisation potential decreases from top to bottom, hence ionisation potential of Rb will be higher than Na.
    Na > Rb.

(b) In a period ionisation potential increases from left to right, i.e., $F^- > O^{--}$

15. Boron. In both cases a valence 2s electron is being removed, but B^+ has fully filled inert configuration.

16. Vertical neighbours = oxygen (O) and selenium (Se), horizontal neighbours = phosphorus (P) and chlorine (Cl). Oxygen and selenium have similar chemical properties.

17. In a group, the metallic character increases with increase in atomic number of elements from top to bottom. This is due to the fact that the number of valence shells and the distance of valence electrons increases from the nucleus. The size also increases from top to bottom. Thus, it is easier to remove electrons from the valence shell of atom of element present at lower position in group.

18. As we move from left to right across a period there is a regular decrease in atomic radii of the representative elements. This is due to the fact that number of energy shells remain the same in a period but nuclear charge increases gradually as the atomic number increases, hence the correct order will be

$Na > Mg > Al > P > Cl$.

19. N and P are in the same group and N is above P. Therefore, the radius of N is smaller than that of P (atomic radius increases as we go down a group). Both Si and P are in the third period, and Si is to the left of P. Therefore, the radius of P is smaller than that of Si (atomic radius decreases as we move from left to right across a period). Thus the order of increasing radius is $N < P < Si$.

Long Answer Questions :

1. (a) K has the electron configuration of $[Ar]4s^1$ and K^+ has the electron configuration of [Ar]. The electron in the 4s orbital makes K much larger than K^+.

(b) These two species have the same electron configuration [Ar] as these two species are isoelectronic. The size of S^2 is larger than Cl^- because generally anions with more negative charge are larger in size. Moreover sulphur is at left to chlorine in periodic table. Therefore periodic trend also favours S^{2-}.

(c) Both Co^{2+} ($[Ar]3d^7$) and Co^{3+} ($[Ar]3d^6$) have valence electrons in the same sublevel, but Co^{2+} is larger because of the additional shielding caused by the extra electron will expand the electron cloud.

2. (a) Magnesium and phosphorus are in the same period. Phosphorus is to the right and has the higher ionization energy.

(b) Boron and chlorine are in different periods, but chlorine is four groups to the right of boron. There is a small decrease in ionization energies on going from the second period to the third period, but this change is small compared to the increase in going from group IIIA to group VIIA. Chlorine has the higher ionization energy. Note that in this case the correct answer is clear. It is more difficult to compare closely positioned elements from different periods, such as boron and silicon, from the general trends among elements.

(c) The ions K^+ and Ca^{2+} are isoelectronic, but calcium is present in right with respect to potassium in IVth period thus according to general periodic trends Ca^{2+} the higher ionization energy.

(d) (I) Cl > Ge (II) F > B (III) $Al^{3+} > Na^+$

3. (a) Oxygen and sulfur are members of Group VIA. They have the same valence electron configuration (ns^2np^4), but the 3p electron in sulfur is farther from the nucleus and experiences less nuclear attraction than the 2p electron in oxygen. Thus, following the general rule that the ionization energy of elements decreases as we move down a periodic group, we predict that sulfur should have a smaller first ionization energy.

(b) For both lithium and beryllium, the first electron is ionized from the 2s sublevel. As we know that fully or exactly half filled configurations are extra stable. Thus while removing first electron is case of Beryllium 2s sublevel is completely filled therefore being more stable in comparison to partially filled 2s subshell of lithium. Thus first ionisation energy is more for beryllium. While removing second electron in case of lithium it has to be removed from 1s subshell which acquire extra stability due to completely filled configuration. Thus require more energy in comparison to 1s subshell of beryllium.

(c) (I) Group IA (II) Group VA (III) Group VIIIA

4. (a) The atomic number of this element is obtained by adding all the electrons present in its electronic configuration.

$\therefore$ Atomic number = 2 + 8 + 7 = 17

(b) The electronic configuration of the given element

= 2, 8, 7

Valence electron in its atom = 7

This element will be chemically similar to that element which is the same as valence electron (7)

(i) N (7) : 2, 5 (5 valance electrons)

(ii) F (9) : 2, 7 (7 valance electrons)

(iii) P (15) : 2, 8, 5 (5 valance electrons)

(iv) Ar (18) : 2, 8, 8 (8 valance electrons)

Clearly F (9) has 7 valence electrons just like that of the

given element. Hence, the given element of atomic number 17 will be chemically similar to the element fluorine (F) of atomic number 9.

Exercise 2

Multiple Choice Questions :

1. (c) According to Dobereneir's triad the atomic mass of Br will be average of the atomic masses of Cl & I

$$= \frac{35.5 + 127}{2} = 81.25$$

2. (c)
3. (c)
4. (d)
5. (b)
6. (c)
7. (c) The element is halogen and has one less electron than inert gas configuration, hence can be represented as M^- ion.
8. (d) In a period, the value of ionisation potential increases from left to right with breaks where the atoms have some what stable configurations hence the correct order will be
   $F > N > O > C$
9. (a)
10. (d)
11. (b)
12. (b)
13. (b) Except $Be(OH)_2$ which is amphoteric in nature, other $M(OH)_2$ are basic in nature. The basic strength increases from $Be(OH)_2$ to $Ba(OH)_2$.
14. (d) On moving along a period atomic radii decreases.
15. (b) The metallic character decreases as we move from left to right in a period.
16. (b)
17. (a)
18. (a)
19. (b)
20. (b)
21. (a)
22. (c)
23. (c)
24. (d)
25. (b)
26. (b) The VII group contains 5 elements
    F, Cl, Br, I and At
27. (a) Group I B contain Cu, Ag and Au.
28. (d) Non-metallic character decreases in the groups from top to bottom, hence Iodine will be least non-metallic.
29. (b)
30. (c). There are about 110 known elements listed in the periodic table. As scientists continue to experiment with particle accelerators and cyclotrons, they will be able to make more elements. Many of the manmade elements on the periodic table only last for a few milliseconds before they break apart.
31. (a). The elements of the periodic table are organized by atomic number. The atomic number represents the number of protons and electrons in a neutral atom.
32. (c). You will not get information about the atomic radius of an atom. The square for each element will have the atomic number, atomic weight, name, and symbol for each element
33. (b). Dmitri Mendeleev is credited with designing the modern periodic table. Joseph Priestly and Antoine Lavoisier were both chemists. Albert Einstein developed theories on photoelectric effect
34. (a)
35. (c)
36. (d)
37. (b)
38. (d) On moving down in a group from top to bottom ionisation potential decreases from top to bottom.
39. (a)
40. (a) Due to fully filled stable orbitals ionisation energy of Be is more than B.
41. (d) Lithium has smallest size hence a large amount of energy is needed for removal of an electron.
42. (d) On moving along a period electronegativity increases and on moving down in a group electronegativity decreases the correct order of electronegativity is

$$\underset{4.0}{F} > \underset{3.5}{O} > \underset{3.0}{N} = \underset{3.0}{Cl}$$

46. (a) 47. (b) 48. (a)
49. (b) 50. (c) 51. (c)
52. (c) 53. (b) 54. (a)
55. (c) 56. (d) 57. (c)
58. (b)
59. (d). 38 is the atomic no. of stronium (Sr) which is s-block element and all the elements of s-block are metals.
60. (d).
61. (d). Mg, as we move across the period atomic radius decreases.

62. (a) O^{-2} has the highest value of ionic radii as this can be explained on the bases of Z/e $\left\{\frac{\text{Nucleus charge}}{\text{No. of electron}}\right\}$

When Z/e ratio increases, the size decreases and when Z/e ratio decreases the size increases.

63. (d) $Na^+ < F^- < O^{2-} < N^{3-}$

All are isoelectronic, effective nuclear charge is highest for Na^+ so it has smallest size.

64. (a) $Mg^{2+} < Na^+ < F^- < Al$

F^- has bigger size than Mg^{2+} and Na^+

65. (c). Halogens are most elentronegative elements.

66. (b). Ionic radii decreases significantly from left to right in a period among representative elements.

67. (d).

68. (d).

69. (b). Halogens have very high electron affinity. It may be rated that the electron affinity of fluorine is unexpectedly low (< Cl). This may perhaps be due to small size of F atom. The value of electron gain enthalpies for Cl, F, S and O are respectively 349, 333, 200 and 142 kJ/mol hence correct order is Cl > F > S > O.

70. **(a)** Correct order of electron affinity is Se < S < O. In a group electron affinity decreases with increase in atomic number.

71. **(d)** Nuclear charge per electron is greater in P^{5+}. Therefore, its size is smaller.

More Than One Correct :

1. (b, c, d)

2. (a, b)

3. (c, d) $_{20}Ca$, $_{38}Sr$, $_{13}Al$ and $_{31}Ga$ are in same group element so have same property.

4. (a, b, c)

5. (a, b, c, d)

6. (a, c, d)

7. (a, b, c)

8. (a, b)

9. (a, b, c)

10. (a, c, d)

11. (b,d)

12. (b, d)

13. (a, b, d) On moving along a period tendency to lose electrons decreases whereas the tendency to gain electron increases.

14. **(a, b, c).** All of these choices are different types and families of elements. Inert gases can be found on the far right of the periodic table with halogens just to the left. Metals make up the center and left side of the table.

15. (a, b)

16. (a, b, c)

17. (a,b,c,d). If reflects trends in physical and chemical properties of the elements.

18. (a,c,d). I.E. (II) of Na is higher than that of Mg because in case of Na, the second e^- has to be remove from the noble gas core while in case of Mg removal of second e^- gives a noble gas core.

19. (a,b,d).

Fill in the Passage :

I. (1) atomic number, (2) groups, (3) representative, (4) transition (5) metal, (6) noble gas, (7) alkaline earth.

II. (1) top, (2) bottom, (3) right, (4) left, (5) bottom, (6) top, (7) left, (8) right

Passage Based Questions :

1. (a) **2.** (c) **3.** (c) **4.** (b)

Assertion & Reason :

1. (b) Group is elements are known as alkali metals as the hydroxides of these metals are soluble in water and these solutions are highly alkaline in nature.

2. (c) Nitrogen has higher ionisation energy as it has stable half filled orbital structure.

3. (d) According to Mendeleev, periodic properties of elements is a function of their atomic masses.

4. (b)

Multiple Matching Questions :

1. A - (p, q) B - (r, s) C - (p, q, r) D -(q, s)

HOTS Subjective Questions :

1. We are given the chemical symbols for five elements. In order to rank them according to increasing first ionization energy, we need to locate each element in the periodic table. We can then use their relative positions and the trends in first ionization energies to predict their order.

The ionization energy increases we move left period to right across a row. It decreases as we move from the top of a group to the bottom. Because Na, P, and Ar are in the same period of the periodic table, we expect It to vary in the order : Na < P < Ar.

Because Ne is above Ar in group 8A, we expect Ne to exhibit the greater first ionization energy : Ar < Ne

Similarly, K is the alkali metal directly below Na in group IA, so we expect I_1 for K to be less than that of Na : K < Na

From these observations we conclude that the ionization energies follow the order : K < Na < P < Ar < Ne.

2. In comparing ionic radii, it is useful to classify the ions into three categories: (1) isoelectronic ions, (2) ions carrying the same charges and are generated from atoms of the same

periodic group, and (3) ions carrying different charges but are generated from the same atom. In case (1), ions carrying a greater negative charge are always larger; in case (2), ions from atoms having a greater atomic number are always larger; in case (3), ions having a greater positive charge are always smaller.

(a) N^{3-} and F^- are isoelectronic anions. Because N^{3-} has only seven protons and F^- has nine, N^{3-} is larger.

(b) Both Mg and Ca belong to Group 2A (the alkaline earth metals). The Ca^{2+} ion is larger than Mg^{2+} because Ca's valence electrons are in a larger shell (n = 4) than are Mg's (n = 3).

3. (a) $O^{2-} < O < F$

(b) $Si < C < N$

(c) $Sr < Ru < Te$

4. (a) W, density increases with Z (b) Os, density increases with Z (c) Pd, density decreases at the end of a row and (d) Nb, density increases with Z.

5. (a) and (d), (b) and (e) (c) and (f)

6. Group 18 elements have their outermost shell completely filled and the atoms of these elements have no tendency to gain or lose electrons. Thus, the elements of this group are zero valent and almost unreactive.

7. **(a)** Beryllium is in the same period with carbon and is to its left, so it is the larger of the two. Magnesium is below beryllium in the same group and is larger. The order of increasing size is C < Be < Mg.

(b) Rubidium is far to the left in the same period as iodine and is largest; bromine is above iodine and is the smallest: Br < I < Rb.

8. This is an isoelectronic series of ions, with all ions having 18 electrons. In such a series, size decreases as the charge of the ion increases. The atomic numbers of the ions are S (16), Cl (17), K (19) and Ca (20). Thus, the ions decrease in size in the order :
$S^{2-} > Cl^- > K^+ > Ca^{2+}$.

9. $Mg^{2+} < Na^+ < F^- < O^{2-} < N^{3-}$
In a group ionic radii increases as the atomic number increases. In a period ionic radii decreases as we move from left to right.

10. The valence configuration of the alkaline earth metals is ns^2. For the process

$$M(g) + e^- \longrightarrow M^-(g)$$

where M denotes a member of the Group IIA family the extra electron to be added must enter the outer most S-subshell but elements of group IIA have general electronic configuration of ns^2. Thus s-subshell is completely filled means there is no vacancy for electron to be added. Therefore for IIA elements EA = 0.

CHART BASED QUESTIONS :

1.

Name and Symbol of element	Atomic Number	Category of element
Sodium (Na)	11	s'-block element
Boron (B)	5	p'-block element
Magnese (Mn)	25	d' block element
Chlorine (Cl)	17	p'-block element
Calcium (Ca)	20	s' block element